

**VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD**  
**B.TECH. III YEAR**  
**MECHANICAL ENGINEERING**

**V SEMESTER**

**R22**

| Course Code  | Title of the Course                      | L         | T        | P/D      | CH        | C         |
|--------------|------------------------------------------|-----------|----------|----------|-----------|-----------|
| 22PC1ME301   | Thermal Engineering                      | 3         | 1        | 0        | 4         | 4         |
| 22PC1ME302   | Automation and Robotics                  | 3         | 0        | 0        | 3         | 3         |
| 22PC1ME303   | Mechanical Engineering Design            | 3         | 1        | 0        | 4         | 4         |
|              | <b>PROFESSIONAL ELECTIVE – I</b>         |           |          |          |           |           |
| 22PE1ME301   | Non-Conventional Energy Sources          | 3         | 0        | 0        | 3         | 3         |
| 22PE1ME302   | Industrial Engineering and Management    |           |          |          |           |           |
| 22PE1ME303   | Experimental Stress Analysis             |           |          |          |           |           |
| 22PE1ME304   | Microprocessors in Automation            |           |          |          |           |           |
| 22PE1ME305   | Supply Chain Management                  |           |          |          |           |           |
|              | <b>OPEN ELECTIVE – I</b>                 | 3         | 0        | 0        | 3         | 3         |
| 22PC2ME301   | Thermal Engineering Laboratory           | 0         | 0        | 2        | 2         | 1         |
| 22PC2ME302   | Automation and Robotics Laboratory       | 0         | 0        | 2        | 2         | 1         |
| 22HS2EN301   | Advanced Communication Skills Laboratory | 0         | 0        | 2        | 2         | 1         |
| 22MN6HS301   | Ancient Wisdom                           | 2         | 0        | 0        | 2         | 0         |
| <b>Total</b> |                                          | <b>17</b> | <b>2</b> | <b>6</b> | <b>25</b> | <b>20</b> |

L – Lecture      T – Tutorial      P – Practical      D – Drawing      CH – Contact Hours/Week  
 C – Credits      SE – Sessional Examination      CA – Class Assessment      ELA – Experiential Learning Assessment  
 SEE – Semester End Examination      D-D – Day to Day Evaluation      LR – Lab Record  
 CP – Course Project      PE – Practical Examination

**VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD**  
**B.TECH. III YEAR**  
**MECHANICAL ENGINEERING**

**VI SEMESTER**

**R22**

| Course Code  | Title of the Course                                | L         | T        | P/D       | CH        | C         |
|--------------|----------------------------------------------------|-----------|----------|-----------|-----------|-----------|
| 22PC1ME304   | Heat Transfer                                      | 3         | 0        | 0         | 3         | 3         |
| 22PC1ME305   | Additive Manufacturing                             | 3         | 0        | 0         | 3         | 3         |
| 22PC1ME306   | Finite Element Methods                             | 3         | 0        | 0         | 3         | 3         |
|              | <b>PROFESSIONAL ELECTIVE – II</b>                  |           |          |           |           |           |
| 22PE1ME306   | Refrigeration and Air Conditioning                 | 3         | 0        | 0         | 3         | 3         |
| 22PE1ME307   | Operations Research and Decision Sciences          |           |          |           |           |           |
| 22PE1ME308   | Industrial robotics                                |           |          |           |           |           |
| 22PE1ME309   | Mechatronic Systems                                |           |          |           |           |           |
| 22PE1ME310   | Theory of Metal Cutting                            |           |          |           |           |           |
|              | <b>OPEN ELECTIVE – II</b>                          | 3         | 0        | 0         | 3         | 3         |
| 22PC2ME304   | Heat Transfer Laboratory                           | 0         | 0        | 2         | 2         | 1         |
| 22PC2ME305   | Additive Manufacturing Laboratory                  | 0         | 0        | 2         | 2         | 1         |
| 22PC2ME306   | Computer Aided Engineering and Analysis Laboratory | 0         | 0        | 2         | 2         | 1         |
| 22PW4ME301   | Internship                                         | 0         | 0        | 4         | 4         | 2         |
| 22MN6HS302   | Gender Sensitization                               | 2         | 0        | 0         | 2         | 0         |
| <b>Total</b> |                                                    | <b>17</b> | <b>0</b> | <b>10</b> | <b>27</b> | <b>20</b> |

L – Lecture      T – Tutorial      P – Practical      D – Drawing  
C – Credits      SE – Sessional Examination      CA – Class Assessment  
SEE – Semester End Examination      D-D – Day to Day Evaluation  
CP – Course Project      PE – Practical Examination

CH – Contact Hours/Week  
ELA – Experiential Learning Assessment  
LR – Lab Record

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PC1ME301) THERMAL ENGINEERING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 1   | 4 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mathematics, Thermodynamics

#### COURSE OBJECTIVES:

- To understand the working principle, cycle performance and effect of their parameters on the various I.C Engines
- To understand the various combustion, knocking phenomena, emission and their measurement and testing
- To understand the working principle and Performance of power consuming thermal systems (like Compressor, refrigeration, Air conditioning systems etc.) and the psychrometric properties on condition air
- To understand the importance, effect, development of ecofriendly EVs and the energy management techniques

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Distinguish the cycles, working, and Performance of various I.C Engines and Engine systems

**CO-2:** Identify the various, combustion and emission processes, knocking Phenomenon, and energy losses in engines

**CO-3:** Differentiate between the convention's engine and electric vehicle

**CO-4:** Evaluate the performance of thermal systems and the psychrometric properties of air for Air Conditioning system

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 2    | 3    | 2    | 2    | 3    | 1    | 1    | 1     | 2     | 3     | 3                               | 2     |
| CO-2 | 3                     | 3    | 2    | 3    | 2    | 2    | 3    | 1    | 1    | 1     | 2     | 3     | 3                               | 2     |
| CO-3 | 3                     | 3    | 2    | 3    | 2    | 2    | 3    | 2    | 1    | 1     | 2     | 3     | 3                               | 2     |
| CO-4 | 3                     | 3    | 2    | 3    | 2    | 2    | 3    | 2    | 1    | 1     | 2     | 3     | 3                               | 2     |

#### UNIT-I:

**Introduction of IC Engines:** Classification of IC engines, Working of SI & CI, Two and Four Stroke Engines, Valve and Port Timing Diagrams, Comparison of Air Standard Cycles, Air-Fuel Cycles and Actual Cycles.

**Actual Cycles and Their Analysis:** Introduction, Time Loss Factor, Heat Loss Factor, Exhaust Blow Down, Loss due to Gas exchange process, Volumetric Efficiency, Loss due to Rubbing Friction.

#### **UNIT-II:**

**Engine Systems:** Fuel Systems, Simple Carburetor, Solex Carburetor, Fuel Injection Systems; Ignition systems: - Battery ignition, Magneto ignition, Cooling and Lubrication systems.

**Combustion in S. I. Engines:** Homogeneous mixture, Heterogeneous mixture, Stages of combustion, flame front propagation, Factors influencing the flame speed, Rate of pressure rise, Abnormal Combustion, Phenomenon of Knock.

#### **UNIT-III:**

**Combustion in C. I. Engines:** Combustion process, stages of combustion, Delay period and its importance, Factors affecting Delay period, Diesel Engine Knock, Comparison of Knock in C.I and S.I engine.

**Testing and Performance:** Measurement and Testing: Friction power, indicated power, Brake power, Fuel consumption, Air consumption, Emission, Performance parameters: Engine power, Engine efficiencies, Engine performance characteristics, Heat Balance.

#### **UNIT-IV:**

**Electric Vehicles:** History of electric vehicles, social and environmental importance of electric vehicles, Types of Electric Vehicles, Energy Storage Requirements in Electric Vehicles, Energy Management Strategies used in electric vehicles.

**Reciprocating Air Compressors:** Classification, Principle of operation, Work required, Isothermal efficiency, Volumetric efficiency and effect of clearance, Multi stage compression with inter cooling, Saving of work, Minimum work condition for stage compression.

#### **UNIT-V:**

**Refrigeration:** Introduction to Ideal Refrigeration cycle, Air Refrigeration: - Bell Colman Refrigeration Cycle, Vapor Compression Refrigeration Cycle, Vapour Absorption Refrigeration System. Psychrometric properties: Dry bulb temperature, Wet bulb temperature, Dew point temperature, Specific humidity, Relative humidity, Degree of saturation, Specific enthalpy, Psychrometric chart, (Indicating only Psychrometric processes on Chart).

#### **TEXT BOOKS:**

1. Internal Combustion Engines, V. Ganesan, 3<sup>rd</sup> Edition, Tata McGraw-Hill, 2007
2. Thermal Engineering, Mahesh M. Rathore, 1<sup>st</sup> Edition, Tata McGraw-Hill, 2010
3. Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives, C. Mi, M. A. Masrur and D. W. Gao, John Wiley & Sons, 2011

#### **REFERENCES:**

1. A Course in Internal Combustion Engines, M. L. Mathur and R. P. Sharma, 1<sup>st</sup> Edition, Dhanpat Rai Publications, 1976

2. Internal Combustion Engines Fundamentals, John B. Heywood, 1<sup>st</sup> Edition, Tata McGraw-Hill, 1988
3. Refrigeration and Air Conditioning, C. P. Arora, 2<sup>nd</sup> Edition, Tata McGraw-Hill, 2000
4. Thermal Engineering, Er. R. K. Rajput, 11<sup>th</sup> Edition, Lakshmi Publications, 2020
5. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design, M. Ehsani, Y. Gao, S. E. Gay and A. Emadi, CRC Press, 2004

**NPTEL COURSE LINKS:**

1. <https://archive.nptel.ac.in/courses/112/103/112103262/>
2. <https://archive.nptel.ac.in/courses/112/105/112105129/>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PC1ME302) AUTOMATION AND ROBOTICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To analyze the concepts of automation and advanced automation functions
- To study the different types of directional control valves, actuators and their control, and the working of PLCs
- To analyze different types of robot grippers, drives and sensors
- To analyze robot kinematics, trajectory planning and its programming for industrial applications

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply principles & strategies of automation, and advanced automation functions

**CO-2:** Develop pneumatic & hydraulic circuits and PLC ladder programs

**CO-3:** Analyze and select robot grippers and sensors for industrial applications

**CO-4:** Develop robot programs and plan its trajectory

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 1                     | 2    | 3    | 2    | 3    | 2    | -    | -    | 2    | -     | -     | 2     | 2                               | 2     |
| CO-2 | 1                     | 2    | 3    | 2    | 3    | 2    | -    | -    | 2    | -     | -     | 2     | 2                               | 2     |
| CO-3 | 1                     | 2    | 3    | 2    | 3    | 2    | -    | -    | 2    | -     | -     | 2     | 2                               | 2     |
| CO-4 | 1                     | 2    | 3    | 2    | 3    | 2    | -    | -    | 2    | -     | -     | 2     | 2                               | 2     |

#### UNIT-I:

**Automation:** Definition and fundamentals of automation, reasons for automating, basic elements of an automated system, safety, maintenance & repair diagnosis, error detection and recovery, Levels of automation, Automation principles and strategies: USA principle, ten strategies of automation and production system, automation migration strategy.

#### UNIT-II:

**Pneumatic and Hydraulic Systems:** Air treatment, Basic elements of hydraulic circuit, Directional and pressure control valves, flow control valves for speed control, time delay valve, control of single and double acting cylinders, solenoid valves, latching relay circuits, pilot and automatic operation, sequential circuits, cascade control.

### **UNIT-III:**

**Programmable Logic Controllers (PLCs):** Introduction, Parts of a PLC, Fundamentals of Logic, AND, OR and NOT functions, latching, internal relays, Timers, Counters, Boolean Algebra, PLC Ladder Programs(basic), advantage & disadvantages, Applications.

### **UNIT-IV:**

**Robot Classification, Terminology, and Grippers:** History of robots, Classification of robots, Basic components of robotic system. Basic terminology-. End effectors, Grippers-different methods of gripping, types of grippers, gripper selection and design considerations; Specifications of robot.

**Robot Drive Systems and Sensors:** Drive system- hydraulic, pneumatic, and electric systems (AC and DC servo motors, stepper motors), Sensors in robot.

**Manipulator Kinematics:** Robot coordinate systems, D-H Notations, Forward and Inverse kinematics, Problems (2P/2R).

### **UNIT-V:**

**Trajectory Planning:** Joint space scheme, Cubic polynomial fit, Avoidance of obstacles, Types of motions: Slew motion, Joint interpolated motion, Straight line motion, Problems.

**Robot Programming, Languages, and Applications:** Lead through programming, Motion programming, Motion interpolation, Robot programming language, interlock and sensor commands, Simulation and Off-Line programming and robot applications in material Handling and transfer, welding, assembly, and inspection applications.

### **TEXT BOOKS:**

1. Industrial Robotics, M. P. Groover, McGraw-Hill Education, 2017
2. Pneumatic and Hydraulic Systems, W. Bolton, Elsevier Science & Technology Books, 1997
3. Robotics & Control, Mittal R. K. & Nagrath, Tata McGraw-Hill, 2003
4. Programmable Logic Controllers, W. Bolton, Newnes, 2015

### **REFERENCES:**

1. Robot Dynamics and Control, M. W. Sponge and M. Vidyasagar, Wiley, 2008
2. Robotics, K. S. Fu, McGraw-Hill, 2017
3. Robotic Engineering, Richard Klaffer, Prentice Hall India, 1993
4. Programmable logic controller, Frank D. Petrusella, McGraw-Hill, 2023
5. Introduction to Programmable Logic Controllers, Gary Dunning, Delmar Thomson Learning, 2001

### **ONLINE RESOURCES:**

1. [https://onlinecourses.nptel.ac.in/noc19\\_me74/](https://onlinecourses.nptel.ac.in/noc19_me74/)
2. <https://robotics.umich.edu/academics/courses/>

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. V Semester

### (22PC1ME303) MECHANICAL ENGINEERING DESIGN

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 1   | 4 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mechanics of Solids, Kinematics of Machinery, and Dynamics of Machinery.

#### COURSE OBJECTIVES:

- To understand overview of phases in design process and general design consideration for selection of materials
- To acquire knowledge on application of various theories of failure under static and fluctuating loading conditions of machine components
- To develop skills on use of tools and techniques and make logical conclusions while building analytical models under certain assumptions and learn systematic engineering methods to solve the practical design engineering problems
- To analyse design criteria for selection of various standardized machine elements such as fasteners, bearings, welded joints etc.
- To identify the failure modes and selection criteria for different power transmission elements like gears, belt drives etc.

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Predict modes of failure in materials or machine elements under the action of various loads during its operation

**CO-2:** Implement the knowledge of design procedures to design the shafts and fasteners

**CO-3:** Apply the knowledge for the design and selection of bearings and engine parts

**CO-4:** Analyze the belt drives and gear drives for the specific application

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 2    | 2    | 2    | 1    | 1    | 2    | 1    | 2     | -     | 3     | 3                               | -     |
| CO-2 | 3                     | 3    | 2    | 2    | 2    | 1    | 1    | 2    | 1    | 2     | -     | 3     | 3                               | -     |
| CO-3 | 3                     | 2    | 2    | 2    | 1    | 1    | 2    | 1    | 1    | 3     | -     | 3     | 3                               | -     |
| CO-4 | 3                     | 2    | 2    | 2    | 2    | 1    | 2    | 1    | 1    | 3     | -     | 3     | 3                               | -     |

#### UNIT-I:

**Engineering Materials and Design Considerations:** Mechanical Properties of Engineering Materials, Design Phase / Methodology-identification of need, Evaluation



and Presentation. Reliability and Product Liability. Overall Design Considerations, Factor of Safety, Preferred Numbers. Standards and Codes, use of Design Data Handbook.

**Design Against Static Load:** Theories of Failure (Yield and Fracture Criteria) - Maximum Normal Stress theory, Maximum Shear Stress theory, Maximum Distortion Energy theory, comparison of various theories of failure, Direct loading, and Combined loading.

#### **UNIT-II:**

**Design Against Fluctuating Load:** Stress Concentration, Stress Concentration Factors, Reduction of Stress Concentration, Fluctuating Stresses. Fatigue strength, Endurance Limit, fatigue test, S-N diagrams for different structural materials. Low cycle and High cycle fatigue, Notch Sensitivity. Gerber Theory Soderberg and Goodman lines for Fatigue Strength. Modified Goodman Theory.

**Design of Shafts:** Transmission shafts, Design of solid and hollow shafts based on strength and rigidity.

#### **UNIT-III:**

**Design of Fasteners:** Temporary Fasteners (Bolted and Screwed Fasteners) Bolted Joints, bolted joint under initial loading, eccentrically loaded Bolted Joints under different static load conditions.

**Permanent Fasteners:** Riveted Joints, eccentrically loaded Riveted Joints, Design of Boiler Riveted joints,

#### **UNIT-IV:**

**Design of Belt Drives:** Introduction, classification of belts, belt materials, design of flat (rectangular) belts, ratio of belt tensions, condition for maximum power, V-Belts, power transmitted through V-Belts, design of V-Belts.

**Design of Spur and Helical Gear Drives:** Spur gears- Helical gears – Load concentration factor – Dynamic load factor. Surface compressive strength – Bending strength – Design analysis of spur gears – Estimation of Centre distance, module, and face width. Check for Plastic Deformation. Check for Dynamic and Wear Considerations.

#### **UNIT-V:**

**Design of Bearings:** Types of Journal bearings – Lubrication – Bearing Modulus – Full and partial bearings – Clearance ratio – Heat dissipation of bearings, bearing materials – journal bearing design – Ball and roller bearings – Static loading of ball and roller bearings, bearing life, Bearing selection.

**Design of Engine Parts:** Pistons, Forces acting on piston – Construction, design and proportions of piston, Connecting Rod, thrust in connecting rod – stress due to whipping action on connecting rod ends.

#### **TEXT BOOKS:**

1. Design of Machine Elements, V. B. Bhandari, Tata McGraw-Hill, 2020
2. Machine Design: An Integrated Approach, R. L. Norton, Pearson Education, 2018

#### **REFERENCES:**

1. Mechanics of Materials (SI Units), Beer & Johnson, McGraw Hill-2020
2. Data Books: P.S.G. College of Technology, 2019

3. Mechanical Engineering Design, J. E. Shigley and C. Mischke, 9<sup>th</sup> Edition, McGraw-Hill, 2011
4. Machine Design, R. S. Khurmi & J. S. Gupta, Sultan Chand, 2020
5. Machine Design (SI Units), Schaums Outline Series; McGraw Hill, 2017

**ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/112/105/112105125/#>
2. <https://archive.nptel.ac.in/courses/112/106/112106137/#>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PE1ME301) NON-CONVENTIONAL ENERGY SOURCES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Thermal Engineering

#### COURSE OBJECTIVES:

- To impart knowledge of solar energy, its principles, and various applications
- To expose the students with various non-conventional energy sources and their conversion techniques
- To keep them updated with the latest emerging technologies in the field of non-conventional energy sources

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basic concepts and fundamentals

**CO-2:** Demonstrate energy conversion techniques

**CO-3:** Analyze the environmental aspects, prospects, and limitations

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | -    | -    | -    | -    | 1    | -    | 1    | -    | -     | -     | -     | 1                               | -     |
| CO-2 | 3                     | 2    | -    | -    | -    | 1    | -    | 1    | -    | -     | -     | -     | 1                               | -     |
| CO-3 | -                     | -    | -    | -    | -    | 3    | -    | 3    | -    | -     | -     | -     | 1                               | -     |

#### UNIT-I:

**Principles of Solar Radiation:** The sun as source of energy, Earth radiation spectrum, Extraterrestrial and terrestrial radiations, Spectral power distribution of solar radiation, Depletion of solar radiation, Scattering, Measurement of solar radiation, Pyranometer, Pyrliometer, Sunshine recorder, Solar radiation data, Solar radiation geometry, Angle of latitude, Declination, Hour angle, Inclination angle, Zenith angle, Solar azimuth angle, Slope, Surface azimuth angle, Angle of Incidence

**Solar Energy Collection:** Classification and types of solar collectors, working principle of Liquid and air flat plate collector.

**Solar Applications:** Solar water heater, Solar passive space heating and cooling systems, Solar Dryer, Solar Distillation, Solar Pond electric power plant

## UNIT-II:

**Wind Energy:** Horizontal axis wind turbine: Main components, Types of rotors, Teetering of rotor, Upwind and downwind machines, Yaw control system, Pitch control system,

**Vertical Axis Wind Turbine:** Main components, Types of rotors, Wind energy conversion systems, Fixed speed drive scheme, Variable speed drive scheme.

## UNIT-III:

**Geothermal Energy:** Types of geothermal resources, Hydrothermal resources: Vapour dominated (dry steam) system, Liquid dominated (wet steam) system, Hot water system, Geo-pressured resources, Hot dry rock resources, Magma resources.

**Biogas:** Biogas production from waste biomass, Fuel properties of biogas, Advantages of anaerobic digestion, Classification of biogas plants, Operational parameters of biogas plant, working principle of floating drum type biogas plant and fixed dome type biogas plant.

## UNIT-IV:

**Ocean Energy:** Ocean Thermal Energy Conversion Technology: open cycle OTEC systems, closed or Anderson OTEC cycle, Prospects of OTEC in India.

**Tidal and Wave Energy:** Origin and Nature of Tidal Energy, Limitations of Tidal Energy, Tidal Energy Conversion Schemes, Present status, Wave energy technology: Heaving Float Type, Pitching Type, Heaving and Pitching Float Type, Oscillating Water Column Type, Present Status.

## UNIT-V:

**Emerging Technologies:** Fuel cell, Hydrogen Energy and Batteries

**Fuel Cell:** Potential Applications, Classification of fuel cells, Phosphoric Acid Fuel Cell, Polymer Electrolyte Membrane Fuel Cell, Fuels for fuel cells.

**Hydrogen:** Hydrogen as energy carrier, Properties of hydrogen, hydrogen production by Electrolysis of water, Storage, Delivery, Conversion, Applications, Safety issues.

**Batteries:** Electrochemical Energy Storage: Secondary Battery Storage, Lithium polymer battery and lithium-ion battery.

## TEXT BOOKS:

1. Non-Conventional Energy Resources, B. H. Khan, 3<sup>rd</sup> Edition, McGraw-Hill Education, 2017
2. Non-Conventional Energy Sources, G. D. Rai, Khanna Publishers, 2008

## REFERENCES:

1. Renewable Energy Sources, Twidell & Weir, 2015
2. Solar Energy Solar Energy - Principles of Thermal Collection and Storage, Sukhatme, Tata McGraw-Hill, 2008
3. Solar Power Engineering, B. S. Magal, Frank Kreith & J. F. Kreith, 3<sup>rd</sup> Edition, 2015
4. Principles of Solar Energy, Frank Krieth & John F. Kreider, 2<sup>nd</sup> Edition, 2000
5. Non-Conventional Energy, Ashok V. Desai, Wiley Eastern, 1990

## ONLINE RESOURCES:

1. <https://archive.nptel.ac.in/courses/121/106/121106014/>
2. <https://archive.nptel.ac.in/noc/courses/noc20/SEM1/noc20-ge06/>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22PE1ME302) INDUSTRIAL ENGINEERING AND MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Knowledge of Manufacturing, General Management, Business Economics, Mathematics & Statistics.

#### COURSE OBJECTIVES:

- To perform as industry leaders in the global marketplace, capable of successfully planning, controlling, and implementing large-scale projects
- To flourish and work effectively in diverse, multicultural environments emphasizing the application of teamwork and communication skills
- To understand and apply the principles of Management, science, technology, engineering, and mathematics involving industry-relevant problems
- To maintain high standards of professional and ethical responsibility, contribute to the profitable growth of industrial economic sectors by using IE analytical tools, effective computational approaches, and systems thinking methodologies
- To understand project management along with the project cost analysis using mathematical methods and statistical tools

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply the concept of Management to industrial situations

**CO-2:** Identify the various manufacturing systems in an industry

**CO-3:** Understand the concepts associated with Industry 4.0

**CO-4:** Interpret on the recent marketing trends and project management concepts

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 3    | 3    | 3    | 2    | 3    | 3    | 3     | 3     | 3     | 3                               | 3     |
| CO-2 | 3                     | 2    | 2    | 2    | 2    | 3    | 3    | 3    | 3    | 2     | 2     | 3     | 2                               | 2     |
| CO-3 | 3                     | 2    | 2    | 2    | 2    | 3    | 3    | 2    | 3    | 2     | 2     | 3     | 2                               | 2     |
| CO-4 | 3                     | 3    | 2    | 2    | 3    | 2    | 2    | 3    | 3    | 2     | 3     | 3     | 3                               | 3     |

#### UNIT-I:

**Introduction to Industrial Engineering and Management:** Concept of Industrial Engineering (I.E), Role of I.E, Applications of I.E, Production Management vs I.E, Principles of Management, Functions of Management, Taylor's Scientific

Management, Maslow's Theory of Human Needs, Leadership Styles, Principles of Organization, Types of Organization Structures their Merits and Demerits, Entrepreneurship.

#### **UNIT-II:**

**Production and Operations Management:** Plant Location, Principles of Plant Layout, Different Types of Plant Layouts, different Types of Production Systems.

**Plant Maintenance:** Objectives and Types, Equipment Selection, Maintenance Planning.

**Materials Handling:** Principles, Concept of Unit Load, Containerization, Selection of Material Handling Equipment, Applications of Belt Conveyors, Cranes, Forklift Trucks in Industry.

#### **UNIT-III:**

**Work Study:** Concept of Productivity, Method Study - Basic steps in Method Study, Process Charts, Principles of Motion Economy, Micro Motion Study, Therbligs, SIMO Chart, Work Measurement – Stopwatch Procedure of Time Study, Performance Rating, Allowances, Work Sampling.

**Materials Management:** Introduction, Purchasing, Objectives of Purchasing Department, Buying Techniques, Purchase Procedure, Stores and Material Control, Inventory Control.

#### **UNIT-IV:**

**Quality Control:** Deming's Contribution to Quality. Basic Concept of Total Quality Management and Taguchi's Quality Engineering. Value Analysis and Value Engineering.

**Marketing:** Functions of Marketing, Marketing Mix, Product Life Cycle. Channels of Distribution and Sales Management, Services Marketing, Consumer Behavior, Recent Trends in Marketing.

**Manufacturing Planning:** MRP, Manufacturing Resource Planning (MRP-II), JIT, CIM, Digital Transformation, Basics of Digital twins.

**Fundamentals of Industry 4.0:** Industry 4.0 Basic Concept, Industrial revolution: Phases, Evolution of Industry4.0, Components, Principles, Technology, Smart Business Perspective, Challenges, Impact of Industry 4.0, Pillars of Industry4.0 and Application of INDUSTRY 4.0

#### **UNIT-V:**

**Human Resources Management (HRM):** Concepts of HRM, Basic Functions of HR Manager: Manpower Planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Global Human Resource Management & Future Issues.

**Project Management – Network Analysis:** Network analysis; Programme evaluation review technique - PERT (probability of completing the project within given time); Critical path method - CPM (Identifying critical path).

#### **TEXT BOOKS:**

1. Management Science, Aryasri, 4<sup>th</sup> Edition, Tata McGraw-Hill, 2012

2. Industrial Engineering and Management, O. P. Khanna, Revised Enlarged Edition 1999 and 17<sup>th</sup> Reprint, 2012

**REFERENCES:**

1. Principles of Management, Harold Koontz and Cyril O'Donnell, 1<sup>st</sup> Edition, Tata McGraw-Hill, 2004
2. Production and Operations Management, Everette Adam and Ronald Ebert, 5<sup>th</sup> Edition, Prentice-Hall, 1992
3. Operations Management, John O. McClain, L. Joseph Thomas, Joseph B. Mazzol, 3<sup>rd</sup> Edition, Prentice Hall, 1992
4. Industrial Engineering and Production Management, Martand Telsang, 2<sup>nd</sup> Revised Edition, S. Chand and Co., 2006
5. Handbook of Industry 4.0 and SMART Systems, Diego Galar Pascual, Pasquale Daponte, Uday Kumar, CRC Press, 2019

**ONLINE RESOURCE:**

1. <https://nptel.ac.in/courses/112107292>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PE1ME303) EXPERIMENTAL STRESS ANALYSIS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mechanics of Solids, Theory of Elasticity

#### COURSE OBJECTIVES:

- To understand experimental techniques
- To learn the strain analysis of circuits
- To explore the different types of coatings and ndt

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the various experimental techniques involved for measuring displacements and stresses

**CO-2:** Learn strain analysis of measuring circuits

**CO-3:** Expertise in photo elasticity tools and techniques

**CO-4:** Explore the different types of coatings and NDT

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 2    | 2    | 1    | 1    | 2    | 1    | 2     |       | 3     | 3                               |       |
| CO-2 | 2                     | 2    | 2    | 2    | 2    | 1    | 1    | 2    | 1    | 2     |       | 3     | 3                               |       |
| CO-3 | 3                     | 2    | 2    | 2    | 1    | 1    | 2    | 1    | 1    | 3     |       | 2     | 2                               |       |
| CO-4 | 2                     | 2    | 2    | 2    | 1    | 1    | 2    | 1    | 1    | 3     |       | 2     | 2                               |       |

#### UNIT-I:

**Extensometers and Displacement Sensors:** Principles of Measurements, Accuracy, Sensitivity and Range of Measurements, Mechanical, Optical, Acoustical and Electrical Extensometers and Their Uses, Advantages and Disadvantages, Laser Displacement Sensors.

#### UNIT-II:

**Electrical Resistance Strain Gauges:** Principle of Operation and Requirements, Types and Their Uses, Materials for Strain Gauges, Wheatstone Bridge And Potentiometer Circuits For Static And Dynamic Strain Measurements, Strain Indicators, Rosette Analysis, Stress Gauges, Load Cells.



**UNIT-III:**

**Photo Elasticity:** Two-Dimensional Photo Elasticity, Photo Elastic Materials, Concept of Light – Photo elastic Effects, Stress Optic Law. Transmission Photo Elasticity, introduction to Three-Dimensional Photo Elasticity.

**UNIT-IV:**

**Brittle Coating and Moire Techniques:** Relation Between Stresses in Coating and Specimen, Use of Failure Theories in Brittle Coating, Moire Method of Strain Analysis.

**UNIT-V:**

**Non – Destructive Testing:** Fundamentals of NDT, Acoustic Emission Technique, Radiography, Thermography, Ultrasonics, Eddy Current Testing, and Fluorescent Penetrate Testing.

**TEXT BOOKS:**

1. Experimental Stress Analysis, Dally J. W. and Riley W. F., McGraw-Hill Inc., 1998
2. Experimental Stress Analysis, L. S. Srinath, M. R. Raghava, K. Lingaiah, G. Garagesha, B. Pant, and K. Ramachandra, Tata McGraw-Hill, 1984

**REFERENCES:**

1. The Strain Gage Primer, C. C. Perry and H. R. Lissner, McGraw-Hill, 2000
2. Experimental Stress Analysis, Abdul Mubeen, Dhanpat Rai and Sons, 2001
3. Moire Fringes in Strain Analysis, P. S. Theocaris, Pergammon Press, 2002

**ONLINE RESOURCE:**

1. <https://nptel.ac.in/courses/112106068>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PE1ME304) MICROPROCESSORS IN AUTOMATION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Basics of Automation

#### COURSE OBJECTIVES:

- To introduce the architecture and operation of Microprocessor
- To familiarize with the programming and interfacing of microcontrollers, PLC SCADA, Embedded system and DCS
- To provide strong foundation for designing real world applications using microcontrollers

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Identify microcontroller for controlling different actuators

**CO-2:** Solve microcontroller and its programming

**CO-3:** Relate PLC and do simulation using PLC for different industrial applications

**CO-4:** Apply the distributed control system for automation

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 2    | 3    | 3    | 3    |      |      |      | 3    | 3     |       | 3     | 2                               | 3     |
| CO-2 | 3                     | 2    | 3    | 3    | 3    |      |      |      | 3    | 2     |       | 2     | 2                               | 2     |
| CO-3 | 3                     | 2    | 3    | 3    | 2    |      |      |      | 3    | 2     |       | 3     | 3                               | 3     |
| CO-4 | 2                     | 3    | 3    | 3    | 2    |      |      |      | 2    | 3     |       | 3     | 2                               | 3     |

#### UNIT-I:

**Microprocessor:** Number Systems, codes, digital electronics: Logic Gates, combinational circuits design, Flip-flops, Sequential logic circuits design: Counters, Shift registers. Introduction to 8085 Functional Block Diagram, Registers, ALU, Bus systems, Timing and control signals. Machine cycles, instruction cycle and timing states, instruction timing diagrams, Memory interfacing.

#### UNIT-II:

**Microprocessors and Microcontrollers:** Introduction, Comparing Microprocessors and Microcontrollers, 4Bit, 8Bit, 16 Bit and 32 Bit Micro Controllers. Micro Controllers Architecture: 8051 Architecture, 8051 Micro Controller Hardware, Input/Output Pins,

Ports and Circuits, External Memory, Counter and Timers, Serial Data Input/Output, Simple programming exercises- Application to automation systems.

### **UNIT-III:**

#### **PLC:**

**Programmable Logic Controllers:** Introduction – Parts of PLC – Principles of operation – PLC sizes – PLC hardware components – I/O section Analog I/O Section Analog I/O modules – digital I/O modules CPU processor memory module – Programming devices – PLC programming Simple instructions - Latching relays PLC ladder diagram, Applications of PLC - Simple materials handling applications, Automatic control of warehouse door, Automatic lubrication of supplier Conveyor belt, Motor control.

### **UNIT-IV:**

**SCADA:** Introduction to SCADA. TAG's – Types – Analog – Digital – Strings – Memory – I/O tags. PC-SCADA- Memory analog/digital/string tag. PLC-SCADA – I/O analog/digital/string tag. Features of SCADA – Dynamic process graphic – Real-time and historical trending – Alarms – Recipe management – Security – Device connectivity – Script for logic development – Database connectivity. Case studies.

### **UNIT-V:**

**Distributed Control System:** Introduction to DCS – Centralized versus Distributed control system. Components of DCS – Field Control Station – Operator Station – Communication Bus. Human Interface Station, Engineering Station – Communication Gateway – Bus converter. Programming and Simulation.

### **TEXT BOOKS:**

1. Digital Electronics: An Introduction to Theory and Practice, William H. Gothmann, PHI
2. Digital Computer Electronics: An Introduction to Microcomputers, Albert Paul Malvino, Tata McGraw-Hill
3. Microprocessor Architecture, Programming, and Applications with the 8085, Ramesh Gaonkar, PENRAM International

### **REFERENCES:**

1. Digital Control Systems, Benjamin C. Kuo, Oxford University Press, 2<sup>nd</sup> Indian Edition, 2007
2. Microcomputer Experimentation with the Intel SDK-85, Lance A. Leventhal, Prentice Hall

### **ONLINE RESOURCE:**

1. <https://nptel.ac.in/courses/117104072>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PE1ME305) SUPPLY CHAIN MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To provide knowledge on logistics and supply chain management
- To enable them in designing the distribution network
- To train in knowing the supply chain Analysis
- To impart knowledge on Dimensions of logistics
- To know the recent trends in supply chain management

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the importance of supply chain management in industries

**CO-2:** Determine the applications of SCM in business administration

**CO-3:** Apply the knowledge of supply chain analysis to industrial scenarios

**CO-4:** Analyse the business processes for successful SCM implementation

**CO-5:** Categorize the recent trends in supply chain management

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 2    | 3    |      | 1    | 3    | 1    | 2    | 1    | 3     | 2     | 2     | 3                               | 2     |
| CO-2 | 2                     | 3    | 3    |      | 1    | 3    | 1    | 2    | 1    | 3     | 2     | 2     | 2                               | 2     |
| CO-3 | 3                     | 3    | 3    | 2    | 3    | 3    | 1    | 2    | 2    | 3     | 2     | 3     | 3                               | 3     |
| CO-4 | 2                     | 2    | 3    | 2    | 3    | 3    |      | 3    | 2    | 3     | 3     | 3     | 3                               | 3     |
| CO-5 | 3                     | 2    | 3    | 2    | 2    | 3    |      | 2    | 2    | 3     | 3     | 3     | 3                               | 3     |

#### UNIT-I:

**Introduction to Supply Chain Management:** Supply chain - objectives - importance - decision phases - process view -competitive and supply chain strategies - achieving strategic fit – supply chain drivers - obstacles – framework - facilities-inventory-transportation-information-sourcing-pricing.

#### UNIT-II:

**Designing the Distribution Network:** Role of distribution - factors influencing distribution - design options - e-business and its impact– distribution networks in practice –network design in the supply chain - role of network -factors affecting the network design decisions modelling for supply chain. Role of transportation – modes and their

performance – transportation infrastructure and policies - design options and their trade-offs tailored transportation.

#### **UNIT-III:**

**Supply Chain Analysis:** Sourcing - In-house or Outsource - 3rd and 4th PLs - supplier scoring and assessment, selection -design collaboration - Procurement process - Sourcing planning and analysis. Pricing and revenue management for multiple customers, perishable products, seasonal demand, bulk and spot contracts.

#### **UNIT-IV:**

**Dimensions of Logistics:** A macro and micro dimension - logistics interfaces with other areas - approach to analysing logistics systems - logistics and systems analysis - techniques of logistics system analysis factors affecting the cost and importance of logistics. Demand Management and Customer Service Outbound to customer logistics systems - Demand Management –Traditional Forecasting- CPFRP - customer service - expected cost of stock outs - channels of distribution.

#### **UNIT-V:**

**Recent Trends in Supply Chain Management:** Introduction, New Developments in Supply Chain Management, Outsourcing Supply Chain Operations, Co-Maker ship, The Role of Ecommerce in Supply Chain Management, Green Supply Chain Management, Distribution Resource Planning, World Class Supply Chain Management. Role of Computer / IT in Supply Chain Management, CRM Vs. SCM, Value Addition in SCM-concept of demand chain management. Fundamentals of block chain management, digital supply chain management

#### **TEXT BOOKS:**

1. Supply Chain Management – “Strategy, Planning and Operation”, Sunil Chopra and Peter Meindl, 3<sup>rd</sup> Edition, Pearson/PHI, 2007
2. Supply Chain Management, Janat Shah, Pearson Publication 2008

#### **REFERENCES:**

1. A Logistic Approach to Supply Chain Management, Coyle, Bardi, Longley, 1<sup>st</sup> Edition, Cengage Learning, 2009
2. Supply Chain Logistics Management, Donald J. Bowersox, Dand J. Closs, M. Bixby Coluper, 2<sup>nd</sup> Edition, Tata McGraw-Hill, 2008
3. Principles of Supply Chain Management a Balanced Approach, Wisner Keong Leong and Keah-Choon Tan, 1<sup>st</sup> Edition, Cengage Learning, 2004
4. Designing and Managing the Supply Chain: Concepts, Strategies, and Case Studies – David Simchi-Levi, Philip Kaminsky, Edith Simchi-Levi, Ravi Shankar, 4<sup>th</sup> Edition, McGraw-Hill, 2022

#### **ONLINE RESOURCE:**

1. <https://nptel.ac.in/courses/110107074>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PC2ME301) THERMAL ENGINEERING LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

**COURSE PRE-REQUISITES:** Thermodynamics, Thermal Engineering

#### COURSE OBJECTIVES:

- To measure the performance parameters and draw its characteristic curve for a SI and CI Engines
- To measure the performance parameters of reciprocating compressor
- To measure the performance COP of refrigeration and air conditioning systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Determine the Performance parameters and various energy losses of engines (like: BP, IP, FP, bsfc, etc.)

**CO-2:** Calculate the performance parameters of reciprocating compressor

**CO-3:** Evaluate the performance parameters of cooling systems (like: refrigeration, air conditioning, etc.)

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 2    | 3    | 2    | 2    | 2    | 1    | 2    | 1     | 2     | 2     | 3                               | 2     |
| CO-2 | 3                     | 3    | 2    | 3    | 2    | 2    | 2    | 1    | 2    | 1     | 2     | 2     | 3                               | 2     |
| CO-3 | 3                     | 3    | 2    | 3    | 2    | 2    | 2    | 1    | 2    | 1     | 2     | 2     | 3                               | 2     |

#### LIST OF EXPERIMENTS:

**Note: ANY 10 EXPERIMENTS can be conducted from the following:**

1. Draw the Valve Timing Diagram on Single Cylinder Four Stroke Water Cooled Diesel Engine Cut Section Model.
2. Draw the Port Timing Diagram on Single Cylinder Two Stroke Air Cooled Petrol Engine Cut Section Model.
3. Performance Test on Single Cylinder Four Stroke Water Cooled Diesel Engine Test Rig.
4. Heat Balance Test on Single Cylinder Four Stroke Water Cooled Diesel Engine Test Rig.

5. Measurement of Optimum Cooling Water Temperature and Flow Rate on Single Cylinder Four Stroke Water Cooled Diesel Engine Test Rig.
6. Evaluation of Engine Frictional Power by Conducting Morse Test on Four Stroke Multi Cylinder Water Cooled Petrol Engine Test Rig.
7. Performance Test on Four Stroke Multi Cylinder Water Cooled Petrol Engine Test Rig.
8. Heat Balance Test on Four Stroke Multi Cylinder Water Cooled Petrol Engine Test Rig.
9. Economic speed test on Four Stroke Multi Cylinder Water Cooled Petrol Engine Test Rig.
10. Performance Test on Four Stroke Twin Cylinder Water Cooled Diesel Engine Test Rig.
11. Performance Test on Single Cylinder Two Stroke Air Cooled Petrol Engine Test Rig.
12. Performance Test on Computerized VCR Water Cooled Multi Fuel Engine Test Rig.
13. Performance Test on Air Cooled Reciprocating Air – Compressor Test Rig.
14. Performance Test on Refrigeration Test Rig.
15. Performance Test on Air Conditioning Test Rig.
16. Performance Test on Water Cooler Test Rig.
17. Performance Test on Domestic Refrigeration Test Rig.

#### **TEXT BOOKS:**

1. Internal Combustion Engines, V. Ganesan, 3<sup>rd</sup> Edition, Tata McGraw-Hill, 2007
2. Thermal Engineering, Mahesh M. Rathore, 1<sup>st</sup> Edition, Tata McGraw-Hill, 2010

#### **REFERENCES:**

1. A Course in Internal Combustion Engines, M. L. Mathur and R. P. Sharma, 1<sup>st</sup> Edition, Dhanpat Rai Publications, 1976
2. Internal Combustion Engines Fundamentals, John B. Heywood, 1<sup>st</sup> Edition, Tata McGraw-Hill, 1988
3. Refrigeration and Air Conditioning, C. P. Arora, 2<sup>nd</sup> Edition, Tata McGraw-Hill, 2000
4. Thermal Engineering, Er. R. K. Rajput, 11<sup>th</sup> Edition, Lakshmi Publications, 2020

#### **ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/112/103/112103262/#>
2. <https://archive.nptel.ac.in/courses/112/105/112105129/#>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22PC2ME302) AUTOMATION AND ROBOTICS LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the working of hydraulic, pneumatic, electric and electronic controls used in automation
- To understand the concepts of PLC's, microprocessor in automation and robotics
- To learn the simulation of simple mechanisms using MSC Adams software
- To understand the characteristics of sensors in automation and robotics

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Design hydraulic and pneumatic circuits

**CO-2:** Understand PLC ladder logic

**CO-3:** Simulate kinematic motions of robot

**CO-4:** Understand the working of various sensors

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 2    | 3    | 2    |      |      |      | 3    | 2     |       | 3     | 3                               | 2     |
| CO-2 | 2                     | 3    | 2    | 3    | 3    |      |      |      | 3    | 2     |       | 3     | 3                               | 2     |
| CO-3 | 3                     | 2    | 2    | 3    | 2    |      |      |      | 3    | 2     |       | 3     | 2                               | 3     |
| CO-4 | 2                     | 2    | 3    | 3    | 3    |      |      |      | 2    | 2     |       | 3     | 2                               | 3     |

#### LIST OF EXPERIMENTS:

**Note: Any TEN Experiments are to be performed from the following list of experiments**

1. To develop Pneumatic circuits to control Single acting and Double acting cylinders by manually operated and solenoid valves.
2. To study and control the hydraulic circuit for double acting cylinder by mechanical lever operated and solenoid valves.
3. To study Torque vs Speed characteristics of 3 phase Squirrel Cage Induction Motor using VVVF Drive.
4. Ladder programming for doorbells and traffic lights.
5. Ladder programming on Logic gates, Timers & Counters.
6. Programs using Arduino boards.



7. To calibrate and study the working of water level indicator using capacitive transducer.
8. To study behavior of closed loop and open loop control system for temperature.
9. To develop and simulate pneumatic & hydraulic circuits using Automation studio.
10. Simulation and analysis of PID controller using SIMULINK.
11. Draw & Simulate Meter-in and Meter-out pneumatic circuits.
12. Kinematic analysis of a RR Type 2 link robot Using Msc Adams Software.
13. To develop wireless controlled Robot with Arduino.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22HS2EN301) ADVANCED ENGLISH COMMUNICATION SKILLS LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

#### INTRODUCTION:

Advanced English Communication Skills is an activity-based Practical/Lab course. With career readiness as its goal, students are trained in spoken and written communication to showcase their skills, articulate their goals, and highlight their contributions. The course helps students hone their LSRW skills, enhances their collaboration skills, equips them with strategies to present ideas effectively, craft compelling career documents, and communicate ethically and persuasively. Students will be provided with opportunities to deliver presentations and actively participate in group discussions. Mock interviews prepare them to handle challenging questions with poise and professionalism. The course aims to enable students to become confident and effective communicators

**COURSE PRE-REQUISITES:** A basic proficiency of Listening, Speaking, Reading, Writing (LSRW) Skills

#### COURSE OBJECTIVES:

- To equip students with the tools to cultivate self-awareness and develop professional communication skills through diverse strategies and training in verbal and non-verbal communication
- To enable students to create dynamic career documents and personal statements that effectively convey their skills and aspirations
- To strengthen student proficiency in academic writing and preparation of clear, concise reports adhering to standard conventions and ethical practices
- To train students to confidently utilize written, spoken, and visual communication tools for professional presentations and emails
- To help students enhance their ability to actively participate and effectively communicate in group discussions and interviews across various settings

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Deliver effective self-introductions tailored to varying professional contexts and audiences, demonstrating clarity, conciseness, and confidence

**CO-2:** Showcase strong employability skills by creating resumes, cover letters, and video resumes, and effective statements of purpose that convey their skills and aspirations besides highlighting personal and professional achievements

**Co-3:** Understand the need for adhering to academic ethics, apply paraphrasing skills to review relevant literature for academic writing, and prepare well-structured technical reports using effective writing strategies

**CO-4:** Compose professional emails adhering to proper etiquette, formatting, and tone, and make impactful presentations with clarity and confidence effectively using AV aids as required

**CO-5:** Actively engage in group discussions, demonstrating critical thinking, collaborative skills, and the ability to structure arguments, express opinions, and participate in mock interviews exhibiting quick thinking, clear communication, professionalism, and responsiveness to questions

**COURSE ARTICULATION MATRIX:**

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 1                     | 1    | 1    | -    | -    | 1    | 1    | -    | 2    | 3     | 1     | 1     | -                               | -     |
| CO-2 | 1                     | 2    | 1    | 1    | 1    | 2    | 1    | -    | 1    | 3     | 1     | 2     | 1                               | 1     |
| CO-3 | 1                     | 1    | 1    | 1    | -    | 2    | 2    | 1    | 1    | 3     | 1     | 1     | 1                               | 1     |
| CO-4 | 1                     | 1    | 1    | 1    | 1    | 2    | 2    | 1    | 2    | 3     | 2     | 2     | 1                               | 1     |
| CO-5 | 2                     | 3    | 3    | 3    | -    | 2    | 2    | 1    | 3    | 3     | 3     | 2     | 1                               | 1     |

**UNIT-I:**

- Introducing Oneself in Professional Contexts
- Non-verbal Communication: Significance of non-verbal cues in communication; body language and its interpretation; proxemics and personal space; tone and vocal cues
- Story Telling Strategies for Communication
- SWOC Analysis for self-awareness and self-assessment

**UNIT-II:**

- Resume and Cover letter: Structure and Formats; Relevant keywords and action verbs; Writing a persuasive cover letter to complement the resume
- Video Resume: Tools and strategies
- Statement of Purpose (SOP): Purpose and significance in academic contexts; Key components and structure

**UNIT-III:**

- Paraphrasing - Purpose- avoiding plagiarism; Techniques for writing effective paraphrase; Awareness about Citation
- Technical Report Writing: Purpose and Significance; Structure of a Report- various sections- Course Project

**UNIT-IV:**

- E-correspondence - Significance, Purpose, and Structure of professional e-mails; Etiquette, tone, and formatting conventions; Writing clear and concise subject lines and messages.
- Presentations Skills - Structure, Organization of Ideas, Effective use of slides/audio-video

**UNIT-V:**

- Group Discussion: Types of GD, Conventions of formal GD; etiquette; organizing ideas; expressing opinions, agreeing and disagreeing; tips and strategies for successful discussions
- Interview Skills: Telephonic, Video Conferencing, and Face-to-Face

**TEXT BOOKS:**

1. Technical Writing Essentials, Suzan Last, University of Victoria, 2019
2. Technical Communication: A Practical Approach, William S. Pfeiffer, 7<sup>th</sup> Edition, Longman, 2012
3. Technical Communication: A Reader-Centered Approach, Anderson, Paul V. Reports In Paul V. Anderson's, 5<sup>th</sup> Edition, Heinle, 2003

**REFERENCES:**

1. Communication in the workplace: What can NC State Students Expect? Professional Writing Program, J. Swartz, S. Pigg, J. Larsen, J. Helo Gonzalez, R. De Haas, and E. Wagner, North Carolina State University, 2018 [Online]  
Available:<https://docs.google.com/document/d/1pMpVbDRWIN6HssQQQ4MeQ6U-oB-sGUrtRswD7feuRB0/edit> ↵
2. Technical Communication, Burnett Rebecca, 5<sup>th</sup> Edition, Heinle, 2001
3. Technical Writing Process and Product, Gerson Sharon J. and Steven Gerson, 3<sup>rd</sup> Edition, Prentice Hall 1999
4. Technical Communication: Situations and Strategies, Markel Mike, 8<sup>th</sup> Edition, 2006-2007
5. [https://kupdf.net/download/learner-english-pdf-pdf\\_59beb5ec08bbc55c18686ee6\\_pdf](https://kupdf.net/download/learner-english-pdf-pdf_59beb5ec08bbc55c18686ee6_pdf)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22MN6HS301) ANCIENT WISDOM

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 2               | 0   | 0 |

| EVALUATION SCHEME |       |     |       |
|-------------------|-------|-----|-------|
| SE-I              | SE-II | SEE | TOTAL |
| 50                | 50    | -   | 100   |

#### COURSE OBJECTIVES:

- To provide students a fundamental understanding of ancient wisdom from various world civilisations and its relevance to STEAM fields about certain aspects of culture, heritage, and history as sources of knowledge and application for the present and future
- To explore the historical context and contributions of ancient Indian and other global civilisations to STEAM disciplines and to foster scientific temperament using the approaches that stood the tests of time
- To help students develop critical thinking and logical reasoning by understanding ancient texts, philosophies, and practices, and by exploring their applicability for responsible consumption and sustainable development
- To help students create a karmic, sustainable, humane, and responsible future for the world with scientific temperament and a sense of belongingness and ownership to our heritage as humans

**COURSE OUTCOMES:** After completion of the course the students should be able to

**CO-1:** Describe important contributions of ancient Indian and other world civilisations to STEM disciplines, fostering a scientific temperament

**CO-2:** Conceptually analyse ancient texts and philosophies to identify aspects of sociocultural relevance to the present day with a focus on sustainable development

**CO-3:** Apply principles, concepts, approaches, and ideas of ancient wisdom to improve human life/ society through STEM professional activities and management

**CO-4:** Know the importance of learning from the past and developing systems thinking to look at the big picture, leading to interdependence that promotes interdisciplinary approaches to holistic development and ethical future

**ACTIVITIES:** Dialogue-based lectures, Videos, Debates and discussions, collaborative learning activities and group work including presentations and posters

#### MODULE 1:

##### Introduction and Motivation

**Introduction:** How ancient is ancient? Is all ancient wisdom fundamentally religious or spiritual? Does ancient wisdom still have anything important to remember and consider relevant today? Why should we look back while living in the present and building for the future? How can we separate religion, spirituality and philosophy?

**The Indian Story:** Was India really a great country in the past? What has changed in India since, and what has not? How is culture and heritage important or relevant for the future that is not like the past at all? Was knowledge really codified in literature? Is

Hinduism a way of life or a religion and should we care? Is the caste system or varna system divisive?

**History of Sciences and History of Arts:** What has humanity achieved so far in Sciences and in Arts? Understanding the geography of thought and pursuit of knowledge; Leading from the emerging future through the history of humanity

**Definitions and classifications:** Eastern, Arabic, and Western philosophical approaches; Epistemology, Phenomenology, Hermeneutics, Metaphysics, tarka, nyāya, mīmāṃsa, ikigai, oubaitori, eudaimonia, gestell

## **MODULE 2:**

### **Deep Dive into the India Story**

**Ancient India:** Indian Ethos, Concepts of dharma, karma, chaturvidha purusharthā (dharma, artha, kāma, moksha), triguna (sattva, rajas, tamas) – relevance and interpretation for contemporary living; Major Indian contributions: To Education, Language, Governance, and Academic Disciplines with particular focus on STEAM (STEM and Arts)

**Documented and Undocumented Knowledge:** Vedas, Vedāṅgas, Upanishats, Epics and Mythology: Ramayana, Mahabharata and Bhagavad-Gita, Puranas including Bhagavata; Examples of codified knowledge from various written verses: did they actually mean what we now interpret them to be?

**Medieval India:** Advancements and developments in education, governance, technology, and society

Modern India: Constitutional values of India influenced by Ancient and Medieval Indian history and culture

**Summary of Values in Indian Ethos:** Family system, the karma principle and the idea of nishkaama karma, practising and swadharma, protecting dharma, Endurance and Patience (sahanam), Motivation and Determination (sankalpam), Attention and Devotion (Sraddha), Industriousness and Exertion (Srama), Understanding the Self and Ego, Peace (Saanti), Proactive non-violence (ahimsa), Empathy, Altruism, Focus on the big picture, Inquisitiveness and Pursuit of Knowledge, Harmony, Oneness of Humanity (vasudhaika kutumbam), Learning from the Nature and concern towards other lifeforms

**Looking beyond the “proud claims”:** Myths and facts about Indian ancient wisdom: Pursuing Truth, in the Indian way; Can India be the “Viswa Guru”? If yes, how? If not, can/shall we still work towards it?

## **MODULE 3:**

### **Ancient Civilisations and Wisdom from other parts of the World**

**Overview:** Indus Valley and other ancient cultures of the Indian subcontinent – their influence on Southeast Asia; Civilisations outside of Indian subcontinent: Mesopotamian, Egyptian, Far East, European, Middle East and African, North, Central and South American, Oceanic cultures and their indigenous peoples

**Advancements of the Ancient World:** Advances in governance, arts, medicine and health, architecture, metallurgy, weaponsmithy, mechanisation, cosmology, other science and technology, mathematics, astronomy, language, trade and travel, philosophy, other cultural aspects

**Sources of Wisdom, Religion, and Spirituality:** Overview of major Abrahamic religions and their holy books: The Holy Qur'an, Hadith, and Sharia for Islam; The Old and New

Testaments for Christians; Overview of other world religions and their tenets; Overview of Ancient Greek and Ancient Chinese philosophies; Philosophies of Abrahamic religions and non-Abrahamic religions of Indian origin

#### **MODULE 4: Applicability of Ancient Wisdom**

Understanding limitations of ancient wisdom; Examining ancient wisdom to find logical relevance and applicability to the present world; Applying tarka Sāstra principles—dialectics, debates, analyses, discussions—and understanding the spirit of scholars of tarka Sāstra, such as Adi Sankara, Aristotle, Ramanuja, Plato; Identifying core principles, approaches, and ideologies in formulation, design, strategy, and execution in various STEAM domains

#### **MODULE 5: Lessons to Learn from Ancient Wisdom**

Understanding Bhagavad Gita as a text of introspection and motivation, for personality development, team management, and organisational excellence; Exploring ancient books for strategy and management: The Art of War (Sun Tzu), Artha Sastra (Kautilya), parts of āmukta malyada (Krishna Deva Raya) and Mahabharata (Vyasa/ Tikkana)

**The Way Forward:** Developing a systems thinking approach; Developing a humanitarian and scientific temperament based on cultural and constitutional values; Leveraging education to resolve societal problems and creating impact with focus on responsible consumption and sustainable development; reinstalling “human” at the centre of development and using technology for enablement instead of enslaving; Understanding the value of interdependence and Presencing

#### **TEXT BOOKS:**

1. The Wonder That Was India: Volume I, A. L. Basham, Picador India, 2019 (ISBN: 978-9389109344)
2. Indian Science and Technology in the Eighteenth Century, Dharampal, Rashtrottana Sahitya, 2021 (ISBN: 978-8175310933)
3. A History of Science in World Cultures: Voices of Knowledge, Scott L. Montgomery and Alok Kumar, Routledge, 2015 (ISBN: 978-0415639842)
4. Sapiens: A Brief History of Humankind, Harper, Yuval Noah Harari, 2015 (ISBN: 978-0099590088)
5. Think on These Things, Jiddu Krishnamurti, Jaico Publishing House, 2024 (ISBN: 978-8119153794)

#### **REFERENCES:**

1. The Geography of Thought: How Asians and Westerners Think Differently and Why, Richard E. Nisbett, Nicholas Brealey Publishing, 2004 (ISBN: 978-1529309416)
2. The Question Concerning Technology, and Other Essays, Martin Heidegger, Harper Perennial Modern Classics, 2013 (ISBN: 978-0062290700)
3. Arise Awake & Don't Stop!, Swami Vivekananda, Namaskar Books, 2022 (ISBN: 978-9355717221)
4. The Hindu Way: An Introduction to Hinduism, Shashi Tharoor, Aleph Book Company, 2019 (ISBN: 978-9388292856)
5. Eastern Religions and Western Thought, Sarvepalli Radhakrishnan, Oxford University Press, 1997 (ISBN: 978-0195624564)

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VI Semester

### (22PC1ME304) HEAT TRANSFER

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Basic Integral and Differential Calculus, Thermodynamics

#### COURSE OBJECTIVES:

- To measure the conduction mode of heat transfer in physical environment and to derive general mathematical equation
- To measure the heat transfer through Homogeneous slabs, hollow cylinders, sphere, extended surfaces, and fins
- To measure convective mode of heat transfer
- To measure heat transfer during radiation, boiling and condensation and through different types of heat exchangers

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** apply conduction equation to solve practical problems

**CO-2:** Apply convective equation to solve practical problems

**CO-3:** Evaluate radiation heat transfer for engineering problems

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | -    | 3    | -    | -    | -    | -    | 2     | 1     | 2     | 3                               | 2     |
| CO-2 | 3                     | 3    | 2    | -    | 3    | -    | -    | -    | -    | 2     | 1     | 2     | 3                               | 2     |
| CO-3 | 3                     | 3    | 3    | -    | 2    | -    | -    | -    | -    | 2     | 1     | 2     | 3                               | 2     |

#### UNIT-I:

**Introduction:** Modes and mechanisms of heat transfer - Basic laws of heat transfer - Simple general discussion about applications of heat transfer. Conduction Heat Transfer: Fourier heat conduction equation - General heat conduction equation in Cartesian, Cylindrical coordinates - simplification and forms of the field equation steady and unsteady heat transfer.

#### UNIT-II:

**One Dimensional Conduction Heat Transfer:** Homogeneous slabs, hollow cylinders, and sphere - overall heat transfer coefficient - electrical analogy - Critical radius of insulation, extended surfaces and fins. One Dimensional Transient Conduction Heat Transfer: Systems with negligible internal resistance



### **UNIT-III:**

**Convective Heat Transfer:** Classification of systems based on causation of flow, condition of flow, configuration of flow and medium of flow - Dimensional analysis as a tool for experimental investigation - Concepts about hydrodynamic and thermal boundary layers, Forced Convection: Flat plates and horizontal pipes. Free Convection: Vertical plates and pipes.

### **UNIT-IV:**

**Heat Transfer with Phase Change:** Heat transfer with boiling - pool boiling and film boiling - boiling curve for pool boiling - simple correlations for pool boiling - Condensation plates heat transfer: film wise and drop wise condensation - film condensation on vertical and horizontal cylinders using empirical correlations. Heat Exchangers: Classification of heat exchangers - overall and fouling resistance - problems using LMTD NTU methods for parallel flow and counter flow.

### **UNIT-V:**

**Radiation Heat Transfer:** Emission characteristics and laws of black-body radiation - incident radiation - total and monochromatic quantities - laws of Planck, Wien, Kirchhoff, Lambert, Stefan and Boltzmann - heat exchange between two black bodies - concepts of shape factor - emissivity - heat exchange between grey bodies - radiation shields - electrical analogy for radiation networks.

### **TEXT BOOKS:**

1. Fundamentals of Engineering Heat and Mass Transfer, Sachdeva R. C., 5<sup>th</sup> Edition, New Age International, 2017
2. Fundamentals of Heat & Mass Transfer, Incopera Dewitt, Wiley, 2006
3. Fundamentals of Heat and Mass Transfer, Thirumaleshwar M., Pearson, 2006

### **REFERENCES:**

1. Heat Transfer: A Basic Approach, Ozsik, McGraw-Hill International Edition, 1985
2. Heat Transfer, Holman J. P., McGraw-Hill, 1990
3. A Textbook on Heat Transfer, Sukhatme S. P., 3<sup>rd</sup> Edition, Sangam Books
4. Heat and Mass Transfer, Kumar D. S., Katson Books, 2013
5. Heat and Mass Transfer Fundamental and Applications, Yunus A. Cengel, Afshin Ghajar, McGraw-Hill Education

### **ONLINE RESOURCE:**

1. <https://archive.nptel.ac.in/courses/112/108/112108149/>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PC1ME305) ADDITIVE MANUFACTURING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** CAD

#### COURSE OBJECTIVES:

- To know the need for and importance of Additive Manufacturing in product development and its applications
- To understand the different types of Additive Manufacturing Processes and Rapid Tooling Methods
- To be familiar with the CAD formats, post-processing & Inspection Techniques involved in Additive Manufacturing

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Summarize the knowledge of Additive Manufacturing in product development and its applications

**CO-2:** Explain the different types of Additive Manufacturing Processes and Rapid Tooling Methods

**CO-3:** Apply the knowledge of CAD Formats and Plan post-processing techniques and perform inspection techniques involved in Additive Manufacturing

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 3    | 3    | -    | -    | -    | 3    | 2     | -     | 3     | 3                               | 2     |
| CO-2 | 3                     | 3    | 3    | 3    | 3    | -    | -    | -    | 3    | 2     | -     | 3     | 3                               | 2     |
| CO-3 | 3                     | 3    | 3    | 3    | 3    | -    | -    | -    | 3    | 2     | -     | 3     | 3                               | 2     |

#### UNIT-I:

**introduction:** Introduction, Need, Classification, Additive Manufacturing Technology in product development, Materials for Additive Manufacturing Technology.

**Applications:** Brief overview of applications in Design, Engineering, Manufacturing and Tooling, Aerospace, Automotive, Biomedical, Jewelry, Coin and Tableware Industry.

## **UNIT-II:**

**Additive Manufacturing Processes – I:** Introduction to Material Extrusion Process - Principles, Process, Advantages and Applications of Fused Deposition Modeling (FDM).

Introduction to Sheet Lamination based AM Systems - Principles, Process, Advantages and Applications of Laminated Object Manufacturing (LOM).

## **UNIT-III:**

**Additive Manufacturing Processes – II:** Introduction to Liquid based AM Process - Principles, Process, Advantages and Applications of Stereolithography (SLA).

Introduction to Three-Dimensional Jetting Based AM Processes - Principles, Process, Advantages and Applications of Material Jetting and Binder Jetting.

## **UNIT-IV:**

**Additive Manufacturing Processes – III:** Introduction to Powder based AM Processes - Principles, Process, Advantages and Applications of Selective Laser Sintering (SLS), Laser Engineered Net Shaping (LENS) and Electron Beam Melting (EBM) and Overview of Direct Energy Deposition AM Processes.

**Rapid Tooling-** Introduction to Rapid Tooling (RT), Indirect Methods of Rapid Tooling: Investment Cast Tooling, 3D Keltool, Spray Metal Tooling. Direct Methods of Rapid Tooling: SLS, EBM Tooling, Direct Metal Laser Sintering (DMLS) and Laminated Tooling.

## **UNIT-V:**

### **CAD, Post Processing & Inspection For AM**

**CAD:** Basic concept, Digitization techniques, CAD data formats, CAD model preparation, Part orientation and support generation, Data loss & repair.

**Post Processing:** Introduction, Post Processing Techniques like Support material removal, Cleaning, Sanding and Polishing for FDM Printed Parts.

**Inspection:** Traditional inspection techniques, non-Destructive defect detection techniques and flatness or warp check.

## **TEXT BOOKS:**

1. Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing, Ian Gibson, David W. Rosen, Brent Stucker, Springer, 2010
2. Rapid Prototyping, Gebhardt A., Hanser Gardener Publications, 2003

## **REFERENCES:**

1. Rapid Prototyping: Principles and Applications, Chua C. K., Leong K. F., and Lim C.S., 3rd Edition, World Scientific, 2010
2. Rapid Prototyping and Engineering Applications: A Toolbox for Prototype Development, Liou L. W. and Liou F. W., CRC Press, 2007
3. Rapid Prototyping: Theory and Practice, Kamrani A. K. and Nasr E. A., Springer, 2006
4. Rapid Tooling: Technologies and Industrial Applications, Hilton P. D. and Jacobs P. F., CRC Press, 2000

## **ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/112/103/112103306/>

2. [https://onlinecourses.nptel.ac.in/noc22\\_me130/preview](https://onlinecourses.nptel.ac.in/noc22_me130/preview)
3. <https://archive.nptel.ac.in/courses/112/104/112104265/>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PC1ME306) FINITE ELEMENT METHODS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mathematics, Engineering Mechanics, Mechanics of Solids

#### COURSE OBJECTIVES:

- To understand the concept of Finite Element Method, Identify the areas of application of FEM and study the procedure
- To solve linear problems by numerical methods and differentiate them
- To apply the concept of FEM to solve basic non-linear problems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply the concepts of FEM to one Dimensional structural systems

**CO-2:** Analyze two dimensional and three-dimensional cases

**CO-3:** Formulate Scalar field variables

**CO-4:** Analyze time dependent systems to determine Eigen vectors

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 1    | 2    | 2    | -    | -    | -    | -     | -     | 3     | 3                               | 1     |
| CO-2 | 3                     | 3    | 2    | 1    | 2    | 2    | -    | -    | -    | -     | -     | 3     | 3                               | 1     |
| CO-3 | 3                     | 3    | 2    | 1    | 2    | 2    | -    | -    | -    | -     | -     | 3     | 3                               | 1     |
| CO-4 | 3                     | 3    | 2    | 1    | 2    | 2    | -    | -    | -    | -     | -     | 3     | 3                               | 1     |

#### UNIT-I:

**Introduction to FEM:** Historical background, application of FEM, general description, comparison of FEM with other methods, variational approach, Galerkin Methods.

**Basic concepts:** Co-ordinates, basic element shapes, treatment of boundary conditions.

**One Dimensional Problems:** Formulation of Stiffness Matrix for a Bar Element by the Principle of Minimum Potential Energy, Properties of Stiffness Matrix, Characteristics of Shape Functions, Quadratic shape functions.

#### UNIT-II:

**Analysis of Trusses:** Derivation of Stiffness Matrix for Trusses, Stress and strain Calculations, Calculation of reaction forces and displacements.

**Analysis of Beams:** Derivation of Stiffness matrix for linear beam element, Load Vector, Deflection, Stresses, Shear force and Bending moment, Problems on uniform and stepped beams for different types of loads applied on beams.

#### **UNIT-III:**

**Finite Element – Formulation of 2D Problems:** Derivation of Element stiffness matrix for two-dimensional Constant Strain Triangular (CST) Element, Derivation of shape functions for CST Element.

**Finite Element – Formulation of 3D Problems:** Concept of Element stiffness matrix for Tetrahedron Element, Properties of Shape functions for 3D Tetrahedral Element, Stress-Strain Analysis for 3D Element, Strain Displacement for Relationship Formulation. Problems only on Axisymmetric analysis.

#### **UNIT-IV:**

**Steady State Heat Transfer Analysis:** One Dimensional Finite Element analysis of fin and composite slabs. Two-dimensional steady state heat transfer problems: Derivation of Thermal Stiffness matrix for 2D heat transfer problems-CST, Derivation of thermal force vector for 2D heat transfer problems.

#### **UNIT-V:**

**Dynamic Analysis:** Formulation of mass matrices for uniform bar, and 2D triangular Elements using lumped and consistent mass methods, Evaluation of Eigen values and Eigen vectors for a stepped bar Problems. Isoparametric Elements and Numerical integration.

#### **TEXT BOOKS:**

1. Introduction to Finite Elements in Engineering, Chandrupatla and Belegundu, 5<sup>th</sup> Edition, Cambridge University Press, 2022
2. A First Course in Finite Element Method, Logan Deryl L., 5<sup>th</sup> Edition, Thomson Brook/Cole, 2022
3. Concepts and Applications of Finite Element Analysis, Cook R. D., 4<sup>th</sup> Edition, Wiley

#### **REFERENCES:**

1. Finite Element Procedures in Engineering Analysis, Bathe K. J., Cliffs, N. J. PHI Learning, Eastern Economy Editions, 1982
2. The Finite Element Methods in Engineering, S. S. Rao, 4<sup>th</sup> Edition, Pergamon, 2004
3. An Introduction to Finite Element Methods, J. N. Reddy, 4<sup>th</sup> Edition, McGraw-Hill, 2018
4. The Finite Element Method: Its Basis and Fundamentals, O. C. Zienkiewicz, 7<sup>th</sup> Edition, McGraw-Hill, 2013
5. Fundamentals of Finite Element Analysis, Hutton D. V., 1<sup>st</sup> Edition, McGraw-Hill, 2003

#### **ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/112/104/112104193/>
2. <https://archive.nptel.ac.in/courses/112/105/112105308/>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PE1ME306) REFRIGERATION AND AIR CONDITIONING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Thermodynamics, Heat Transfer

#### COURSE OBJECTIVES:

- To apply the fundamentals of Thermodynamics and its relative laws and effect on the system
- To evaluate performance of various thermodynamic cycles used in RAC
- To evaluate the performance of vapour compression and vapour absorption system
- To design a real-time air-conditioning system

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Develop and analyze a refrigeration system

**CO-2:** Apply the basic principles of Thermodynamics to solve an engineering problem related to Refrigeration and Air conditioning

**CO-3:** Select suitable Refrigeration cycle and apply the concept of Heat Transfer

**CO-4:** Develop and evaluate the performance of air conditioning system

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 1    | 1    | -    | -    | 1    | -    | -    | 2     | -     | 1     | 3                               | 1     |
| CO-2 | 3                     | 2    | 1    | 1    | -    | -    | 2    | -    | -    | -     | -     | 1     | 3                               | 1     |
| CO-3 | 3                     | 3    | 1    | 1    | -    | -    | 1    | -    | -    | 2     | 2     | 2     | 3                               | 1     |
| CO-4 | 3                     | 3    | 2    | 2    | -    | -    | 1    | -    | -    | 2     | 3     | 1     | 3                               | 1     |

#### UNIT-I:

**Introduction to Refrigeration:** Necessity and applications – unit of refrigeration and C.O.P. Refrigerants: – Desirable properties – classification refrigerants used – Nomenclature – Ozone Depletion– Global Warming.

**Vapour Compression Refrigeration:** Working principle and essential components of the plant – simple Vapour compression refrigeration cycle – COP – Representation of cycle on T-S and p-h charts – effect of sub cooling and super heating – cycle analysis — Use of p-h charts – numerical Problems.

## **UNIT-II:**

**Vapor Absorption System:** Calculation of max COP – description and working of NH<sub>3</sub> – water system and Li Br –water system. Principle of operation Three Fluid absorption system, salient features.

**Steam Jet Refrigeration System:** Working Principle and Basic Components. Principle and operation of (i) Thermoelectric refrigerator (ii) Vortex tube or Hilsch tube.

## **UNIT-III:**

**Air Refrigeration:** Bell Coleman cycle and Brayton Cycle, Open and Dense air systems – Actual air refrigeration system problems – Refrigeration needs of Air crafts.

**System Components:** Compressors – General classification – Multistage or Compound Compression – comparison – Advantages and Disadvantages. Condensers – classification – Working Principles Evaporators – Multi Operator System, Cascade System – classification – Working Principles Expansion devices – Types – Working Principles.

## **UNIT-IV:**

**Psychrometry:** Classification of Air- Conditioning Systems, ASHRAE Nomenclature, Applications of Air-Conditioning, Psychrometry - Air-water vapor mixtures, Psychrometric Properties, Psychrometric or Air Conditioning processes, Psychrometric chart. Cooling, heating humidification and dehumidification, filters, grills and registers, fans and blowers.

## **UNIT-V:**

**HVAC Systems:** Heating: Heat Pump – Heat sources – different heat pump circuits. Need for Ventilation. Characterization of Sensible and latent heat loads — Consideration of Infiltration – Load concepts of RSHF, GSHF- Concept of ESHF and ADP. Requirements of human comfort and concept of effective temperature - Comfort chart –Comfort Air conditioning – Requirements of Industrial air - conditioning, Air-conditioning Load Calculations.

## **TEXT BOOKS:**

1. Refrigeration and Air Conditioning, C. P. Arora, 3<sup>rd</sup> Edition, Tata McGraw-Hill, 2017
2. A Course in Refrigeration and Air Conditioning, S. C. Arora & Domkundwar, 7<sup>th</sup> Edition, Dhanpat Rai, 2008

## **REFERENCES:**

1. Refrigeration and Air Conditioning, S. N. Sapali, 2<sup>nd</sup> Edition, PHI Learning, 2016
2. Principles of Refrigeration, Dossat, 4<sup>th</sup> Edition, Pearson Education, 2002
3. Basic Refrigeration and Air-Conditioning, Ananthanarayanan, 4<sup>th</sup> Edition, Tata McGraw-Hill, 2013
4. Refrigeration and Air Conditioning, R. S. Khurmi & J. K. Gupta, 5<sup>th</sup> Edition, S. Chand, Eurasia Publishing, 2020

## **ONLINE RESOURCE:**

1. <https://archive.nptel.ac.in/courses/112/107/112107208/>



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PE1ME307) OPERATIONS RESEARCH AND DECISION SCIENCES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mathematics

#### COURSE OBJECTIVES:

- To analyze linear programming models in practical and their practical use
- To apply the transportation, assignment and sequencing models and their solution methodology for solving problems
- To apply the theory of games, replacement, inventory and queuing models and their solution methodology for solving problems
- To evaluate the dynamic programming and simulation models

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Evaluate the allocation problems using linear programming

**CO-2:** Select the concepts of sequencing and queuing for industrial situations

**CO-3:** Apply the concepts of game theory in various practical applications

**CO-4:** Solve the dynamic programming problem and simulation models in practical situations

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 2    | 3    | 3    | -    | 2    | 2    | 2     | 2     | 3     | 3                               | 3     |
| CO-2 | 3                     | 3    | 3    | 2    | 3    | 3    | -    | 2    | 2    | 2     | 2     | 3     | 2                               | 2     |
| CO-3 | 3                     | 3    | 2    | 3    | 3    | 3    | -    | 2    | 2    | 2     | 3     | 3     | 3                               | 3     |
| CO-4 | 3                     | 3    | 3    | 3    | 3    | 3    | -    | 2    | 2    | 2     | 2     | 3     | 2                               | 2     |

#### UNIT-I:

**Introduction:** Origin, Development-Definition-Characteristics and Phases-Types of OR models- applications, limitations. Importance of Decision Sciences and role of quantitative techniques in decision making.

**Allocation:** Linear Programming Problem Formulation- Graphical Solution-Simplex Method-Artificial variables technique-Two phase method, Big-M Method-Duality Principle.

## **UNIT-II:**

**Transportation Problem:** Formulation-Optimal solution-unbalanced transportation problem-Degeneracy. Assignment problem-Formulation-Optimal Solution-Variations of Assignment Problem- Travelling Salesman Problem.

**Sequencing:** Introduction-Flow Shop sequencing-n jobs through two machines-n jobs through three machines-Job shop sequencing-two jobs through m machines.

## **UNIT-III:**

**Replacement:** Introduction-Replacement of items that deteriorate with time-when money value is not counted and counted-Replacement of items that fail completely, group replacement.

**Theory of Games:** Introduction-Minimax (maximin) - criterion and optimal strategy-Solution of games with saddle points-rectangular games without saddle points-principles of dominance-M X 2 and 2 X n games-graphical method.

## **UNIT-IV:**

**Waiting Lines:** Introduction-Single channel-Poisson arrivals-exponential service times-with infinite population and finite population models-Multichannel-Poisson arrivals-exponential service times with infinite population single channel Poisson arrivals.

**Inventory:** Introduction-Single Item-Deterministic Models-Purchase inventory models with one price break and multiple price breaks-shortages not allowed-Stochastic models-demand may be discrete variable or continuous variable-Instantaneous production, Instantaneous demand and continuous demand and no set up cost-single period model.

## **UNIT-V:**

**Evolutionary Algorithms:** An overview of evolutionary algorithms, Introduction to Intelligent Optimization, Types of reproduction operators, crossover & mutation, Simulated Annealing Algorithm, Genetic algorithm, Particle swarm optimization algorithm.

**Simulation:** Definition-Types of simulation models-phases of simulation-applications of simulation- Inventory and Queuing Problems-Advantages and Disadvantages-Brief Introduction of Simulation Languages.

## **TEXT BOOKS:**

1. Operations Research: Theory and Applications, J. K. Sharma, 2<sup>nd</sup> Edition, McMillan India, 2003
2. Operations Research, R. Pannarselvam, 2<sup>nd</sup> Edition, Prentice Hall International, 2016

## **REFERENCES:**

1. Decision Sciences Theory and Practice, Raghu Nandan Sengupta, Joydeep Dutta, Aparna Gupta, 1<sup>st</sup> Edition, CRC Press, 2016
2. Operations Research, A. M. Natarajan, P. Balasubramani, A. Tamilarasi, 1<sup>st</sup> Edition, Pearson Education, 2006
3. Operations Research: Methods and Problems, Maurice Saseini, Arthur Yaspan and Lawrence Friedman, 1<sup>st</sup> Edition, Wiley, 1962
4. Operations Research an Introduction, Taha, 8<sup>th</sup> Edition, Prentice Hall International, 2007

**ONLINE RESOURCES:**

1. <https://nptel.ac.in/courses/112106134>
2. <https://nptel.ac.in/courses/112103301>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PE1ME308) INDUSTRIAL ROBOTICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automation and Robotics

#### COURSE OBJECTIVES:

- To understand the concepts of robotics and control systems
- To develop skills in transformations, kinematic and dynamic analysis of robot manipulator
- To introduce knowledge in sensors and programming used in robotics
- To develop knowledge of Industrial applications using robotics

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Classify mechanical structures of industrial robots and their workspace characteristics

**CO-2:** Apply the concepts of coordinate transformations for forward and inverse kinematics model for given serial manipulator

**CO-3:** Develop the algorithms for robotic controller and its programming

**CO-4:** Design the robotic manipulator integrated industrial application

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 2                     | 3    | 2    | 3    | 2    | -    | -    | -    | 2    | 2     | -     | 3     | 2                               | 2     |
| CO-2 | 3                     | 2    | 3    | 3    | 2    | -    | -    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-3 | 2                     | 3    | 3    | 3    | 2    | -    | -    | -    | 3    | 2     | -     | 3     | 2                               | 3     |
| CO-4 | 3                     | 2    | 3    | 3    | 2    | -    | -    | -    | 2    | 2     | -     | 3     | 2                               | 2     |

#### UNIT-I:

**Introduction:** Automation and Robotics, Historical Development, Definitions, Basic Structure of Robots, Robot Anatomy, Complete Classification of Robots, Fundamentals about Robot Technology, Factors related to use Robot Performance, Basic Robot Configurations and their Relative Merits and Demerits, the Wrist & Gripper Subassemblies. Concepts about Basic Control System, Control Loops of Robotic Systems, Different Types of Controllers-Proportional, Integral, Differential, PID controllers

## **UNIT-II:**

**Kinematics of Robot Manipulator:** Introduction, General Mathematical Preliminaries on Vectors & Matrices, Direct Kinematics problem, Geometry Based Direct kinematics problem, Co-ordinate and vector transformation using matrices, Rotation matrix, Inverse Transformations, Problems, Composite Rotation matrix, Homogenous Transformations, Robotic Manipulator Joint Co-Ordinate System, Euler Angle & Euler Transformations, Roll-Pitch-Yaw (RPY) Transformation. D H Representation & Displacement Matrices for Standard Configurations, Jacobian Transformation in Robotic Manipulation. Introduction, Trajectory Interpolators.

## **UNIT-III:**

**Dynamics of Robotic Manipulators:** Introduction, Preliminary Definitions, Generalized Robotic Coordinates, Jacobian for a two link Manipulator, Euler Equations, and The Lagrangian Equations of motion. Application of Lagrange-Euler (LE) Dynamic Modeling of Robotic Manipulators: - Velocity of Joints, Kinetic Energy T of Arm, Potential Energy V of Robotic Arm, The Lagrange L, Two Link Robotic Dynamics with Distributed Mass.

## **UNIT-IV:**

**Robot Teaching:** Introduction, Various Teaching Methods, Task Programming, Survey of Robot Level Programming Languages, A Robot Program as a Path in Space, Motion Interpolation, WAIT, SIGNAL & DELAY Commands, Branching, Robot Language Structure, various Textual Robot Languages Such as VAL II, RAIL and their Features, Typical Programming Examples such as Palletizing, Loading a Machine Etc.

## **UNIT-V:**

**Robot Sensing & Vision:** Various Sensors and their Classification, Use of Sensors and Sensor Based System in Robotics, Machine Vision System, Description, Sensing, Digitizing, Image Processing and Analysis and Application of Machine Vision System, Robotic Assembly Sensors and Intelligent Sensors. Industrial Applications: Objectives, Automation in Manufacturing, Robot Application in Industry, Task Programming, Robot Intelligence and Task Planning, Modern Robots.

## **TEXT BOOKS:**

1. Robot Dynamics and Control, Mark W. Spong & M. Vidyasagar, John Wiley & Sons
2. Robotics and Control, R. K. Mittal and I. J. Nagrath, Tata McGraw-Hill, 2003
3. Introduction to Robotics, S. K. Saha, McGraw-Hill Education, 2008

## **REFERENCES:**

1. Introduction to Robotics: Mechanics and Control, J. Craig, 2<sup>nd</sup> Edition, Addison-Wesley, 1989
2. Industrial Robotics, M. P. Groover, M. Weiss, R. M. Gnagel and N. G. Ordrey, Tata McGraw-Hill, 2012
3. Robotics: Control, Sensing, Vision, and Intelligence, Fu K. S., Gonzalez R. C. and Lee C. S. G., McGraw-Hill, 1987

## **ONLINE RESOURCE:**

1. [https://onlinecourses.nptel.ac.in/noc23\\_me143/preview](https://onlinecourses.nptel.ac.in/noc23_me143/preview)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PE1ME309) MECHATRONIC SYSTEMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Analog and Digital Electronics, Control Power Systems Instrumentation and Control System

#### COURSE OBJECTIVES:

- To design mechatronics system and simulation for ergonomics and safety
- To learn theoretical and practical aspects of computer interfacing, real time data acquisition and control
- To design motion control, motion converter and temperature control
- To realize the concepts of real time interfacing and data acquisition

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Classify the basics and key elements of Mechatronics design process

**CO-2:** Develop basic system modeling

**CO-3:** Identify the concepts of engineering system and dynamic response of the system

**CO-4:** Design and Develop the Mechatronics systems

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 2    | 2    | 3    | 2    | -    | -    | -    | 3    | 2     | -     | 2     | 2                               | 2     |
| CO-2 | 2                     | 3    | 3    | 2    | 2    | -    | -    | -    | 2    | 3     | -     | 2     | 3                               | 3     |
| CO-3 | 3                     | 2    | 2    | 3    | 2    | -    | -    | -    | 3    | 2     | -     | 3     | 2                               | 3     |
| CO-4 | 2                     | 3    | 3    | 3    | 2    | -    | -    | -    | 2    | 2     | -     | 3     | 2                               | 3     |

#### UNIT-I:

**Introduction to Design of Mechatronics System:** Key elements – Mechatronics design process – design parameters – mechatronics and traditional design – Advanced approaches in mechatronics design – Introduction to industrial design, modelling, simulation, and analysis – Ergonomics and safety.

#### UNIT-II:

**Basic System Modelling:** Introduction – model categories – model development – Simulation using software's – verification and validation – Mathematical modelling: Basic system modelling – mechanical electrical, fluid, and thermal.

### **UNIT-III:**

**Mechatronic System Modelling:** Engineering systems: Rotational – translational, electro-mechanical, pneumatic- mechanical, hydraulic-mechanical, micro electromechanical system – Dynamic responses of system: first order, second order system – Performance measures

### **UNIT-IV:**

**Real Time Interfacing:** Introduction – Selection of interfacing standards- elements of data acquisition and control systems – Overview of I/O process – general purpose I/O cards and its installation – Data conversion process – Application software's – Man machine interface

### **UNIT-V:**

**Robotic Vision System - Image Acquisition:** Vidicon, charge coupled device (CCD) and charge injection device (CID) cameras. Image processing techniques: histogram processing: sliding, stretching, equalization and thresholding. Case studies of **Mechatronics systems:** Automatic camera, bar code reader, pick and place robot, automatic car park barrier system, automobile engine management system.

### **TEXT BOOKS:**

1. Mechatronics System Design, Devdas Shetty, Richard A. Kolk, Cengage Learning, 2010
2. Mechatronic Systems: Modeling and Simulation with HDL's, Georg Pelz, 2<sup>nd</sup> Edition, John Wiley and Sons, 2003

### **REFERENCES:**

1. Mechatronics Handbook, Bishop, Robert H., CRC Press, 2002
2. Mechatronics: Electronics in Products and Processes, Bradley D. Dawson, N. C. Burd and A. J. Loader, CRC Press, 2010
3. Mechatronics: A Foundation Course, De-Silva, Indian Reprint, Taylor & Francis, 2013

### **ONLINE RESOURCE:**

1. [https://onlinecourses.nptel.ac.in/noc21\\_me27/preview](https://onlinecourses.nptel.ac.in/noc21_me27/preview)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PE1ME310) THEORY OF METAL CUTTING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Production Technology, Machine Tools

#### COURSE OBJECTIVES:

- To evaluate the relations between the machining parameters and temperatures associated with the orthogonal cutting based on principles of metal cutting
- To evaluate the cutting tools geometry and their areas of application as per metal cutting theory
- To analyze tool wear, replacement strategy, cutting fluids, tool materials and their properties
- To analyze the economics of machining

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Analyze the relations for chip reduction coefficient, shear angle, shear strain, forces, power, specific energy and temperatures associated with orthogonal cutting

**CO-2:** Select a cutting tool, its geometry and material in cutting processes.

**CO-3:** Identify the thermal zones and select suitable cutting tools.

**CO-4:** Apply the methods to calculate the economics of machining.

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 2                     | 3    | 1    | -    | 2    | -    | -    | 1    | -    | -     | -     | 1     | 1                               | 2     |
| CO-2 | 3                     | 2    | 2    | 1    | 2    | -    | -    | -    | -    | -     | -     | 2     | 2                               | 1     |
| CO-3 | 2                     | 3    | 2    | 1    | 1    | 1    | 1    | -    | -    | -     | 1     | 2     | 2                               | 2     |
| CO-4 | 2                     | 2    | 3    | -    | 2    | 1    | -    | -    | 1    | -     | -     | 2     | 1                               | 1     |

#### UNIT-I:

**Introduction:** Classification of Manufacturing Process, Mechanics of Metal Cutting: Mechanism of chip formation, types of chips, built-up edge, Orthogonal and Oblique cutting, Determination of shear plane angle, forces on the chips, forces in orthogonal cutting, merchant circle diagram and analysis, co-efficient of friction, power and energy relationship, velocity relationship, shear-strain relationship, problems.



## **UNIT-II:**

**Measurement of Cutting Forces:** Reasons for measuring cutting forces, Classification of cutting force dynamometers – mechanical, piezoelectric, and strain gage type dynamometers, Calibration of dynamometers.

**Tool Wear and Tool Life:** Mechanisms of tool wear, Sudden, gradual wear, crater wear, flank wear, tool life criteria and machinability index, machinability criteria, tool life equations, effect of process parameters on tool life.

## **UNIT-III:**

**Geometry of Cutting Tools:** Single point and multipoint cutting tools, tool point reference systems, tool angles specifications–ISO and ASA systems, conversion from one system to another. Effect of cutting parameters on tool geometry.

**Tool Materials and their Properties:** Characteristics of tools materials, types of tool materials – carbon tool steels, high speed steels, cast alloys, cemented carbides, ceramics, diamonds, sialon, CBN, UCON, recommended cutting speeds for the above tools.

## **UNIT-IV:**

**Thermal Aspects in Metal Cutting:** Heat sources in metal cutting, temperature in chip formation, temperature distribution, and experimental determination of tool temperatures.

**Cutting Fluids:** Basic actions of cutting fluids, properties of cutting fluids, selection of cutting fluids, green cutting fluids, application of cutting fluids, recommended cutting fluids.

## **UNIT-V:**

**Economics of Machining:** Introduction, elements of total production cost, optimum cutting speed and tool life for minimum cost, optimum cutting speed and tool life for maximum production, problems.

## **TEXT BOOKS:**

1. Principles of Metal Cutting, G. C. Sen and A. Bhattacharya, New Central Book Agency, 2011
2. Metal Cutting Principles, M. C. Shaw, Oxford and IBH Publications, 2012

## **REFERENCES:**

1. Fundamentals of Metal Machining and Machine Tools (Mechanical Engineering, 198), Geoffrey Boothroyd, Winston A. Knight, Taylor and Francis, 2005
2. Metal Cutting Theory and Cutting Tool Design, V. Arshinov & G. Alekseev, Mir Publishers
3. Fundamentals of Metal Cutting and Machine Tools, B. L. Juneja, G. S. Sekhom & Nitin Seth, New Age International, 2017
4. Principles of Metal Cutting, G. Kuppaswamy, Universities Press, 1996

## **ONLINE RESOURCE:**

1. <https://archive.nptel.ac.in/courses/112/105/112105233/>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PC2ME304) HEAT TRANSFER LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

**COURSE PREREQUISITES:** Heat and Mass Transfer, Thermodynamics

#### COURSE OBJECTIVES:

- To analyze various modes of heat transfer experimentally
- To measure heat transfer through conduction
- To measure heat transfer through natural and forced convection
- To measure heat transfer through radiation

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Identify and analyze the mode of heat transfer

**CO-2:** Evaluate thermal conductivity of composite wall, lagged pipe, metal bar and insulating powder

**CO-3:** Evaluate heat transfer coefficient for natural, forced convection and unsteady heat transfer

**CO-4:** Evaluate emissivity and Stefan Boltzmann constant of the given metal

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    |      | 3    |      |      |      |      | 2     | 1     | 2     | 3                               | 2     |
| CO-2 | 3                     | 3    | 2    |      | 3    |      |      |      |      | 2     | 1     | 2     | 3                               | 2     |
| CO-3 | 3                     | 3    | 3    |      | 2    |      |      |      |      | 2     | 1     | 2     | 3                               | 2     |
| CO-4 | 3                     | 3    | 3    |      | 2    |      |      |      |      | 2     | 1     | 2     | 3                               | 2     |

#### LIST OF EXPERIMENTS:

**Any 10 experiments to be conducted from the following:**

1. Determination of thermal conductivity of given metal rod.
2. Determination of Stefan Boltzmann constant.
3. To find out critical heat flux.
4. Determination of overall heat transfer coefficient of composite wall.
5. Determination of thermal conductivity of lagged Pipe.
6. Determination of heat transfer coefficient in forced convection apparatus.
7. Determination of heat transfer coefficient in natural convection apparatus.

8. Determination of thermal conductivity of insulating powder.
9. Determination of effectiveness of heat exchanger.
10. Measurement of emissivity of given test plate.
11. Heat transfer in dropwise and film wise condensation.
12. Determination of heat transfer coefficient and instantaneous heat transfer rate for transient heat conduction.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22PC2ME305) ADDITIVE MANUFACTURING LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

**COURSE PRE-REQUISITES:** Rapid Prototyping, Additive Manufacturing Processes

#### COURSE OBJECTIVES:

- To demonstrate the knowledge of 3D Printing Technology
- To understand about the CAD model conversion to STL file format and rectification of errors in STL file
- To explain about build parameters, support structures and part orientation involved in 3D Printing
- To understand about the post processing, inspection, and testing methods

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Develop CAD models and convert them to STL files for 3D Printing

**CO-2:** Apply the knowledge on CAD Files and be able to rectify the STL errors

**CO-3:** Decide the print parameters, part orientations and its support structures involved in 3D Printing

**CO-4:** Build a 3D printed engineering component and be in a position to perform inspection and post processing

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 3    | 3    |      |      |      | 3    | 2     |       | 3     | 3                               | 2     |
| CO-2 | 3                     | 3    | 3    | 3    | 3    |      |      |      | 3    | 2     |       | 3     | 3                               | 2     |
| CO-3 | 3                     | 3    | 3    | 3    | 3    |      |      |      | 3    | 2     |       | 3     | 3                               | 2     |
| CO-4 | 3                     | 3    | 3    | 3    | 3    |      |      |      | 3    | 2     |       | 3     | 3                               | 2     |

#### LIST OF EXPERIMENTS:

##### Overview of 3D Printing and introduction to FDM

##### ❖ 3D Printing:

- Introduction to CAD modeling and preparation of CAD models of simple mechanical engineering components.
- Conversion of CAD model to STL file format and correction of errors.
- Definition of build parameters for 3D printing including support structures.

- Working with build orientation for optimum printing time.
  - Verification of 3D mesh model before printing.
  - 3D printing of modeled components.
  - Inspection of 3D printed models – visual inspection and dimensional measurements using digital caliper.
  - Post-processing of 3D printed models – support removal, sanding, gap filling, polishing.
- ❖ Demonstration of simple team design project.

**SOFTWARES:** CATIA, Autodesk Netfabb, Flash Print Open Source Softwares

**REFERENCES:**

1. CATIA V5 Help Manual
2. Autodesk Netfabb User Manual
3. Flash Print User Manual

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22PC2ME306) COMPUTER AIDED ENGINEERING AND ANALYSIS LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

**COURSE PRE-REQUISITES:** Mechanics of solids, Finite element Method (Co-learning)

#### COURSE OBJECTIVES:

- To introduce fundamentals of the analysis software, its features and applications
- To learn and apply the basic element types in finite element analysis
- To know the concept of discretization of continuum, loading and boundary conditions
- To analyze the mechanical systems using pre-processor and postprocessor conditions

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Analyze the structural components with one dimensional, two dimensional and three-dimensional elements

**CO-2:** Evaluate the thermal components with one dimensional and two dimensional elements

**CO-3:** Examine the dynamic cases with one dimensional and two dimensional elements

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 3    | 3    | 1    | -    | -    | 1    | 2     | 2     | 2     | 3                               | 1     |
| CO-2 | 3                     | 3    | 3    | 3    | 3    | 1    | -    | -    | 1    | 2     | 2     | 2     | 3                               | 1     |
| CO-3 | 3                     | 3    | 3    | 3    | 3    | 1    | -    | -    | 1    | 2     | 2     | 2     | 3                               | 1     |

#### LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

**FEA software - ANSYS may be used.**

**Any 10 experiments can be conducted.**

1. Analysis of bar with various cross sections and materials under static, thermal, and combined loading.
2. Analysis of Plane Truss & Spatial Truss with various cross sections and materials under static, thermal, and combined loading.

3. Analysis of beam with emphasize on sections, materials, loads (forces and moments) and supports.
  - a) ( Use of MATLAB for finding B matrix, stiffness matrix and loading matrices of beam/in plane/solid elements).
4. Static analysis of plate (isotropic and orthotropic material) with a hole to determine the deformations, the stresses to study the failure behavior and Stress Concentration factor.
5. Structural analysis in Plane stress, plane strain and axisymmetric loading on the in-plane members within plane loading.
6. Static analysis of connecting rod with tetrahedron and brick elements.
7. Static Analysis of flat and curved shell due to internal pressure and moments to estimate the strains, stresses and reaction forces and moments with different boundary conditions.
8. Steady state heat transfer analysis of one-dimensional heat transfer through fins.
9. Steady state heat transfer analysis of cross section of chimney and transient heat transfer analysis of solidification of castings.
10. Modal analysis of beams, plates and shells for natural frequencies and mode shapes.
11. Harmonic analysis of a Shaft subjected to periodic force and transient analysis of plate subjected to stepped and ramped loading with varying time.
12. Coupled field analysis (Structural & Thermal).
13. Flow analysis of pipes with different fluids/gasses/air for velocity and pressure gradients.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22PW4ME301) INTERNSHIP

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 4   | 2 |

| EVALUATION SCHEME |     |       |
|-------------------|-----|-------|
| CIE               | SEE | TOTAL |
| -                 | 100 | 100   |

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Gain exposure to the current technological developments relevant to the subject area of training

**CO-2:** Apply the technical knowledge in real industrial situations

**CO-3:** Promote academic, professional and/or personal development

**CO-4:** Demonstrate effective communication skills through oral presentation

**CO-5:** Engage in effective written communication through internship report

#### **COURSE OUTLINE:**

- A student shall undergo an internship at a reputed industry/research organization/academic institution.
- Student shall register for this course immediately after IV semester (II year II semester) End Examinations and pursue it during summer vacation/semester break & during III year without affecting the regular course work.
- Evaluation shall be done by a committee consisting of an external examiner, Head of the Department, internal supervisor and a senior faculty member of the department.
- A student shall submit a detailed report and present it before the committee for evaluation.



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22MN6HS302) GENDER SENSITIZATION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 2               | 0   | 0 |

| EVALUATION SCHEME |       |     |       |
|-------------------|-------|-----|-------|
| SE-I              | SE-II | SEE | TOTAL |
| 50                | 50    | -   | 100   |

#### COURSE DESCRIPTION:

This course offers an introduction to Gender Studies, an interdisciplinary field that asks critical questions about the meanings of sex and gender in society. The primary goal of this course is to familiarize students with key issues, questions and debates in Gender Studies, both historical and contemporary. It draws on multiple disciplines – such as literature, history, economics, psychology, sociology, philosophy, political science, anthropology and media studies – to examine cultural assumptions about sex, gender, and sexuality. This course integrates analysis of current events through student presentations, aiming to increase awareness of contemporary and historical experiences of women, and of the multiple ways that sex and gender interact with race, class, caste, nationality and other social identities. This course also seeks to build an understanding and initiate and strengthen programs combating gender-based violence and discrimination. The course also features a number of exercises and reflective activities designed to examine the concepts of gender, gender-based violence, sexuality, and rights. It will further explore the impact of gender-based violence on education, health and development

#### ACTIVITIES:

Classes will consist of a combination of activities: dialogue-based lectures, discussions, collaborative learning activities, group work and in-class assignments

#### COURSE OBJECTIVES:

- To sensitize students on issues of gender in contemporary India
- To provide a critical perspective on the socialization of men and women
- To expose the students to debates on the politics and economics of work
- To enable students to reflect critically on gender violence
- To expose students to more egalitarian interactions between men and women

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand important issues related to gender in contemporary India

**CO-2:** Attain a finer grasp of how gender discrimination works in our society and how to counter it

**CO-3:** Acquire insight into the gendered division of labour and its relation to politics and economics

**CO-4:** Respond to put an end to gender violence

**CO-5:** Equipped to work with the other gender treating them as equals

**COURSE ARTICULATION MATRIX:**

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 |                       |      |      |      |      |      |      |      |      |       |       |       |                                 |       |
| CO-2 |                       |      |      |      |      |      |      |      |      |       |       |       |                                 |       |
| CO-3 |                       |      |      |      |      |      |      |      |      |       |       |       |                                 |       |
| CO-4 |                       |      |      |      |      |      |      |      |      |       |       |       |                                 |       |
| CO-5 |                       |      |      |      |      |      |      |      |      |       |       |       |                                 |       |

**MODULE 1: Introduction to Gender**

- Definition of Gender
- Basic Gender Concepts and Terminology
- Exploring Attitudes towards Gender Social
- Construction of Gender

**MODULE 2: Gender Roles and Relations**

- Types of Gender Roles
- Gender Roles and Relationships Matrix
- Gender-based Division and Valuation of Labour

**MODULE 3: Gender Development Issues**

- Identifying Gender Issues
- Gender Sensitive Language
- Gender, Governance and Sustainable Development Gender and
- Human Rights
- Gender and Mainstreaming

**MODULE 4: Gender-based Violence**

- The concept of violence
- Types of Gender-based violence
- The relationship between gender, development and violence
- Gender-based violence from a human rights perspective

**MODULE 5: Gender and Culture**

- Gender and Film
- Gender and Electronic Media Gender
- Advertisement Gender
- Popular Literature
- Gender and the Structure of Knowledge
- Questions for Historians and Others, Reclaiming a Past, Writing Other Histories

**TEXT BOOK:**

1. Towards a World of Equals: A Bilingual Textbook on Gender, A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu, Telugu Akademi, Telangana Government, 2015

**REFERENCES:**

1. More than One Million Women are Missing, Sen, Amartya, New York Review of Books 37.20 (20 December 1990), Print 'We Were Making History' Life Stories of Women in the Telangana People's Struggle, New Delhi: Kali for Women, 1989
2. By the Numbers: Where Indian Women Work, Women's Studies Journal (14 November 2012), Tripti Lahiri, Available online at: <http://blogs.wsj.com/India/real-time/2012/11/14/by-the-numbers-where-india-women-work/>&gt
3. I Fought For My Life and Won, Abdulali Sohaila, Available online at: <http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal>
4. The Violence of Development: The Politics of Identity, Gender and Social Inequalities in India, K. Kapadia, London: Zed Books, 2002
5. Just Development: Beyond Adjustment with a Human Face, T. Banuri and M. Mahmood, Karachi: Oxford University Press, 1997

# **OPEN ELECTIVE TRACKS**

**OE TRACKS BASED ON MEZZANINE TECHNOLOGIES:**

| <b>OE TRACKS<br/>(Parent<br/>Department)</b>               | <b>V SEMESTER</b>                                                                                                                                   | <b>VI SEMESTER</b>                                                      | <b>VII SEMESTER</b>                                                          | <b>VIII SEMESTER</b>                                                            |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Smart Cities<br/>(CE)</b>                               | Smart Cities Planning and Development<br><b>(22OE1CE301)</b>                                                                                        | Green Building Technology<br><b>(22OE1CE302)</b>                        | Smart Materials and Structures<br><b>(22OE1CE401)</b>                        | Intelligent Transportation System<br><b>(22OE1CE402)</b>                        |
| <b>Waste Management<br/>(CE)</b>                           | Solid Waste Management<br><b>(22OE1CE303)</b>                                                                                                       | Hazardous Waste Management<br><b>(22OE1CE304)</b>                       | Waste to Energy<br><b>(22OE1CE403)</b>                                       | Intelligent Waste Management System and Recycling System<br><b>(22OE1CE404)</b> |
| <b>Green Energy<br/>(EEE)</b>                              | Renewable Energy sources<br><b>(22OE1EE301)</b>                                                                                                     | Renewable Energy Technologies<br><b>(22OE1EE302)</b>                    | Energy Storage Technologies<br><b>(22OE1EE401)</b>                           | Energy Management and Conservation<br><b>(22OE1EE402)</b>                       |
| <b>3D Printing &amp; Design<br/>(ME)</b>                   | Elements of CAD<br><b>(22OE1ME301)</b>                                                                                                              | Introduction to 3D Printing<br><b>(22OE1ME302)</b>                      | 3D Printing - Tooling and Systems<br><b>(22OE1ME401)</b>                     | Reverse Engineering<br><b>(22OE1ME402)</b>                                      |
| <b>Internet of Things<br/>(ECE)</b>                        | Sensors Transducers and Actuators<br><b>(22OE1EC301)</b>                                                                                            | Introduction to Microcontrollers and Interfacing<br><b>(22OE1EC302)</b> | Fundamentals of Internet of Things<br><b>(22OE1EC401)</b>                    | Wireless Sensor Networks<br><b>(22OE1EC402)</b>                                 |
| <b>Augmented Reality (AR) / Virtual Reality (VR) (ECE)</b> | Introduction to Signal Processing<br><b>(22OE1EC303)</b>                                                                                            | Introduction to Image and Video Processing<br><b>(22OE1EC304)</b>       | Fundamentals of Augmented Reality and Virtual Reality<br><b>(22OE1EC403)</b> | Virtual Reality Engineering<br><b>(22OE1EC404)</b>                              |
| <b>Microelectronics<br/>(ECE)</b>                          | Principles of Microelectronics<br>(22OE1EC305) / Fundamentals of Microelectronics<br>(22OE1EC309)                                                   | Programmable Logic Devices for Microelectronics<br>(22OE1EC306)         | Fundamentals of Physical Design for VLSI<br>(22OE1EC405)                     | Trends in Microelectronics<br>(22OE1EC406)                                      |
| <b>Artificial Intelligence<br/>(CSE)</b>                   | Applied Mathematics in Artificial Intelligence<br><b>(22OE1MT301)</b> / Mathematical Foundations for Artificial Intelligence<br><b>(22OE1MT302)</b> | Fundamentals of Artificial Intelligence<br><b>(22OE1CS303)</b>          | Machine Learning Techniques<br><b>(22OE1CS401)</b>                           | Essentials of Deep Learning<br><b>(22OE1CS402)</b>                              |
| <b>Blockchain Technologies<br/>(CSE)</b>                   | Fundamentals of Computer Networks<br><b>(22OE1CS301)</b> / Relational Data Base Management Systems<br><b>(22OE1CS302)</b>                           | Cryptography and Network Security<br><b>(22OE1IT301)</b>                | Distributed Data Bases<br><b>(22OE1CS403)</b>                                | Blockchain Technology<br><b>(22OE1CS404)</b>                                    |
| <b>Robotics<br/>(EIE)</b>                                  | Fundamentals of Robotics<br><b>(22OE1EI301)</b>                                                                                                     | Kinematics of Robots<br><b>(22OE1EI302)</b>                             | Drive Systems for Robots<br><b>(22OE1EI401)</b>                              | Intelligent Systems for Robots<br><b>(22OE1EI402)</b>                           |
| <b>Cyber Security<br/>(IT)</b>                             | Fundamentals of Computer Networks<br><b>(22OE1CS301)</b> / Relational Data Base Management Systems<br><b>(22OE1CS302)</b>                           | Cryptography and Network Security<br><b>(22OE1IT301)</b>                | Essentials of Cyber Security<br><b>(22OE1IT401)</b>                          | Computer Forensics<br><b>(22OE1IT402)</b>                                       |
| <b>Data Sciences / Big Data &amp; Analytics<br/>(IT)</b>   | Statistical Methods for Data Science<br><b>(22OE1MT303)</b>                                                                                         | Computational Thinking using Python<br><b>(22OE1IT302)</b>              | Fundamentals of Data Mining<br><b>(22OE1IT403)</b>                           | Data Analysis and Visualization<br><b>(22OE1IT404)</b>                          |

|                                  |                                                             |                                                       |                                                                |                                                          |
|----------------------------------|-------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------|
| <b>Autonomous Vehicles (AME)</b> | Principles of Automobile Engineering<br><b>(22OE1AE301)</b> | Modern Automotive Technologies<br><b>(22OE1AE302)</b> | Electric, Hybrid and Fuel Cell Vehicles<br><b>(22OE1AE401)</b> | Connected and Autonomous Vehicles<br><b>(22OE1AE402)</b> |
|----------------------------------|-------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------|

#### GENERAL POOL OF OE COURSES:

| <b>OE TRACKS<br/>(Parent Departments)</b> | <b>COURSES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General-Computing<br/>(CSE / IT)</b>   | <ul style="list-style-type: none"> <li>• Programming Through JAVA <b>(22OE1IT303)</b></li> <li>• Relational Data Base Management Systems <b>(22OE1CS302)</b></li> <li>• Computational Thinking using Python <b>(22OE1IT302)</b></li> <li>• Introduction to Data Analytics <b>(22OE1IT304)</b></li> <li>• Fundamentals of Computer Algorithms <b>(22OE1CS305)</b></li> <li>• Introduction to Front End Technologies <b>(22OE1CS306)</b></li> <li>• Concepts of Generative AI <b>(22OE1CS307)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>General<br/>(H&amp;S)</b>              | <ul style="list-style-type: none"> <li>• Professional Ethics and Human Values <b>(22OE1HS301)</b></li> <li>• Entrepreneurship <b>(22OE1HS302)</b></li> <li>• Foreign Language-French <b>(22OE1HS401)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>General</b>                            | <ul style="list-style-type: none"> <li>• Smart Cities <b>(22OE1CE305)</b></li> <li>• Trends in Energy Sources for Sustainable Development <b>(22OE1EE303)</b></li> <li>• 3D Printing and Design <b>(22OE1ME303)</b></li> <li>• Advanced Materials <b>(22OE1ME304)</b></li> <li>• 5G/6G Technologies <b>(22OE1EC307)</b></li> <li>• Fundamentals of VLSI <b>(22OE1EC308)</b></li> <li>• AI for Beginners <b>(22OE1CS304)</b></li> <li>• Blockchain Technology Essentials <b>(22OE1CS309)</b></li> <li>• Fundamentals of Robotics and Drones <b>(22OE1EI303)</b></li> <li>• Introduction to Cloud Computing <b>(22OE1IT305)</b></li> <li>• Fundamentals of Computer Vision <b>(22OE1IT306)</b></li> <li>• Introduction to Advanced Vehicle Technologies <b>(22OE1AE303)</b></li> <li>• Green Hydrogen <b>(22OE1AE403)</b></li> <li>• Introduction to Application Development with C# <b>(22OE1CS310)</b></li> <li>• Introduction to Application Development with Java <b>(22OE1CS311)</b></li> <li>• Introduction to Application Development with Python <b>(22OE1CS312)</b></li> <li>• Nanoscience and Technology for Engineers <b>(22OE1PH301)</b></li> <li>• Essentials For Quantum Computing <b>(22OE1PH302)</b></li> <li>• Introduction to Climate Change <b>(22OE1CH301)</b></li> <li>• Sustainable Chemistry for Engineering <b>(22OE1CH302)</b></li> <li>• Storytelling For Effective Communication <b>(22OE1ENH301)</b></li> <li>• Film Analysis and Critical Appreciation <b>(22OE1EN302)</b></li> <li>• Creative Writing <b>(22OE1EN303)</b></li> <li>• Cross-Cultural Communication Through World Literature <b>(22OE1EN401)</b></li> <li>• Business Communication <b>(22OE1EN402)</b></li> <li>• Fundamentals of Management <b>(22OE1MG301)</b></li> <li>• Personal Finance and Tax Planning <b>(22OE1MG302)</b></li> <li>• Corporate Finance <b>(22OE1MG303)</b></li> <li>• Cost And Management Accountancy <b>(22OE1MG304)</b></li> <li>• Human Resource Management <b>(22OE1MG305)</b></li> <li>• Marketing Management <b>(22OE1MG306)</b></li> <li>• Numerical Methods and Linear Programming <b>(22OE1MT304)</b></li> <li>• Optimization Techniques <b>(22OE1MT305)</b></li> </ul> |

# **OPEN ELECTIVE TRACKS**

# **SMART CITIES**





Offered by:

**CIVIL ENGINEERING**

**Courses in the OE Track:**

| OE Tracks           | V Sem<br>(OE-I)                             | VI Sem<br>(OE-II)               | VII Sem<br>(OE-III)                     | VIII Sem<br>(OE-IV)                     |
|---------------------|---------------------------------------------|---------------------------------|-----------------------------------------|-----------------------------------------|
| <b>Smart Cities</b> | Smart Cities<br>Planning and<br>Development | Green<br>Building<br>Technology | Smart<br>Materials<br>and<br>Structures | Intelligent<br>Transportation<br>System |

## SMART CITIES

In the twenty-first century, engineers are being tasked with solving ever more complex and subtle societal challenges – from climate change to unprecedented urbanisation that is materially affecting the lives of many urban populations. As engineers become ever more interdisciplinary and the boundaries of disciplines soften, they need to reflect as a community as to the appropriateness of the engineering paradigm to address these needs. Currently the engineering community is pointing to the digital technologies and the 'smart city' as a deliverer of efficiency and resilience without fully acknowledging the intricate socio-political context in which it is situated.

The domain of EIE was developed to modernise and automate these operations using the technological advancements in the realm of electronics. Even outside the industry, common household appliances — such as washing machine, air-conditioner, geyser, and microwave oven — cannot attract customers without features such as auto cut-off after certain time or temperature, which is again an example of instrumentation. The field of Instrumentation Engineering is also core to the recent advances such as smart home appliances, smart cities and automobiles. It is thus not far from the truth to claim that the fourth industrial revolution.

The world population is continuously growing and reached a significant evolution of the society, where the number of people living in cities surpassed the number of people in rural areas. This puts national and local governments under pressure because the limited resources, such as water, electricity, and transports, must thus be optimized to cover the needs of the citizens. Therefore, different tools, from sensors to processes, service, and artificial intelligence, are used to coordinate the usage of infrastructures and assets of the cities to build the so-called smart cities.

Different definitions and theoretical models of smart cities are given in literature. However, smart city can usually be modelled by a layered architecture, where communication and networking layer plays a central role. In fact, smart city applications lay on collecting field data from different infrastructures and assets, processing these data, taking some intelligent control actions, and sharing information in a secure way. Thus, a two-way reliable communications layer is the basis of smart cities. This chapter introduces the basic concepts of this field and focuses on the role of communication technologies in smart cities. Potential technologies for smart cities are discussed, especially the recent wireless technologies adapted to smart city requirements.

### ***What is the concept of a smart city?***

There is no universally accepted definition for a smart city because people can interpret different meanings for it. Hence, it means different things to different people. Here, you will get a basic definition that captures the essence of what a smart city is and what it does. While the concept varies from area to area depending on the resources, the basic idea behind it remains the same. A smart city aims to bring various components together to live harmoniously and attempts to do with the least environmental damage or impact. In other words, a smart city is a place with high standards of living, which survives and thrives on eco-friendly means. The size and amenities within a smart city vary according to geography, resources available, geopolitical scenario and investment received.

Growth in Global population continues to drive citizens from rural areas to cities. With rapid expansion of urban areas, cities need to become intelligent to handle this large scale urbanization. This is driving city operators to look at smarter ways to manage complexities, increase efficiencies and improve quality of life. Today we need cities that monitor & integrate infrastructure to better optimize resources while maximizing service to its citizens. So to meet all the needs we need our cities to be smarter which brings a concept "**Smart cities**" Smart cities optimize the use of technology in the design & operation of infrastructure and buildings in such a way which meets the current and future needs of their citizens. To be truly smart they also require consideration of governance & growth, urban development and infrastructure, the environment & natural resources, society and community.

Smart city programs provide a range of technologies that can be applied to solve infrastructure problems associated with ageing infrastructure and increasing demands. The potential for infrastructure and urban improvement remains unrealized, however, due to technical, financial, and social constraints and criticisms that limit the implementation of smart cities concepts for infrastructure management. The discussion presented here provides a review of smart technologies including sensors, crowdsourcing and citizen science, actuators, data transmission, Internet of Things, big data analytics, data visualization, and blockchain, which can be used for infrastructure management. Smart infrastructure programs are reviewed to explore how enabling technologies have been applied across civil engineering domains, including transportation systems, water systems, air quality, energy infrastructure, solid waste management, construction engineering and management, structures, and geotechnical systems.

Making cities "smarter" by efficient management of resources and infrastructure, greener environment, and smart governance resulting in a better quality of living of its citizens. This can be enabled by the effective use of information and communication technologies (ICTs) tools, which have the ability to provide eco-friendly and economically viable solutions for cities.

Setting up a smart city is more than improving the old system with technology by simply adding sensors, remote supervision, and control to essential city services. It should be a complete shift of a paradigm in daily life when using new technologies, especially new ICT leading to smart outcomes.

### **Smart solutions**

Another important feature of smart cities is that they will provide smart solutions to modern problems. These include:

- Public information systems
- Redressal of grievances
- Electronic service delivery
- Maximum engagement of citizens
- Reduced energy and fuel usage
- Reduces the development of wastes
- Smart water monitoring
- Treatment of wastewater
- Sustainable monitoring water quality



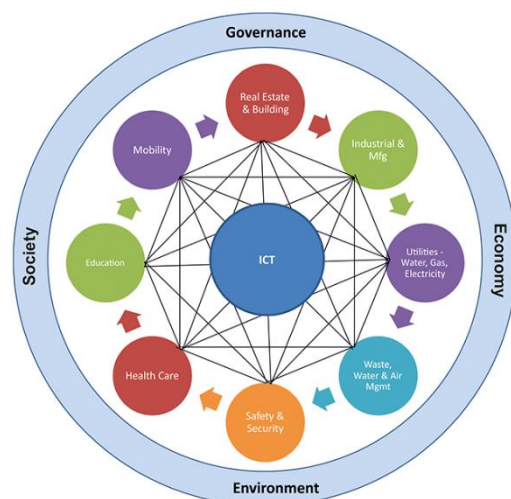
- Maximum utilization of renewable energy sources
- Usage of green building techniques
- Smart parking to reduce clutter
- Intelligent traffic management system.

### **Advantages of a smart cities,**

1. Promotion of mixed land usage resulting in higher efficiency and reduced wastage of land.
2. Expanded housing opportunities.
3. Reduced congestion, air pollution and resource depletion.
4. Helps to boost local economies by promoting localized trade and interactions.
5. Efficient use of public transport to reduce fuel wastage.
6. Safe and secure localities.
7. Preservation of open spaces.
8. Reduction in urban heating.
9. Promotion of transit-oriented development.
10. Making governance more people-friendly and cost-effective.

Here's a look at some projects that have taken inspiration from the concepts used for the design of smart cities. These projects will help you build energy-efficient systems that will help heal the world.

1. **Home Automation using IoT**
2. **Smart Irrigation System**
3. **Smart Building using IoT**
4. **Smart Energy Meter using GSM**
5. **Solar and Smart Energy Systems**
6. **Smart Water Monitoring**
7. **Automated Street Lighting**
8. **Automated Railway Crossing**
9. **Intelligent Transportation Systems**
10. **Smart Sewage Maintenance Systems.**



To develop new smart cities and to transform our cities into smart cities the engineers in particular are stepping up as leaders.

**Civil & Environmental Engineers** are working to harness the potential of latest technologies and data for our urban infrastructure, which is among the most complex system in the world. They provide sustainable, resilient and advanced means of transportation system, green building, better water management system and better waste management system. This not only develop physical infrastructure but also develop institutional & social infrastructure that enable our societies to function. Modelling these systems of systems will require managing data at an unprecedented scale.

To support them Computer and **Electronics & Communication Engineers** help in creating future cities that are digital, build and operate cities ICT landscape across application and infrastructure like IOT (Internet of Things), e-payment, e-market, the latest communication devices etc which is leveraging next generation technologies.

They create a platform for conveyance of different city services, leverage big data analytics to manage city performance and proactive crisis management.

**Electrical Engineers** developing new renewable source of energy to meet ever increasing power demands. They also develop methods of effective power transmission with minimum losses which is more economical and safer. They also work on developing microchips to micro sensors which are helping in making our households, institution efficient and safer.

### **Conclusion**

It is clear that dreaming of a smart city without active contribution of engineers is a myth. So, there will always be demand of Engineers and because of which even after crises in the placement scenario still the maximum science students choose Engineering as their first career choice in hope of a better future.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CE301) SMART CITIES PLANNING AND DEVELOPMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To Introduce smart city basic concepts, global standards and Indian context of smart cities
- To understand smart community, smart transportation and smart buildings
- To understand energy demand, green approach to meet energy demand and their capacities
- To identify smart transportation technologies in cities and concepts towards smart city

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Recognize smart city concepts and their international and national standards

**CO-2:** Recognize smart community, transportation and building concepts

**CO-3:** Develop and calibrate energy demand and their capacity limits

**CO-4:** Predict the various smart urban transportation systems and the transition from existing city towards a smart city

#### UNIT-I:

**Introduction to Smart Urban Infrastructures and Smart Cities:** Introduction to City Planning - Understanding Smart Cities - Dimensions of Smart Cities - Global Experience of Smart Cities – Global Standards and Performance Benchmarks, Practice Codes - Indian scenario - India “100 Smart Cities” Policy and Mission.

#### UNIT-II:

**Smart Cities Planning and Development:** Introduction to Smart Community - Smart community concepts: Concept of Smart Community - Smart Transportation - Smart Building and Home Device - Smart Health - Smart Government - Smart Energy and Water – Cyber Security, Safety, and Privacy - Internet of Things, Blockchain, Artificial Intelligence, Alternate Reality, Virtual Reality.

#### UNIT-III:

**Smart Urban Energy Systems:** Conventional vs. Smart, City components, Energy demand, Green approach to meet Energy demand, Index of Indian cities towards smartness – A statistical analysis -Meeting energy demand through direct and indirect solar resources, Effectiveness in responsive environment in smart city; Smart communication using green resources, Introduction to PV technology- Policies of Solar PV in smart domains - Smart Grid- Indian Perspective- Advantage & limitation.

**UNIT-IV:**

**Smart Urban Transportation Systems:** Smart Transportation Technologies - Driverless and connected vehicles - Ride sharing solutions - The "improve" pathway - The "shift" pathway – Smart Roads and Pavement systems.

**UNIT-V:**

**Towards Smart Cities:** The transition of legacy cities to Smart -. Right transition process- The benefit of citizens, cities to adopt effective management and governance approaches - Factors in the transition phase of legacy cities to smart cities and their managerial implications.

**TEXT BOOKS:**

1. Internet of Things in Smart Technologies for Sustainable Urban Development, G. R. Kanaga Chidambaresan, R. Maheswar, V. Manikandan, K. Ramakrishnan, Springer, 2020
2. Society 5.0: A People-centric Super-smart Society, Hitachi-U, Tokyo Laboratory (H-U Tokyo Lab), Springer, 2020
3. The Routledge Companion to Smart Cities, Katharine S. Willis, Alessandro Aurigi, Routledge International Handbooks, 2020

**REFERENCES:**

1. Smart Cities in Asia: Governing Development in the Era of Hyper-Connectivity Yu-min Joo, Yu-Min Joo, Teck-Boon Tan, Edward Elgar Pub, 2020
2. Urban Systems Design: Creating Sustainable Smart Cities in the Internet of Things Era, Yoshiki Yamagata, Perry P. J. Yang, Elsevier, 2020
3. Smart Cities and Artificial Intelligence: Convergent Systems for Planning, Design, and Operations, Christopher Grant Kirwan, Zhiyong Fu, Elsevier, 2020

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1CE302) GREEN BUILDING TECHNOLOGY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Smart Cities Planning and Development

#### **COURSE OBJECTIVES:**

- To expose to green buildings, their features and importance in the present context of sustainable development
- To introduce various sustainable building materials for green buildings
- To acquire knowledge on various design concepts and construction aspects of green buildings
- To learn the various policies and incentives for green buildings and also different green building rating systems and codes

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the importance, features and requisites of a green building

**CO-2:** Identify suitable sustainable building materials for construction of green building

**CO-3:** Apply the fundamental knowledge on design and construction techniques for green buildings

**CO-4:** Implement the acquired knowledge of policies and rating systems for green buildings

#### **UNIT-I:**

**Introduction to Green Buildings:** Sustainable Development - Definition and Importance – United Nations Sustainable Development Goals (UN SDG's) - Definition of green buildings - Typical features of green buildings - Benefits of green buildings - Key requisites for constructing green buildings - Important sustainable features for green buildings - Climate responsive buildings - Carbon footprint and Eco footprints of buildings.

#### **UNIT-II:**

**Green Building Materials:** Introduction to sustainable building materials - Sustainable concrete - Partial replacements in concrete - Use of industrial waste such as fly-ash, GGBS, silica fume, rice husk ash, construction & demolition waste - Natural building materials - Bio materials - Mycelium - Engineered wood - Structural Insulated Panels (SIPs) - Natural fiber – Non-toxic materials: low VOC paints, organic paints, coating and adhesives - Use of waste materials such as paper, cellulose, glass bottles, tires, shipping containers.



### **UNIT-III:**

**Design of Green Buildings:** Indoor environmental quality requirement and management: Thermal comfort - HVAC - Visual perception - Illumination requirement - Auditory requirement - Energy Efficiency - Lighting and day lighting - Steady and non-steady heat transfer through the glazed window and the wall - Indoor air quality - Local climatic conditions - temperature, humidity, wind speed and direction.

### **UNIT-IV:**

**Construction of Green Buildings:** IoT integrated automated building systems - Synthetic roof underlayment - Green roofs - Grid hybrid system - Passive solar - Greywater plumbing systems - Electrochromic glass - Solar thermal cladding - Structural 3D printing - Self-healing concrete - Bird friendly design - Landscaping for parking lot runoff - Composting toilets - Proactive maintenance - Green cleaning.

### **UNIT-V:**

**Green Building Policies and Incentives:** Green products and material certification - parameters making products green - products transparency movement - Cradle to cradle certification - Product emission testing - Carbon trust - Carbon credit - Returns on investments - Savings - Policies towards electrical power in India - Tax credits and Grants.

**Green Building Rating Systems:** Overview of green building rating systems: BREAM, LEED and GRIHA.

### **TEXT BOOKS:**

1. Green Building Handbook: Volume 1, A Guide to Building Products and their Impact on the Environment, Tom Woolley, Sam Kimmins, Rob Harrison, Paul Harrison, Taylor & Francis
2. Sustainable Construction: Green Building Design and Delivery, 5th Edition by Charles J. Kibert, Wiley Publishing

### **REFERENCES:**

1. **Green Building** Fundamentals: A Practical Guide to Understanding and Applying Fundamental Sustainable Construction Practices and the LEED System, Pearson, USA
2. Sustainable Construction and Design-II, Regina Leffers, Pearson / Prentice Hall, 2009

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22OE1CE401) SMART MATERIALS AND STRUCTURES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Smart Cities Planning and Development, Green Building Technology

#### COURSE OBJECTIVES:

- To introduce various smart materials and their working principles
- To acquire knowledge on different measuring techniques
- To learn about various smart sensors, actuators and their application in structural health monitoring
- To acquire knowledge on different smart composite materials and their modelling concepts
- To learn about the data acquisition and processing and their application in engineering domain

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the different smart materials and their principles

**CO-2:** Explain and understand different measuring techniques

**CO-3:** Identify suitable smart sensors and actuators for a specific engineering application

**CO-4:** Gain the knowledge on data acquisition and processing and advantages in smart materials and smart structures

#### UNIT-I:

**Introduction:** Introduction to Smart Materials and Structures – Instrumented structures functions and response – Sensing systems – Self-diagnosis – Signal processing consideration – Actuation systems and effectors.

#### UNIT-II:

**Measuring Techniques:** Measuring techniques: Strain Measuring Techniques using Electrical strain gauges, Types – Resistance – Capacitance – Inductance – Wheatstone bridges – Pressure transducers – Load cells – Temperature Compensation – Strain Rosettes.

#### UNIT-III:

**Sensors & Actuators:** Sensing Technology – Types of Sensors – Physical Measurement using Piezo Electric Strain measurement – Inductively Read Transducers – LVDT – Fiber optic Techniques- Application of Smart Sensors for Structural Health Monitoring (SHM), System Identification using Smart Sensors

**Actuators:** Actuator Techniques – Actuator and actuator materials – Piezoelectric and Electrostrictive Material – Magneto structure Material – Shape Memory Alloys – Electro rheological fluids – Electromagnetic actuation – Role of actuators and Actuator Materials - IPMC and Polymeric Actuators, Shape Memory Actuators

**UNIT-IV:**

**Signal Processing and Control Systems:** Data Acquisition and Processing – Signal Processing and Control for Smart Structures – Sensors as Geometrical Processors – Signal Processing – Control System – Linear and Non-Linear

**UNIT-V:**

**Advances in Smart Structures & Materials:** Self-Sensing Piezoelectric Transducers, Energy Harvesting Materials, Autophagous Materials, Self-Healing Polymers, Intelligent System Design, Emergent System Design

**TEXT BOOKS:**

1. Smart Materials and Structures, Gandhi M. V. and Thompson B. S., Chapman & Hall, 1992
2. Dynamics and Control of Structures, Meirovitch L., John Wiley, 1992

**REFERENCES:**

1. Smart Structures: Analysis and Design, A. V. Srinivasan, D. Michael McFarland, Cambridge University Press, 2009
2. Smart Materials and Technologies: For the Architecture and Design Professions, Michelle Addington and Daniel L. Schodek, Routledge, 2004
3. Smart Structures and Materials, Brian Culshaw, Artech House – Borton, 1996

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1CE402) INTELLIGENT TRANSPORTATION SYSTEM

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Smart Cities Planning and Development, Green Building Technology, Smart Materials and Structures

#### COURSE OBJECTIVES:

- To understand ITS architecture and standards
- To apply appropriate ITS technology depending upon site specific conditions
- To design and implement ITS components
- To understand concept and application of Automated Highway Systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Differentiate different ITS user Services

**CO-2:** Apply ITS for road user safety

**CO-3:** Interpret importance of AHS in ITS

**CO-4:** Extend future research and special project

#### UNIT-I:

**Introduction To ITS:** System Architecture, Standards, Database – Tracking Database – Commercial Vehicle Operations – Intelligent Vehicle Initiative - Metropolitan ITS – Rural ITS – ITS for Rail network.

#### UNIT-II:

**ITS Travel Management:** Autonomous Route Guidance System – Infrastructure based systems – Telecommunications – Vehicle – Roadside communication – Vehicle Positioning System – Electronic Toll Collection – Electronic Car Parking

#### UNIT-III:

**ITS Designs:** Modeling and Simulation Techniques - Peer – to – Peer Program – ITS for Road Network – System Design – Mobile Navigation Assistant – Traffic Information Center – Public Safety Program.

#### UNIT-IV:

**Automated Highway Systems:** Evolution of AHS and Current Vehicle Trends - Vehicles in Platoons – Aerodynamic Benefits - Integration of Automated Highway Systems – System Configurations - Step by Step to an Automated Highway System- Spacing and Capacity for Different AHS Concepts – Communication Technologies for AHS - The Effects of AHS on the Environment – Regional Mobility - Impact Assessment of Highway Automation.

**UNIT-V:**

**Implementation of ITS:** ITS programs globally- overview of ITS in developed countries and developing countries – ITS at Toll Plazas – Parking lots – Highways.

**TEXT BOOKS:**

1. Intelligent Transport Systems Handbook: Recommendations for World Road Association (PIARC), Kan Paul Chen, John Miles, 2000
2. Intelligent Transport Systems – Cases and Policies, Roger R. Stough, Edward Elgar, 2001
3. Intermodal Freight Transport, David Lowe, Elsevier Butterworth-Heinemann, 2005

**REFERENCES:**

1. Positioning Systems in Intelligent Transportation Systems, Chris Drane and Chris Rizos, Artech House Publishers, 2000
2. Perspectives on Intelligent Transport Systems, Joseph M. Sussman, Springer, 2000
3. Intelligent Transport System, Intelligent Transportation Primer, 2001

# **WASTE MANAGEMENT**



## WASTE MANAGEMENT

Offered by:

**CIVIL ENGINEERING**

Courses in the OE Track:

| OE Tracks                   | V Sem<br>(OE-I)           | VI Sem<br>(OE-II)                | VII Sem<br>(OE-III) | VIII Sem<br>(OE-IV)                                              |
|-----------------------------|---------------------------|----------------------------------|---------------------|------------------------------------------------------------------|
| <b>Waste<br/>Management</b> | Solid Waste<br>Management | Hazardous<br>Waste<br>Management | Waste to<br>Energy  | Intelligent<br>Waste<br>Management<br>and<br>Recycling<br>System |

## **WASTE MANAGEMENT**

The courses such as solid waste management (SWM), hazardous waste management (HWM), waste to energy (WTE) and intelligent waste management and recycling system (IWM&RS) are the courses available in the waste management track stream which having a potential syllabus content to meet out the industrial and research needs.

Solid waste management is an interesting track course which actual highlights the day-to-day problems where everybody is facing due to the improper management of industrial, domestic and household waste. Further, the enthusiastic aspects involved in the track courses such as: awareness on its impact over on environment, formal or scientific way of handling and management of waste and disposal scenarios.

In hazardous waste management course, handling and management of nuclear waste at national and international level have been highlighted. Further, the content enlightens about the legal process of state, central and industrial responses toward any emergency situations arise by hazardous waste. Finally, it deals about natural resource damage assessment and restoration.

Waste to energy is a pioneering course available in the track; it is one of the interesting and mindboggling course in the track which highlights the importance of converting the waste materials into wealth. It gives enough space to understand the basic process technologies in a theoretical and industrial way such as: thermal, chemical and biological conversion process. From the above, biological conversion process is in its embryonic state and having potential to expands its technological wings in the near future and having enormous scope of industrial applications where students can be benefited. Finally, conversion devices is an innovative module have been framed to explore the young minds in the line of designing and creating a demand based conversion device products which even lays an entrepreneurial pathway to them.

First of its kind, even at both international and national level a dedicated and extensive course for intelligent waste management and recycling system have been framed with conventional and advanced modules. It is really an interesting course where a student can apply his/her innovative creations to solve the existing and futuristic problems in a smart way with the help of smart tools. Optimistic modules such as: life cycle assessment and carbon-footprint-based IWMS, principles of systems engineering and regulatory frameworks have been incorporated to meet out the international requirements.

In the pathway of exploring the fundamentals and basic knowledges about the course, the six units of all the courses have been formulated keeping in the mind that the students can be able to competitive among the international community at the end of semester. In this context, comprehensive theoretical and industrial processes have been incorporated in each and every module of courses. Further, it is highly believed that the framed syllabus modules having 100% industrial applications which can make the students to feel motivated, satisfied and confidence to compete with the international community.



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CE303) SOLID WASTE MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn about different types of solid and hazardous wastes, their sources, classification systems, and regulatory framework for management
- To understand techniques for waste sampling, characterization, and minimization through reduce-reuse-recycle concepts
- To study various storage, collection, and transportation systems for efficient and safe handling of different waste streams
- To impart knowledge on biological, thermal, and other technologies for processing and treatment of solid and hazardous wastes

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Classify different types of wastes and interpret regulatory policies on their handling

**CO-2:** Select appropriate waste sampling and characterization methods and design waste minimization programs

**CO-3:** Plan and design efficient storage, collection, and transportation systems for municipal solid wastes

**CO-4:** Compare applicable biological and thermal waste treatment technologies for energy and resource recovery and disposal

#### UNIT-I:

**Sources, Classification and Regulatory Framework:** Types and Sources of solid and hazardous wastes - Need for integrated solid and hazardous waste management – Salient features of Indian legislations on management and handling of municipal solid wastes, hazardous wastes, biomedical wastes, e-wastes, plastic wastes, construction & demolition wastes, nuclear wastes - Elements of circular economy and integrated waste management - Roles and responsibilities of stakeholders – Financing and Public Private Partnerships for waste management.

#### UNIT-II:

**Waste Characterization and Minimization:** Waste generation rates and variation - Composition, physical, chemical, and biological properties of solid wastes – Hazardous Characteristics and TCLP tests – tools and software for waste sampling and characterization - Waste minimization techniques - Extended Producer Responsibility - Recycling, recovery, and reuse strategies - Concept of zero waste.

**UNIT-III:**

**Storage, Collection and Transportation of Wastes:** Onsite handling, storage, and segregation - Storage and collection of municipal solid wastes – Analysis of collection systems - Transfer stations design, operation, and optimization - Compatibility considerations, labelling, manifest system and safety transportation of hazardous wastes.

**UNIT-IV:**

**Waste Processing Technologies:** Objectives of waste processing – material separation and processing technologies – biological and chemical conversion technologies – methods and controls of Composting - thermal conversion technologies and energy recovery – incineration – solidification and stabilization of hazardous wastes.

**UNIT-V:**

**Waste Disposal:** Waste disposal options – Disposal in landfills - Landfill Classification, types and methods – site selection - design and operation of sanitary landfills, secure landfills – leachate and landfill gas management – landfill closure and environmental monitoring – Rehabilitation of open dumps – landfill remediation

**TEXT BOOKS:**

1. Integrated Solid Waste Management, George Tchobanoglous, Hilary Theisen and Samuel A. Vigil, McGraw-Hill International Edition, 1993
2. Handbook of Solid Waste Management, George Tchobanoglous and Frank Kreith, McGraw- Hill, 2002
3. Solid Waste Management, K. Sasi Kumar & S. Gopi Krishna, Prentice-Hall, 2009

**REFERENCES:**

1. Management of Municipal Solid Waste, T. V. Ramachandra, The Energy and Resources Institute, TERI, 2009
2. Municipal Solid Waste Management in India, Subhrabaran Das and Korobi Gogoi, VDM Verlag, 2010
3. Solid Waste Engineering, William A. Worrell and P. Arne Vesilind, 2<sup>nd</sup> Edition, Cengage Learning, 2000
4. CPHEEO, Manual on Municipal Solid Waste Management, Central Public Health and Environmental Engineering Organization, Government of India, New Delhi, 2000

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22OE1CE304) HAZARDOUS WASTE MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Solid Waste Management

#### **COURSE OBJECTIVES:**

- To understand the concepts of hazardous waste management
- To understand the principle of waste characterization, storage, transport and processing
- To understand the principles of nuclear waste and Hazardous Management (HM) and emergency Response
- To understand the principle and process of landfills and natural resource Damage Assessment & Restoration

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply the fundamental concepts of hazardous waste management

**CO-2:** Apply the knowledge to resolve the problems on storage, transport and processing

**CO-3:** Apply the knowledge to resolve the practical problems on nuclear waste and HM & emergency response

**CO-4:** Impart the gained knowledge and skills to resolve the practical problems on landfills and natural resource damage assessment & restoration on field

#### **UNIT-I:**

**Introduction to Hazardous waste management:** Need for hazardous waste management – Sources of hazardous wastes – Effects on community – classification – Storage and collection of hazardous wastes – Problems in developing countries – Protection of public health and the environment.

#### **UNIT-II:**

**Waste Characterization, Storage, Transport and Processing:** Hazardous Waste Characterization– Hazardous waste inventory- Source reduction of hazardous wastes - Handling and storage of Hazardous wastes –Waste Compatibility Chart – – Hazardous waste treatment technologies – Physical, chemical and thermal treatment of hazardous waste – Solidification – Chemical fixation – Encapsulation – Incineration.

#### **UNIT-III:**

**Nuclear Waste:** Characteristics – Types – Nuclear waste – Uranium mining and processing – Power reactors – Refinery and fuel fabrication wastes – spent fuel –

Management of nuclear wastes – Decommissioning of Nuclear power reactors – Health and environmental effects.

#### **UNIT-IV:**

**Management of Hazardous Wastes:** Identifying a hazardous waste – methods – Quantities of hazardous waste generated – Components of a hazardous waste management plan – Hazardous waste minimization – Disposal practices in Indian Industries – Future challenges - Emergency Response - National Response Team and Regional Response Teams; National Contingency Plan and Regional Contingency Plans; National Response Center; State, Local and Industry Response Systems, Secured Landfill.

#### **UNIT-V:**

**Natural Resource Damage Assessment and Restoration:** Natural Resource Damage Assessment Laws and Regulations - Central and State government agencies - Damage Assessment and Restoration Procedures - Groundwater Hydrology and Contamination Processes - Groundwater Contamination Detection, Analysis and Monitoring - Hazard Ranking System - Remedial Investigations and Feasibility Studies.

#### **TEXT BOOKS:**

1. Hazardous Waste Management, Charles A. Wentz., 2nd Edition, McGraw-Hill International, 1995
2. Standard Handbook of Hazardous Waste Treatment and Disposal, Harry M. Freeman, McGraw-Hill, 1997

#### **REFERENCES:**

1. Hazardous Waste (Management and Transboundary Movement) Rules, Ministry of Environment and Forests, Government of India
2. Guidelines and Criteria for Hazardous Waste Landfills and Hazardous Waste Treatment Disposal Facilities, Central Pollution Control Board, 2010
3. Hazardous Waste Management, Prof. Anjaneyulu
4. Hazardous Waste Management, M. LaGrega and others, McGraw-Hill Publication

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22OE1CE403) WASTE TO ENERGY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Solid Waste Management, Hazardous Waste Management

#### COURSE OBJECTIVES:

- To learn about different types of waste as potential fuel and related conversion technologies for energy recovery
- To understand fundamental principles and systems involved in thermal waste processing methods like incineration, gasification etc.
- To study various chemical processes and technologies for conversion of biomass and waste to fuel and other products
- To impart knowledge on biological waste treatment methods and biomass energy technologies

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Ability to identify and characterize suitable waste derived fuels and assess environmental impacts of waste-to-energy systems

**CO-2:** Ability to analyze key parameters and design considerations for thermal waste processing systems

**CO-3:** Ability to select appropriate chemical conversion methods for production of hydrocarbons, biodiesel etc. from biomass waste

**CO-4:** Ability to understand kinetics and design bio-digestion systems for organic waste treatment and energy recovery

#### UNIT-I:

**Introduction to Energy from Waste:** Classification of waste as fuel – Agro based, forest residue, industrial waste, MSW – conversion devices – incinerators, gasifiers, digesters, Environmental monitoring system for land fill gases, Environmental impacts; Measures to mitigate environmental effects due to incineration.

#### UNIT-II:

**Thermal Conversion Technologies:** Fundamentals of thermal processing – combustion system – pyrolysis system – gasification system – environmental control system – energy recovery system – incineration. Briquetting technology: Production of RDF and briquetted fuel.

#### UNIT-III:

**Chemical Conversion Technologies:** Acid & Alkaline hydrolysis – hydrogenation; solvent extraction of hydrocarbons; solvolysis of wood; biocrude; biodiesel production via chemical process; catalytic distillation; transesterification methods; Fischer Tropsch

diesel: chemicals from biomass - various chemical conversion processes for oil, gas, cellulose acetate.

#### **UNIT-IV:**

**Biological Conversion Technologies:** Nutritional requirement for microbial growth – types of microbial metabolism – types of microorganisms – environmental requirements – aerobic biological transformation – anaerobic biological transformation – aerobic composting – low solid anaerobic digestion – high solid anaerobic digestion – development of anaerobic digestion processes and technologies for treatment of the organic fraction of MSW – Biodegradation and biodegradability of substrate; biochemistry and process parameters of bio methanation - other biological transformation processes.

#### **UNIT-V:**

**Biomass Energy Technologies:** Biomass energy resources – types and potential; Energy crops - Biomass characterization (proximate and ultimate analysis); Biomass pyrolysis and gasification; Biofuels – biodiesel, bioethanol, Biobutanol; Algae and biofuels; Pellets and bricks of biomass; Biomass as boiler fuel; Social, economic, and ecological implications of biomass energy.

#### **TEXT BOOKS:**

1. Integrated Solid Waste Management, George Tchobanoglous, Hilary Theisen and Samuel A, Vigil, McGraw-Hill International Edition, 1993
2. Energy from Waste - An Evaluation of Conversion Technologies, C. Parker and T. Roberts, Elsevier Applied Science, London, 1985

#### **REFERENCES:**

1. Introduction to Biomass Energy Conversion, Capareda S., CRC Press, 2013
2. Thermo-chemical Processing of Biomass: Conversion into Fuels, Chemicals and Power, Brown RC and Stevens C, Wiley and Sons, 2011
3. Biomass Conversion Processes for Energy and Fuels, Sofer, Samir S. (Ed.), Zaborsky, R. (Ed.), Plenum Press, 1981
4. Energy Recovery from Municipal Solid Waste Thermal Conversion Technologies, P. Jayarama Reddy, CRC Press, Taylor & Francis Group, 2016

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22OE1CE404) INTELLIGENT WASTE MANAGEMENT SYSTEM AND RECYCLING SYSTEM

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Solid Waste Management, Hazardous Waste Management, Waste to Energy

#### COURSE OBJECTIVES:

- To understand the concepts of IWMS Tools
- To understand applications of IWMS Tools in SWM
- To Understand the concepts of Life Cycle Assessment and Carbon-Footprint-Based IWMS
- To understand the applications of IoT, ML, DL, BC and LCA & Carbon Foot Print (CFP) based SWM

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the concepts of IWMS tools

**CO-2:** Acquire the knowledge to resolve the practical problems with the help of IWMS Tools.

**CO-3:** To analyze Life Cycle Assessment and Carbon-Footprint-Based IWMS

**CO-4:** Apply the knowledge of IoT, ML, DL, BC and LCA & CFP to resolve the practical problems in SWM

#### UNIT-I:

**Introduction to IWMS Tools:** Introduction – Need of the IWMS – functional elements of IWMS – Ultrasonic Sensor, Arduino Board, GSM Module, Bread Board, Power Supply (Battery) – Jump Wires - Navigation system – Cloud Services - Zero Waste Principle.

#### UNIT-II:

**Applications of IOT in Intelligent Waste Management System:** Introductory Applications of IoT, Machine Learning, Deep Learning and Block Chain Technology in Waste Characterization and Source Reduction, Storage, Collection and Transport of Wastes, Waste Processing Technologies and Waste Disposal.

#### UNIT-III:

**Life Cycle Assessment and Carbon-Footprint-Based IWMS:** Phases of Life Cycle Assessment: Goal and Scope Definition - Life Cycle Inventory - Life Cycle Impact Assessment – Interpretation - LCA Waste Management Software - Umberto Software.

#### UNIT-IV:

**Sima Pro Software - LCA Assessment Methodology:** Life Cycle Inventory Analysis - Life Cycle Impact Assessment – Interpretation - Sensitivity Analysis - Carbon-Footprint-

Based SWM - The Global-Warming Potential Impact - GHG Accounting - GWP Assessment for Solid Waste Management.

#### **UNIT-V:**

**Engineering Principles and Tools for SWM:** Systems Engineering Principles and Tools for SWM - Planning Regional Material Recovery Facilities - Optimal Planning for Solid Waste Collection, Recycling, and Vehicle Routing – Multi attribute Decision Making with Sustainability Considerations - Decision Analysis for Optimal Balance between Solid Waste Incineration and Recycling Programs - Environmental Informatics for Integrated Solid Waste Management - Future Perspectives.

#### **TEXT BOOKS:**

1. Sustainable Solid Waste Management - A Systems Engineering Approach, Ni-Bin Chang and Ana Pires, IEEE & John Wiley, Hoboken, 2015
2. Integrated Solid Waste Management, George Tchobanoglous, Hilary Theisen and Samuel A. Vigil, McGraw-Hill International Edition, 1993

#### **REFERENCES:**

1. Manual on Municipal Solid Waste Management, CPHEEO, Central Public Health and Environmental Engineering Organization, Government of India, 2014
2. Smart Waste Management- Nutshell, Vishal Gupta, Amazon.com Services LLC, 2017
3. Recyclable Household Waste Management System for Smart Home in IOT, Manpreet Kaur & Dr. Kamaljit Singh Saini, Independently, 2018
4. GoI, Ministry of Environment and Forest and Climate Change, Various Recent Laws and Rules of Solid Waste Management



**GREEN ENERGY**



Offered by:

**ELECTRICAL AND  
ELECTRONICS ENGINEERING**

Courses in the OE Track:

| OE Tracks               | V Sem<br>(OE-I)                | VI Sem<br>(OE-II)                   | VII Sem<br>(OE-III)               | VIII Sem<br>(OE-IV)                         |
|-------------------------|--------------------------------|-------------------------------------|-----------------------------------|---------------------------------------------|
| <b>Green<br/>Energy</b> | Renewable<br>Energy<br>Sources | Renewable<br>Energy<br>Technologies | Energy<br>Storage<br>Technologies | Energy<br>Management<br>and<br>Conservation |

## 1. RENEWABLE ENERGY SOURCES

### What we are studying?

The climate landscape is changing rapidly, and new technologies and solutions keep arising to respond to global and local challenges.

Renewable energy sources course makes you discover how Solar Thermal Energy conversion system works. It makes you understand how a Solar Photo voltaic generation system generates electricity. Scope of the course also includes wind energy generation. It also navigates you through Biomass and geo thermal energy generation systems.

### Job opportunities:

When it comes to the hottest and most buzzing careers in the 21st century, the majority of people think of hardcore technical domains such as data science, machine learning & artificial intelligence. Few people might also come up with biotechnology (or biosciences). But, quite often people forget about one of the dark horses – the Renewable Energy sector. Even [Bill Gates lobbied for the Energy sector as one of the top three career choices for making an impactful career.](#)

### Reference:

<https://www.stoodnt.com/blog/careers-in-renewable-energy-job-opportunities-fields-of-study-and-top-universities/>

## 2. RENEWABLE ENERGY TECHNOLOGIES

Within Crisis, there are seeds of opportunity..! We are at the wedge of fossil fuel end. After few years you can witness fuel crisis all over the world, as an engineer one must aware of the solution. To design sustainable systems those last for decades, one must use renewable energy as main or auxiliary source of energy. The application may be electrical or mechanical or chemical, one must convert energy from renewable source into electricity for ease of use.

Renewable Energy Technologies course will introduce you to Different types of Solar PV systems and their characteristics. Students will know the functionality of Power Converters such as Inverters etc., through block diagram approach. Fuel cell technology, which is one of the solutions for energy crisis will be discussed in detail. Course will conclude by discussing impact of PV panel production on environment and disposal of it.

### Job Opportunities:

Green jobs in the renewable energy sector are expected to touch new figures with 6 digit monthly income. Following link may describe the interesting interdisciplinary careers for budding engineers.

### Reference:

<https://www.businessinsider.in/slideshows/miscellaneous/21-high-paying-careers-for-people-who-want-to-save-the-planet-and-also-have-job-security/slidelist/70677782.cms#slideid=70677804>

### **3. ENERGY STORAGE TECHNOLOGIES**

Battery technology is an essential skill for every engineer in present scenario. Course on energy storage technologies will enable student to, Design storage system Residential loads integrated to Renewable and storage systems for Electric Vehicles. It will make student to understand various electrochemical storages such as Lead acid, Li Ion cell etc. and their characteristics. The course enables student to compare non-electric, electric storage systems and analyze application of them to various domains.

#### **Job opportunities:**

Upon successful completion of course student will enhance the chances of getting into EV industry, which almost open fact. Job Profiles include

- i. Battery algorithms engineer
- ii. Battery management engineer
- iii. Battery modeling expert
- iv. Design engineer – EV

### **4. ENERGY MANAGEMENT AND CONSERVATION**

Energy Management And Conservation course is mainly intended to monitor Energy consumption of industries and to manage energy systems. This course also deals with methods of improving efficiency of electric machinery and to design a good illumination system. It also teaches student calculate pay back periods for energy saving equipment.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1EE301) RENEWABLE ENERGY SOURCES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the role of various renewable energy sources
- To know the solar thermal & PV system conversions
- To understand the energy conversion principles of wind & biomass
- To learn energy extraction process of ocean & geo thermal resources

**COURSE OUT COMES:** After completion of the course, the student should be able to

**CO-1:** Understand Solar Thermal Energy conversion principles

**CO-2:** Understand Solar Photo voltaic system

**CO-3:** Analyze wind & biomass energy conversion system

**CO-4:** Understand the geo thermal & ocean energy conversion principles

#### UNIT-I:

**Principles of Solar Radiation:** Role and potential of new and renewable source, the solar energy option, Environmental impact of solar power, physics of the sun, The apparent motion of the sun, the solar constant, extraterrestrial and terrestrial solar radiation, solar radiation on tilted surface, instruments for measuring solar radiation and sunshine, solar radiation data.

#### UNIT-II:

**Solar Thermal Energy Conversion:** Types of solar thermal collectors-Evaluation of the performance of solar collectors- Selective coatings for collectors and glazing, Solar heating systems -Individual and collective solar water heaters.

Concentric Solar Power Plants- Concentrating systems- Components for production of heat and conversion into electricity.

#### UNIT-III:

**Solar PV Conversion:** The PV Cell-Crystalline Solar cells-Thin film solar cell, Module, Array, Equivalent Electrical circuit, Open circuit voltage and Short circuit current, I-V, P-V Curves, Array design- Sun angle- effect of Temperature-Sun tracking, PV system components

#### UNIT-IV:

**Wind & Bio-Mass Energy:** Basic Principle of wind energy conversion, Nature of wind. Wind turbine types & construction. Power in the wind- Betz criteria, performance characteristics. Speed Control strategies of wind turbine-Yaw control & Pitch control. working of Induction generator (Principle only).

Sources of Biomass & usable forms, Principles of Bio-Conversion-Photosynthesis, Biomass conversion - Anaerobic Fermentation & various stages of digestion process. Bio-gas plants-Floating drum, Fixed dome types.

#### **UNIT-V:**

**Geothermal & Ocean Energy:** Resources, types of wells, methods of harnessing the energy-Dry steam, wet steam-single Flashed steam-binary cycle power plants. OTEC-conversion technology-Claude cycle, Andreson cycle. Tidal energy-Origin and nature, tidal energy conversion schemes- single basin, double basin. wave energy: wave conversion devices-Heaving float type, pitching type, oscillating water column. Micro-hydro power generation-Layout.

#### **TEXT BOOKS:**

1. Non-Conventional Energy Sources, B. H. Khan, Mc Graw Hill Publishers
2. Renewable Energies, John Claude Sabbonedere, ISTE & John Wiley, 2007
3. Renewable Energy Resources, Twidell & Wier, CRC Press (Taylor & Francis), 2016

#### **REFERENCE:**

1. Wind & Solar Power Systems, Mukund R. Patel, CRC Press, 2003

#### **ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/121/106/121106014/>
2. [https://onlinecourses.nptel.ac.in/noc21\\_ch11/preview](https://onlinecourses.nptel.ac.in/noc21_ch11/preview)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1EE302) RENEWABLE ENERGY TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Renewable Energy Sources

#### COURSE OBJECTIVES:

- To provide necessary knowledge about the modeling, design and analysis of various PV systems
- To show that PV is an economically viable, environmentally sustainable alternative to the world's energy supplies
- To understand the power conditioning of PV and WEC system's power output

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Model, analyze and design various photovoltaic systems

**CO-2:** Know the feasibility of PV systems as an alternative to the fossil fuels

**CO-3:** Design efficient stand alone and grid connected PV and WEC power systems

#### UNIT-I:

**Behavior of Solar Cells-Basic Structure and Characteristics:** Types - equivalent circuit-modeling of solar cells including the effects of temperature, irradiation and series/shunt resistances on the open-circuit voltage and short-circuit current-Solar cell arrays- PV modules-PV generators- shadow effects and bypass diodes- hot spot problem in a PV module and safe operating area.

#### UNIT-II:

**Types of PV Systems:** Grid connected PV systems- Net-metering- Estimation of actual a.c. output power from PV systems.

Stand-alone system- Approach to designing an off-grid PV system with battery- with battery and diesel generator- Stand-alone solar water pumping system- Sizing/designing PV water pumping system- Procedure for decommissioning of PV plant.

#### UNIT-III:

**Power Converters for PV and Wind:** Basic switching devices, AC-DC Rectifier, DC-AC inverter (Basic operation), DC DC converter - Buck, Boost converters Basic operation, Battery charger (Basic operation), grid interface requirements in Renewable energy integration

#### UNIT-IV:

**Maximum Power Point Tracking:** Various Sources of Losses in PV system, Charge Control in Battery Backed PV Systems, Maximum Power Point Tracking (MPPT)- Role of DC-DC

converter in MPP tracking- Perturb and Observe Method-pseudo program for P&O method, Advanced Issues & Algorithms- search steps-variable step size algorithm.

#### **UNIT-V:**

**Fuel Cell Technology:** History of Fuel cells, Fuel Cell Vehicle Emissions, Hydrogen safety factors, Principle of Operation- Fuel cell Model- cell voltage, Power and efficiency of fuel cell, Various types of fuel cells, Various storage systems for Hydrogen, Applications.

#### **TEXT BOOKS:**

1. Handbook of Renewable Energy Technology, Ahmed F. Zobaa, World Scientific Publishing Company, 2011
2. Wind and Solar Power Systems Design, Analysis, and Operation, Patel M. R., 2nd Edition, CRC Press, New York, 2005
3. Practical Handbook of Photovoltaics - Fundamentals and Applications, Augustin McEvoy, Tom Markvart, T. Markvart, L. Castaner, Elsevier Science, 2003

#### **REFERENCE:**

1. Electric Powertrain - Energy Systems, Power Electronics & Drives for Hybrid, Electric & Fuel Cell Vehicles, Goodarzi, Gordon A., Hayes, John G., John Wiley & Sons, 20



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22OE1EE401) ENERGY STORAGE TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Renewable Energy Sources, Renewable Energy Technologies

#### COURSE OBJECTIVES:

- To understand Techno economic analysis of various storage systems
- To know Feasibility of different storage technologies
- To learn the operation of electrochemical storage systems
- To comprehend the functionality of non-electric storage system

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand techno-economic aspects of energy storage technologies

**CO-2:** Appreciate electrochemical storage systems and evaluate their characteristics

**CO-3:** Estimate the energy storage requirements and investigate the feasibility

**CO-4:** Compare non-electric and electric storage systems

#### UNIT-I:

**Techno-economic Analysis of Various Energy Storage Technologies:** Electrical Energy Storage (EES)-Definition-Role, Energy storage components, Applications and Technical support, Financial Benefits of EES, Techno economic analysis, Classification of Energy Storage systems, Comparison

#### UNIT-II:

**Energy Storage Estimation and Feasibility:** Background-Solar Power - Wind Power (Brief discussion), Estimation of daily residential load-daily available solar energy- daily available wind energy-Importance, Estimation of Storage, Sizing: Steps for Storage sizing, case study, Grid connected residential- PV-grid connected residential- Wind-hybrid system, Feasibility analysis of Storage systems- Various Terms involved- Case study of comparison between Off grid and grid connected systems.

#### UNIT-III:

**Advanced Electrochemical Storage Systems:** Standard Batteries: Lead Acid- VRLA - Ni-cd, Modern Batteries: Ni MH- Li Ion, Flow Batteries: Br<sub>2</sub> Zn-Vanadium Redox, Battery composition, construction, Principle of operation, Types, Advantages and disadvantages of above batteries.

#### UNIT-IV:

**Fuel Cells:** Introduction, working and types of fuel cell, low, medium and high temperature fuel cell, liquid and methanol types, proton exchange membrane fuel cell solid oxide, hydrogen fuel cells, Hydrogen storage technology – pressure cylinders,

liquid hydrogen and metal hydrides, methods of hydrogen production, green hydrogen

#### **UNIT-V:**

**Non-Electric Storage and Emerging Applications:** Flywheel, Energy Relations, Flywheel System Components, Benefits of Flywheel over Battery, Superconducting Magnet Energy Storage, Compressed Air Energy storage, Overview of Thermal Energy Storage. Capacitor bank storage, Comparison of storage Technologies -Emerging applications of Energy storage

#### **TEXT BOOKS:**

1. Energy Storage Technologies and Applications, Ahmed Faheem Zobaa, InTech Publishers, 2013
2. Lithium Batteries and Other Electrochemical Storage Systems, Christian Glaize, Sylvie Geniès, ISTE & John Wiley, 2013
3. Wind and Solar Power Systems, Mukund R. Patel, 2<sup>nd</sup> Edition, CRC Press, 2006

#### **REFERENCE:**

1. Rechargeable Batteries Applications Handbook, EDN Series for Design Engineers, Elsevier
2. Fuel Cells: Principles and Applications, Viswanathan B. and Scibioh Aulice M, University Press, 2006

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1EE402) ENERGY MANAGEMENT AND CONSERVATION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Renewable Energy sources, Renewable Energy Technologies, Energy Storage Technologies

#### COURSE OBJECTIVES:

- To understand the necessity of conservation of Energy
- To Know the methods of Energy management
- To identify the factors to increase the efficiency of electrical equipment
- To know the benefits of carrying out energy Audits

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Conduct Energy Audit of industries

**CO-2:** Manage energy Systems

**CO-3:** Specify the methods of improving efficiency of electric motor

**CO-4:** Improve power factor and to design a good illumination system

**CO-5:** Calculate pay back periods for energy saving equipment

#### UNIT-I:

**Basic Principles of Energy Audit:** Energy audit- definitions, concept, types of audits, energy index, cost index, pie charts, Sankey diagrams, load profiles, Energy conservation schemes- Energy audit of industries- energy saving potential, building energy audit

#### UNIT-II:

**Energy Management:** Principles of energy management, organizing energy management program, initiating, planning, controlling, promoting, monitoring, reporting- Energy manager, Qualities and functions, Questionnaire - check list for top management

#### UNIT-III:

**Energy Efficient Motors:** Energy efficient motors-Minimizing Watts Loss in Motors- Technical aspects of energy efficient motors, factors affecting efficiency - power supply quality, voltage unbalance, motor loading, Concept of variable frequency drive-Need for VFD, principles of VFD, motor energy audit with numerical example

#### UNIT-IV:

**Lighting System:** Basic parameters & terms in lightning system, Zonal cavity method for indoor lighting calculations, General energy saving opportunities, energy efficient

lighting controls. Energy Instruments- watt meter, data loggers, thermocouples, pyrometers, flux meters.

#### **UNIT-V:**

**Economic Aspects and Analysis:** Economics Analysis-Depreciation Methods, Financial analysis techniques-payback period, Net present value method, Return on Investment, Internal rate of return with advantage and limitations, replacement analysis, life cycle costing analysis.

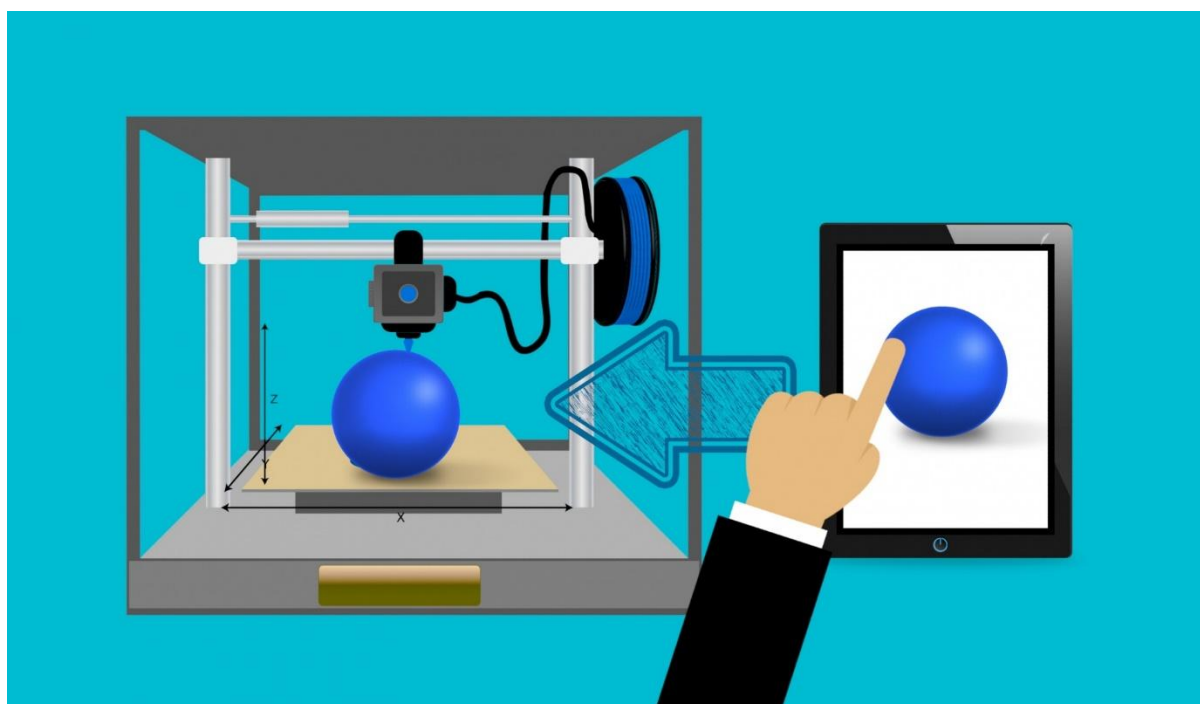
#### **TEXT BOOKS:**

1. Energy Management, W. R. Murphy & G. Mckay, Butterworth-Heinemann Publications
2. Energy Management, Paul o' Callaghan, 1<sup>st</sup> Edition, McGraw Hill Book Company, 1998

#### **REFERENCES:**

1. Energy Efficient Electric Motors, John C. Andreas, 2<sup>nd</sup> Edition, Marcel Dekker Inc. Ltd., 1995
2. Energy Management Handbook, W. C. Turner, John Wiley and Sons
3. Energy Management and Good Lighting Practice: Fuel Efficiency Booklet12-EEO

# **3D PRINTING AND DESIGN**



## 3D PRINTING AND DESIGN

Offered by:

**MECHANICAL ENGINEERING**

Courses in the OE Track:

| OE Tracks                         | V Sem<br>(OE-I)    | VI Sem<br>(OE-II)              | VII Sem<br>(OE-III)                                | VIII Sem<br>(OE-IV)    |
|-----------------------------------|--------------------|--------------------------------|----------------------------------------------------|------------------------|
| <b>3D Printing<br/>and Design</b> | Elements of<br>CAD | Introduction<br>to 3D Printing | 3D Printing -<br>Machines,<br>Tooling &<br>Systems | Reverse<br>Engineering |

## **3D PRINTING AND DESIGN**

3D Printing is a process for making a physical object from a three-dimensional digital model by laying down many successive thin layers of a material. It brings a digital CAD model into its physical form by adding layer by layer of materials. Thus called 'Additive Manufacturing'. It is the opposite of subtractive manufacturing i.e., removing material from an object using a mechanical machine. It enables to produce complex shapes using less material than traditional manufacturing methods. There are several different techniques to 3D print an object. It saves time through prototyping and is also responsible for manufacturing impossible shapes. Due to these, it has many applications in different fields like consumer products (eyewear, footwear, design, furniture, industrial products (manufacturing tools, prototypes, functional end-use parts, dental products, prosthetics, architectural scale models, reconstructing fossils, replicating ancient artefacts, reconstructing evidence in forensic pathology etc.

3D printing has good prospects from career perspective. Various positions that could be available are CAD designers, engineers, technical developers, software developers, electronics engineers, etc.

This OE track consists of 04 courses and is designed with an objective to provide an overview of all the constituents of 3D Printing starting from elements of CAD that are needed to create CAD models, followed by basics of 3D Printing required for setting the parameters, then the machines and tools used in 3D Printing for thorough understanding of systems and processes and finally the reverse engineering of 3D printing models from actual objects.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1ME301) ELEMENTS OF CAD

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basics of CAD and the devices used
- To know the various types of modeling used in CAD
- To appreciate the concept of feature-based modeling and geometric transformations
- To comprehend the assembly modeling procedure and data exchange formats

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Use the fundamentals of CAD and its devices in design applications

**CO-2:** Identify the types of CAD modeling techniques and utilize them

**CO-3:** Evaluate the objects or models using geometric transformations and manipulations

**CO-4:** Perform the assembly modeling and assess the various data exchange formats

#### UNIT-I:

**Fundamentals of CAD:** Introduction to Computer-Aided Design (CAD), Design process, Application of Computers for Design and Manufacturing, Benefits of CAD, Overview of CAD / CAM Hardware, Display devices, Hard copy devices. Design Cycle, Raster scan graphics, coordinate systems.

#### UNIT-II:

**Geometric Modeling:** Introduction to Geometric Model, Types of modeling, Curve representation.

**Wireframe Modeling:** Introduction, advantages, limitations and applications, Wireframe entities-analytic and synthetic, Basic definitions of Cubic, Bezier, and B-spline curves.

#### UNIT-III:

**Surface Modeling:** Introduction, advantages, limitations and applications, surface entities, Basic definitions of analytic surfaces - planar surface, ruled surface, tabulated cylinder, surface of revolution; Basic definitions of synthetic surfaces - Bezier surface, B-spline surface.

#### UNIT-IV:

**Solid Modeling:** Introduction, advantages, limitations and applications, Solid Entities, Solid Representation schemes – Boundary Representation (B-Rep) scheme, Constructive Solid Geometry (CSG) scheme. Feature-based Modeling: Introduction,



Feature entities, Feature representation, 3D Sketching, Parameter, Relations and Constraints.

**Product Data Exchange:** Introduction, Graphics Standards, Types of translators, Importance of formats in 3D Printing, Data exchange formats - IGES, STEP and STL.

**UNIT-V: Geometric Transformations:** Introduction to 2D & 3D transformations, Brief treatment on Translation, Scaling, Reflection, and Rotation using Homogeneous and concatenated transformations Manipulations: Displaying, Segmentation, Trimming, Intersection, Projection. Assembly Modeling: Introduction, Assembly modeling, Assembly Tree, Mating Conditions, Bottom-up and Top-down approach.

**TEXT BOOKS:**

1. CAD/CAM Theory and Practice, Ibrahim Zeid, Tata McGraw-Hill. 2009
2. CAD/CAM: Concepts and Applications, Alavala, Prentice Hall International, 2018
3. Mastering CAD/CAM, Ibrahim Zeid, Tata McGraw-Hill.2018

**REFERENCES:**

1. CAD/CAM-Computer Aided Design and Manufacturing, Mikell P. Groover, E. W. Zimmers, Pearson Education/Prentice Hall.2018
2. CAD/CAM Principles and Applications, P. N. Rao, Tata McGraw-Hill, 2017
3. CAD / CAM / CIM, Radhakrishnan and Subramanian, Pearson Education, 2000

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1ME302) INTRODUCTION TO 3D PRINTING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Elements of CAD

#### COURSE OBJECTIVES:

- To understand the fundamental concepts & classification of 3D Printing technologies
- To analyze the process chain involved in 3D Printing and
- To know about the Material extrusion, sheet lamination, Liquid and jetting based techniques
- To know about the data formats and post processing methods involved in 3D Printing

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Summarize the importance of 3D Printing

**CO-2:** Explain the process chain involved in 3D Printing

**CO-3:** Explain about Material extrusion, sheet lamination, Liquid and jetting based techniques

**CO-4:** Apply the knowledge gained in data formats and post-processing methods

#### UNIT-I:

**Introduction to 3D Printing:** Introduction to 3D Printing, 3D Printing evolution, Classification of 3D Printing, Distinction between 3D Printing & CNC Machining, Advantages of 3D Printing, Artificial Intelligence and IOT in 3D Printing.

#### UNIT-II:

**Generalized 3D Printing Process Chain:** Process chain, Materials for 3D Printing, Design for 3D Printing and Overview of Medical Modeling & Reverse Engineering.

#### UNIT-III:

**Material Extrusion and Sheet Lamination Techniques:** Introduction, basic principles, Fused Deposition Modeling, Laminated Object Manufacturing (LOM), Advantages and Applications

#### UNIT-IV:

**Liquid and Three-Dimensional Jetting Techniques:** Stereolithography (SLA), Material Jetting, Binder Jetting, Advantages and Applications.

**UNIT-V:**

**3D Printing Data Formats and Post Processing:** STL Format, STL File Problems, Brief Overview of other translations - IGES File, HP/GL File and CT data.

Post-Processing: Introduction, Support Material Removal, Surface Texture Improvements, Accuracy Improvements, Aesthetic Improvements.

**TEXT BOOKS:**

1. Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing, Ian Gibson, David W Rosen, Brent Stucker, Springer, 2010
2. Rapid Prototyping: Principles & Applications, Chuah Chee Kai, Leong Kah Fai, World Scientific, 2010

**REFERENCES:**

1. Rapid Prototyping: Theory and Practice, Ali K. Karmani, Emand Abouel Nasr, Springer, 2006
2. Understanding Additive Manufacture: Rapid Prototyping, Rapid Tooling and Rapid Manufacture, Andreas Gebhardt, Hanser Publishers, 2013
3. Rapid Manufacturing: Advanced Research in Virtual and Rapid Prototyping, Hopkinson, N. Haque, and Dickens, Taylor and Francis, 2007

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22OE1ME401) 3D PRINTING - TOOLING AND SYSTEMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Elements of CAD, Introduction to 3D Printing

#### **COURSE OBJECTIVES:**

- To understand the need of prototyping
- To understand about the liquid and solid based 3D printing systems
- To know about the Laser based 3D Printing Systems
- To know about the Rapid Tooling and applications of 3D Printing

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Summarize the importance of Prototyping

**CO-2:** Explain the process involved in liquid and solid based 3D printing systems

**CO-3:** Explain the laser-based 3D Printing Systems

**CO-4:** Adapt the knowledge gained in rapid tooling and applications of 3D Printing

#### **UNIT-I:**

**Introduction:** Prototype Fundamentals, Types of Prototypes, Roles of Prototypes, Phases of Development Leading to Rapid Prototyping, Fundamentals of Rapid Prototyping.

#### **UNIT-II:**

**Liquid Based 3D Printing Systems:** Introduction, Principles, Processes and Applications of Solid Ground Curing, Material Jetting & Binder Jetting

#### **UNIT-III:**

**Solid Based 3D Printing Systems:** Introduction, Principles, Processes and Applications of Paper Lamination Technology (PLT) and Laminated Object Manufacturing (LOM)

#### **UNIT-IV:**

**Laser Based 3D Printing Systems:** Selective Laser Sintering (SLS)-Principle, Process and Applications, Three-Dimensional Printing- Principle, Process and Applications, Laser Engineered Net Shaping (LENS)- Principle, Process and Applications, Overview of Direct Energy Deposition.

#### **UNIT-V:**

**Rapid Tooling and Applications of 3D Printing:** Introduction and need for Rapid Tooling, Overview of Indirect and Direct Processes, Applications

**Applications of 3D Printing:** Brief overview of Applications in Design, Engineering, Aerospace Industry, Automotive Industry and Biomedical Industry

**TEXT BOOKS:**

1. Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing, Ian Gibson, David W Rosen, Brent Stucker, Springer, 2010
2. Rapid Prototyping: Principles & Applications, Chuah Chee Kai, Leong Kah Fai, World Scientific, 2010

**REFERENCES:**

1. Rapid Prototyping: Theory and Practice, Ali K. Karmani, Emand Abouel Nasr, Springer, 2006
2. Understanding Additive Manufacture: Rapid Prototyping, Rapid Tooling and Rapid Manufacture, Andreas Gebhardt, Hanser Publishers, 2013
3. Rapid Manufacturing: Advanced Research in Virtual and Rapid Prototyping, Hopkinson, N. Haque, and Dickens, Taylor and Francis, 2007

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

### (22OE1ME402) REVERSE ENGINEERING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Elements of CAD, Introduction to 3D Printing, 3D Printing Machines, Tooling & Systems

#### COURSE OBJECTIVES:

- To understand the Reverse Engineering (RE) and its methodologies
- To comprehend Data Acquisition Techniques for Reverse Engineering
- To understand Integration Between Reverse Engineering and Additive manufacturing
- To know the applications of reverse engineering

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Basic understanding of Reverse Engineering and its methodologies

**CO-2:** Understanding the data acquisition techniques for reverse engineering

**CO-3:** Understanding of amalgamation Between Reverse Engineering and Additive manufacturing

**CO-4:** Adapt the knowledge gained in reverse engineering for various applications

#### UNIT-I:

**Introduction to Reverse Engineering:** Scope and tasks of RE, Process of duplicating, Definition and use of RE, The Generic Process, History of Reverse Engineering, Overview of Applications

#### UNIT-II:

**Methodologies and Techniques:** Potential for Automation with 3-D Laser Scanners, Computer-aided (Forward) Engineering, Computer-aided Reverse Engineering, Computer Vision and Reverse Engineering

#### UNIT-III:

**Data Acquisition Techniques:** Contact Methods: Point-to-point sensing with touch-trigger probes, analogue sensing with scanning probes, Coordinate Measurement Machine (CMM)-CNC machines and CMM- Articulated Arms, Noncontact Methods: Triangulation, Structured Light, and Destructive Method

#### UNIT-IV:

**Integration Between Reverse Engineering and Additive manufacturing:** Modeling Cloud Data, Integration of RE and AM for Layer-based Model Generation, Adaptive Slicing Approach for Cloud Data Modeling.

**UNIT-V:****Applications:**

**Automotive:** Workflow for Automotive Body Design, Reverse Engineering for Better Quality

**Aerospace:** RE in Aerospace—A Work in Progress, Reducing Costs of Hard Tooling  
Medical: Orthodontics, Hearing Instruments, Knee Replacement

**TEXT BOOKS:**

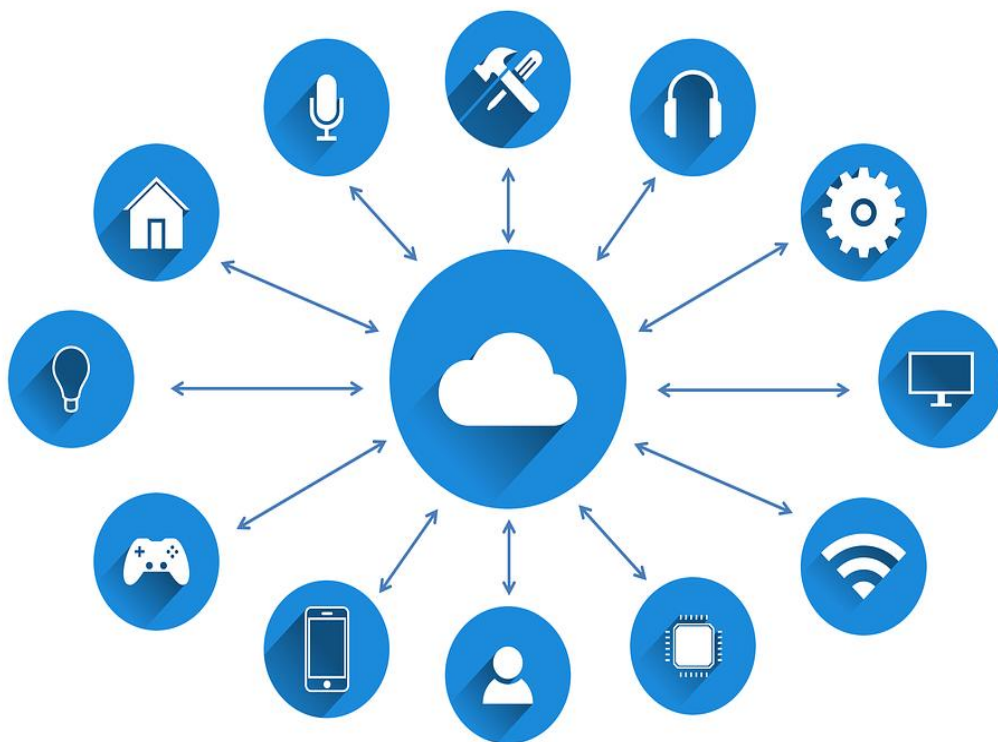
1. Reverse Engineering: An Industrial Perspective, V. Raja and K. Fernandes, Springer-Verlag
2. Reverse Engineering, K. A. Ingle, McGraw-Hill
3. Rapid Prototyping, Ali Kamrani, Emad Nasr, Springer, 2006

**REFERENCES:**

1. Smart Product Engineering, Michael Abramovici, Rainer stark, Springer Berlin Heidelberg
2. Product Design: Techniques in Reverse Engineering and New Product Development, K. Otto and K. Wood, Prentice Hall, 2001

# **INTERNET OF THINGS**





## INTERNET OF THINGS

Offered by:

**ELECTRONICS AND  
COMMUNICATION  
ENGINEERING**

Courses in the OE Track:

| OE Tracks                 | V Sem<br>(OE-I)                            | VI Sem<br>(OE-II)                                      | VII Sem<br>(OE-III)                      | VIII Sem<br>(OE-IV)            |
|---------------------------|--------------------------------------------|--------------------------------------------------------|------------------------------------------|--------------------------------|
| <b>Internet of Things</b> | Sensors<br>Transducers<br>and<br>Actuators | Introduction to<br>Microcontrollers<br>and Interfacing | Fundamentals<br>of Internet of<br>Things | Wireless<br>Sensor<br>Networks |

## INTERNET OF THINGS

**Internet of Things:** The IoT creates opportunities for more direct integration of the physical world into computer-based systems, resulting in efficiency improvements, economic benefits, and reduced human exertions. **IoT is changing how we live, work, travel, and do business. It is even the basis of a new industrial transformation, known as Industry 4.0, and key in the digital transformation of organizations, cities, and society overall.** The IoT track helps students to learn about how to

- Learn different protocols and connectivity technologies used in IOT.
- Expose the various sensors and transducers for measuring mechanical quantities.
- Develop simple applications using 8051 microcontrollers.
- Understand the key routing protocols for sensor networks and their design issues.

**Some of the more common career paths in the Internet of Things path are**

- IoT Developer. ...
- IoT Architect...
- IoT Embedded Systems Designer...
- IoT Solutions Engineer...
- Professional in Sensors and Actuators...
- Embedded Programs Engineer...
- Safety Engineer...

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EC301) SENSORS TRANSDUCERS AND ACTUATORS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To expose the students to various sensors and transducers for measuring mechanical quantities
- To make the students familiar with the specifications of sensors and transducers
- To make the students identify for various sensors and transducers for various applications
- To expose the students to various actuators

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Familiar with classification and characteristics of various sensors and transducers

**CO-2:** Familiar with the principle and working of various sensors and transducers

**CO-3:** Familiar with the principle and working of various actuators

**CO-4:** Able to select proper Transducer / Sensor for a specific measurement application

**CO-5:** Able to select proper Actuator for a specific measurement application

#### UNIT-I:

**Primary Sensing Elements and Transducers:** Mechanical devices as primary detectors, mechanical spring devices, pressure sensitive primary devices, flow rate sensing elements, Transducers-electrical Transducers, classification of Transducers, characteristics and choice of Transducers, factors influencing the choice of Transducers.

#### UNIT-II:

**Electric Transducers:** Resistive transducers, Potentiometers, Strain gauges, Types of Strain gauges, Resistance thermometers, Thermistors, Thermocouples, variable Inductance Transducers, Linear Variable Differential Transformer, Synchros, Resolvers, Capacitive Transducers, Piezo electric Transducers.

#### UNIT-III:

**Magnetic & Optical Transducers:** Hall Effect Transducers, Magneto resistors, Magneto-Elastic and Magneto-Strictive Transducers, Opto electronic Transducers, Digital Encoding Transducers, Photo Optic Transducers.

#### UNIT-IV:

**Smart Sensors & Applications:** Introduction, Primary Sensors, Excitation, Amplification, Filters, Converters, Compensation, Information Coding/Processing, Data Communication, Standards for Smart Sensor Interface, the Automation. Sensors

Applications: Introduction, On-board Automobile Sensors (Automotive Sensors), Home Appliance Sensors, Aerospace Sensors, Sensors for Manufacturing, Sensors for environmental Monitoring.

#### **UNIT-V:**

**Pneumatic and Hydraulic Actuators:** Pneumatic and Hydraulic Actuation Systems-Actuation systems, Pneumatic and hydraulic systems, Rotary actuators.

**Mechanical and Electrical Actuators:** Mechanical Actuation Systems-Types of motion, Kinematic chains, Cams, Gears, Ratchet and pawl, Belt and chain drives, Electrical Actuation Systems, Electrical systems, Mechanical switches, Solid-state switches, Solenoids, D.C. Motors, A.C. Motors, Stepper motors.

#### **TEXT BOOKS:**

1. A Course in Electrical and Electronic Measurements and Instrumentation, A. K. Sawhney, Puneet Sawhney, 19<sup>th</sup> Edition, 2011
2. Sensors and Transducers, D. Patranabis, 2<sup>nd</sup> Edition, PHI Learning Private Limited, 2013
3. Mechatronics, W. Bolton, 7<sup>th</sup> Edition, Pearson Education Limited, 2018

#### **REFERENCES:**

1. Sensors and Actuators, Patranabis, 2<sup>nd</sup> Edition, PHI, 2013

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22OE1EC302) INTRODUCTION TO MICROCONTROLLERS AND INTERFACING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Sensors Transducers and Actuators

#### COURSE OBJECTIVES:

- To familiarize fundamental concepts of digital systems
- To understand programming concepts of Microcontrollers
- To gain knowledge for developing Embedded applications

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand basic computing concepts

**CO-2:** Know the architecture and internal peripherals of 8051 microcontroller

**CO-3:** Develop knowledge for programming 8051 microcontroller

**CO-4:** Interface peripherals to ATMEGA328 microcontroller

**CO-5:** Understand the basic principles and techniques of embedded system design

#### UNIT-I:

**Introduction to Computing:** Numbering and Coding Systems: Binary, Decimal, Octal, Hexadecimal and conversions, Binary and Hexadecimal Arithmetic, Complements, Binary codes, Alphanumeric codes, Inside the Computer

#### UNIT-II:

**8051 Microcontroller:** Microprocessors Vs Microcontrollers, Architecture and Programming Model of 8051, Special Function Register formats, Memory Organization, Timers and Counters- Operating modes, Serial port, Interrupts

#### UNIT-III:

**8051 Programming in C:** Data types, software delay generation, Logical operations, Accessing code and data space in 8051, I/O port programming, Timer/counter programming.

Serial IO modes and their programming in C, interrupts programming in C: serial, timer and external interrupts.

#### UNIT-IV:

**Introduction to Arduino:** Features of Arduino, Arduino components and IDE, Pin Configuration of ATMEGA328 microcontroller.

**Interfacing:** Seven Segment Display, Pulse Width Modulation, Analog to Digital Converter, Stepper Motor, Wireless connectivity to Arduino.

Case study: WiFi Controlled Arduino Robot Car.

**UNIT-V:**

**Embedded System Design:** Embedded system - Definition, Characteristics of embedded computing applications, Design challenges, Requirements, Specification, Architecture design, Designing hardware and software components, system integration, Design example: Model train controller.

**TEXT BOOKS:**

1. The 8051 Microcontroller and Embedded Systems: Using Assembly and C, Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D. McKinlay, 2nd Edition, 2005.
2. Computers as Components: Principles of Embedded Computing System Design, Wayne Wolf, Morgan kaufmann Publishers, 2001

**REFERENCES:**

1. Digital Design, Morris Mano, PHI, 3<sup>rd</sup> Edition, 2006
2. The 8051 Microcontroller: Programming, Architecture, Ayala & Gadre, 3<sup>rd</sup> Edition, Cengage Publications, 2008
3. Embedded Systems: Architecture, Programming and Design, 2<sup>nd</sup> Edition, TMH

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22OE1EC401) FUNDAMENTALS OF INTERNET OF THINGS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Sensors Transducers and Actuators, Introduction to Microcontrollers and Interfacing

#### COURSE OBJECTIVES:

- To understand the fundamentals of the Internet of Things
- To explore IoT enabling technologies
- To study IoT platform design and development methodology
- To explore applications of IoT

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the key concepts of the Internet of Things

**CO-2:** Gain knowledge of the enabling technologies and IoT levels

**CO-3:** Comprehend the design methodology for IoT platform

**CO-4:** Understand onboard components for IoT

**CO-5:** Apply Internet of Things to different applications in the real world

#### UNIT-I:

**Introduction to Internet of Things:** Definition and Characteristics of IoT, Physical Design of IoT.

**Logical Design of IoT:** IoT Functional Blocks, IoT Communication Models, IoT Communication API's.

#### UNIT-II:

**IoT enabling Technologies:** Wireless Sensor Networks, Cloud Computing, Big data analytics, Communication protocols, Embedded Systems, IoT Levels and Deployment Templates.

**IoT and M2M:** Introduction, M2M, Difference between IoT and M2M, SDN and NFV for IoT.

#### UNIT-III:

**IoT Platforms Design Methodology:** Introduction, IoT design methodology- Purpose & Requirements specification, process specification, Domain Model specification, Information Model specification, Service Specifications, IoT Level Specification, Functional view specification, Operational View specification, Device & Component Integration, Application Development.

#### UNIT-IV:

**IoT Physical Devices & End Points:** Basic building blocks of an IoT Device, Exemplary device, Serial, SPI and I2C interfaces.

**Programming:** controlling LED, interfacing LED and switch, interfacing LDR

**UNIT-V:**

**Applications of IoT:** Home automation, Cities, Retail, Logistics, Agriculture, Industry and health and lifestyle.

**TEXT BOOKS:**

1. Internet of Things: A Hands-On Approach, Vijay Madisetti, ArshdeepBahga, Universities Press, 2015
2. The Internet of Things – Key applications and Protocols, Olivier Hersent, David Boswarthick, Omar Elloumi, Wiley, 2012

**REFERENCES:**

1. Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, River Publishers, 2013
2. Menno Huisman, Building the Internet of Things, Sara Cordoba, Wimer Hazenberg, BIS Publishers. 2011
3. Designing the Internet of Things, Adrian Mcewen, Hakin Cassimally, John Wiley and Sons, 2015



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

### (22OE1EC402) WIRELESS SENSOR NETWORKS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Sensors Transducers and Actuators, Introduction to Microcontrollers and Interfacing, IoT Protocols and its applications

#### COURSE OBJECTIVES:

- To understand basics of Wireless Sensor Networks
- To study of medium access control protocols
- To distinguish key routing protocols used in sensor networks
- To learn transport layer protocols used in for sensor networks

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the fundamental concepts of Wireless Sensor Networks

**CO-2:** Analyze the concepts of MAC layer in Wireless Sensor Networks

**CO-3:** Identify issues in Routing Protocols

**CO-4:** Recognize concerns of Transport Layer for Ad Hoc Wireless Networks

**CO-5:** Address the security threats in Wireless Sensor Networks

#### UNIT – I:

**Overview of Wireless Sensor Networks:** Key definitions of sensor networks, Advantages of Sensor Networks, Unique constraints and challenges, Driving Applications, Sensor Node Architecture.

**Networking Technologies:** Physical Layer and Transceiver Design Considerations, Personal Area Networks (PANs), Hidden node and Exposed node problem.

#### UNIT – II:

**MAC Protocols for Wireless Sensor Networks:** Issues in Designing a MAC protocol for Ad Hoc Wireless Networks, Design goals, Classifications of MAC Protocols.

#### UNIT – III:

**Routing Protocols:** Introduction, Issues in Designing a Routing Protocol for Ad Hoc Wireless Networks, Classification of Routing Protocols, Table-Driven Routing Protocols, On-Demand Routing Protocols.

#### UNIT – IV:

**Transport Layer Protocols:** Introduction, Issues in Designing a Transport Layer Protocol for Ad Hoc Wireless Networks, Design Goals of a Transport Layer Protocol for Ad Hoc Wireless Networks, Classification of Transport Layer Protocols.

## **UNIT – V:**

**Security in Wireless Sensor Networks:** Security in Ad Hoc Wireless Networks, Network Security Requirements, Issues and Challenges in Security Provisioning, Network Security Attacks, Key Management, Secure Routing in Ad Hoc Wireless Networks.

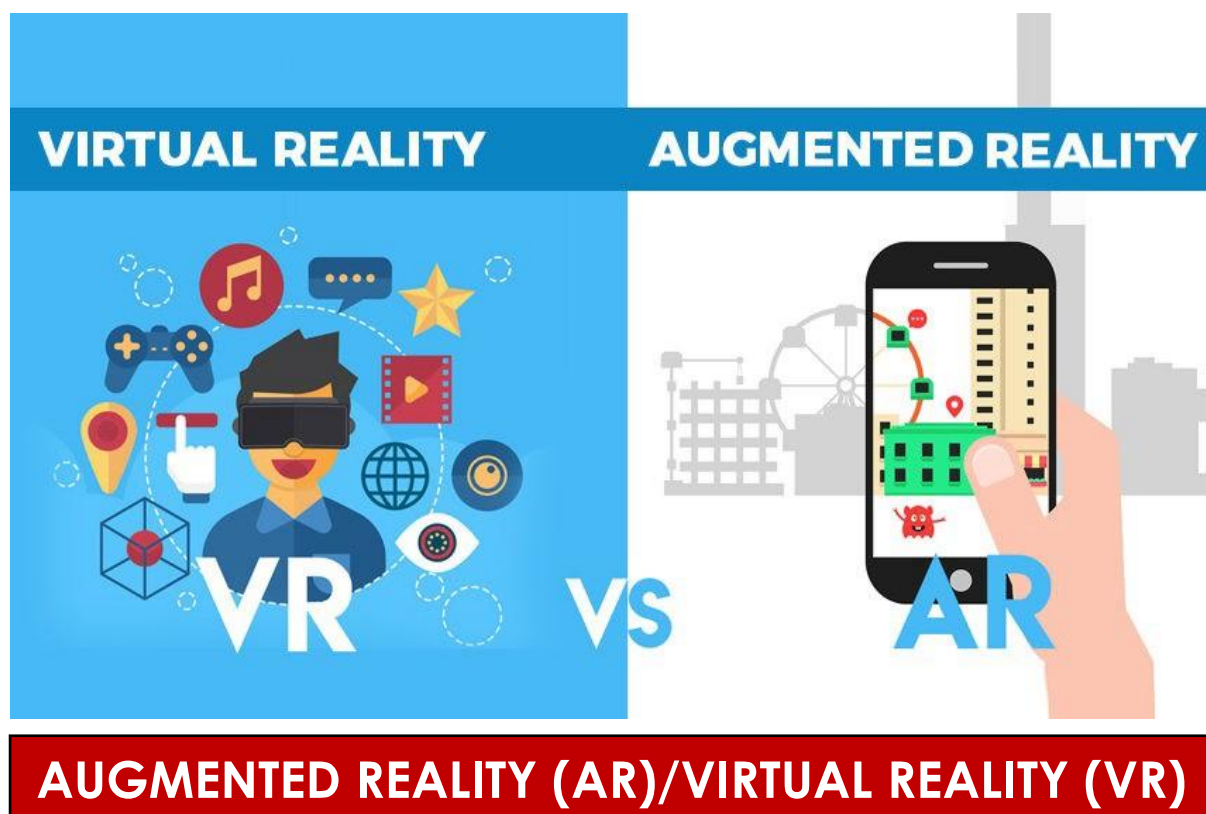
### **TEXT BOOKS:**

1. Adhoc Wireless Networks: Architectures and Protocols, C. Siva Ram Murthy and B. S. Manoj, PHI, 2004
2. Wireless Adhoc and Sensor Networks: Protocols, Performance and Control, Jagannathan Sarangapani, CRC Press, 2007

### **REFERENCES:**

1. Wireless Sensor Networks - Technology, Protocols, and Applications, Kazem Sohraby, Daniel Minoli, & Taieb Znati, John Wiley, 2007
2. Wireless Sensor Networks - An Information Processing Approach, Feng Zhao & Leonidas J. Guibas, Elsevier, 2007
3. Adhoc Mobile Wireless Networks: Protocols & Systems, C. K. Toh, 1st Edition, Pearson Education, 2015
4. Wireless Sensor Networks, C. S. Raghavendra, Krishna M. Sivalingam, Taieb Znati, Springer, 2004
5. Protocols and Architectures for Wireless Sensor Networks, Holger Karl & Andreas Willig, John Wiley, 2005

**AUGMENTED  
REALITY  
(AR)/VIRTUAL  
REALITY (VR)**



Offered by:

**ELECTRONICS AND  
COMMUNICATION  
ENGINEERING**

Courses in the OE Track:

| OE Tracks                                               | V Sem<br>(OE-I)                         | VI Sem<br>(OE-II)                                 | VII Sem<br>(OE-III)                            | VIII Sem<br>(OE-IV)            |
|---------------------------------------------------------|-----------------------------------------|---------------------------------------------------|------------------------------------------------|--------------------------------|
| <b>Augmented Reality (AR)/<br/>Virtual Reality (VR)</b> | Introduction<br>to Signal<br>Processing | Introduction<br>to Image &<br>Video<br>Processing | Augmented<br>Reality and<br>Virtual<br>Reality | Virtual Reality<br>Engineering |

## **OE TRACK :: AUGMENTED REALITY (AR) / VIRTUAL REALITY (VR)**

Augmented reality (AR) and Virtual Reality (VR) bridge the digital and physical worlds. They allow you to take in information and content visually, in the same way you take in the world. AR dramatically expands the ways our devices can help with everyday activities like searching for information, shopping, and expressing yourself. VR lets you experience what it's like to go anywhere from the front row of a concert to distant planets in outer space.

### **Job Roles in Augmented reality and virtual reality (AR & VR) Track**

- Design Architect. ...
- Software Designer. ...
- System Validation Engineers. ...
- Software Developer. ...
- 3D Artist...

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EC303) INTRODUCTION TO SIGNAL PROCESSING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand various fundamental characteristics of signals and systems
- To analyze signals in frequency domain
- To know principles of signal transmission through systems
- To understand fundamentals of digital signal

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Classify and Analyze signals and Systems

**CO-2:** Convert the signals from continuous-time to discrete-time

**CO-3:** Transform signals from one-domain to another domain

**CO-4:** Apply the transformation tools on signals and systems and analyze their significance and applications

**CO-5:** Design the structures of different types of digital filters

#### UNIT-I:

**Representation of Signals:** Continuous time and Discrete Time signals, Classification of Signals – Periodic and aperiodic, even and odd, energy and power signals, deterministic and random signals, causal and non-causal signals, complex exponential and sinusoidal signals. Concepts of standard signals. Various operations on Signals.

#### UNIT-II:

**Representation of Systems:** Classification of discrete time Systems, impulse response, Concept of convolution in time domain and frequency domain, response of a linear system, System function, Signal bandwidth, system bandwidth. Ideal filter characteristics.

#### UNIT-III:

**Sampling Theorem:** Representation of continuous time signals by its samples - Sampling theorem – Reconstruction of a Signal from its samples, aliasing

**Z –Transform:** Basic principles of z-transform, region of convergence, properties of ROC, Inverse z-transform using Partial fraction.

#### UNIT-IV:

**Introduction to Digital Signal Processing:** Applications of Z-Transforms- Solution of Linear Constant Coefficient Difference equations (LCCD), System function, Frequency Response of the system.

**Discrete Fourier Transforms:** Circular convolution, Comparison between linear and circular convolution, Computation of DFT.

**UNIT-V:**

**IIR Digital Filters:** Design of IIR Digital filters ( $H(s)$  to be given) - Impulse invariance transformation techniques, Bilinear transformation method.

**FIR Digital Filters:** Characteristics of linear phase FIR filters and its frequency response, Comparison of IIR and FIR filters. Design of FIR filters using Fourier Method and Windowing Technique (only Hanning).

**Realization of IIR and FIR Filters:** Direct and Cascade forms.

**TEXT BOOKS:**

1. Signals, Systems and Communications, B.P. Lathi, BS Publications, 2009
2. Signals and Systems, Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab, 2<sup>nd</sup> Edition, PHI
3. Digital Signal Processing: Principles, Algorithms and Applications, John G. Proakis, D.G. Manolakis, 4<sup>th</sup> Edition, Pearson/PHI, 2009

**REFERENCES:**

1. Signals and Systems, Simon Haykin and Barry Van Veen, 2nd Edition, John Wiley
2. Signals, Systems and Transforms, C.L. Philips, J. M Parr and Eve A. Riskin, 3rd Edition, Pearson, 2004
3. Signals and Systems, Schaum's Outlines, Hwei P. Hsu, Tata Mc Graw Hill, 2004
4. Digital Signal Processing – A Practical Approach, Emmanuel C. Ifeachor, Barrie W. Jervis, 2nd Edition, Pearson Education

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22OE1EC304) INTRODUCTION TO IMAGE AND VIDEO PROCESSING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Introduction to Signal Processing

#### **COURSE OBJECTIVES:**

- To introduce fundamentals of digital image processing
- To demonstrate feature extraction, feature classification and object recognition from images
- To understand the basic concepts of video formation and motion estimation
- To learn image processing applications

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basic operations on digital images

**CO-2:** Apply various image representation and description techniques to an image

**CO-3:** Implement object recognition techniques to an image

**CO-4:** Apply image processing to real world problems

**CO-5:** Comprehend video information

#### **UNIT-I:**

**Basics of Image processing:** Introduction, steps involved in image processing, Image acquisitions, Color models, Gray-Level Mapping, Non-linear Gray-Level Mapping, Histogram processing, Image segmentation and compression

#### **UNIT-II:**

**Image Representation and Description:** Representation, Boundary descriptors-simple descriptors, shape numbers, Regional Descriptors-Some Simple Descriptors, Topological Descriptors, Texture, Moment Invariants, Relational Descriptors.

#### **UNIT-III:**

**Object recognition:** Patterns and Pattern Classes, Recognition Based on Decision Theoretic Methods, Matching, Optimum Statistical Classifiers, Neural Networks, Structural Methods, Matching Shape Numbers, String Matching.

#### **UNIT-IV:**

**Applications of image processing:** Image water marking, Fingerprint recognition, Face recognition, Edutainment Game, Coin sorting using a robot.



**UNIT-V:**

**Basics of Video Processing:** Analog video, Digital Video Time varying Image Formation models: 3D motion models, Geometric Image formation, Photometric Image formation, sampling of video signals, Introduction to motion estimation in video.

**TEXT BOOKS:**

1. Digital Image Processing, Gonzalez and Woods, 4th Edition, Pearson, 2018
2. Digital Video Processing, M. Tekalp, 2nd edition, Prentice Hall International, 2015
3. Digital Image Processing, S.Jayaraman, S.Esakkirajan, T.Veerakumar, Tata Mc Graw Hill, 2009

**REFERENCES:**

1. Video Processing and Communication, Yao Wang, Joem Ostarmann and Ya – Quin Zhang, 1st Edition, PHI, 2001
2. Introduction to video and image processing: Building real system and applications, Thomas B. Moeslund, Springer London Ltd; 2012th edition, 23 Jan 2012
3. Fundamentals of Digital Image Processing, A Practical Approach with Examples in Matlab, Chris Solomon, Toby Breckon, 1st edition, John Wiley & Sons, Dec 2010

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22OE1EC403) AUGMENTED REALITY AND VIRTUAL REALITY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Introduction to Image & Video Processing

#### COURSE OBJECTIVES:

- To provide a foundation to the fast growing field of AR and make the students aware of the various AR concepts
- To give historical and modern overviews and perspectives on virtual reality. It describes the fundamentals of sensation, perception, technical and engineering aspects of virtual reality systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain augmented reality, its history, displays, and tracking technologies

**CO-2:** Implement tracking and interaction techniques

**CO-3:** Analyze virtual reality, human physiology, perception, and geometric models

**CO-4:** Understand Visual perception and rendering in VR

**CO-5:** Understand the interaction, auditory perception and rendering in VR

#### UNIT-I:

**Introduction to Augmented Reality:** Augmented Reality - Defining augmented reality, history of augmented reality, Examples, Related fields

**Displays:** Multimodal Displays, Visual Perception, Requirements and Characteristics, Spatial Display Model, Visual Displays.

**Tracking:** Tracking, Calibration, and Registration, Coordinate Systems, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors

#### UNIT-II:

**Computer Vision for Augmented Reality:** Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Outdoor Tracking.

**Interaction:** Output Modalities, Input Modalities, Tangible Interfaces, Virtual User Interfaces on Real Surfaces, Augmented Paper, Multi-view Interfaces, Haptic Interaction

**Software Architectures:** AR Application Requirements, Software Engineering Requirements, Distributed Object Systems, Dataflow, Scene Graphs

#### UNIT-III:

**Introduction to Virtual Reality:** Defining Virtual Reality, History of VR, Human Physiology and Perception.

**The Geometry of Virtual Worlds:** Geometric Models, Axis-Angle Representations of Rotation, Viewing Transformations

**Light and Optics:** Basic Behavior of Light, Lenses, Optical Aberrations, The Human Eye, Cameras, Displays

#### **UNIT-IV:**

**The Physiology of Human Vision:** From the Cornea to Photoreceptors, From Photoreceptors to the Visual Cortex, Eye Movements, Implications for VR

**Visual Perception:** Visual Perception - Perception of Depth, Perception of Motion, Perception of Color

**Visual Rendering:** Visual Rendering -Ray Tracing and Shading Models, Rasterization, Correcting Optical Distortions, Improving Latency and Frame Rates, Immersive Photos and Videos

#### **UNIT-V:**

**Motion in Real and Virtual Worlds:** Velocities and Accelerations, The Vestibular System, Physics in the Virtual World, Mismatched Motion and Vection

**Interaction:** Motor Programs and Remapping, Locomotion, Social Interaction.

**Audio:** The Physics of Sound, The Physiology of Human Hearing, Auditory Perception, Auditory Rendering

#### **TEXT BOOKS:**

1. Augmented Reality: Principles & Practice by Schmalstieg / Hollerer, Pearson Education India; First edition (12 October 2016), ISBN-10: 9332578494
2. Virtual Reality, Steven M. LaValle, Cambridge University Press, 2016

#### **REFERENCES:**

1. AR Game Development II, Allan Fowler, 1st Edition, A press Publications, 2018, ISBN 978-1484236178
2. Understanding Virtual Reality: Interface, Application and Design, William R Sherman and Alan B Craig, (The Morgan Kaufmann Series in Computer Graphics)". Morgan Kaufmann Publishers, San Francisco, CA, 2002
3. Developing Virtual Reality Applications: Foundations of Effective Design, Alan B Craig, William R Sherman and Jeffrey D Will, Morgan Kaufmann, 2009
4. Designing for Mixed Reality, Kharis O'Connell Published by O'Reilly Media, Inc., 2016, ISBN: 9781491962381
5. Theory and applications of marker-based augmented reality, Sanni Siltanen, Julkaisija –Utgivare Publisher, 2012. ISBN 978-951-38-7449-0

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

### (22OE1EC404) VIRTUAL REALITY ENGINEERING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understanding the fundamentals of VR/AR modeling and programming
- To analyzing the engineering, scientific, and functional aspects of VR/AR

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the fundamentals and features of Virtual Reality (VR) and Augmented Reality (AR)

**CO-2:** Understand modeling and visual perception in VR/AR environments

**CO-3:** Analyze light, the human optical system, and tracking systems for effective VR/AR design

**CO-4:** Implement rendering and various Sensory Modalities for VR/AR experiences

**CO-5:** Acquire knowledge about the applications of VR/AR technologies

#### UNIT-I:

**Introduction to Virtual and Augmented Reality:** Introduction to Virtual Reality (VR) and Augmented Reality (AR). Fundamentals, features, and the chronicle order of VR and AR. Differences between AR and VR. History of VR/AR.

#### UNIT-II:

**Fundamentals of VR/AR Modeling and Perception:** Geometric Modeling and Transforms: Mathematical modeling for 3D objects. Transforms, rotations, and quaternions in VR systems.

**Study of Perception and Sensation:** Human perception and sensory systems. Perceptual engineering and its importance in VR/AR.

**Visual Perception:** Depth and motion perception. Integration into immersive VR/AR environments

#### UNIT-III:

**Human Systems and Interaction Light and Optics:** Principles of light and its interaction with materials. Understanding the human optical system for VR displays. Human Visual Physiology: Anatomy and functionality of the human visual system. Implications for AR/VR headset design.

**Tracking Systems:** Pose tracking and motion tracking technologies. Applications of pose tracking in VR/AR systems.

**UNIT-IV:**

**Rendering and Programming Visual Rendering:** Rasterization and shading techniques. CUDA programming for VR/AR rendering.

**Auditory Sensation and Perception:** Basics of auditory perception and rendering 3D audio.

**Haptics and Other Sensory Modalities:** Haptic feedback and stereognosis. Rendering tactile sensations and exploring other sensory inputs in VR/AR.

**UNIT-V:**

**Applications, Interfaces, and Challenges Interfaces for VR/AR:** Locomotion, manipulation, and social interaction in VR/AR.

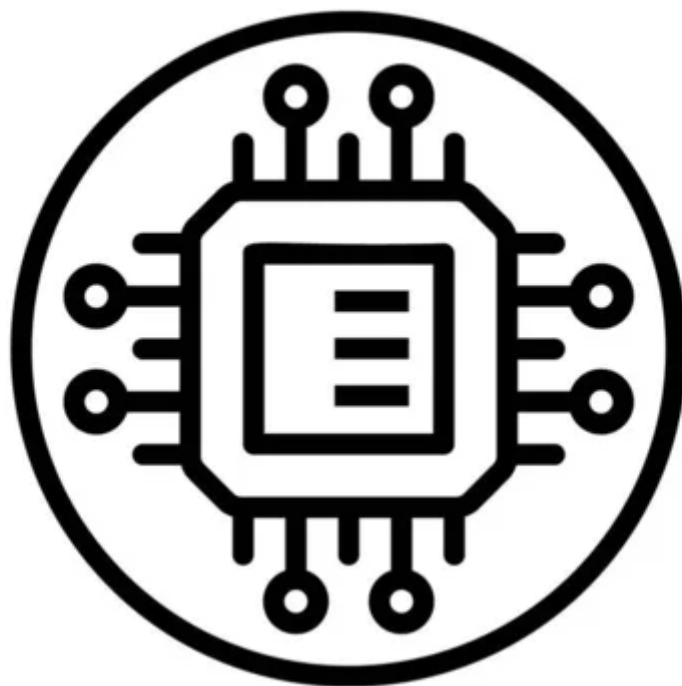
**Applications of VR/AR:** Current and potential uses across various industries.

**Challenges and Evaluation:** Methods for evaluating VR/AR systems. Design challenges in effective VR/AR implementation.

**TEXT BOOKS:**

1. "Virtual Reality Technology" by Grigore C. Burdea and Philippe Coiffet, John Wiley & Sons, 2003
2. [https://archive.nptel.ac.in/content/syllabus\\_pdf/121106013.pdf](https://archive.nptel.ac.in/content/syllabus_pdf/121106013.pdf)

# **MICROELECTRONICS**



## MICROELECTRONICS

Offered by:

**ELECTRONICS AND  
COMMUNICATION  
ENGINEERING**

Courses in the OE Track:

| OE Tracks                                           | V Sem<br>(OE-I)                                                   | VI Sem<br>(OE-II)                               | VII Sem<br>(OE-III)                      | VIII Sem<br>(OE-IV)        |
|-----------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------|------------------------------------------|----------------------------|
| <b>Augmented Reality (AR)/ Virtual Reality (VR)</b> | Principles of Microelectronics / Fundamentals of Microelectronics | Programmable Logic Devices for Microelectronics | Fundamentals of Physical Design For VLSI | Trends in Microelectronics |

## **MICROELECTRONICS**

The microelectronics open elective track covers various aspects of microelectronics and Very Large Scale Integration (VLSI) design.

Principles of Microelectronics course covers the fundamental principles and concepts of microelectronics, including digital electronics, electronic devices and basic circuit design.

Programmable Logic Devices for Microelectronics course focuses on the design and application of programmable logic devices (PLDs) in microelectronics, emphasizing Field Programmable Gate Arrays (FPGAs) and Complex Programmable Logic Devices (CPLDs).

Fundamentals of Physical Design for VLSI course covers the physical design process in VLSI, including placement, routing, and design rules.

Recent Trends in Microelectronics course explores the latest advancements and emerging trends in the field of microelectronics like nanotechnology, Micro-Electro-Mechanical Systems (MEMS), 3D ICs etc,

These courses collectively provide a comprehensive understanding of microelectronics and VLSI design, equipping students with the knowledge and skills to tackle contemporary challenges and innovations in the field.



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EC305) PRINCIPLES OF MICROELECTRONICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the concepts of Diode and BJT
- To study the fundamentals of combinational circuits
- To learn electrical properties of MOS transistors
- To learn the basic principles of VLSI and fabrication steps
- To know the design of digital circuits using MOS logic

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Acquire knowledge of PN Junction diode and BJT

**CO-2:** Realize combinational logic circuits using Boolean algebra and Karnaugh maps

**CO-3:** Interpret the electrical properties of MOS transistors

**CO-4:** Understand the fabrication steps of integrated circuits

**CO-5:** Realize Boolean expressions using MOS logic

#### UNIT-I:

**Electronic Devices:** PN junction diode and its operation, V-I characteristics of PN Junction Diode, Operation of Half wave and full wave rectifiers. BJT structure and operation, BJT configurations.

#### UNIT-II:

**Digital Fundamentals:** Types of number representations and conversions, Boolean Algebra, Canonical and Standard forms, Karnaugh maps (Three and Four Variables, Digital Logic gates, realization of Boolean functions, NAND and NOR implementations.

#### UNIT-III:

**Basic Electrical Properties of MOS Transistors:** MOSFET types, MOSFET structure and operation, I-V characteristics of MOSFETs,  $I_{DS}$  versus  $V_{DS}$  relationship, MOS transistor threshold voltage, Transconductance and Output conductance (Qualitative analysis only).

#### UNIT-IV:

**Microelectronics:** Introduction to Microelectronics and Integrated Circuit Technology, Evolution of Integrated Circuits, Moore's Law, VLSI design flow, NMOS, PMOS and CMOS fabrication steps.

#### UNIT-V:

**MOS Logic Circuits:** CMOS Inverter, CMOS logic gates design, Realizing Boolean expressions using NMOS and CMOS logic, Pass transistor logic, CMOS transmission gates.

**TEXT BOOKS:**

1. Essentials of VLSI Circuits and Systems, Kamran Eshraghian, Douglas and A. Pucknell, PHI, 2005
2. Electronic Devices and Circuit Theory, Robert L. Boylestad and Louis Nashelsky
3. Digital Design, M. Morris Mano, 3rd Edition, Pearson Education/PHI, 2003

**REFERENCES:**

1. CMOS Digital Integrated Circuits Analysis and Design – Sung-Mo Kang, Yusuf Leblebici, , 3<sup>rd</sup> Edition, TMH, 2011
2. CMOS VLSI design: A Circuits and Systems Perspective, N.H.E. Weste and D.M. Harris, 4<sup>th</sup> Edition, Pearson Education India, 2011
3. The Design and Analysis of VLSI Circuits, L. Glaser and D. Dobberpuhl, Addison Wesley, 1985

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EC309) FUNDAMENTALS OF MICROELECTRONICS (EEE, EIE)

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn the fabrication process of integrated circuits
- To learn electrical properties of MOS transistors
- To know the design of digital circuits using MOS logic
- To study the concepts of stick diagrams and layouts with the knowledge of MOS layers
- To understand basic circuit concepts and new trends in microelectronics

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand IC Fabrication process steps required for various MOS transistors

**CO-2:** Interpret the electrical properties of MOS transistors

**CO-3:** Design the digital circuits using various logic styles

**CO-4:** Comprehend the design rules and scaling factors of MOS Transistors

**CO-5:** Know the circuit concepts of ICs and new trends in microelectronics

#### UNIT-I:

**Microelectronics:** Introduction to Integrated Circuit Technology, The Integrated Circuit Era, MOS and Related Technology, Basic MOS Transistors, Operation of Enhancement and Depletion Mode Transistors.

**Fabrication Process:** Silicon Wafer Preparation, Epitaxial Growth, Oxidation, Photolithography, Diffusion, Ion implantation, Metallization, Assembly Processing and Packaging, Encapsulation, nMOS and pMOS fabrication, CMOS fabrication using p-Well, n-Well and Twin Tub processes.

#### UNIT-II:

**Basic Electrical Properties of MOS circuits:**  $I_{DS}$  versus  $V_{DS}$  relationship, MOS transistor threshold voltage, Transconductance and Output conductance (Qualitative analysis only), MOS transistor Figure of Merit, NMOS Inverter, Alternative forms of pull-up.

#### UNIT-III:

**MOS Logic Circuits:** Characteristics of CMOS Inverter, Design of logic gates using NMOS and CMOS, Realizing Boolean expressions using NMOS and CMOS logic. Operation Pass Transistor Logic (PTL) and CMOS Transmission Gates (TGs), Logic gates design with PTL and TGs

**UNIT-IV:**

**VLSI Circuit Design Processes:** VLSI Design Flow, MOS Layers, Stick Diagrams- nMOS and CMOS design styles, Design Rules and Layouts - nMOS Lambda based design rules, Contact cuts, Layout Diagram for nMOS and CMOS inverters.

**Scaling of MOS circuits** - Scaling models and scaling factors, scaling factors for device parameters.

**UNIT-V:**

**Basic Circuit Concepts:** Sheet Resistance and its concept applied to MOS transistors and inverters, Area Capacitance of layers and its calculations, delay unit, Inverter Delays, Wiring Capacitances, Choice of layers.

**Trends in Microelectronics:** Introduction to CNTFET, FinFET, multi-gate FET and GNR-FET.

**TEXT BOOKS:**

1. Essentials of VLSI Circuits and Systems, Kamran Eshraghian, Douglas and A. Pucknell, PHI, 2005
2. CMOS Digital Integrated Circuits Analysis and Design – Sung-Mo Kang, Yusuf Leblebici, , 4<sup>th</sup> Edition, TMH, 2019

**REFERENCES:**

1. CMOS VLSI design: A Circuits and Systems Perspective, N.H.E. Weste and D.M. Harris, 4<sup>th</sup> Edition, Pearson Education India, 2011
2. Introduction to VLSI Systems: A Logic, Circuit and System Perspective, Ming-BO Lin, CRC Press, 2011
3. Digital Integrated Circuits–A Design Perspective, Jan M. Rabaey, Anantha Chandrakasan, Borivoje Nikolic, 2nd Edition, PHI, 2016

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1EC306) PROGRAMMABLE LOGIC DEVICES FOR MICROELECTRONICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Principles of Microelectronics

#### COURSE OBJECTIVES:

- To introduce digital design concepts of combinational and Sequential Circuits
- To learn digital design concepts through Programmable Logic Devices
- To understand the design flow of digital circuits
- To learn physical design steps of FPGA
- To know the architectures of CPLD and FPGAs

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basics of digital circuits

**CO-2:** Realize digital circuits using PLDs

**CO-3:** Simulate and synthesize digital circuits

**CO-4:** Analyze physical design flow of FPGA

**CO-5:** Implement digital circuits using PLDs

#### UNIT-I:

**Digital Design:** Binary adders and subtractors, multiplexers, demultiplexers, Realization of logic expressions using Multiplexers, decoders, encoders, Latches and flip-flops, shift registers, synchronous binary counter and Up-Down Counter (Qualitative analysis only).

#### UNIT-II:

**Programmable Logic Devices:** Classification of memories, PROM, Programmable Logic Arrays and Programmable Array Logic.

#### UNIT-III:

**Simulation and Synthesis:** Simulation, Simulation types, Synthesis, Synthesis Methodologies, Translation, Mapping, Optimization.

**Introduction to FPGA:** Features and Applications of FPGAs, Advantages of FPGA, General Architecture of FPGA, Programming Technologies.

#### UNIT-IV:

**Physical Design flow of FPGAs:** FPGA Design flow, Physical Design cycle for FPGAs, Partitioning, placement, Segmented and Non-Segmented Routing techniques.

**UNIT-V:**

**CPLD and FPGA Architectures:** CPLD architecture- Altera series – Max 7000, General Design Issues, Commercially Available FPGAs, XILINX FPGA- Artix-7, Actel ACT1, ACT2 and ACT3 FPGA architectures.

**Case study:** Full adder implementation using FPGA.

**TEXT BOOKS:**

1. Modern Digital Electronics, R. P. Jain, 4<sup>th</sup> Edition, Tata McGraw-Hill, 2009
2. Digital Design, M. Morris Mano, 3<sup>rd</sup> Edition, Pearson Education/PHI
3. Application Specific Integrated Circuits, M.J.S .Smith, Pearson Education, 2004

**REFERENCES:**

1. Field-Programmable Gate Arrays, Stephen D. Brown, Springer, 1992
2. Algorithms for VLSI Physical Design Automation, Naveed Sherwani, 3<sup>rd</sup> Edition, Springer International Edition, 2005
3. Field Programmable Gate Array Technology, Stephen M. Trimberger, Springer International Edition, 1994

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

### (22OE1EC405) FUNDAMENTALS OF PHYSICAL DESIGN FOR VLSI

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Principles of Microelectronics

#### COURSE OBJECTIVES:

- To know the basics of VLSI Physical Design Automation
- To learn system design packages
- To study different algorithms of physical design

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the automation process for VLSI System design

**CO-2:** Analyze the different packaging techniques

**CO-3:** Comprehend the partitioning algorithms and computational techniques

**CO-4:** Design the systems using various floor planning algorithms

**CO-5:** Apply various Routing algorithms in VLSI system design

#### UNIT-I:

**Introduction to Physical Design:** VLSI Design Cycle, New Trends in VLSI Design Cycle - Design Representation- Y- Chart, VLSI Design Styles - Full-Custom, Standard Cell, Gate Arrays and Field Programmable Gate Arrays.

#### UNIT-II:

**System Packaging Styles:** Die Package Styles, Package and Die Attachment Styles, Printed Circuit Boards, Multichip Modules, Wafer Scale Integration, Comparison of Different Packaging Styles

#### UNIT-III:

**Partitioning:** Constraints and objectives of the partitioning process - System Level, Board Level and Chip Level Partitioning, Problem Formulation, Classification of Partitioning Algorithms.

#### UNIT-IV:

**Floor planning:** Factors considered for floor planning, Problem Formulation, Classification of Floor planning Algorithms. Introduction to Pin Assignment and placement.

#### UNIT-V:

**Global Routing:** Introduction to Global Routing, general routing problem, objective of the routing problem, Problem Formulation- Grid Graph, Checker Board and Channel Intersection Graph Models, Classification of Global Routing Algorithms.

**Detailed Routing:** Introduction to detailed Routing, Routing Considerations, Routing Models, Channel Routing Problems, Switchbox Routing Problems.

**TEXT BOOKS:**

1. Algorithms for VLSI Physical Design Automation, Naveed Sherwani, 3<sup>rd</sup> Edition, 2005
2. Algorithms for VLSI Design Automation, S. H. Gerez, Wiley Student Edition, John Wiley & Sons (Asia) Pvt. Ltd., 1999

**REFERENCES:**

1. Computer Aided Logical Design with Emphasis on VLSI, Hill & Peterson, Wiley, 1993
2. Modern VLSI Design: Systems on Silicon, Wayne Wolf, 2<sup>nd</sup> Edition, Pearson Education Asia, 1998
3. Physical Design Automation of VLSI Systems, Bryan, Lorenzetti, Michael T
4. Digital VLSI Design, Ajay Kumar Singh, Published by PHI Learning, 2011



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1EC406) TRENDS IN MICROELECTRONICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the concepts of MOSFETs
- To learn the fabrication process of integrated circuit
- To study the ASIC design flow
- To Understand System on chip design and its constituents
- To know the different SOC packaging techniques

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basics of IC technology and transistor characteristics and operation

**CO-2:** Apply the fabrication concepts for IC design,

**CO-3:** Apply ASIC concepts in designing VLSI systems

**CO-4:** Develop VLSI systems using SOC principles

**CO-5:** Use different SoC packaging technologies in VLSI design

#### UNIT-I:

**Introduction to MOS Technology:** Introduction to Integrated Circuit Technology, The Integrated Circuit Era, Trends in VLSI- Complexity, Speed of Operation, Die Size, Design Methodology, VLSI Design Flow.

MOS and Related Technology, Basics of MOS, Bi-CMOS Transistors, Operation of Enhancement and Depletion Mode Transistors.

#### UNIT-II:

**Fabrication Process:** Silicon Wafer Preparation, Epitaxial Growth, Oxidation, Photolithography, Diffusion, Ion implantation, Metallization, Assembly Processing and Packaging, Encapsulation, nMOS and pMOS fabrication, CMOS fabrication using p-Well, n-Well and Twin Tub processes.

#### UNIT-III:

**Introduction to ASICs:** Types of ASICs, ASIC Design flow, EDA tools for ASIC Design, ASIC cell libraries, Combinational Logic Cell, Sequential logic cell, I/O cell

#### UNIT-IV:

**System on Chip:** Overview of System on Chip (SOC), VLSI Circuit to System on Chip, SOC Design Flow, SOC Constituents- Embedded Processor Subsystem for System on Chip, Embedded Memories, Protocol Blocks, Mixed Signal Blocks, RF Control Blocks, Analog Blocks, Third-Party IP Cores, System Software.

**UNIT-V:**

**SOC Packaging:** Introduction to VLSI SOC Packaging, Classification of Packages, Criteria for Selection of Packages, Package Components, Package Assembly Flow, Packaging Technology, Flip-Chip Packages, Typical Packages, Package Performance, System Integration.

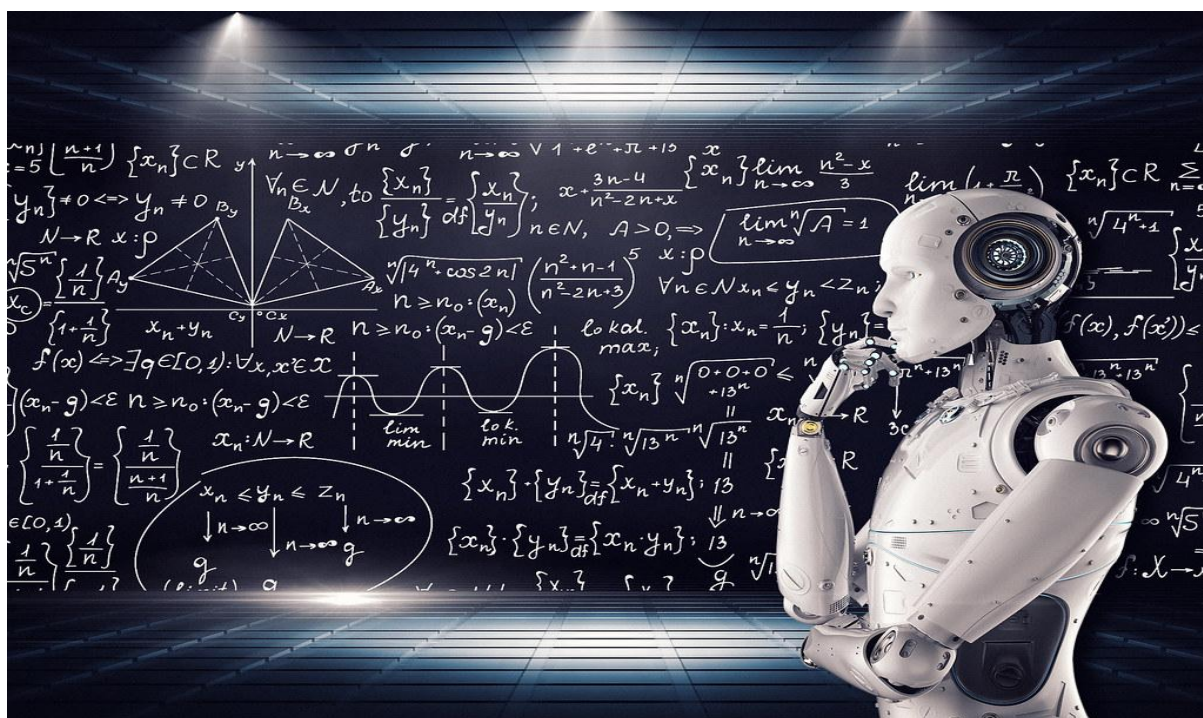
**TEXT BOOKS:**

1. Essentials of VLSI Circuits and Systems, Kamran Eshraghian, Douglas and A. Pucknell, PHI, 2005
2. Application Specific Integrated Circuits, J. Smith, Addison-Wesley, 2010
3. A practical approach to VLSI system on chip (SoC) design, Chakravarthi, Veena, Springer International Publishing, 2020.

**REFERENCES:**

1. Introduction to VLSI Systems: A Logic, Circuit and System Perspective, Ming-BO Lin, CRC Press, 2011
2. VLSI Fabrication Principles, S. K. Gandhi, John Wiley & Sons, 1983
3. System on Chip-Next Generation Electronics, B. Al Hashimi, The IET, 2006
4. VLSI Design- K. Lal Kishore, V. S. V. Prabhakar, I.K International, 2009
5. Introduction to VLSI Circuits and systems, John P. Uyemura, John Wiley & Sons, 2011

# **ARTIFICIAL INTELLIGENCE**



## ARTIFICIAL INTELLIGENCE

Offered by:

**COMPUTER SCIENCE AND  
ENGINEERING**

Courses in the OE Track:

| OE Tracks                          | V Sem<br>(OE-I)                                                                                                               | VI Sem<br>(OE-II)                             | VII Sem<br>(OE-III)               | VIII Sem<br>(OE-IV)               |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------|-----------------------------------|
| <b>Artificial<br/>Intelligence</b> | Applied<br>Mathematics<br>in Artificial<br>Intelligence<br>/<br>Mathematical<br>Foundations<br>for Artificial<br>Intelligence | Fundamentals<br>of Artificial<br>Intelligence | Machine<br>Learning<br>Techniques | Essentials of<br>Deep<br>Learning |

## **ARTIFICIAL INTELLIGENCE**

Artificial Intelligence (AI) is a cognitive science with highly research activities in the major areas like Machine Learning, Robotics, Natural Language Processing and image processing. This track will cover basic foundations of artificial intelligence it will make the students industry-ready for artificial intelligence and data science job roles. Artificial intelligence is used in wide range of industrial applications such as healthcare, transportation, entertainment, insurance, transport and logistics, and customer service.

Future applications of AI would be utilized in automated transportation, cyborg technology, solving problems associated with climate change, deep-sea and space exploration.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1MT301) APPLIED MATHEMATICS IN ARTIFICIAL INTELLIGENCE (CE, ME & AE)

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Basic knowledge of probability, regression and Calculus

#### COURSE OBJECTIVES:

- To learn multi-dimensional generalization of a univariate normal random variable
- To learn concept of multivariable linear regression model
- To applying knowledge of vectors, inner products, and linear transformations for solving real world situations
- To constructing mathematical models for solving real-world problems
- To understanding optimization techniques like convex and constrained optimization

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Model the linear relationship between the explanatory (independent) Variables and response(dependent)variables

**CO-2:** Make better decisions using linear regression techniques

**CO-3:** Recognize and use basic properties of subspaces and vector spaces

**CO-4:** Identify the use of matrix theory to solve the system of linear equations and apply in various engineering problems

**CO-5:** Analyze and solve the complexity of a given problem with suitable optimization techniques

#### UNIT-I:

**Multivariate Normal Distribution & Linear Regression Model:** Multivariate Normal Distribution Functions, Conditional Distribution and its relation to regression model, Standard multiple regression models with emphasis on detection of collinearity, outliers.

#### UNIT-II:

**Multivariate Regression:** Assumptions of Multivariate Regression Models, Parameter estimation, Multivariate Analysis of variance and covariance

#### UNIT-III:

**Vector Space:** Orthogonal vectors, normal and orthonormal vectors, linear combination, linear span, linear independence of vectors, Vector space; Subspace, Dimension; Basis; Orthogonality; Projections; Gram-Schmidt orthogonalization and QR decomposition.

**UNIT-IV:**

**Applications of Matrices:** Solution of Tri-diagonal system of equations using LU decomposition Method; Applications of linear systems- Network Flows and Mechanical Systems.

Review of eigen values and eigen vectors, intuition, significance, Principal component analysis-concept, properties, applications, Singular value decomposition-concept, properties, applications

**UNIT-V:**

**Multivariate Calculus:** Review of gradient and its properties. Hessian, Gradient of vector valued function, Gradient of matrices. Local/global maxima and minima, convex sets, convex functions, gradient descent algorithms- Learning rate, momentum, stochastic. Constrained optimization (Lagrange Multiplier method), convex optimization.

**TEXTBOOKS:**

1. An introduction to Multivariate Statistical Analysis, T.W.Anderson
2. Mathematics for Machine Learning, Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, Published by Cambridge University Press., 2020
3. Linear algebra and it's applications, David C. Lay, third edition, Pearson publications

**REFERENCES:**

1. Math for Machine Learning: Open Doors to Data Science and Artificial Intelligence, Richard Han, Paperback-2018
2. Artificial Intelligence Engines: A Tutorial Introduction to the Mathematics of Deep Learning by James V Stone
3. Advanced Engineering Mathematics, Erwin Kreyszig, 9<sup>th</sup> Edition, John Wiley & Sons, 2006

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. V Semester

### (22OE1MT302) MATHEMATICAL FOUNDATIONS FOR ARTIFICIAL INTELLIGENCE (EEE & EIE)

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Permutations, Combinations, Differentiation & Integration

#### COURSE OBJECTIVES:

- To understanding elementary ideas in basic probability, distribution functions
- To demonstrating various statistical methods and linear relationship between the given variables
- To applying knowledge of vectors, inner products, and linear transformations for solving real world situations
- To constructing mathematical models for solving real-world problems
- To understanding optimization techniques like convex and constrained optimization

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explore and demonstrate practical approaches related to implementation of the AI algorithms using probability concepts and distributions

**CO-2:** Use and fit a linear regression model to data and use it for prediction

**CO-3:** Recognize and use basic properties of subspaces and vector spaces

**CO-4:** Identify the use of matrix theory to solve the system of linear equations and apply in various engineering problems

**CO-5:** Analyze and solve the complexity of a given problem with suitable optimization techniques

#### UNIT-I:

**Probability:** Basic rules and axioms, events, sample space, frequentist approach, dependent and independent events, conditional probability, Random variables, continuous and discrete, expectation, variance, distributions- joint and conditional, Bayes' Theorem, Popular distributions- Bernoulli, binomial, Poisson, Normal.

#### UNIT-II:

**Descriptive Statistics & Linear Regression:** Classification and tabulation of univariate data, graphical representation, Frequency curves. Descriptive measures - Central tendency and Dispersion. Bivariate data, Summarization, marginal and conditional distribution. Correlation and Regression, Simple Linear Regression Models.

#### UNIT-III:

**Vector Space:** Orthogonal vectors, normal and orthonormal vectors, linear combination, linear span, linear independence of vectors, Vector space; Subspace,



Dimension; Basis; Orthogonality; Projections; Gram-Schmidt orthogonalization and QR decomposition.

#### **UNIT-IV:**

**Applications of Matrices:** Solution of Tri-diagonal system of equations using LU decomposition Method; Applications of linear systems- Network Flows and Mechanical Systems.

Review of eigen values and eigen vectors, intuition, significance, Principal component analysis-concept, properties, applications, Singular value decomposition-concept, properties, applications

#### **UNIT-V:**

**Multivariate Calculus:** Review of gradient and its properties. Hessian, Gradient of vector valued function, Gradient of matrices. Local/global maxima and minima, convex sets, convex functions, gradient descent algorithms- Learning rate, momentum, stochastic. Constrained optimization (Lagrange Multiplier method), convex optimization.

#### **TEXTBOOKS:**

1. Mathematics for Machine Learning, Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, Published by Cambridge University Press., 2020
2. Linear algebra and it's applications, David C. Lay, third edition, Pearson publications
3. 3. Probability and Statistics for Engineers, Richard. A.Johanson, 5<sup>th</sup> Edition, Prentice-Hall, 1995

#### **REFERENCES:**

1. Math for Machine Learning: Open Doors to Data Science and Artificial Intelligence, Richard Han, Paperback–2018
2. Artificial Intelligence Engines: A Tutorial Introduction to the Mathematics of Deep Learning by James V Stone
3. Advanced Engineering Mathematics, Erwin Kreyszig, 9<sup>th</sup> Edition, John Wiley & Sons, 2006

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1CS303) FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mathematics for Artificial Intelligence

#### COURSE OBJECTIVES:

- To understand and analyze the importance and basic concepts of artificial intelligence and the use of agents
- To identify, explore the complex problem-solving strategies and approaches
- To analyze the concepts of basic concepts of neural networks and learning process
- To explore and analyze the methodology used in machine learning

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basic concepts of artificial intelligence and the use of agents into the real-world scenario

**CO-2:** Design and formulate complex problem solutions with the use of various searching techniques

**CO-3:** Estimate the skill for representing knowledge using the appropriate technique for a given problem

**CO-4:** Apply AI techniques to solve problems of game playing, and machine learning

#### UNIT-I:

**Introduction to AI:** Foundations of AI – History of AI - Applications of AI, Intelligent Agents – Agents and Environments – Nature of Environments – Structure of Agents – Problem solving Agents – Problem formulation – Example Problems.

#### UNIT-II:

**Searching Techniques:** Uninformed Search Strategies – Breadth first search – Depth first search – Depth limited search - Bidirectional search – comparison – Search with partial information - Heuristic search – Greedy best first search – A\* search – Memory bounded heuristic search - Heuristic functions - Local search- Hill climbing – Simulated annealing search - Local beam search, Genetic algorithms.

#### UNIT-III:

**Constraint Satisfaction Problems:** Backtracking search for CSP's - local search for constraint satisfaction problem. Adversarial search – Games - Minimax algorithm, Alpha beta pruning, cutting-off search.

**UNIT-IV:**

**Knowledge Representation and Reasoning:** Propositional Logic, Rules of Inference, First Order Logic (FOL) Syntax, Semantics, Entailment.

**Classical Planning:** Definition of Classical Planning, Algorithms for Planning with State Space Search, Planning Graphs, other Classical Planning Approaches, Analysis of Planning approaches.

**UNIT-V:**

**Planning and Acting in the Real World:** Time, Schedules, and Resources, Hierarchical Planning, Planning and Acting in Nondeterministic Domains, Multi agent Planning.

**TEXT BOOKS:**

1. Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, 3<sup>rd</sup> Edition, Prentice Hall, 2010
2. Machine Learning, Tom M. Mitchell, McGraw Hill Publications
3. Neural Networks A Comprehensive Foundation, Simon Haykin, Pearson Education, 2<sup>nd</sup> Edition, 2004

**REFERENCES:**

1. Artificial Intelligence, Elaine Rich & Kevin Knight, 2<sup>nd</sup> Edition, TMH
2. Artificial Intelligence-A New Synthesis, Nils J. Nilsson, Elsevier
3. Artificial Neural Networks, Yegnanarayana B., PHI

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22OE1CS401) MACHINE LEARNING TECHNIQUES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mathematics for Artificial Intelligence, Fundamentals of Artificial Intelligence

#### COURSE OBJECTIVES:

- To understand applications in computational learning theory
- To analyse the pattern comparison techniques

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand and Familiarize the basics concept, notations used in machine learning and mathematics behind machine learning algorithms

**CO-2:** Demonstrate different types of machine learning algorithms

**CO-3:** Apply the suitable machine learning techniques and construct a machine learning model to solve real world applications

**CO-4:** Evaluate model accuracy and familiarize with advanced learning algorithms

#### UNIT-I:

**Introduction to Machine Learning:** Perspectives and issues in machine learning, Goals and applications of machine learning. Aspects of developing a learning system: training data, concept representation, function approximation.

**Regression:** Regression types, gradient descent; features of Over fitting and complexity; training, validation, test data, Logistic regression and applications.

#### UNIT-II:

**Supervised Learning:** Classification, decision boundaries; nearest neighbor methods, Decision Tree Learning – Introduction, decision tree representation, appropriate problems for decision tree learning, Linear classifiers Bayes' Rule and Naive Baye's classification.

**Instance-Based Learning:** Introduction, k-nearest neighbour algorithm, locally weighted regression, radial basis functions, case-based reasoning, remarks on lazy and eager learning.

#### UNIT -III:

**Unsupervised Learning:** Clustering, k-means, hierarchical, partition-based clustering, overlapping clustering, Support vector machines, Support vector regression.

**UNIT -IV:**

**Reinforcement Learning:** Introduction to Reinforcement learning, the learning task, rewards and actions, temporal difference learning, generalizing from examples, relationship to dynamic programming.

**UNIT-V:**

**Neural Networks:** Introduction to neural networks, neural network representation, appropriate problems for neural network learning, perceptions, multilayer networks and Convolution neural networks.

**TEXT BOOKS:**

1. Machine Learning, Tom M. Mitchell, McGraw-Hill
2. Neural Networks and Learning Machines, S. Haykin, Pearson, 2008

**REFERENCES:**

1. Machine Learning: An Algorithmic Perspective, Stephen Marshland, Taylor & Francis
2. Machine Learning: The Art and Science of Algorithms that make Sense of Data, Peter Flash, Cambridge, University Press
3. Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, MIT Press, 2012

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1CS402) ESSENTIALS OF DEEP LEARNING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Mathematics for Artificial Intelligence, Fundamentals of Artificial Intelligence, Machine Learning Techniques

#### COURSE OBJECTIVES:

- To introduce the foundations of deep learning
- To acquire the knowledge on Deep Learning Concepts

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Identify and select appropriate learning network models required for real world problems

**CO-2:** Design an efficient model with various deep learning techniques

**CO-3:** Implement deep learning algorithms and solve real-world problems

**CO-4:** Apply optimization strategies necessary for problem solving required for large scale applications

#### UNIT-I:

**Introduction to Deep Learning:** History of Deep Learning, Deep Learning Success Stories, Biological Neuron, Idea of computational units, McCulloch Pitts Neuron, Thresholding Logic, Perceptrons, Perceptron Learning Algorithm and Convergence.

#### UNIT-II:

**Feedforward Networks:** Multilayer Perceptron, Gradient Descent, Back-propagation, Kohonen Self-Organizing Feature Maps, Learning Vector Quantization, Counter Propagation Networks, Adaptive Resonance Theory Networks.

#### UNIT-III:

**Regularization for Deep Learning:** Parameter norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise Robustness, Semi-Supervised learning, Multi-task learning, Early Stopping, Parameter Typing and Parameter Sharing, Sparse Representations, Bagging and other Ensemble Methods, Dropout, Adversarial Training, Tangent Distance, tangent Prop and Manifold, Tangent Classifier.

#### UNIT-IV:

**Optimization for Training Deep Models:** Challenges in Neural Network Optimization, Basic Algorithms, Parameter Initialization Strategies, Algorithms with Adaptive Learning Rates, Approximate Second-Order Methods, Optimization Strategies and Meta-Algorithm.

**UNIT-V:**

**Convolutional Neural Networks:** LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, Markov Networks, Object Detection, RCNN, Fast RCNN, Faster RCNN, YOLO, Auto-Encoders: Regularization in auto-encoders, De-noising auto-encoders, Sparse auto-encoders, Contractive auto-encoders,

**TEXT BOOKS:**

1. Deep Learning: An MIT Press Book, Ian Goodfellow and Yoshua Bengio and Aaron Courville
2. Neural Networks and Learning Machines, Simon Haykin, 3rd Edition, Pearson Prentice Hall

**REFERENCES:**

1. Neural Networks: A Systematic Introduction, Raúl Rojas, 1996
2. Pattern Recognition and Machine Learning, Christopher Bishop, 2007

# **BLOCKCHAIN TECHNOLOGIES**





## BLOCKCHAIN TECHNOLOGIES

Offered by:

**COMPUTER SCIENCE AND  
ENGINEERING**

Courses in the OE Track:

| OE Tracks                          | V Sem<br>(OE-I)                                                                              | VI Sem<br>(OE-II)         | VII Sem<br>(OE-III)                     | VIII Sem<br>(OE-IV)      |
|------------------------------------|----------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------|--------------------------|
| <b>Blockchain<br/>Technologies</b> | Fundamentals<br>of Computer<br>Networks /<br>Relational<br>Database<br>Management<br>Systems | Distributed<br>Data Bases | Cryptography<br>and Network<br>Security | Blockchain<br>Technology |

## **BLOCKCHAIN TECHNOLOGIES**

The blockchain is one of the fastest growing skills in the IT sector today. This track will help the students to gain knowledge in blockchain technology, it has taken quite a turn in the industry given its popularity in providing safe and secured online transactions. Most individuals and organizations have started adopting blockchain because of the many benefits it offers to the industry today. It is used in many industry applications such as banking sector, voting, health care, real estate, the legal industry and government.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1CS301) FUNDAMENTALS OF COMPUTER NETWORKS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce the fundamental various types of computer networks
- To demonstrate the TCP/IP and OSI models with merits and demerits
- To explore the various layers of OSI model
- To introduce UDP and TCP models
- To have the concept of different routing techniques for data communications

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand and explore the basics of Computer Networks reference models and the functionalities of physical layer

**CO-2:** Learn major concepts, principles involved in Data Link Layer and Network Layer

**CO-3:** Analyze how to maintain QoS in Network and maintaining of Congestion Control

**CO-4:** Demonstrate the Application Layer functionalities and importance of Security in the Network

#### UNIT-I:

**Introduction to Networks:** Internet, Protocols and Standards, The OSI Model, Layers in OSI Model, TCP/IP Suite, Addressing.

**Physical Layer:** Multiplexing, Transmission Media, Circuit Switched Networks, Datagram Networks, and Virtual Circuit Networks.

#### UNIT-II:

**Data Link Layer:** Introduction, Checksum, Framing, Flow and Error Control, Noiseless Channels, Noisy Channels, Random Access Controlled Access, Channelization, IEEE Standards, Ethernet, Giga-Bit Ethernet, Wireless LANs, SONET-SDH, Frame Relay and ATM.

#### UNIT-III:

**Network Layer:** Logical Addressing, Internetworking, Tunneling, Address Mapping, ICMP, IGMP, Forwarding, Routing-Flooding, Bellman Ford, Disjkstra's routing protocols, RIP, OSPF, BGP and Multicast Routing Protocols. Connecting Devices-Passive Hubs, Repeaters, Active Hubs, Bridges, Routers.

#### UNIT-IV:

**Transport Layer:** Process to Process Delivery, UDP, TCP and SCTP Protocols, Congestion, Congestion Control, Quality of Service.

**Application Layer:** Domain Name Space, DNS in Internet, Electronic Mail.

**UNIT-V:**

File Transfer Protocol, WWW, HTTP, SNMP, Multimedia.

**Network Security:** Security services, mechanisms and attacks, IPSec, SSL, VPN, Firewall. Bluetooth, Zigbee, IPv4, IPv6.

**TEXT BOOKS:**

1. Data Communications and Networking, Behrouz A. Forouzan, 4<sup>th</sup> Edition, McGraw Hill Education, 2006
2. Computer Networks, Andrew S. Tanenbaum, 4<sup>th</sup> Edition, Pearson Education
3. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose, K. W. Ross, 3<sup>rd</sup> Edition, Pearson Education

**REFERENCES:**

1. Data Communications and Networks, William Stallings
2. Data Communication and Networks, Bhusan Trivedi, Oxford University Press, 2016
3. An Engineering Approach to Computer Networks, S. Keshav, 2<sup>nd</sup> Edition, Pearson Education
4. Understanding Communications and Networks, 3<sup>rd</sup> Edition, W.A. Shay, Cengage Learning

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS302) RELATIONAL DATABASE MANAGEMENT SYSTEMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basic concepts and the applications of database systems
- To master the basics of SQL and construct queries using SQL
- To understand the relational database design principles
- To become familiar with the basic issues of transaction processing and concurrency control
- To become familiar with database storage structures and access techniques

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Demonstrate the basic concepts of database systems and design E-R models to represent simple database application scenarios

**CO-2:** Formulate SQL queries on the data

**CO-3:** Improve the database design by normalization

**CO-4:** Apply and relate the concept of transaction, concurrency control and recovery in database

**CO-5:** Familiar with basic database storage structures and access techniques using Indexing, hashing including B tree methods

#### UNIT-I:

**Introduction:** Database System Applications, Purpose of Database Systems, View of Data, Database Languages – DDL, DML, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Data Mining and Information Retrieval, Specialty Databases, Database Users and Administrators, History of Database Systems.

**Introduction to Database Design:** Database Design and ER diagrams, Entities, Attributes and Entity sets, Relationships and Relationship sets, Additional features of ER Model, Conceptual Design with the ER Model, Conceptual Design for Large enterprises.

**Relational Model:** Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity constraints, Querying relational data, Logical database Design: ER to Relational, Introduction to Views, Destroying /Altering Tables and Views.

#### UNIT-II:

**Relational Algebra and Calculus:** Preliminaries, Relational Algebra, Relational calculus – Tuple relational Calculus, Domain relational calculus, Expressive Power of Algebra and calculus.

**SQL:** Queries, Constraints, Triggers: Form of Basic SQL Query, UNION, INTERSECT, and EXCEPT, Nested Queries, Aggregate Operators, NULL values Complex Integrity Constraints in SQL, Triggers and Active Databases, Designing Active Databases.

#### **UNIT-III:**

**Schema Refinement and Normal Forms:** Introduction to Schema Refinement, Functional Dependencies - Reasoning about FDs, Normal Forms, Properties of Decompositions, Normalization, Schema Refinement in Database Design, Other Kinds of Dependencies.

#### **UNIT-IV:**

**Transaction Management:** Transactions, Transaction Concept, A Simple Transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Serializability, Transaction Isolation and Atomicity Transaction Isolation Levels, Implementation of Isolation Levels.

**Concurrency Control:** Lock-Based Protocols, Multiple Granularity, Timestamp-Based Protocols, Validation-Based Protocols, Multiversion Schemes.

Recovery System-Failure Classification, Storage, Recovery and Atomicity, Recovery Algorithm, Buffer Management, Failure with loss of nonvolatile storage, Early Lock Release and Logical Undo Operations, Remote Backup systems.

#### **UNIT-V:**

**Storage and Indexing:** Overview of Storage and Indexing: Data on External Storage, File Organization and Indexing, Index Data Structures, Comparison of File Organizations.

**Tree-Structured Indexing:** Intuition for tree Indexes, Indexed Sequential Access Method (ISAM), B+ Trees: A Dynamic Index Structure, Search, Insert, Delete.

**Hash- Based Indexing:** Static Hashing, Extendible hashing, Linear Hashing, Extendible vs. Linear Hashing.

#### **TEXT BOOKS:**

1. Database Management Systems, Raghu Ramakrishnan, Johannes Gehrke, 3<sup>rd</sup> Edition, McGraw Hill Education (India) Private Limited
2. Database System Concepts, A. Silberschatz, Henry. F. Korth, S. Sudarshan, 6<sup>th</sup> Edition, McGraw Hill Education (India) Private Limited,
3. Database Systems, R. Elmasri, Shamkant B. Navathe, 6<sup>th</sup> Edition, Pearson Education

#### **REFERENCES:**

1. Database System Concepts, Peter Rob & Carlos Coronel, Cengage Learning
2. Introduction to Database Management, M. L. Gillenson and others, Wiley Student Edition
3. Database Development and Management, Lee Chao, Auerbach Publications, Taylor & Francis Group
4. Introduction to Database Systems, C. J. Date, Pearson Education

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22OE1IT301) CRYPTOGRAPHY AND NETWORK SECURITY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fundamentals of Computer Networks, Distributed Data Bases

#### COURSE OBJECTIVES:

- To outline security concepts, threats, attacks, services and mechanisms
- To describe various cryptosystems- symmetric key cryptography, public key cryptography
- To apply authentication services and Secure hash functions
- To discuss the concepts of IP Security, web security, viruses and firewalls

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Analyze the basics of security attacks, services, goals and mechanism

**CO-2:** Apply a variety of cryptographic algorithms, Hash Functions and protocols underlying network security and authentication applications

**CO-3:** Understand and analyze the key management requirements and Hash functions

**CO-4:** Examine and analyze various Email and Web Security mechanisms

**CO-5:** Understand the System Level Security issues

#### UNIT-I:

**Security Attacks:** Security Attacks (Active and Passive), Security Services (Confidentiality, Authentication, Integrity, Non-repudiation, Access Control and Availability) and Mechanisms, A model for Internetwork security, Buffer overflow & format string vulnerabilities, TCP session hijacking, ARP attacks, route table modification, UDP hijacking, and man-in-the-middle attacks.

#### UNIT-II:

**Conventional Encryption:** Classical Encryption techniques(Rail fence, Hill Cipher, Caesar cipher, Vigenere cipher, Play Fair Cipher), Fiestel Cipher Structure, Data Encryption Standard, Block Cipher Design Principles and Modes of Operation, Triple DES, RC-4, AES Cipher, Placement of Encryption Function.

**Public Key Cryptography and Authentication:** Confidentiality using Symmetric Encryption – Principles of Public key Cryptosystems, RSA algorithm,

#### UNIT-III:

**Key Management:** Diffie-Hellman key Exchange, Elliptic Curve Cryptography.

**Hash Functions:** Hash Functions, Security of Hash Functions and MACs, Secure Hash Algorithm, HMAC, Digital Signatures, Authentication Protocols, Digital Signature Standard, Authentication Applications: Kerberos, X.509 Authentication Service

**UNIT-IV:**

**Email Security and Web Security:** Electronic Mail Security – PGP/ SMIME, IP security- Architecture, Authentication Header, Encapsulating Security Payload, Key Management, Web Security- Secure Socket Layer, Transport Layer Security and Secure Electronic Transaction

**UNIT-V:**

**System Level Security:** Intrusion detection – password management – Viruses and related Threats – Virus Counter measures – Firewall Design Principles – Trusted Systems.

**TEXT BOOKS:**

1. Cryptography and Network Security – Principles and Practices, William Stallings, Prentice Hall of India, 4th Edition, 2005
2. Hack Proofing Your Network, Ryan Russell, Dan Kaminsky, Rain Forest, Puppy, Joe Grand, David Ahmad, Hal Flynn Ido Dubrawsky, Steve W. Manzuik and Ryan Permech, Wiley Dreamtech

**REFERENCES:**

1. Network Security Essentials: Applications and Standards, William Stallings Prentice Hall, 1999, ISBN 0130160938
2. Security in Computing, Charles B. Pfleeger, Shari Lawrence Pfleeger, 3<sup>rd</sup> Edition, Pearson Education, 2003



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22OE1CS403) DISTRIBUTED DATABASES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Database Management Systems, Computer Networks

#### COURSE OBJECTIVES:

- To introducing distributed database architecture and fragmentation methodologies
- To describe conceptual data models to design and access of fragmented data
- To demonstrate optimized query access strategies
- To enumerate the concepts of distributed transaction processing for reliable data

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Analyze issues related to distributed database design

**CO-2:** Apply Partitioning techniques to databases

**CO-3:** Design and develop query processing strategies

**CO-4:** Describe transaction processing and concurrency control in distributed database

**CO-5:** Derive reliable recovery techniques

#### UNIT-I:

**Introduction:** Features of Distributed versus Centralized Databases,

**Levels of Distribution Transparency:** Reference Architecture for Distributed Databases, Types of Data Fragmentation, Distribution transparency for Read – only Applications, Distribution transparency for update Applications, Distributed database Access primitives, Integrity Constraints in Distributed Databases.

#### UNIT-II:

**Distributed Database Design:** A framework, the design of database fragmentation, the allocation of fragments.

**Translation of Global Queries to Fragment Queries:** Equivalence Transformations for Queries, Transforming Global Queries into Fragment Queries, Distributed Grouping and Aggregate Function Evaluation, Parametric Queries.

#### UNIT-III:

**Query Optimization:** Framework for Query Optimization, Join Queries, General Queries.

**Management of Distributed Transactions:** A Framework for Transaction Management, Supporting Atomicity of Distributed Transactions, Concurrency Control for Distributed Transactions, Architectural aspects of Distributed Transactions

**UNIT-IV:****Concurrency Control:**

Foundation of Distributed Concurrency Control, Distributed Deadlocks, Concurrency Control based on Timestamps, Optimistic Methods for Distributed Concurrency Control.

**UNIT-V:****Reliability:**

Basic Concepts, Nonblocking Commitment Protocols, Reliability and concurrency Control, Determining a Consistent View of the Network, Detection and Resolution of Inconsistency, Checkpoints and Cold Restart.

**TEXT BOOKS:**

1. Principles of Distributed Database Systems, M. Tamer OZSU and Patuck Valduriez,
2. Pearson Education Asia, 2001
3. Distributed Databases, Stefano Ceri and Willipse Pelagatti, McGraw Hill

**REFERENCES:**

1. Database System Concepts, Henry F. Korth, A. Silberchatz and Sudershan, MGH
2. Database Management Systems, Raghuramakrishnan and Johhanes Gehrke, MGH

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

### (22OE1CS404) BLOCKCHAIN TECHNOLOGY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fundamentals of Computer Networks, Distributed Data Bases, Cryptography and Network Security

#### COURSE OBJECTIVES:

- To get the terminologies and overview of blockchain technologies
- To study the concepts and foundation of blockchain technology
- To understand security mechanism and consensus in blockchain
- To design use cases and architecture blockchain technology

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the basic concepts and characteristics of Blockchain technology

**CO-2:** Demonstrate key mechanisms like Decentralization, Transparency and trust, Immutability, High availability, highly secure and different types of Blockchain

**CO-3:** Apply the concept of Hash Function and Related Hash Algorithm to provide the security and analyze the various types of blockchains

**CO-4:** Understand the Crypto Currency and implement, the applications using Blockchain Technology

#### UNIT-I:

**Introduction to Blockchain Part I:** Introduction to Centralized, Decentralized and Distributed system, History of Blockchain, Various technical definitions of Blockchain.

**Introduction to Blockchain Technology Part II:** Generic elements of a blockchain: Block, Transaction, Peer-to-peer network, Node, Smart contract, Why It's Called "Blockchain", Characteristics of Blockchain Technology, Advantages of blockchain technology.

#### UNIT-II:

**Concept of Blockchain Technology Part I:** Cryptography, Hashing, Nonce, Distributed database, Consensus, Smart Contract, Component of block, Structure of Block chain, Technical Characteristics of the Blockchain.

**Concept of Blockchain Technology Part II:** Applications of blockchain technology, Tiers of blockchain technology Blockchain 0, Blockchain 1, Blockchain 2, Blockchain 3, Generation of Blockchain X.

#### UNIT-III:

**Technical Foundations Part I:** Cryptography, Confidentiality, Integrity, Authentication, Cryptographic primitives, Public and private keys, RSA, Discrete logarithm problem,

**Hash Function:** Message Digest (MD), Secure Hash Algorithms (SHAs), Design of Secure

Hash Algorithms (SHA), SHA-256, Design of SHA3, Elliptic Curve Digital signature algorithm.

**Technical Foundations Part II:** Consensus algorithm: Proof of work (PoW), Proof-of-Stake (PoS), Byzantine Fault Tolerance (BFT)

#### **UNIT-IV:**

**Types of Blockchain:** Public blockchains, Private blockchains, Semi-private blockchains, Side chains, Permissioned ledger, Distributed ledger, Shared ledger, Fully private and proprietary blockchains, Tokenized blockchains, Tokenless blockchains, CAP theorem and blockchain

#### **UNIT-V:**

Financial markets and trading, Trading, Exchanges, Trade life cycle, Order anticipators, Market manipulation.

**Crypto Currency:** Bitcoin, Bitcoin definition, Keys and addresses, Public keys in Bitcoin, Private keys in Bitcoin, Bitcoin currency units

**Implementation Platforms:** Hyperledger as a protocol, Reference architecture, Hyperledger Fabric, Transaction Flow, Hyperledger Fabric Details, Fabric Membership, Fabric Membership

#### **TEXT BOOKS:**

1. Mastering Blockchain, Imaran Bashir, Second Edition, Packt
2. Blockchain Basic, Daniel Drescher, A Press

#### **REFERENCES:**

1. Blockchain For Dummies®, IBM Limited Edition, John Wiley & Sons Inc.

**ROBOTICS**



## ROBOTICS

Offered by:

**ELECTRONICS AND  
INSTRUMENTATION  
ENGINEERING**

Courses in the OE Track:

| OE Tracks       | V Sem<br>(OE-I)             | VI Sem<br>(OE-II)      | VII Sem<br>(OE-III)            | VIII Sem<br>(OE-IV)                  |
|-----------------|-----------------------------|------------------------|--------------------------------|--------------------------------------|
| <b>Robotics</b> | Fundamentals<br>of Robotics | Kinematics of<br>Robot | Drive<br>Systems for<br>Robots | Intelligent<br>Systems for<br>Robots |

**ROBOTICS**

Robotics is a field of study that involves the design, construction and operation of robots. This field overlaps with electronics, computer science, mechatronics and artificial intelligence. Robotic companies are booming all over the world and are seeking engineers with skills for implementing **Next -Level Automation**. This Open Elective Track for Robotics consists of four courses and is intended for making students industry ready in the field of robotics.

The First course in this track "Fundamentals **of Robotics**" introduces various physical aspects of building a robot, exploring topics like how a robot perceives its environment using Sensors and how it interacts with its environment through various Actuators & Grippers. This course also inspects a variety of robot applications in different domains. Second Course in this track "Kinematics **& Dynamics of robots**" delves a level deeper discussing analysis and control of robots. It establishes strong mathematical foundation for describing and controlling robot movement. In this course students will learn in detail about Forward Kinematics, Inverse Kinematics, Workspace Analysis and Trajectory planning for robots.

Third Course in the Robotics track "**Drives and Control System for Robots**" explores in detail various Drive Mechanisms used in robotics such as Hydraulic, Pneumatic & Electric drives. After completing this course students will be able to analyze operational aspects of a drive system for a given robotic application. Fourth Course in the track "**Robot Programming and Intelligent Control System**" expands on Robot Programming, discussing various aspects of Robot Programming Languages and their functions. This course also dives deep into advanced topics like Artificial Intelligence, Neural Networks and Fuzzy control for robots.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1EI301) FUNDAMENTALS OF ROBOTICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basic components of a Robot
- To learn different types of Robot sensors and actuators used in Robotics
- To identify different types of Robot grippers and their applications
- To acquire basic Knowledge on Robot kinematics
- To expose to various application fields of Robotics

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Gain knowledge about basic concepts of robots

**CO-2:** Appreciate the usage of different sensors and actuators in Robotics

**CO-3:** Select appropriate Gripping mechanism for a particular application

**CO-4:** Analyze the direct and the inverse kinematic problems

**CO-5:** Appreciate robot design deference's for various applications

#### UNIT-I:

**Basic Concepts:** An overview of Robotics, classification of Robots, Robot Components, Robot degrees of freedom, Robot Joints, Robot Coordinates, Robot reference frames, Programming modes, Robot Characteristics.

#### UNIT-II:

**Sensors:** Sensor characteristics, Position sensors, Velocity sensors, Acceleration sensors, Force and Pressure sensors, Torque sensors, Microswitches, Light and infrared sensors, Touch and tactile sensors, Proximity sensors, Range finders.

#### UNIT-III:

**Actuators:** Characteristics of actuating system, Comparison of actuating systems, Hydraulic actuators, Pneumatic devices, Electric motors, Magneto-strictive actuators, Shape-Memory Metals, Electro-active Polymer Actuators.

**Grippers:** Classification of Grippers, Drive system for Grippers, Mechanical Grippers, Magnetic Grippers, Vacuum Grippers, Adhesive Grippers, Hooks and Scoops, Gripper Force analysis and design, Active and Passive Grippers.

#### UNIT-IV:

**Kinematics:** Robots as Mechanisms, Matrix Representation, Homogeneous Transformation Matrices, Representation of Transformations, Inverse of Transformation Matrices, Forward and Inverse Kinematics with Equations.



**UNIT-V:**

**Applications:** Industrial applications, material handling, processing, assembly application, inspection application, application planning, justification of robots, non-industrial applications, Robot safety.

**TEXT BOOKS:**

1. Introduction to Robotics: Analysis, Control, Applications, Saeed B. Niku, 2<sup>nd</sup> Edition, Wiley
2. Robotics Technology and Flexible Automation, Deb S. R., John Wiley
3. Robotics and Control, R. K. Mittal, I. J. Nagrath, McGraw-Hill Education

**REFERENCES:**

1. Industrial Robotics, Technology programming and Applications, Mikell P. Groover, Nicholas G. Odrey, Mitchel Weiss, Roger N. Nagel, Ashish Dutta, McGraw-Hill, 2012
2. Robotics-Control, Sensing, Vision and Intelligence, K. S. Fu, R. C. Gonzalez, C. S. G. Lee, McGraw-Hill International Edition
3. Robotic Engineering–An Integrated Approach, Klafter R. D., Chimielewski T. A., Negin M., Prentice Hall of India, 2009

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22OE1EI302) KINEMATICS OF ROBOTS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basics of robot coordinate frames and their representation
- To obtain knowledge about direct kinematics and inverse kinematics for a robot manipulator
- To examine techniques for planning robot motion in a workspace
- To understand various methods for developing dynamic models for manipulator
- To learn control techniques applied to robot manipulators

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Mathematically represent a Robot system

**CO-2:** Calculate robot hand position and orientation for specific joint angles

**CO-3:** Calculate joint angles to achieve a particular hand position

**CO-4:** Plan trajectories for robot tool to do meaningful tasks

**CO-5:** Analyze different controlling techniques used for robot manipulators

#### UNIT-I:

**Introduction:** Introduction, position and orientation of objects, objects coordinate frame Rotation matrix, Euler angles Roll, pitch and yaw angles coordinate Transformations, Joint variables and position of end effector, Dot and cross products.

#### UNIT-II:

**Direct Kinematics:** Coordinate frames, Rotations, Homogeneous coordinates, Link coordinates D-H Representation, The ARM equation. Direct kinematic analysis for Four axis SCARA Robot and three, five and six axis Articulated Robots.

#### UNIT-III:

**Inverse Kinematics:** The inverse kinematics problem, General properties of solutions. Tool configuration, Inverse kinematics of four axis SCARA robot and three and five axis Articulated robot.

#### UNIT-IV:

**Workspace Analysis and Trajectory Planning:** Workspace Analysis, work envelope of a Four axis SCARA robot and five axis articulated robot workspace fixtures, the pick and place operations, Joint space technique - continuous path motion, Interpolated motion, straight line motion and Cartesian space technique in trajectory planning.

**UNIT-V:**

**Robot Control:** The Control Problem, State Equations: one axis robot; three axis SCARA robot, Constant solutions, Linear Feedback Systems, Single Axis PID Control, PD-Gravity Control.

**TEXT BOOKS:**

1. Fundamentals of Robotics: Analysis & Control, Robert J. Schilling, Prentice Hall of India
2. Robotics and Control, R. K. Mittal, I. J. Nagrath, McGraw Hill Education

**REFERENCES:**

1. Robotic Engineering–An Integrated Approach, Klafter. R.D, Chimielewski. T.A, Negin M, Prentice Hall of India, New Delhi, 2009
2. Industrial Robotics, Technology Programming and Applications, Mikell P. Groover & Nicholas G. Odrey, Mitchel Weiss, Roger N. Nagel, Ashish Dutta, Tata McGraw-Hill Education, 2012
3. Robotics-Control, Sensing, Vision and Intelligence, K.S. Fu, R.C. Gonzalez, C.S.G Lee, McGraw-Hill International Edition

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22OE1EI401) DRIVE SYSTEMS FOR ROBOTS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To get acquainted with different robot drive mechanisms
- To understand in detail, working of hydraulic and pneumatic drives used in robotics
- To learn working principles of various electric drive systems for robotics
- To acquire basic Knowledge on servo systems for robot control

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Categorize various drive systems for robot movement

**CO-2:** Select appropriate drive system for a particular application

**CO-3:** Inspect different Hydraulic and Pneumatic drives and their applications in robotics

**CO-4:** Inspect different electric drives and their applications in robotics

#### UNIT-I:

**Introduction:** Objectives, motivation, open loop control, closed loop control with velocity and position feedback, Types of drive systems. Functions of drive system.

#### UNIT-II:

**Robot Drive Mechanism:** Lead Screws, Ball Screws, Chain & linkage drives, Belt drives, Gear drives, Precision gear boxes, Harmonic drives, Cyclo speed reducers.

#### UNIT-III:

**Hydraulic Drives:** Introduction, Requirements, Hydraulic piston and transfer valve, hydraulic circuit incorporating control amplifier, hydraulic fluid considerations, hydraulic actuators Rotary and linear actuators. Hydraulic components in robots.

#### UNIT-IV:

**Pneumatic Drives:** Introduction, Advantages, pistons-Linear Pistons, Rotary pistons, Motors-Flapper motor, Geared motor, Components used in pneumatic control. Pneumatic proportional controller, pneumatically controlled prismatic joint.

#### UNIT-V:

**Electric Drives:** Introduction, Types, DC electric motor, AC electric motor, stepper motors, half step mode operation, micro step mode. Types of stepper motors, Direct drive actuator.

**TEXT BOOKS:**

1. Engineering Foundation of Robotics, Francis N-Nagy Andras Siegler, Prentice Hall Inc.
2. Robotics Engineering - An Integrated Approach, Richard D. Klafter, Thomas. A, Chri Elewski, Michael Negin, PHI Learning, 2009

**REFERENCES:**

1. Industrial Robotics, Technology Programming and Applications, Mikell P. Groover & Nicholas G. Odrey, Mitchel Weiss, Roger N. Nagel, Ashish Dutta, Tata McGraw-Hill Education, 2012
2. Industrial Robotics, Bernard Hodges, 2nd Edition, Jaico Publishing House, 1993
3. Fundamentals of Robotics Analysis and Control, Robert J. Schilling, PHI Learning, 2009
4. Foundations of Robotics Analysis and Control, Tsuneo Yohikwa, MIT Press, 2003
5. Introduction to Robotics Mechanics and Control, John J. Craig, 3rd Edition, Pearson, 2008

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1EI402) INTELLIGENT SYSTEMS FOR ROBOTS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the fundamentals of robot programming
- To learn robot textual languages that are in common use
- To expose to artificial intelligence in robotics
- To acquire basic Knowledge on neural networks in robotics
- To acquire basic Knowledge on fuzzy logic in robotics

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Gain knowledge about different methods of robot programming

**CO-2:** Examine various robot language elements and their functions

**CO-3:** Analyze different AI techniques employed in robotics

**CO-4:** Design basic neuro-controller for robot motion control

**CO-5:** Apply fuzzy logic to robot control systems

#### UNIT-I:

**Robot Programming:** Methods of robot programming, leadthrough programming methods, robot program as a path in space - defining position in space, speed control, motion interpolation, WAIT, SIGNAL, DELAY commands, Branching.

#### UNIT-II:

**Robot Languages:** Textual robot language, generations of robot languages, robot language structure, operating systems, Robot language Elements and functions, constraints and variables, aggregates and location variables.

#### UNIT-III:

**Basic Commands and Operations:** Motion commands- move and related statements, speed control, points in workspace, paths and frames. End effector and sensor commands- end effector operation, sensor operation, REACT statement. Computations and operation. Program control and subroutines. Communications and data processing. Monitor mode commands.

#### UNIT-IV:

**AI for Robotics:** Introduction to Artificial Intelligence, goals of AI research, AI techniques- knowledge representation, problem representation, search techniques. LISP programming. AI and Robotics. LISP in the factory. Robotic Paradigms.

**UNIT-V:**

**Neural Network & Fuzzy Logic Approach in Robotics:** Introduction, Connectionist Models, Learning Principles and Learning Rules: Supervised, unsupervised, reinforcement learning. Neural Network in Robotics: Control of robot hands by neural network. Introduction to fuzzy logic, Fuzzy sets, Operation of Fuzzy sets, Fuzzy rule formation, Control rules, Fuzzy algorithm in robotics, Robot obstacle avoidance using fuzzy logic.

**TEXT BOOKS:**

1. Industrial Robotics Technology, Programming and Applications, Mikell. P. Groover, McGraw Hill, 2012
2. Robotics Technology and Flexible Automation, Deb S. R., Tata McGraw Hill Publishing Company Limited

**REFERENCES:**

1. Design and Control of Intelligent Robotic Systems, (Studies in Computational Intelligence 177) M. Begum, F. Karray (auth.), Dikai Liu, Lingfeng Wang, Kay Chen Tan (eds.), Springer
2. Neural Networks in Robotics, Edited by George Bekey, Kenneth Y. Goldberg, Springer US, 2012
3. Neural Networks, Fuzzy Logic, Genetic Algorithm - Synthesis and Applications, Rajasekharan and Rai, PHI Publications
4. Introduction to Neural Networks using MATLAB 6.0, S. N. Sivanandam, S. Sumathi, S. N. Deepa, TMH, 2006

**CYBER SECURITY**





## CYBER SECURITY

Offered by:

**INFORMATION TECHNOLOGY**

**Courses in the OE Track:**

| OE Tracks             | V Sem<br>(OE-I)                                                            | VI Sem<br>(OE-II)               | VII Sem<br>(OE-III)          | VIII Sem<br>(OE-IV) |
|-----------------------|----------------------------------------------------------------------------|---------------------------------|------------------------------|---------------------|
| <b>Cyber Security</b> | Fundamentals of Computer Networks / Relational Database Management Systems | Cryptography & Network Security | Essentials of Cyber Security | Computer Forensics  |

## **CYBER SECURITY**

**Cybersecurity is important** because it incorporates everything that relates to protecting our sensitive data, personally identifiable information (PII), protected health information (PHI), personal information, intellectual property, data, and governmental and **industry** information systems from stealing and destruction endeavoured. The cyber security track helps students to learn about how to Defend networks and data from unapproved access.

Enhanced information security and business endurance supervision.

Upgraded stakeholder confidence in your information security preparations.

Developed company authorizations with the correct security controls in place.

### **Some of the more common career paths in the cyber security path are**

- Chief Information Security Officer. ...
- Forensic Computer Analyst. ...
- Information Security Analyst. ...
- Penetration Tester. ...
- Security Architect. ...
- IT Security Engineer. ...
- Security Systems Administrator. ...
- IT Security Consultant.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS301) FUNDAMENTALS OF COMPUTER NETWORKS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce the fundamental various types of computer networks
- To demonstrate the TCP/IP and OSI models with merits and demerits
- To explore the various layers of OSI model
- To introduce UDP and TCP models
- To have the concept of different routing techniques for data communications

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand and explore the basics of Computer Networks reference models and the functionalities of physical layer

**CO-2:** Learn major concepts, principles involved in Data Link Layer and Network Layer

**CO-3:** Analyze how to maintain QoS in Network and maintaining of Congestion Control

**CO-4:** Demonstrate the Application Layer functionalities and importance of Security in the Network

#### UNIT-I:

**Introduction to Networks:** Internet, Protocols and Standards, The OSI Model, Layers in OSI Model, TCP/IP Suite, Addressing.

**Physical Layer:** Multiplexing, Transmission Media, Circuit Switched Networks, Datagram Networks, and Virtual Circuit Networks.

#### UNIT-II:

**Data Link Layer:** Introduction, Checksum, Framing, Flow and Error Control, Noiseless Channels, Noisy Channels, Random Access Controlled Access, Channelization, IEEE Standards, Ethernet, Giga-Bit Ethernet, Wireless LANs, SONET-SDH, Frame Relay and ATM.

#### UNIT-III:

**Network Layer:** Logical Addressing, Internetworking, Tunneling, Address Mapping, ICMP, IGMP, Forwarding, Routing-Flooding, Bellman Ford, Disjkstra's routing protocols, RIP, OSPF, BGP and Multicast Routing Protocols. Connecting Devices-Passive Hubs, Repeaters, Active Hubs, Bridges, Routers.

#### UNIT-IV:

**Transport Layer:** Process to Process Delivery, UDP, TCP and SCTP Protocols, Congestion, Congestion Control, Quality of Service.

**Application Layer:** Domain Name Space, DNS in Internet, Electronic Mail.

**UNIT-V:**

File Transfer Protocol, WWW, HTTP, SNMP, Multimedia.

**Network Security:** Security services, mechanisms and attacks, IPSec, SSL, VPN, Firewall. Bluetooth, Zigbee, IPv4, IPv6.

**TEXT BOOKS:**

1. Data Communications and Networking, Behrouz A. Forouzan, 4<sup>th</sup> Edition, McGraw Hill Education, 2006
2. Computer Networks, Andrew S. Tanenbaum, 4<sup>th</sup> Edition, Pearson Education
3. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose, K. W. Ross, 3<sup>rd</sup> Edition, Pearson Education

**REFERENCES:**

1. Data Communications and Networks, William Stallings
2. Data Communication and Networks, Bhusan Trivedi, Oxford University Press, 2016
3. An Engineering Approach to Computer Networks, S. Keshav, 2<sup>nd</sup> Edition, Pearson Education
4. Understanding Communications and Networks, 3<sup>rd</sup> Edition, W.A. Shay, Cengage Learning

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS302) RELATIONAL DATABASE MANAGEMENT SYSTEMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basic concepts and the applications of database systems
- To master the basics of SQL and construct queries using SQL
- To understand the relational database design principles
- To become familiar with the basic issues of transaction processing and concurrency control
- To become familiar with database storage structures and access techniques

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Demonstrate the basic concepts of database systems and design E-R models to represent simple database application scenarios

**CO-2:** Formulate SQL queries on the data

**CO-3:** Improve the database design by normalization

**CO-4:** Apply and relate the concept of transaction, concurrency control and recovery in database

**CO-5:** Familiar with basic database storage structures and access techniques using Indexing, hashing including B tree methods

#### UNIT-I:

**Introduction:** Database System Applications, Purpose of Database Systems, View of Data, Database Languages – DDL, DML, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Data Mining and Information Retrieval, Specialty Databases, Database Users and Administrators, History of Database Systems.

**Introduction to Database Design:** Database Design and ER diagrams, Entities, Attributes and Entity sets, Relationships and Relationship sets, Additional features of ER Model, Conceptual Design with the ER Model, Conceptual Design for Large enterprises.

**Relational Model:** Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity constraints, Querying relational data, Logical database Design: ER to Relational, Introduction to Views, Destroying /Altering Tables and Views.

#### UNIT-II:

**Relational Algebra and Calculus:** Preliminaries, Relational Algebra, Relational calculus – Tuple relational Calculus, Domain relational calculus, Expressive Power of Algebra and calculus.

**SQL:** Queries, Constraints, Triggers: Form of Basic SQL Query, UNION, INTERSECT, and EXCEPT, Nested Queries, Aggregate Operators, NULL values Complex Integrity Constraints in SQL, Triggers and Active Databases, Designing Active Databases.

#### **UNIT-III:**

**Schema Refinement and Normal Forms:** Introduction to Schema Refinement, Functional Dependencies - Reasoning about FDs, Normal Forms, Properties of Decompositions, Normalization, Schema Refinement in Database Design, Other Kinds of Dependencies.

#### **UNIT-IV:**

**Transaction Management:** Transactions, Transaction Concept, A Simple Transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Serializability, Transaction Isolation and Atomicity Transaction Isolation Levels, Implementation of Isolation Levels.

**Concurrency Control:** Lock-Based Protocols, Multiple Granularity, Timestamp-Based Protocols, Validation-Based Protocols, Multiversion Schemes.

Recovery System-Failure Classification, Storage, Recovery and Atomicity, Recovery Algorithm, Buffer Management, Failure with loss of nonvolatile storage, Early Lock Release and Logical Undo Operations, Remote Backup systems.

#### **UNIT-V:**

**Storage and Indexing:** Overview of Storage and Indexing: Data on External Storage, File Organization and Indexing, Index Data Structures, Comparison of File Organizations.

**Tree-Structured Indexing:** Intuition for tree Indexes, Indexed Sequential Access Method (ISAM), B+ Trees: A Dynamic Index Structure, Search, Insert, Delete.

**Hash- Based Indexing:** Static Hashing, Extendible hashing, Linear Hashing, Extendible vs. Linear Hashing.

#### **TEXT BOOKS:**

1. Database Management Systems, Raghu Ramakrishnan, Johannes Gehrke, 3<sup>rd</sup> Edition, McGraw Hill Education (India) Private Limited
2. Database System Concepts, A. Silberschatz, Henry. F. Korth, S. Sudarshan, 6<sup>th</sup> Edition, McGraw Hill Education (India) Private Limited,
3. Database Systems, R. Elmasri, Shamkant B. Navathe, 6<sup>th</sup> Edition, Pearson Education

#### **REFERENCES:**

1. Database System Concepts, Peter Rob & Carlos Coronel, Cengage Learning
2. Introduction to Database Management, M. L. Gillenson and others, Wiley Student Edition
3. Database Development and Management, Lee Chao, Auerbach Publications, Taylor & Francis Group
4. Introduction to Database Systems, C. J. Date, Pearson Education

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1IT301) CRYPTOGRAPHY AND NETWORK SECURITY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fundamentals of Computer Networks, Distributed Data Bases

#### COURSE OBJECTIVES:

- To outline security concepts, threats, attacks, services and mechanisms
- To describe various cryptosystems- symmetric key cryptography, public key cryptography
- To apply authentication services and Secure hash functions
- To discuss the concepts of IP Security, web security, viruses and firewalls

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Analyze the basics of security attacks, services, goals and mechanism

**CO-2:** Apply a variety of cryptographic algorithms, Hash Functions and protocols underlying network security and authentication applications

**CO-3:** Understand and analyze the key management requirements and Hash functions

**CO-4:** Examine and analyze various Email and Web Security mechanisms

**CO-5:** Understand the System Level Security issues

#### UNIT-I:

**Security Attacks:** Security Attacks (Active and Passive), Security Services (Confidentiality, Authentication, Integrity, Non-repudiation, Access Control and Availability) and Mechanisms, A model for Internetwork security, Buffer overflow & format string vulnerabilities, TCP session hijacking, ARP attacks, route table modification, UDP hijacking, and man-in-the-middle attacks.

#### UNIT-II:

**Conventional Encryption:** Classical Encryption techniques(Rail fence, Hill Cipher, Caesar cipher, Vigenere cipher, Play Fair Cipher), Fiestel Cipher Structure, Data Encryption Standard, Block Cipher Design Principles and Modes of Operation, Triple DES, RC-4, AES Cipher, Placement of Encryption Function.

**Public Key Cryptography and Authentication:** Confidentiality using Symmetric Encryption – Principles of Public key Cryptosystems, RSA algorithm,

#### UNIT-III:

**Key Management:** Diffie-Hellman key Exchange, Elliptic Curve Cryptography.

**Hash Functions:** Hash Functions, Security of Hash Functions and MACs, Secure Hash Algorithm, HMAC, Digital Signatures, Authentication Protocols, Digital Signature Standard, Authentication Applications: Kerberos, X.509 Authentication Service

**UNIT-IV:**

**Email Security and Web Security:** Electronic Mail Security – PGP/ SMIME, IP security- Architecture, Authentication Header, Encapsulating Security Payload, Key Management, Web Security- Secure Socket Layer, Transport Layer Security and Secure Electronic Transaction

**UNIT-V:**

**System Level Security:** Intrusion detection – password management – Viruses and related Threats – Virus Counter measures – Firewall Design Principles – Trusted Systems.

**TEXT BOOKS:**

1. Cryptography and Network Security – Principles and Practices, William Stallings, Prentice Hall of India, 4th Edition, 2005
2. Hack Proofing Your Network, Ryan Russell, Dan Kaminsky, Rain Forest, Puppy, Joe Grand, David Ahmad, Hal Flynn Ido Dubrawsky, Steve W. Manzuik and Ryan Permech, Wiley Dreamtech

**REFERENCES:**

1. Network Security Essentials: Applications and Standards, William Stallings Prentice Hall, 1999, ISBN 0130160938
2. Security in Computing, Charles B. Pfleeger, Shari Lawrence Pfleeger, 3<sup>rd</sup> Edition, Pearson Education, 2003



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22OE1IT401) ESSENTIALS OF CYBERSECURITY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fundamentals of Computer Networks, Cryptography and Network Security

#### COURSE OBJECTIVES:

- To identify the key components of cyber security in network
- To describe various security levels and categories, operating system security
- To define authentication issues and network security
- To describe memory management and protection measures

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand Information security and categorize cyber-crimes

**CO-2:** Demonstrate cyber offenses, security levels and models with objects and access control

**CO-3:** Understand Security implications and threats on Mobile devices

**CO-4:** Analyze tools and methods used in cybercrime

**CO-5:** Understand Organizational Implications and security risks

#### UNIT-I:

**Introduction to Cybercrime:** Introduction, Cybercrime, and Information Security, Who are Cybercriminals, Classifications of Cybercrimes, And Cybercrime: The legal Perspectives and Indian Perspective, Cyber crime and the Indian ITA2000, A Global Perspective on Cybercrimes.

#### UNIT-II:

**Cyber Offenses:** How Criminals Plan Them: Introduction, How Criminals plan the Attacks, Social Engineering, Cybers talking, Cyber Café and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector, Cloud Computing.

#### UNIT-III:

**Cybercrime:** Mobile and Wireless Devices: Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication service Security, Attacks on Mobile/Cell Phones, **Mobile Devices:** Security Implications for Organizations, Organizational Measures for Handling Mobile, Organizational Security Policies and Measures in Mobile Computing Era, Laptops.

**UNIT-IV:**

**Tools and Methods Used in Cybercrime:** Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Key loggers and Spywares, Virus and Worms, Trojan Horse And Backdoors, Steganography, DoS and DDoS attacks, SQL Injection, Buffer Overflow.

**UNIT-V:**

**Cyber Security:** Organizational Implications

Introduction, Cost of Cybercrimes and IPR issues, Web threats for Organizations, Security and Privacy Implications.

**Social Media Marketing:** Security Risks and Perils for Organizations, Social Computing and the associated challenges for Organizations.

**TEXTBOOK:**

1. Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Nina Godbole and SunilBelapure, WileyIndia

**REFERENCES:**

1. Cyber Security Essentials, James Graham, Richard Howard and Ryan Otson, CRC Press
2. Introduction to Cyber Security, Chwan-Hwa (John)Wu, J.DavidIrwin, CRC Press T&F Group

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1IT402) COMPUTER FORENSICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fundamentals of Computer Networks, Cryptography and Network Security, Essentials of Cyber Security

#### COURSE OBJECTIVES:

- To provide an understanding of computer forensics fundamentals
- To analyze various computer forensics technologies and to provide computer forensics systems
- To identify methods for data recovery
- To apply the methods for preservation of digital evidence

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Define and discuss the concepts of computer forensics

**CO-2:** Explain and apply the concepts of computer investigations

**CO-3:** Discuss the Computer Forensics Analysis and Validation and apply current practices for processing crime and incident scenes

**CO-4:** Select and apply current computer forensics tools

**CO-5:** Analyze the Windows Operating systems and DOS

#### UNIT –I:

**Computer Forensics Fundamentals:** What is Computer Forensics? Use of Computer Forensics in Law Enforcement, Computer Forensics Assistance to Human Resources/Employment Proceedings, Computer Forensics Services, Benefits of Professional Forensics Methodology, Steps taken by Computer Forensics Specialists.

**Types of Computer Forensics Technology:** Types of Military Computer Forensic Technology, Types of Law Enforcement — Computer Forensic Technology — Types of Business Computer Forensic Technology Computer Forensics Evidence and Capture: Data Recovery Defined—Data Back-up and Recovery—The Role of Back-up in Data Recovery—The Data-Recovery Solution.

#### UNIT-II:

**Evidence Collection and Data Seizure:** Why Collect Evidence? Collection Options—Obstacles — Types of Evidence — The Rules of Evidence — Volatile Evidence — General Procedure—Collection and Archiving—Methods of Collection—Artifacts—  
**Collection Steps—Controlling Contamination:** The Chain of Custody Duplication and Preservation of Digital Evidence: Preserving the Digital Crime Scene—Computer Evidence Processing Steps—Legal Aspects of Collecting and Preserving Computer

Forensic Evidence Computer Image Verification and Authentication: Special Needs of Evidential Authentication—Practical Consideration—Practical Implementation.

#### **UNIT-III:**

**Computer Forensics Analysis and Validation:** Determining what data to collect and analyze, validating forensic data, addressing data-hiding techniques, performing remote acquisitions Network Forensics: Network forensics overview, performing live acquisitions, developing standard procedures for network forensics, using network tools, examining the honey net project.

**Processing Crime and Incident Scenes:** Identifying digital evidence, collecting evidence in private-sector incident scenes, processing law enforcement crime scenes, preparing for a search, securing a computer incident or crime scene, seizing digital evidence at the scene, storing digital evidence, obtaining a digital hash, reviewing a case.

#### **UNIT-IV:**

**Current Computer Forensic Tools:** Evaluating computer forensic tool needs, computer forensic software tools, computer forensics hardware tools, validating and testing forensics software E-Mail Investigations: Exploring the role of e-mail in investigation, exploring the roles of the client and server in e-mail, investigating e-mail crimes and violations, understanding e-mail servers, using specialized e-mail forensic tools.

**Cell Phone and Mobile Device Forensics:** Understanding mobile device forensics, understanding acquisition procedures for cell phones and mobile devices.

#### **UNIT-V:**

**Working with Windows and DOS Systems:** understanding file systems, exploring Microsoft File Structures, Examining NTFS disks, Understanding whole disk encryption, windows registry, Microsoft startup tasks, MS-DOS startup tasks, virtual machines.

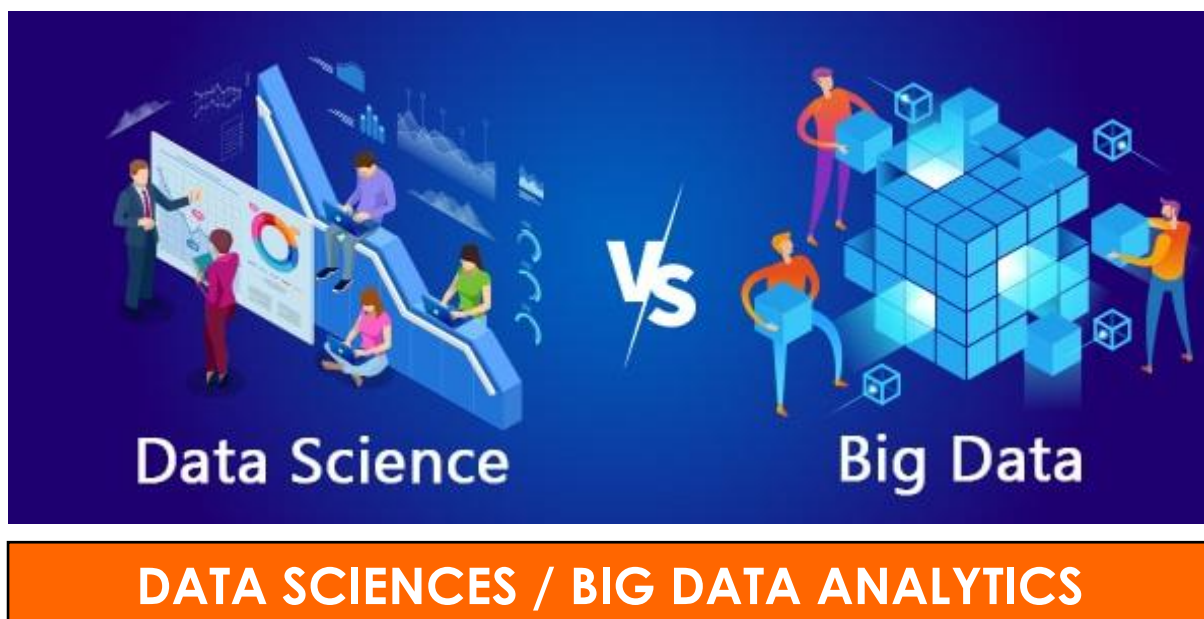
#### **TEXTBOOKS:**

1. Computer Forensics, Computer Crime Investigation, John R.Vacca, Firewall Media, New Delhi
2. Computer Forensics and Investigations, Nelson, Phillips Enfinger, Steuart, CENGAGE Learning
3. Real Digital Forensics, Keith J.Jones, Richard Bejtich, Curtis W.Rose, Addison Wesley, Pearson Education

#### **REFERENCES:**

1. Forensic Compiling, A Practitioners Guide, Tony Sammes and Brian Jenkinson, Springer International Edition
2. Computer Evidence Collection & Presentation, Christopher L.T.Brown, Firewall Media
3. Home land Security, Techniques & Technologies, Jesus Mena, Firewall Media
4. Software Forensics Collecting Evidence from the Scene of a Digital Crime, Robert M. Slade,TMH 2005
5. Windows Forensics, Chad Steel, Wiley India Edition

# **DATA SCIENCES / BIG DATA AND ANALYTICS**



Offered by:

**INFORMATION TECHNOLOGY**

**Courses in the OE Track:**

| OE Tracks                                                   | V Sem<br>(OE-I)                               | VI Sem<br>(OE-II)                         | VII Sem<br>(OE-III)               | VIII Sem<br>(OE-IV)                   |
|-------------------------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------------------------------|---------------------------------------|
| <b>Data<br/>Sciences /<br/>Big Data &amp;<br/>Analytics</b> | Statistical<br>Methods for<br>Data<br>Science | Computational<br>Thinking using<br>Python | Fundamentals<br>of Data<br>Mining | Data Analysis<br>and<br>Visualization |

## DATA SCIENCES / BIG DATA AND ANALYTICS

**Data science** helps in risk evaluation and observing, possible deceitful comportment, payments, customer analysis, and experience, among much other exploitation. The capability to make **data**-driven choices generates a steadier financial situation and **data scientists** make the strength of the **industry**.

As such, **data science** track helps students to apply business concepts in banking, finance, manufacturing, transport, e-commerce, education, etc. that use **data science**. As a consequence, there are numerous **Data Science** Applications associated to it

### Job Roles in Data Science Track

- [Data Analyst](#)
- [Data Engineers](#)
- [Database Administrator](#)
- [Machine Learning Engineer](#)
- [Data Scientist](#)
- [Data Architect](#)
- [Statistician](#)
- [Business Analyst](#)
- [Data and Analytics Manager](#)

**Big Data analytics** track helps the students to learn the process of gathering, establishing and examining large sets of **data** (called **Big Data**) to determine patterns and other beneficial information. Analysts occupied with **Big Data** characteristically want the acquaintance that comes from investigating the **data**.

Big data analytics is the practice of mining useful information by examining different **types** of big data sets. Big data analytics is utilized to determine concealed patterns, market developments and consumer favorites, for the advantage of organizational decision making.

### Job responsibilities in a Big Data Analytics Track are

- To gather and accumulate data from disparate sources, clean it, organize it, process it, and analyse it to extract valuable insights and information.
- To identify new sources of data and develop methods to improve data mining, analysis, and reporting.
- To create data definitions for new database files or alterations made to the already existing ones for analysis purposes.
- To present the findings in reports (in table, chart, or graph format) to help the management team in the decision-making process.
- To apply statistical analysis methods for consumer data research and analysis purposes.
- To keep track of the trends and correlational patterns among complex data sets.
- To perform routine analysis tasks to support day-to-day business functioning and decision making.
- To collaborate with Data Scientists to develop innovative analytical tools.
- To work in close collaboration with both the IT team and the business management team to accomplish company goals.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1MT303) STATISTICAL METHODS FOR DATA SCIENCE

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PREREQUISITES:** Calculus and probability theory

#### COURSE OBJECTIVES:

- To provide insights about the basic roles of various statistical methods in building computer applications
- To develop problem-solving skills using R programming
- To understanding the importance of Data Visualization for univariate, bivariate techniques
- To inferences about the population parameters using sample data
- To understanding regression techniques

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Analyze an extremely large data set and perform exploratory data analysis to extract meaningful insights

**CO-2:** Examine a real-world problem and solve the same with the Knowledge gained from various distributions study through R Programming

**CO-3:** Develop various visualizations of the data in hand and communicate results of analysis effectively (visually and verbally)

**CO-4:** Use and fit a linear regression model to data and use it for prediction

**CO-5:** Fit a polynomial regression model to data and use it for prediction

#### UNIT-I:

**Introduction to Statistics and Descriptive Statistics:** Definition of Statistics, Basic objectives. Applications in various branches of science with examples, Collection of Data: Internal and external data, Primary and secondary Data, Population and sample, Representative sample.

**Descriptive Statistics:** Classification and tabulation of univariate data, graphical representation, Frequency curves. Descriptive measures - Central tendency and Dispersion Bivariate data. Summarization, Marginal and Conditional frequency distribution.

#### UNIT-II:

**Introduction to R:** Introduction, Installing R and Data Types in R,

**Programming using R:** Operators, Conditional Statements, Looping, Scripts, Function creation. Creating list, List operations, Recursive list, creating a data frame, operations on data frames.



### **UNIT-III:**

**Data Visualization using R:** Import-Export of data, Measures of central Tendency and Measures of Dispersion, Data Visualization – Scatter plot, Pie chart, Histogram, Bar chart, Box plot, Absolute and Relative frequencies, frequency distribution.

### **UNIT-IV:**

**Correlation & Linear Regression:**

**Correlation:** Types of correlation

**Linear and Multiple Regression:** Introduction, the Regression Model, Multiple linear regression, Interval estimation, Estimation of parameters of  $\beta_0$  and  $\beta_1$ , Estimation of  $\sigma^2$ .

### **UNIT-V:**

**Non Linear Regression:** Regression of second degree polynomial (Non-linear least square method for polynomial function), Power function, Exponential, Estimation of coefficients, Linear and Polynomial Regressions in R.

### **TEXT BOOKS:**

1. Introductory Statistics by Thomas H Wonnacott & Ronald J Wonnacot, John Wiley & Sons Inc., 1969
2. Applied Statistics and Probability for Engineers by Douglas C. Montgomery, George C. Runger, John Wiley & Sons, Inc. 3<sup>rd</sup> edition, 2003
3. R for Beginners, Sandip Rakshit, McGraw Hill Education, (Unit-II, Unit-III)

### **REFERENCES:**

1. Statistical Methods, Rudolf J. Freund, Donna Mohr, William J. Wilson, 3<sup>rd</sup> Edition, 2010, (Unit-I, Unit-IV, Unit-V)
2. R-The Statistical Programming Language by Dr. Mark Gardner of Wiley India Pvt. Ltd.
3. Introduction to the Theory of Statistics, A. M. Mood, F.A. Graybill and D.C. Boes, McGraw Hill Education
4. A first course in Probability, S. M. Ross, Prentice Hall
5. Statistical Methods, S.P Gupta, 42<sup>nd</sup> revised edition, Sultan chand & sons, 2012

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22OE1IT302) COMPUTATIONAL THINKING USING PYTHON

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Statistical Methods for Data Science

#### COURSE OBJECTIVES:

- To Understand why Python is a useful scripting language for developers
- To create and execute Python programs and to learn how to use lists, tuples and dictionaries in Python programs
- To learn how to build and package Python modules for reusability
- To learn how to design object-oriented programs with Python classes
- To learn how to use exception handling in Python applications for error handling

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Adapt and combine standard algorithms to solve a given problem (includes numerical as well as non-numerical algorithms)

**CO-2:** Adequately use standard programming constructs: repetition, selection, functions, composition, modules, aggregated data (arrays, lists etc.)

**CO-3:** Explain what a given program (in Python) does identify and repair coding errors in a program

**CO-4:** Understand and use object-based software concepts (constructing OO software will be deal with in the course Software Engineering)

**CO-5:** Use library software for (e.g.) building a graphical user interface, web application or mathematical software

#### UNIT-I:

Introduction, History, Features, setting up path, Working with Python, Basic Syntax, Variable and Data Types, Operator, Conditional Statements- If, If-else, Nested If-else Looping for While Nested loops Control Statements Break Continue Pass String Manipulation Accessing Strings Basic Operations String Slices Function.

#### UNIT-II:

**Methods, Lists:** Introduction, accessing list, Operations, Working with lists, Function and Methods,

**Tuple:** Introduction, accessing tuples, Operations, Working Functions and Methods.

**Dictionaries:** Introduction, accessing values in dictionaries, Working with dictionaries, Properties.

#### UNIT-III:

**Functions:** Defining a function, Calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables.

Modules: Creation, Importing module, Math module, Random module, Packages.

**Composition:** Input-Output Printing on screen, reading data from keyboard, Opening and closing file Reading and writing files, Functions.

#### **UNIT-IV:**

**Exception Handling:** Exception, Exception Handling, except clause, Try, Finally, clause, User Defined Exceptions.

**OOPs concept:** Class and object, Attributes, Inheritance, Overloading, Overriding, Data hiding, Regular Expressions-Match function, Search function, Matching vs Searching Modifiers, Patterns.

**Multithreading:** Thread, starting a thread. Threading module, Synchronizing threads.

#### **UNIT-V:**

**CGI:** Introduction, Architecture, CGI environment variable, GET and POST methods, Cookies, File upload.

**Database:** Introduction, Connections, Executing queries, Transactions Handling error. Networking: Socket, Socket Module, Methods, Client and server, Internet modules, Sending email.

#### **TEXT BOOK:**

1. Learning Python, David Ascher and Mark Lutz, O' Relly

#### **REFERENCES:**

1. Python Programming: An Introduction to Computer Science, John M. Zelle, 2<sup>nd</sup> Edition, Kindle Edition
2. Python Essential Reference, David M. Beazley, 4<sup>th</sup> Edition, Developer's Library

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1IT403) FUNDAMENTALS OF DATAMINING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Statistical Methods for Data Science, Computational Thinking using Python

#### COURSE OBJECTIVES:

- To introduce the basic concepts and techniques in building a Data Warehouse
- To apply preprocessing methods for any given raw data
- To develop skills of using recent datamining software for solving practical problems
- To implement and apply basic algorithms for supervised and unsupervised learning

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Assess raw input data and process it to provide suitable input for arrange of data mining algorithms

**CO-2:** Discover and measure interesting patterns from different kinds of databases

**CO-3:** Evaluate and select appropriate data-mining classification algorithms and apply, interpret and report the output appropriately

**CO-4:** Design and implement data – mining applications using sample, realistic data sets and modern tools

**CO-5:** Evaluate and select appropriate data-mining clustering algorithms and apply, interpret and report the output appropriately

#### UNIT-I:

**Data Warehousing & Modelling:** Basic Concepts: Data Warehousing: A multitier Architecture, Data ware house models: Enterprise warehouse, Data mart and virtual warehouse, Extraction, Transformation and loading.

#### UNIT-II:

**Data Cube:** A multidimensional data model, Stars, Snowflakes and Fact constellations: Schemas for multidimensional Data models, Dimensions: The role of concept Hierarchies, Measures: Their Categorization and computation, Typical OLAP Operations.

#### UNIT-III:

**Data Warehouse Implementation & Data Mining:** Data Ware house Architecture, What is data mining, Challenges, From Data Ware housing and Data Mining, Data Mining Tasks, Data Mining Functionalities, Major Issues in Data Mining. Data: Types of Data, Data Quality, Data Pre-processing, Measures of Similarity and Dissimilarity.

**UNIT-IV:**

**Association Analysis:** Association Analysis: Problem Definition, Frequent Item set Generation, Rule generation. Alternative Methods for Generating Frequent Item sets, FP-Growth Algorithm, Evaluation of Association Patterns.

**UNIT-V:**

**Classification:** Decision Trees Induction, Method for Comparing Classifiers, Rule Based Classifiers, Nearest Neighbor Classifiers, Bayesian Classifiers.

**Clustering Analysis:** Overview, K-Means, Agglomerative Hierarchical Clustering, DBSCAN, Cluster Evaluation, Density-Based Clustering, Graph- Based Clustering, Scalable Clustering Algorithms.

**TEXTBOOKS:**

1. Introduction to Data Mining, Pang-NingTan, Michael Steinbach, Vipin Kumar, First Impression, Pearson,2014
2. Data Mining-Concepts and Techniques, JiaweiHan, Micheline Kamber, JianPei, 3<sup>rd</sup> Edition, Morgan Kaufmann, 2012

**REFERENCES:**

1. Data Ware housing in the Real World, Sam Anahory, Dennis Murray, Tenth Impression, Pearson,2012
2. Mastering Data Mining, Michael J.Berry, Gordon S. Linoff, 2<sup>nd</sup> Edition, Wiley, 2012

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1IT404) DATA ANALYSIS AND VISUALIZATION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Statistical Methods for Data Science, Computational Thinking using Python, Fundamentals of Data Mining

#### COURSE OBJECTIVES:

- To introduce concept and characteristics of probability distribution
- To introduce underlying design principles, properties and assumptions of linear and non-linear regression modelling
- To introduce design principles involved in identifying interesting classification and prediction of data patterns
- To introduce properties of time series data and perform time series analysis

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply probability distribution concepts to identify univariate data patterns

**CO-2:** Apply regression modeling to build efficient mathematical models for prediction and classification

**CO-3:** Apply decision and regression trees for supervised learning

**CO-4:** Visualize time series data by applying time series techniques

**CO-5:** Explore case studies and projects

#### UNIT-I:

**Data Definitions and Analysis Techniques:** Elements, Variables, and Data categorization, Introduction to statistical learning, Descriptive Statistics: Measures of central tendency, Measures of location of dispersions.

#### UNIT-II:

**Basic Analysis Techniques:** Basic analysis techniques, Statistical hypothesis generation and testing, Chi-Square test, t-Test Analysis of variance, Correlation analysis, Maximum likelihood test.

#### UNIT-III:

**Data Analysis Techniques:** Regression analysis and visualization, Classification techniques and visualization, Clustering and visualization, Association rules analysis and visualization

#### UNIT-IV:

Time-series Analysis and Forecasting– Time-series components, Variation in Time Series, Cyclic Variation, Seasonal Variation, Irregular Variation.

**UNIT-V:**

**Smoothing Techniques:** A problem involving all four components of time series, Introduction to forecasting, forecasting models, Trend and Seasonal effects, Trend Analysis

**Case-studies and Projects:** Understanding business scenarios, Feature engineering and visualization, Sensitivity Analysis.

**TEXTBOOKS:**

1. Data Mining and Analysis, Mohammed J.Zaki, Wagner Meira, Cambridge, 2012
2. Data Mining: Theories, Algorithms, and Examples, NongYe, CRC Press Taylor & FrancisGroup,2014
3. Statistics for Management, David S.Rubin, Sanjay Rastogi, Masood Husain Siddiqui RichardI.Levin,7<sup>th</sup> Edition, Pearson Learning

**REFERENCES:**

1. Probability & Statistics for Engineers & Scientists, Ronald E.Walpole, RaymondH. Myers, Sharon L.Myersand KeyingYe, 9<sup>th</sup>Edition, PrenticeHallInc.
2. The Elements of Statistical Learning, Data Mining, Inference and Prediction, Trevor Hastie, Robert Tibshirani, Jerome Friedman, 2<sup>nd</sup> Edition, Springer, 2014
3. An Introduction to Statistical Learning Mining Massive Data Sets, A.Rajaramanand J.Ullman, Cambridge University Press, 2012
4. Software for Data Analysis: Programming with R (Statistics and Computing), John M.Chambers, Springer

# **AUTONOMOUS VEHICLES**



# SELF-DRIVING CAR



## AUTONOMOUS VEHICLES

Offered by:

**AUTOMOBILE ENGINEERING**

Courses in the OE Track:

| OE Tracks                  | V Sem<br>(OE-I)                      | VI Sem<br>(OE-II)              | VII Sem<br>(OE-III)                     | VIII Sem<br>(OE-IV)               |
|----------------------------|--------------------------------------|--------------------------------|-----------------------------------------|-----------------------------------|
| <b>Autonomous Vehicles</b> | Principles of Automobile Engineering | Modern Automotive Technologies | Electric, Hybrid and Fuel Cell Vehicles | Connected and Autonomous Vehicles |

## **AUTONOMOUS VEHICLES**

The invention of the wheel marked a large step in the evolution of mankind. With mobility, man experienced a newfound freedom that opened the doors for several other inventions. Automobile engineering or automotive engineering is one of the most challenging careers in the field of engineering with a wide scope. This branch deals with the designing, developing, manufacturing, testing and servicing automobiles such as cars, trucks, motorcycles, scooters, etc. and the related engineering sub systems. For the perfect blend of designing and manufacturing automobiles, automobile engineering uses the features of different elements of engineering such as mechanical, electrical, electronic, instrumentation, civil, software and safety engineering. Exploring the topic from an interdisciplinary perspective is indispensable. Globalization and incredible growth of automobile industry have resulted in numerous opportunities for engineers both in India and abroad.

The 17<sup>th</sup> and 18<sup>th</sup> centuries were mostly about steam-powered vehicles transporting people and goods. While electric cars enjoyed popularity in the 19<sup>th</sup> and early 20<sup>th</sup> centuries, the later period saw the accelerated adoption of the petrol car, due to its advantages of power, mass production, cost and advances in the internal combustion engine. It is only in the 21<sup>st</sup> century that interest in electric cars has come back, given the need for cleaner, greener modes of transport. The modern period is associated with several path breaking technologies. Over the last couple of decades, there has been an explosion of electronics in vehicles. Connected cars that include technology features are ever more popular. These smart cars come with internet access, GPS, wi-fi, superior infotainment, advanced telematics and navigation capabilities. More innovations in in-vehicle infotainment and electronics promise to give car users even more enhanced capabilities in the near future.

Today, safety has become a larger concern than ever before. While entertainment and infotainment have made car driving a pleasure, this has also given rise to a growing tribe of distracted drivers. Add to this, underdeveloped roads, which take a toll on drivers today. Increased distractions and fatigue can also contribute to human fatalities. The future certainly points in the direction of driverless cars, which promise to alleviate concerns of traffic congestion and road safety. Driverless cars, also known as autonomous cars, will usher in a paradigm shift in the evolution of the modern automobile. Self-driving cars can sense the environment and traffic with the help of RADAR, LIDAR, GPS and computer vision and navigate without human intervention. Autonomous cars are claimed to have greater accuracy, reliability and faster reaction time compared to human drivers. This would lead to fewer traffic collisions and less road congestion.

Autonomous driving is a popular subject of today's discussion and automakers are developing complex systems that allow cars to drive themselves. If technology continues on its current course, car will do the concentrating for you. Self-parking, automatic emergency braking, adaptive cruise control and lane keeping are just some of the technologies that have leapt into the market in the past few years. Put them all together, get a picture of driving to assisted driving to fully autonomous cars. The open elective track "Autonomous Vehicles" offered by the department of automobile engineering trains the students to meet the technological challenges and

diverse needs of the industry and society in various areas of automobile engineering and equips them to excel in a truly competitive industry. With thorough knowledge in this field, engineering graduates get opportunity to serve many top-notch automobile companies and IT companies as well.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1AE301) PRINCIPLES OF AUTOMOBILE ENGINEERING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the layout of an automobile and functionalities subsystems
- To provide overview on concepts of engine, cooling, lubrication and fuel systems
- To present constructional features and working of automotive driveline and running systems
- To study the fundamentals and principles of automotive electrical systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the functionalities of automotive systems and subsystems

**CO-2:** Give an overview on engine and engine subsystems

**CO-3:** Describe working of automotive driveline and running systems

**CO-4:** Discuss the concepts of automotive starting, ignition and charging systems

#### UNIT-I:

**Introduction:** History, classification of automobiles, layout of an automobile, automobile sub systems and their role, types of chassis layouts and types of bodies.  
**Engine:** Classification and components of an engine, principle and working of four stroke and two stroke SI and CI engines, air intake and exhaust system, fuel system layout for petrol and Diesel engines.

#### UNIT-II:

**Cooling and Lubrication:** Necessity of cooling, air-cooling, water cooling - thermosyphon and pump cooling, radiator, pump, thermostat, antifreeze solution and radiator fan. Mist, splash and forced lubrication, oil filters and oil pumps.

#### UNIT-III:

**Drive Line:** Clutches, principle, single plate clutch, multi plate clutch and centrifugal clutch. Gear box - Need, sliding mesh, constant mesh and synchromesh gear box. Propeller shaft, universal joint, differential, wheels and tyres.

#### UNIT-IV:

**Running Systems:** Suspension systems – Objective, rigid axle and independent suspension system and torsion bar. Steering system – Layout, steering mechanism, steering geometry and steering gearboxes. Brake system – Principle, stopping distance, types of brakes and actuation.

**UNIT-V:**

**Electrical Systems:** Starting system - Principle, working of different starter drive units and solenoid switches. Ignition system - Conventional ignition system types, ignition advance and retarding mechanisms. Charging system - Alternator principle, construction and working, cut-outs and regulators.

**TEXTBOOKS:**

1. Advanced Vehicle Technology, Heinz Heisler, Butterworth Heinemann Publishers, 2002
2. Automobile Electrical Equipment, Crouse W H, McGraw Hill Book Co., Inc., New York 3<sup>rd</sup> edition, 1986

**REFERENCES:**

1. Motor Vehicle, Garrett T K, Newton K. and Steeds W., Butter Worths & Co. Publishers Ltd., New Delhi, 2001
2. Automotive Electrical Equipment, Kholi P L, Tata McGraw Hill Co., Ltd., New Delhi, 1975
3. Automotive Chassis and Body, Crouse W H, McG raw Hill Book Co., 5th edition, 1976
4. Automotive Mechanics, Giri N K, Khanna Publications, 2006

**ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/107/106/107106088/>
2. <https://www.youtube.com/watch?v=BOP8qLQzhDc>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

### (22OE1AE302) MODERN AUTOMOTIVE TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Principles of Automobile Engineering

#### **COURSE OBJECTIVES:**

- To provide an overview on advanced engine control system concepts
- To understand the interdisciplinary concepts and GPS-enabled applications in automobile
- To present intelligent vehicle technologies like comfort, safety and security systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply advanced engine control system concepts in engineering

**CO-2:** Address the key technologies in automotive navigation

**CO-3:** Appreciate the technological advancements driver assistance systems

#### **UNIT-I:**

**Advanced Engine Controls:** Concept of an electronic engine control system, engine control module, powertrain control module, electronic fuel injection - throttle body fuel injection, multi-point fuel injection, gasoline direct injection, common rail direct injection, electronic ignition control, engine mapping and on-board diagnostics.

#### **UNIT-II:**

**Telematics:** Global positioning system, geographical information systems, navigation system, architecture, automotive vision system, road recognition and infotainment system.

#### **UNIT-III:**

**Comfort Systems:** Adaptive cruise control system, active suspension system, power steering, collapsible and tilttable steering column and power windows.

#### **UNIT-IV:**

**Safety Systems:** Active and passive safety, airbags, seat belt tightening system, forward collision warning systems, child lock, anti-lock braking systems, traction control system and lane departure warning system.

#### **UNIT-V:**

**Security Systems:** Anti-theft technologies – mechanical, electromechanical and electronic immobilizers, alarm system, stolen vehicle tracking system, remote keyless entry, smart card system and number plate coding.

**TEXTBOOKS:**

1. Understanding Automotive Electronics, William B Ribbens, fifth edition, Butter worth Heinemann Woburn, 1998
2. Intelligent Vehicle Technologies: Theory and Applications, Ljubo Vlacic, Michel Parent and Fumio Harashima, Butterworth-Heinemann publications, Oxford, 2001

**REFERENCES:**

1. Automotive Handbook, Robert Bosch, SAE, 5<sup>th</sup> edition, 2000
2. Navigation and Intelligent Transportation Systems – Progress in Technology, Ronald K Jurgen, Automotive Electronics Series, SAE, USA, 1998
3. Understanding Automotive Electronics, Bechhold, SAE, 1998

**ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/107/106/107106088/>
2. <https://www.youtube.com/watch?v=BOP8qLQzhDc>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22OE1AE401) ELECTRIC, HYBRID AND FUEL CELL VEHICLES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Principles of Automobile Engineering, Modern Automotive Technologies

#### COURSE OBJECTIVES:

- To study the concepts and drivetrain configurations of electric and hybrid vehicles
- To understand about electric propulsion system
- To provide various energy storage devices
- To present principle, working and automotive applications of fuel cell and solar technology

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the concepts and drivetrain configurations of electric and hybrid vehicles

**CO-2:** Present various electric motors used in EVs and HEVs

**CO-3:** Control electric motors used in EVs and HEVs

**CO-4:** Discuss various energy storage devices and hybridization

**CO-5:** Describe automotive applications of fuel cell and solar technology

#### UNIT-I:

**Electric and Hybrid Vehicles:** Electric vehicle – layout, components, configurations, advantages and limitations. Hybrid vehicle - concepts of hybrid electric drivetrain based on hybridization and powertrain configuration, architecture of series, parallel and series-parallel hybrid electric drivetrains, modes of operation, merits and demerits.

#### UNIT-II:

**Electric Motors:** Review of technology suited to automotive propulsion, requirements, DC motors, Induction motors, permanent magnet brushless DC motors, permanent magnet synchronous motors and switched reluctance motors.

#### UNIT-III:

**Motor Drives:** Speed and torque control, DC motor - Chopper based four quadrant operations, induction motor, permanent magnet motor and switched reluctance motor.

#### UNIT-IV:

**Energy Storages:** Electromechanical batteries - Types, parameters, lead acid batteries, nickel-based batteries, lithium-based batteries, battery management



system, ultracapacitors, hybridization of energy storages and battery swapping technology.

#### **UNIT-V:**

**Fuel Cell and Solar Vehicles:** Fuel cell vehicle – Operating principle, types of fuel cells, fuel cell options for fuel cell vehicle and fuel cell hybrid vehicle. Solar vehicle - Solar photovoltaic cell, solar array, solar car electrical system and drive train.

#### **TEXTBOOKS:**

1. Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay and Ali Emadi, CRC Press, 2004
2. Electric Vehicle Technology-Explained, James Larminie and John Lory, John Wiley & Sons Ltd., 2003

#### **REFERENCES:**

1. Electric and Hybrid Vehicles – Design Fundamentals, Iqbal Husain, CRC Press, 2010
2. Electric Vehicle Battery Systems, Sandeep Dhameja, Butterworth–Heinemann, 2002
3. Electric and Hybrid – Electric Vehicles, Ronald K. Jurgen, SAE, 2002
4. Light Weight Electric/Hybrid Vehicle Design, Ron Hodkinson and John Fenton, Butterworth–Heinemann

#### **ONLINE RESOURCES:**

1. [https://onlinecourses.nptel.ac.in/noc21\\_ee112/preview](https://onlinecourses.nptel.ac.in/noc21_ee112/preview)
2. <https://elearn.nptel.ac.in/shop/nptel/electric-vehicles-and-renewable-energy>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

#### (22OE1AE402) CONNECTED AND AUTONOMOUS VEHICLES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Principles of Automobile Engineering, Modern Automotive Technologies, Electric, Hybrid and Fuel Cell Vehicles

#### COURSE OBJECTIVES:

- To understand the fundamentals of vehicle communication and networking
- To present the applications of communications technologies in AVs
- To know intra-vehicle and inter-vehicle communication technologies
- To provide various levels of vehicle autonomy and current State of AVs

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the fundamentals of vehicle communication and networking

**CO-2:** Relate the applications of communications technologies in AVs

**CO-3:** Appreciate the intra-vehicle communication technologies

**CO-4:** Express the inter-vehicle communication technologies

**CO-5:** Discuss various levels of vehicle autonomy and current state of AVs

#### UNIT-I:

**Communication Fundamentals and Controller Area Network:** Communication fundamentals – frequency, bandwidth, power measurement, signal to noise ratio, transmission rate constraints, radio frequency spectrum allocation, RADAR operation and types of RADAR. CAN evolution, versions, types of controllers and layered architecture. CAN bus, message frames and error handling.

#### UNIT-II:

**Introduction to Vehicle Communications:** Intra-vehicle communications - communications protocols, systems and sensors (Braking, steering, power train, chassis systems, body electronics, instrument clusters, infotainment systems), inter-vehicle communications - cooperative driving (accident warning, frontal/rear collision prevention, lane change, assistance). Consumer assistance – traffic information, multimedia support and smart parking.

#### UNIT-III:

**Intra-Vehicle Communications:** Wired communication – Network comparison, two tier approach, LIN applications - Localized vehicle area support, general support areas, CAN applications - In vehicle operation, infotainment, wireless communication – Bluetooth vehicle applications, satellite services – satellite radio, vehicle care and traffic status.

#### **UNIT-IV:**

**Inter-Vehicle Communication:** Adhoc communications – applications in vehicle traffic monitoring, collision and congestion avoidance, highway lane reservation, emission control, vehicle frequency utilization – AM radio, Bluetooth, FM radio, GPS, short range RADAR, wireless LAN, intelligent roadway-infrastructure to vehicle and vehicle to vehicle communications. Evolving smart vehicle – ECU, wireless networking, forward RADAR, side RADAR, GPS, cellular transmission and event Recorder.

#### **UNIT-V:**

**Autonomous Vehicles:** Importance, levels of automation, policy making, social costs, safety and crashes, congestion, land use, energy and emissions, costs and disadvantages.

**Current State of Autonomous Vehicles:** Research, challenges, commercial development, sensor systems, sensor suits, environmental challenges, graceful degradation, V2V and V2I communication, sharing the drive, integrity, security, verification and policy implications.

#### **TEXTBOOKS:**

1. Inter and Intra Vehicle Communications, Gilbert Held Auerbach Publications, 2008
2. Autonomous Vehicle Technology-A Guide for Policymakers, James M. Anderson, Nidhi Kalra, Karlyn D. Stanley, Paul Sorensen, Constantine Samaras, Oluwatobi A. Oluwatola, RAND Corporation, Santa Monica, Calif., 2016
3. Autonomous Driving - Technical, Legal and Social Aspects, Markus Maurer, J. Christian Gerdes, Barbara Lenz, Hermann Winner, Editors, Springer, 2016

#### **REFERENCES:**

1. Intelligent Vehicle Technologies: Theory and Applications, Ljubo Vlacic, Michel Parent and Fumio Harashima, Butterworth-Heinemann Publications, Oxford, 2001
2. Navigation and Intelligent Transportation Systems – Progress in Technology, Ronald K. Jurgen, Automotive Electronics Series, SAE, USA, 1998
3. Automotive In-vehicle Networks, J. Gabrielleen, Wiley-Blackwell, 2008
4. In-Vehicle Network Architecture for the Next-Generation Vehicles, Syed Masud Mahmud, IGI
5. Communication Technologies for Vehicles, Mohamed Kassab Springer, 2015

#### **ONLINE RESOURCES:**

1. <https://www.classcentral.com/subject/autonomous>
2. <https://www.youtube.com/watch?v=MX6aEkcFu6U>

# **GENERAL - COMPUTING**

# GENERAL-COMPUTING

**Offered by:**

**COMPUTER SCIENCE AND  
ENGINEERING /  
INFORMATION TECHNOLOGY**

## **Courses in the Pool:**

- Programming through Java
- Relational Data Base Management Systems
- Computational Thinking using Python
- Introduction to Data Analytics
- Fundamentals of Computer Algorithms

## 1. PROGRAMMING THROUGH JAVA

**Java** is an extensively **used** programming language specifically intended for use in the distributed environment of the internet. **Java** help students to create wide-ranging applications that possibly will run on a single workstation or be distributed among servers and clients in a network.

Java is an extremely fruitful language and an upper option for many developers for many years. The motive that it has remained so prevalent is since it still happens the needs of functioning across networks.

### **Students will have different roles and responsibilities by learning Java Programming**

- Designing, implementing, and maintaining Java applications that are often high-volume and low-latency, required for mission-critical systems.
- Delivering high availability and performance.
- Contributing in all phases of the development lifecycle.
- Writing well-designed, efficient, and testable code.

## 2. RELATIONAL DATABASE MANAGEMENT SYSTEMS

A relational database permits you to effortlessly find precise information. It also consents you to sort based on any field and produce reports that comprise only definite fields from each record. With features like, Data Accuracy, Easy Access to Data, Data Integrity, Flexibility, Normalization, High Security, Feasible for Future Modifications

### **By learning RDBMS Students will have different roles in Database environment**

- Data Administrator,
- Database Administrator
- Database Designer
- Application Programmer

## 3. COMPUTATIONAL THINKING USING PYTHON

The **python** language is one of the utmost accessible programming languages available because it has streamlined syntax and not complex, which gives more importance on natural language. Due to its comfort of learning and practice, **python** codes can be readily written and executed much quicker than former programming languages.

Data Science: The libraries and frameworks Python offers, e.g. PyBrain, PyMySQL, and NumPy are one of the big reasons. Another reason is diversity. Python experience allows you to do a lot more than any other language, e.g. you can create scripts to automate stuff, go into web development, and so much more.

### **Students will have various Job Profiles by learning Python**

- Software Engineer.
- Python Developer.
- Research Analyst.
- Data Analyst.
- Data Scientist.
- Software Developer.

#### **4. INTRODUCTION TO DATA ANALYTICS**

**Data** Scientists and Analysts **use data analytics** techniques in their research, and businesses also **use** it to inform their conclusions. **Data analysis** can assistance corporations healthier comprehend their customers, assess their ad-campaigns, personalize gratified, create content approaches and progress products.

**By learning Data Analytics students will get Jobs with different designations**

- IT Systems Analyst. Systems analysts use and design systems to solve problems in information technology. ...
- Healthcare Data Analyst. ...
- Operations Analyst. ...
- Data Scientist. ...
- Data Engineer. ...
- Quantitative Analyst. ...
- Data Analytics Consultant. ...
- Digital Marketing Manager.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1IT303) PROGRAMMING THROUGH JAVA

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce object-oriented programming concepts using the Java language
- To introduce the principles of inheritance and polymorphism; and demonstrates how they relate to the design of abstract classes
- To introduce the implementation of packages and interfaces
- To introduce exception handling, event handling and multithreading

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Develop applications for range of problems using object-oriented programming techniques

**CO-2:** Design simple graphical user interface applications

**CO-3:** Explore the design of graphical user interface using applets and swings

#### UNIT-I:

**Object Oriented Thinking and Java Basics:** Need for OOP Paradigm, Summary of OOP Concepts, Coping with Complexity, Abstraction Mechanisms, A Way of Viewing World – Agents, Responsibility, Messages, Methods, History of Java, Java Buzzwords, Data Types, Variables, Scope and Life Time of Variables, Arrays, Operators, Expressions, Control Statements, Type Conversion and Casting, Simple Java Program, Concepts of Classes, Objects, Constructors, Methods, Access Control, This Keyword, Garbage Collection, Overloading Methods and Constructors, Method Binding, Inheritance, Overriding and Exceptions, Parameter Passing, Recursion, Nested and Inner Classes, Exploring String Class.

#### UNIT-II:

**Inheritance, Packages and Interfaces:** Hierarchical Abstractions, Base Class Object, Subclass, Subtype, Substitutability, Forms of Inheritance- Specialization, Specification, Construction, Extension, Limitation, Combination, Benefits of Inheritance, Costs of Inheritance. Member Access Rules, Super Uses, Using Final with Inheritance, Polymorphism- Method Overriding, Abstract Classes, The Object Class. Defining, Creating and Accessing a Package, Understanding Classpath, Importing Packages, Differences between Classes and Interfaces, Defining an Interface, Implementing Interface, Applying Interfaces, Variables in Interface and Extending Interfaces, Exploring Java.IO.

#### UNIT-III:

**Exception Handling and Multi-threading:** Concepts of Exception Handling, Benefits of Exception Handling, Termination or Resumptive Models, Exception Hierarchy, Usage



of Try, Catch, Throw, Throws and Finally, Built in Exceptions, Creating Own Exception Sub Classes.

String Handling, Exploring Java. Util, Differences between Multi-Threading and Multitasking, Thread Life Cycle, Creating Threads, Thread Priorities, Synchronizing Threads, Interthread Communication, Thread Groups, Daemon Threads. Enumerations, Autoboxing, Annotations, Generics.

#### **UNIT-IV:**

**Event Handling:** Events, Event Sources, Event Classes, Event Listeners, Delegation Event Model, Handling Mouse and Keyboard Events, Adapter Classes.

The AWT Class Hierarchy, User Interface Components- Labels, Button, Canvas, Scrollbars, Text Components, Check Box, Check Box Groups, Choices, Lists Panels – Scrollpane, Dialogs, Menubar, Graphics, Layout Manager – Layout Manager Types – Border, Grid, Flow, Card and Grid Bag.

#### **UNIT-V:**

**Applets:** Concepts of Applets, Differences between Applets and Applications, Life Cycle of an Applet, Types of Applets, Creating Applets, Passing Parameters to Applets. Swing: Introduction, Limitations of AWT, MVC Architecture, Components, Containers, Exploring Swing- Japplet, JFrame and JComponent, Icons and Labels, Text Fields, Buttons – The JButton Class, Check Boxes, Radio Buttons, Combo Boxes, Tabbed Panes, Scroll Panes, Trees, and Tables.

#### **TEXT BOOKS:**

1. Java The Complete Reference, Herbert Schildt, 7<sup>th</sup> Edition, TMH
2. Understanding OOP with Java Updated Edition, T. Budd, Pearson Education
3. An Introduction to Programming and OO Design using Java, J. Nino and F.A. Hosch, John Wiley & Sons

#### **REFERENCES:**

1. Introduction to Java Programming, Y. Daniel Liang, Pearson Education
2. An Introduction to Java Programming and Object-Oriented Application Development, R. A. Johnson, Thomson
3. Core Java 2, Vol. 1 - Fundamentals, Cay. S. Horstmann and Gary Cornell, 8<sup>th</sup> Edition, Pearson Education
4. Core Java 2, Vol. 2 - Advanced Features, Cay. S. Horstmann and Gary Cornell, 8<sup>th</sup> Edition, Pearson Education

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS302) RELATIONAL DATABASE MANAGEMENT SYSTEMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basic concepts and the applications of database systems
- To master the basics of SQL and construct queries using SQL
- To understand the relational database design principles
- To become familiar with the basic issues of transaction processing and concurrency control
- To become familiar with database storage structures and access techniques

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Demonstrate the basic concepts of database systems and design E-R models to represent simple database application scenarios

**CO-2:** Formulate SQL queries on the data

**CO-3:** Improve the database design by normalization

**CO-4:** Apply and relate the concept of transaction, concurrency control and recovery in database

**CO-5:** Familiar with basic database storage structures and access techniques using Indexing, hashing including B tree methods

#### UNIT-I:

**Introduction:** Database System Applications, Purpose of Database Systems, View of Data, Database Languages – DDL, DML, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Data Mining and Information Retrieval, Specialty Databases, Database Users and Administrators, History of Database Systems.

**Introduction to Database Design:** Database Design and ER diagrams, Entities, Attributes and Entity sets, Relationships and Relationship sets, Additional features of ER Model, Conceptual Design with the ER Model, Conceptual Design for Large enterprises.

**Relational Model:** Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity constraints, Querying relational data, Logical database Design: ER to Relational, Introduction to Views, Destroying /Altering Tables and Views.

#### UNIT-II:

**Relational Algebra and Calculus:** Preliminaries, Relational Algebra, Relational calculus – Tuple relational Calculus, Domain relational calculus, Expressive Power of Algebra and calculus.

**SQL:** Queries, Constraints, Triggers: Form of Basic SQL Query, UNION, INTERSECT, and EXCEPT, Nested Queries, Aggregate Operators, NULL values Complex Integrity Constraints in SQL, Triggers and Active Databases, Designing Active Databases.

#### **UNIT-III:**

**Schema Refinement and Normal Forms:** Introduction to Schema Refinement, Functional Dependencies - Reasoning about FDs, Normal Forms, Properties of Decompositions, Normalization, Schema Refinement in Database Design, Other Kinds of Dependencies.

#### **UNIT-IV:**

**Transaction Management:** Transactions, Transaction Concept, A Simple Transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Serializability, Transaction Isolation and Atomicity Transaction Isolation Levels, Implementation of Isolation Levels.

**Concurrency Control:** Lock-Based Protocols, Multiple Granularity, Timestamp-Based Protocols, Validation-Based Protocols, Multiversion Schemes.

Recovery System-Failure Classification, Storage, Recovery and Atomicity, Recovery Algorithm, Buffer Management, Failure with loss of nonvolatile storage, Early Lock Release and Logical Undo Operations, Remote Backup systems.

#### **UNIT-V:**

**Storage and Indexing:** Overview of Storage and Indexing: Data on External Storage, File Organization and Indexing, Index Data Structures, Comparison of File Organizations.

**Tree-Structured Indexing:** Intuition for tree Indexes, Indexed Sequential Access Method (ISAM), B+ Trees: A Dynamic Index Structure, Search, Insert, Delete.

**Hash- Based Indexing:** Static Hashing, Extendible hashing, Linear Hashing, Extendible vs. Linear Hashing.

#### **TEXT BOOKS:**

1. Database Management Systems, Raghu Ramakrishnan, Johannes Gehrke, 3<sup>rd</sup> Edition, McGraw Hill Education (India) Private Limited
2. Database System Concepts, A. Silberschatz, Henry. F. Korth, S. Sudarshan, 6<sup>th</sup> Edition, McGraw Hill Education (India) Private Limited,
3. Database Systems, R. Elmasri, Shamkant B. Navathe, 6<sup>th</sup> Edition, Pearson Education

#### **REFERENCES:**

1. Database System Concepts, Peter Rob & Carlos Coronel, Cengage Learning
2. Introduction to Database Management, M. L. Gillenson and others, Wiley Student Edition
3. Database Development and Management, Lee Chao, Auerbach Publications, Taylor & Francis Group
4. Introduction to Database Systems, C. J. Date, Pearson Education

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1IT302) COMPUTATIONAL THINKING USING PYTHON

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Statistical Methods for Data Science

#### COURSE OBJECTIVES:

- To Understand why Python is a useful scripting language for developers
- To create and execute Python programs and to learn how to use lists, tuples and dictionaries in Python programs
- To learn how to build and package Python modules for reusability
- To learn how to design object-oriented programs with Python classes
- To learn how to use exception handling in Python applications for error handling

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Adapt and combine standard algorithms to solve a given problem (includes numerical as well as non-numerical algorithms)

**CO-2:** Adequately use standard programming constructs: repetition, selection, functions, composition, modules, aggregated data (arrays, lists etc.)

**CO-3:** Explain what a given program (in Python) does identify and repair coding errors in a program

**CO-4:** Understand and use object-based software concepts (constructing OO software will be deal with in the course Software Engineering)

**CO-5:** Use library software for (e.g.) building a graphical user interface, web application or mathematical software

#### UNIT-I:

Introduction, History, Features, setting up path, Working with Python, Basic Syntax, Variable and Data Types, Operator, Conditional Statements- If, If-else, Nested If-else Looping for While Nested loops Control Statements Break Continue Pass String Manipulation Accessing Strings Basic Operations String Slices Function.

#### UNIT-II:

**Methods, Lists:** Introduction, accessing list, Operations, Working with lists, Function and Methods,

**Tuple:** Introduction, accessing tuples, Operations, Working Functions and Methods.

**Dictionaries:** Introduction, accessing values in dictionaries, Working with dictionaries, Properties.

#### UNIT-III:

**Functions:** Defining a function, Calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables.

Modules: Creation, Importing module, Math module, Random module, Packages.

**Composition:** Input-Output Printing on screen, reading data from keyboard, Opening and closing file Reading and writing files, Functions.

#### **UNIT-IV:**

**Exception Handling:** Exception, Exception Handling, except clause, Try, Finally, clause, User Defined Exceptions.

**OOPs concept:** Class and object, Attributes, Inheritance, Overloading, Overriding, Data hiding, Regular Expressions-Match function, Search function, Matching vs Searching Modifiers, Patterns.

**Multithreading:** Thread, starting a thread. Threading module, Synchronizing threads.

#### **UNIT-V:**

**CGI:** Introduction, Architecture, CGI environment variable, GET and POST methods, Cookies, File upload.

**Database:** Introduction, Connections, Executing queries, Transactions Handling error. Networking: Socket, Socket Module, Methods, Client and server, Internet modules, Sending email.

#### **TEXT BOOK:**

1. Learning Python, David Ascher and Mark Lutz, O' Relly

#### **REFERENCES:**

1. Python Programming: An Introduction to Computer Science, John M. Zelle, 2<sup>nd</sup> Edition, Kindle Edition
2. Python Essential Reference, David M. Beazley, 4<sup>th</sup> Edition, Developer's Library

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1IT304) INTRODUCTION TO DATA ANALYTICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To be exposed to conceptual framework of big data
- To understand different techniques of data analysis
- To be familiar with concepts of data streams
- To be exposed to item sets, clustering, frame works and Visualization

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand big data fundamentals

**CO-2:** Learn various data analysis techniques

**CO-3:** Implement various data streams

**CO-4:** Understand item sets, clustering, frame works & Visualizations

#### UNIT –I:

**Introduction to Big Data:** Introduction to Big Data Platform – Challenges of Conventional systems – Web data – Evolution of Analytic scalability, analytic process and tools, Analysis vs Reporting–Modern data analytic tools,

**Statistical Concepts:** Sampling distributions, resampling, statistical inference, prediction error.

#### UNIT-II:

**Data Analysis:** Regression modeling, Multivariate analysis, Bayesian modeling, inference and Bayesian networks, Support vector and Kernel methods

Analysis of Time Series: Linear systems analysis, nonlinear dynamics – Rule induction –

**Neural Networks:** Learning and Generalization, competitive learning, Principal component analysis and neural networks

Fuzzy Logic: extracting fuzzy models from data, fuzzy decision trees, Stochastic search methods.

#### UNIT-III:

**Mining Data Streams:** Introduction to Streams Concepts–Stream data model and architecture–Stream Computing, Sampling data in a stream–Filtering streams–Counting distinct elements in a stream–Estimating moments–Counting oneness in a Window–Decaying window–Real time Analytics Platform (RTAP) applications–case studies–real time sentiment analysis, stock market predictions.

**Frequent Item sets and Clustering:** Mining Frequent item sets– Market based Modeling A priori Algorithm– Handling large datasets in Main Memory–Limited Pass Algorithm–Counting frequent item sets in a Stream–Clustering Techniques –Hierarchical–K-Means.

**UNIT-IV:**

Clustering high dimensional data– CLIQUE and Pro CLUS–Frequent pattern-based clustering methods – Clustering in non-Euclidean space – Clustering for streams and Parallelism.

**UNIT-V:**

**Frameworks and Visualization:** MapReduce–Hadoop, Hive, Map R–Sharding–No SQL Databases – S3 – Hadoop Distributed file systems – Visualizations – Visual data analysis techniques,

**Interaction Techniques:** Systems and Applications

**TEXTBOOKS:**

1. Intelligent Data analysis, Michael Berthold, David J.Hand, Springer, 2007
2. Mining of Massive Datasets, Anand Rajaraman and Jeffrey DavidUllman, Cambridge University Press, 2012

**REFERENCES:**

1. Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics, BillFranks, John Wiley & Sons, 2012
2. Big Data Glossary, Glenn J.Myatt, Making Sense of Data, John Wiley & Sons, PeteWarden, O'Reilly, 2011
3. Data Mining Concepts and Techniques, JiaweiHan, Micheline Kamber, 2ndEdition, Elsevier, 2008

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS305) FUNDAMENTALS OF COMPUTER ALGORITHMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To reinforce algorithms analysis methods
- To ability to analyze running time of an algorithm
- To understand different algorithm design strategies
- To familiarity with an assortment of important algorithms

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the concepts of computing complexity measures of algorithms

**CO-2:** Apply Divide and conquer method to solve engineering problems

**CO-3:** Illustrate the usage of the greedy method for optimization problems

**CO-4:** solve the complex recursive problems with dynamic programming strategy

**CO-5:** Examine the complexity classes, backtracking and branch & bound strategies

#### UNIT-I:

**Introduction:** Characteristics of algorithms. Analysis of algorithms: Asymptotic analysis of complexity bounds – best, average and worst-case behavior; Performance measurements of Algorithm, Time and space trade-offs.

#### UNIT-II:

**Divide and Conquer:** General method, applications-Binary search, Quick sort, Merge sort, Strassen's matrix multiplication. Analysis of recursive algorithms through recurrence relations: Substitution method, Recursion tree method and Masters' theorem.

#### UNIT-III:

**Greedy Method:** General method, applications-Job sequencing with deadlines, 0/1 knapsack problem, Minimum cost spanning trees, Single source shortest path problem, Huffman Codes.

#### UNIT-IV:

**Dynamic Programming:** General method, Principle of optimality, applications-Multistage graphs, Matrix chain multiplication, Optimal binary search trees. 0/1 knapsack problem, All pairs shortest path problem, Traveling salesperson problem, Reliability design.

#### UNIT-V:

**Backtracking:** General method, applications- N-Queen problem, Sum of subsets problem, Graph coloring, Hamiltonian cycles.



**Branch and Bound:** Traveling Salesperson problem, 0/1 Knapsack problem: LC Branch and Bound solution, FIFO Branch and Bound solution.

**NP-Complete and NP-Hard problems:** Basic concepts, non-deterministic algorithms, P, NP, NP-Complete, and NP-Hard classes.

**TEXT BOOKS:**

1. Fundamentals of Computer Algorithms, E. Horowitz et al, Galgotia Publications
2. Introduction to Algorithms, Thomas H. Cormen, Charles E. Lieserson, Ronald L. Rivest and Clifford Stein, 4<sup>th</sup> Edition, MIT Press/McGraw-Hill

**REFERENCES:**

1. Algorithm Design, Jon Kleinberg and EvaTardos, 1<sup>st</sup> Edition, Pearson
2. Algorithm Design: Foundations, Analysis and Internet Examples, Michael T. Goodrich and Roberto Tamassia, Second Edition, Wiley
3. Algorithms – A Creative Approach, Udi Manber, 3<sup>rd</sup> Edition, Addison-Wesley, Reading, MA
4. Introduction to the Design and Analysis of Algorithms, Anany Levitin, 3<sup>rd</sup> Edition, Pearson Publications

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS306) INTRODUCTION TO FRONT END TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To enable participants to develop elegant and responsive Front-end by leveraging latest technologies
- To enable them to learn new technologies by applying foundation paradigms
- To understand how HTML, CSS, javascript and Bootstrap work together
- To use jQuery to create webpages

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply HTML, CSS and JavaScript to develop dynamic web pages

**CO-2:** Use advanced technologies jquery and json to make web responsive and interactive web pages

**CO-3:** Demonstrate Bootstrap to make customized applications

**CO-4:** Apply the concepts of AngularJS in developing Web applications

#### UNIT-I:

**Overview of HTML, CSS :** Fundamentals of HTML, HTML Headings ,HTML Paragraph, HTML Styles, HTML Formatting , HTML Quotations , HTML Computer Code, HTML Comments & Colors, HTML CSS, Links and Images , HTML Lists, HTML Blocks , HTML Classes ,HTML Layout ,HTML Responsive , HTML iframes , Creating Style Sheet, CSS Box,Model- Border properties, Padding properties, Margin properties,CSS Navigation Bar , CSS Dropdowns ,CSS Tooltips , CSS3 Animations , CSS3 Buttons , CSS3 Pagination, CSS3 Multiple Columns .

#### UNIT-II:

**Introduction to JavaScript:** Introduction to JavaScript, Statements, expressions, Working with Web Forms and validating user input, JavaScript functions and events, Error handling in JavaScript, Document Object Model, Programming HTML DOM with JavaScript, Assigning event handlers in JavaScript using DOM object property.

#### UNIT-III:

**Working with jQuery and JSON:** Basics of jQuery, jQuery Events, Benefits of using CDN, jQuery Selectors, jQuery input vs :input, jQuery DOM manipulation methods, jQuery Elements, Working with JSON Objects, JSON Arrays, Nested JSON object, Conversion of JSON object to string, Conversion of string to JSON object.

#### UNIT-IV:

**Working with Bootstrap:** Introduction to Bootstrap, Setting up Bootstrap, Bootstrap Grid System, Bootstrap Image Gallery, Bootstrap Typography, Bootstrap Blockquotes and

Lists, Bootstrap Code Blocks, Bootstrap Table Classes, Bootstrap Button Classes, Bootstrap Dropdown.

#### **UNIT-V:**

**Introduction AngularJs:** Building Single page applications with AngularJS Single page application – introduction, two way data binding, MVC in angular JS, controllers, getting user inputs, loops, Client side routing – accessing URL data, various ways to provide data in angular JS.

#### **TEXT BOOKS:**

1. HTML 5 Black Book, Dt Editorial Services, Dreamtech Press
2. Radford, S. (2015). Learning Web Development with Bootstrap and AngularJS. India: Packt Publishing
3. York, R. (2011). Beginning JavaScript and CSS Development with JQuery. Germany: Wiley

#### **REFERENCES:**

1. Learning Web Development With Bootstrap And Angularjs By Stephen Radford, Packt Publishing
2. Powell, T. A. (2010). HTML & CSS: The Complete Reference, Fifth Edition. Switzerland: McGraw Hill LLC

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS307) CONCEPTS OF GENERATIVE AI

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To Understand what Generative AI is, why it matters
- To know how it's shaping the future of business
- To analyze different AI tools
- To be able to decide about the application of Generative AI in various domains

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Demystify the technical terminology surrounding Generative AI

**CO-2:** Understand the significance of Generative AI in today's digital landscape

**CO-3:** Gain hands-on experience with state-of-the-art AI tools

**CO-4:** Make informed decisions about the application of Generative AI in various domains

#### UNIT-I:

**Introduction and Capabilities of Generative AI:** Introduction to Generative AI, Capabilities of Generative AI, Text Generation using Generative AI.

#### UNIT-II:

**Applications and Tools of Generative AI:** Applications of Generative AI, Economic Potential of Generative AI, Tools for Image Generation, Tools for Audio and Video Generation, Tools for Code Generation

#### UNIT-III:

**Generative AI for your own Industry:** Impacts of generative AI on your own Industry, Use cases of Generative AI, AI and Digital Transformation, Impact of Generative AI over job roles, Developing Internal Policies.

#### UNIT-IV:

**Strategic Planning in the age of AI:** Roles of AI in Strategic Decision-making, Building AI Policies within Your Organization, Build a Plan for Implementing Generative AI in Your Organization

#### UNIT-V:

**Prompt Engineering and Advanced ChatGPT:** Introduction to Advanced ChatGPT, Advanced Techniques for Prompting ChatGPT, Applications of Advanced ChatGPT, Chatbot Development with ChatGPT, Advanced Integration with Other Tools.

**TEXT BOOKS:**

1. Generative Deep Learning, David Foster
2. Hands-On Generative AI with Python, Kirill Dubovikov
3. Research papers from top conferences (NeurIPS, ICML, etc.)

**GENERAL**

## **Professional Ethics and Human Values**

Ethics is a necessary and listed Graduate Attribute for all engineers according to the Washington Accord. As engineers deal with society and provide for society, it is important that the ethical concerns pertaining to technology are well understood and addressed, particularly when technological advancements are still in their infancy and early growth stages but already enjoy a widespread use! Human Values form the basis for all ethics, and ethical theories help resolve professional dilemmas too. This course aims to create an appreciation for normative and applied ethics with special focus on professionalism as well as technology education and practice. Given the diverse set of roles a STEM professional may play in a society, there is an inherent societal need for STEM professionals to be ethical. The formative years of students of engineering are deemed the best time to impress upon them the practical importance and application aspects of ethics. The curriculum is designed to include an inherent appreciation for the Indian Ethos and cover a wide variety of topics with suitable case studies and examples all through. The course is aimed to become a practicable guide that enables the learners to apply the learned content while facing the practical contexts encountered in their individual life and career, even as the times and lives are ever-changing.

As a course designed to ideally educate and empower all engineers, this course content contributes substantially to the attainment of "Ethics" as one of the eleven Graduate Attributes identified in the Washington Accord, and contributes moderately to a few other Graduate Attributes—including but not limited to The Engineer and The World, Lifelong Learning, and Design/Development of Solutions—while also contributing slightly to the attainment of some more Graduate Attributes such as Individual and Collaborative Team Work, Communication, Project Management and Finance.

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1HS301) PROFESSIONAL ETHICS AND HUMAN VALUES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To emphasise on the importance of ethics for STEM professionals
- To help create a personal and practicable toolkit for ethical behaviour in personal and professional settings
- To relate the profession of engineering to sociocultural, ethical, and moral contexts in India and globally
- To develop more socially conscious STEM professionals who create and conceive a better society and a better world without sacrificing or ignoring public good

**COURSE OUTCOMES:** After completion of the course the students should be able to

**CO-1:** Distinguish morals, values, and ethics in Indian and global contexts as well as personal, sociocultural, and professional contexts

**CO-2:** Resolve moral and ethical dilemmas through relevant ethical guidance using ethical inquiries and/or appropriate ethical theories

**CO-3:** Demonstrate understanding of professionalism and be responsible social experimenters as STEM professionals in diverse roles

**CO-4:** Be conscientious about professional contributions that adhere to digital and technology ethics principles in ensuring public welfare and sustainable development

**CO-5:** Extend and apply individual moral/ ethical persona in organisational, business, and global environments

#### UNIT-I:

##### Introduction and Motivation

**Definitions:** Morals, Values, Ethics; Golden Rule; Maslow's Theory of Needs; Diversity, Equity, Inclusivity, and Accessibility (DEIA); Sustainable Development Goals (SDG)

**Human Values:** Universal Human Values, Conventional and Constitutional Values in Indian Ethos, Building Blocks for Moral Persona—Conscientiousness, Integrity, Empathy  
**Morality and Moral Development:** Moral Character, Moral Autonomy; Platform vs. Purpose

**Ethicality:** Civic Virtue for Ideal Society vs. Anomie; Ethics as basis for Legal Framework; Indian Ethos

**Motivation:** Examples and Case Studies of Ethical and Moral Decline in social, business, and professional contexts; Threats to Ethics due to changing technology landscape; Need for protection of Personally Identifiable Information (PII) and Data Privacy; Introduction to Trolley Problem and Discussion on Moral and Ethical Dilemmas; Issues in Decision-making—Value Judgement and Moral Judgement; Ethics in Academics and Research



## **UNIT-II:**

Human Values, Moral Development, and Ethical Theories of Right Action

**Moral Development:** Kohlberg's Theory vs. Gilligan's Theory; Understanding Biases

Ethical Theories: Classification of Ethical Theories—Prescriptive, Descriptive, and Applied Ethics by Character and by Conduct; Deontological Ethics (Immanuel Kant, John Rawls), Virtue Ethics (Aristotelianism), Consequentialism (Act & Rule Utilitarianism); Divine Command Ethics and Ethics through Spirituality; Ethical Pluralism; Overview of Rights Theories

**Ethical Guidance:** Ethical Frameworks and Values for organisations; Practicable guidance from the Bhagavad-Gita and Eastern Philosophies; Ethical Inquiries—Normative, Conceptual, and Factual; Ethical Decision-making Tools; Ethics and Law—Ethics as a Basis for Law, Awareness of Rights of Convicts and Criminals

## **UNIT-III:**

### **Ethics for STEM Professionals**

**Profession and Professionalism:** Definitions; Professional Characters, Traits, Virtues, Roles, Rights and Responsibilities; Overview of Business Ethics and Work Ethics; STEM Professionals as Managers, Leaders, Consultants, Expert Witnesses, Scientists, Researchers and Academicians, and Entrepreneurs

**STEM Profession as Social Experimentation:** Comparison with scientific experiments; Application of Ethical Theories and Ethical Inquiries in professional contexts to gather relevant information; Informed Consent

**Responsibilities of STEM Professionals as Social Experimenters:** Need for being Ethical by Design; Accountability and Answerability in general and in autonomous technology applications; Focus on DEIA, SDG, and Social Innovation; Guidance through Professional Societies

## **UNIT-IV:**

### **Understanding Ethics through STEM Case Studies**

**Safety and Risk:** Understanding Safety and Risk through case studies—Titanic, Chernobyl, Bhopal; Overview of risks implied by digital technologies

**General Aspects:** Environmental Ethics and Carbon Footprint; Non-disclosure Agreements (NDA) and Intellectual Property (IP) as a Professional Right and as exploited by businesses; Copyrights and Ethics in Arts; Social Engineering and Ethics

**Digital Ethics:** Data Ethics; Ethics of AI; Ethics on Social Media; Ethics of Humans as Data Objects; Legal frameworks and policies for Data Management—Content Development, Protection, Integrity, Transparency, and Accountability; Ethics in the context of emerging technology areas such as Large Language Models (LLM), Digital Forensics, Internet of Things (IoT), Robotics and Automation, Cybersecurity & Cybercrime, Healthcare & Clinical Management, Finance, and Personal Tech

**Technology Ethics:** Computer Ethics, Machine and Robot Ethics; Ethics of technology advancements — an ethical urge to create green technologies, steps taken in the EU

## **UNIT-V:**

### **Ethics for Workplace and Societal Wellbeing**

**Ethics in Organizational Behavioral Aspects:** Collegiality, Loyalty, Trust; Workplace Deviances, Occupational Crime, and Culpability; Employee Rights and Relations; Ethical Frameworks and Codes of Ethics and Conduct; Conflicts of Interest, Conflict

Management and Resolution; Creating Ethical Climate, Ethics Education and Training, Grievance Redressal Procedures; Whistleblowing

**Global Context:** Work-Life Balance, DEIA at workplace, Cultural aspects, Concerns for Multinational Companies

**Ethics and Society:** Impact of Wars, Politics, and Religion on society; Understanding Privilege, Marginalisation, and Minorities; Ethico-economical Aspects: Fair Trade, Capitalism vs. Communism, Developed vs. Developing vs. Underdeveloped economies

#### **TEXTBOOKS:**

1. Ethics in Engineering, Mike W. Martin, Roland Schinzinger, McGraw Hill Education, 2017 (ISBN: 978-9339204457)
2. Business Ethics: An Indian Perspective, A.C. Fernando, K.P. Muralidheeran, E.K. Satheesh, Pearson Education, 2019 (ISBN: 978-9353437442)
3. Professional Ethics, R. Subramanian, Oxford University Press, 2017 (ISBN: 978-0199475070)
4. Professional Ethics and Human Values, M. Govindarajan, S. Natarajan, V.S. Senthil Kumar, Prentice Hall India, 2015 (ISBN: 978-8120348165)

#### **REFERENCES:**

1. Engineering Ethics: Concepts & Cases, Charles E. Harris, Jr., Michael S. Pritchard, Michael J. Rabins, Cengage Learning, 2012 (ISBN: 978-8131517291)
2. Classical Indian Ethical Thought: A Philosophical Study of Hindu, Jaina and Bauddha Morals, Kedar Nath Tiwari, Motilal Banarsidass Publishers, 2017 (ISBN: 978-8120816084)
3. Ethics for the Whole World, Dalai Lama, 978-9351360803 Vijay Kumar Ivaturi et al., The Manual for Indian Start-Ups, Penguin Random House India, 2017 (ISBN: 978-0143428527), To Be Human, Jiddu Krishnamurti, Shambhala, 2000 (ISBN: 978-1570625961)
4. On Ethics and Economics, Amartya Sen, Oxford India, 1999 (ISBN: 978-0195627619)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VI Semester

#### (22OE1HS302) ENTREPRENEURSHIP

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To motivate STEM professionals to inculcate the skills thereof in any professional role and to consider intrapreneurship or entrepreneurship as career choices for personal and societal growth
- To impart lean management principles and practices to plan, execute, and convert one's own idea into a sustainable business model using relevant tools and templates
- To gain practical knowledge to design and develop one's own lean startup venture
- To identify and avoid some potential pitfalls in validation and design of an innovative product or service and prepare against the same in production and marketing phases

**COURSE OUTCOMES:** After completion of the course the students should be able to

**CO-1:** Discover societal problems as entrepreneurial opportunities and ideate to develop solutions through systematic and creative approaches to innovation and business strategy

**CO-2:** Apply lean methodology to startup ideas and create a suitable Business Plan

**CO-3:** Use various tools/templates to pitch/present innovative solution/business plan/idea

**CO-4:** Validate ideas, design, production, and marketing systematically using techniques such as 5 Whys, Innovation Accounting, Value and Growth Propositions

**CO-5:** To strategise during ideation, production, market research, marketing and facing competition

#### UNIT-I:

##### Entrepreneurial Skills and Opportunities

**Entrepreneurship as a Career for STEM Professionals:** Motivation and Overview; Ethical and Empathetic Personality and Skill Set needed for an entrepreneur; Current Business Trends; Concepts of Intrapreneurship, Social Entrepreneurship, Technopreneurship, Studentpreneurship; Overview of Innovation—Sources and Types, inspiration from SDG as well as Industry 4.0 and Emerging Technologies; Entrepreneurial Management vs. Corporate Management; Overview of Startup India and list of other Online and Offline Resources for aspiring entrepreneurs including funding opportunities

## **UNIT-II:**

**Introduction to Lean Startup Methodology:** Overview, Principles of Lean Startup, Lean vs. Traditional Startup; Lean Startup Management; Vision-to-Steering, Start-Define-Learn-Experiment, Leap-Test-Measure-Pivot, Build-Measure-Learn

## **UNIT-III:**

**Business Model Concepts and Design Thinking:** Components of Business Plan; Business Model Canvas (BMC); Lean Canvas (LC); Pitch Deck; Elevator Pitch; Minimum Viable Product (MVP), Proof of Concept (PoC), Prototype

**Design Thinking for Business:** Understanding Users and Personas, Customers' Jobs to be done (JTBD); Examples and case studies of product-based, service-based, and social innovation-based startups; Business Modelling through the three lenses of innovation—Desirability, Feasibility, and Viability; Sustainability and Scalability

## **UNIT-IV:**

**Building and Evaluating Your Business Model:** Identifying Early Adopters, Value Proposition, and High-level Concept; Awareness of other resources: Problem-framing Canvas, Users' Bulls-Eye Canvas, Opportunity Canvas, Value Proposition Canvas, Ethics Canvas and Data Ethics Canvas

**Market Research and Analysis:** TAM, SAM, SOM; Three Learning Milestones of Innovation; Pivot or Persevere; The Engines of Growth: Sticky, Viral, and Paid; Root Cause Analysis (RCA) through 5 Whys; Value and Waste

## **UNIT-V:**

**Evaluating and Strengthening Your Business Model:** List of Evaluation tools for Team Management and Project Management; Kan-ban Diagram for Project Planning and Resource Allocation; Awareness of Financial Planning—Projection, Valuation, Stages, Forms of Financing; Analogs and Antilogos; The three A's: Actionable, Accessible, and Auditable Metrics and Vanity Metrics; Why startups still fail? Paralysis by Analysis and Extinct by Instinct; Can you make your startup failproof? Building right teams and partners; Strategy as innovation

## **TEXTBOOKS:**

1. The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses, Eric Ries, Penguin Portfolio, 2015 (ISBN: 978-0670921607)
2. Entrepreneurship Simplified: From Idea to IPO, Ashok Soota, S R Gopalan, Penguin Random House India, 2016 (ISBN: 978-0670088959)
3. Startup Easy - Part 1: The Essentials, Shishir Gupta, StartupLanes.Com, 2017 (ISBN: 978-9386503886)

## **REFERENCES:**

1. Measure What Matters: OKRs: The Simple Idea that Drives 10x Growth, John Doerr, Penguin Portfolio, 2018 (ISBN: 978-0241348482)
2. Entrepreneurship Development and Business Ethics, Abhik Kumar Mukherjee, Shaunae Roy, Oxford University Press, 2019 (ISBN: 978-0199494460)
3. The Manual for Indian Start-Ups, Vijay Kumar Ivaturi et al., Penguin Random House India, 2017 (ISBN: 978-0143428527)

4. Social Entrepreneurship in India: Quarter Idealism and a Pound of Pragmatism, Madhukar Shukla, SAGE Publications India Pvt Ltd, 2020 (ISBN: 978-9353882372)
5. Entrepreneurship: A South Asian perspective, Donald F Kuratko, T.V Rao, Cengage Learning. 2012

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1CE305) SMART CITIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand smart city basic concepts, global standards, and Indian context of smart cities
- To explain smart community, smart transportation and smart buildings
- To understand Energy demand, Green approach to meet Energy demand and their capacities
- To identify Smart Transportation Technologies in cities and concepts towards smart city

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain and elaborate smart city concepts and their international and national standards

**CO-2:** Conceptualize smart community, transportation and building concepts

**CO-3:** Develop and calibrate energy demand and their capacity limits

**CO-4:** Predict the various smart urban transportation systems and the transition from existing city towards a smart city

#### UNIT-I:

**Introduction to Smart Cities:** Introduction to Smart Cities - Understanding Smart Cities - Dimensions of Smart Cities – World urbanization, Global Experience of Smart Cities, Smart City case studies-Indian scenario - India “100 Smart Cities” Policy and Mission. City as a System of Systems- Systems thinking – Developing a smart city approach – Core elements of a smart city – Relevant open data for a smart city – Sustainability – Privacy and Ethics.

#### UNIT-II:

**Smart Cities Planning and Development:** Introduction to Smart Community; Smart community concepts: Concept of Smart Community - Smart Transportation - Smart Building and Home Device - Smart Health - Smart Government - Smart Energy and Water - Cybersecurity, Safety, and Privacy; Internet of Things, Blockchain, Artificial Intelligence, Alternate Reality, Virtual Reality.

#### UNIT-III:

**Smart Urban Energy Systems:** Conventional vs. Smart, City components, Energy demand, Green approach to meet Energy demand, Index of Indian cities towards smartness – a statistical analysis -Meeting energy demand through direct and indirect solar resources- Efficiency of indirect solar resources and its utility, Capacity.

**UNIT-IV:**

**Smart Transportation Systems:** Smart Transportation Technologies - Driverless and connected vehicles - ride sharing solutions - The "improve" pathway - The "shift" pathway – Smart Roads and Pavement systems – Relevant case studies

**UNIT-V:**

**Future of Smart Cities:** The transition of legacy cities to Smart - Right transition process - the benefit of citizens, cities have to adopt effective management and governance approaches-factors in the transition phase of legacy cities to Smart cities and their managerial implications.

**TEXT BOOKS:**

1. Internet of Things in Smart Technologies for Sustainable Urban Development, G. R. Kanagachidambaresan, R. Maheswar, V. Manikandan, K. Ramakrishnan., Springer, 2020
2. Society 5.0: A People-Centric Super-Smart Society, Hitachi-U Tokyo Laboratory (H-U Tokyo Lab), Springer, 2020
3. The Routledge Companion to Smart Cities, Katharine S. Willis, Alessandro Aurigi, Routledge International Handbooks, 2020

**REFERENCES:**

1. Smart Cities in Asia: Governing Development in the Era of Hyper-Connectivity Yu-min Joo, Yu-Min Joo, Teck-Boon Tan, Edward Elgar, 2020
2. Urban Systems Design: Creating Sustainable Smart Cities in the Internet of Things Era, Yoshiki Yamagata, Perry P. J. Yang, Elsevier, 2020
3. Smart Cities and Artificial Intelligence: Convergent Systems for Planning, Design, and Operations, Christopher Grant Kirwan, Zhiyong Fu, Elsevier, 2020

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EE303) TRENDS IN ENERGY SOURCES FOR SUSTAINABLE DEVELOPMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the role of sustainable energy
- To know components of solar PV and wind energy conversion systems
- To understand the principles of Biomass, geo-thermal and wave energy systems
- To learn various energy storage methods

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand various sources for sustainable energy

**CO-2:** Understand Solar Photo voltaic and wind energy systems

**CO-3:** Learnt the harnessing techniques of Biomass, geothermal and ocean energy

**CO-4:** Familiarize with energy storage methods

#### UNIT-I:

**Introduction:** Trends in energy consumption - Conventional and renewable sources, Energy sources and their availability, Energy Conservation status in India -need of new energies for sustainable development.

#### UNIT-II:

**Fundamentals of Solar Radiation:** Introduction-The Sun as Source of Energy, Extraterrestrial and Terrestrial Radiations, Spectral Power Distribution of Solar Radiation, instruments for measuring solar radiation and sunshine recorder.

**Solar PV Conversion:** The PV Cell-Crystalline Solar cells -Thin film and amorphous solar cells, Module, Array, Equivalent Electrical circuit- Open circuit voltage and Short circuit current, I-V, P-V Curves. Developments in efficient non silicon solar cells

#### UNIT-III:

**Wind Energy:** origin of winds-Global (or Planetary) Winds- Local Winds-Factors Affecting the Distribution of Wind Energy on the Surface of Earth, Wind Turbine – Types, construction of HAWT, VAWT, performance characteristics, Betz criteria.

#### UNIT-IV:

**Bio-Mass:** Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Biogas digesters, combustion characteristics of bio-gas, utilization for cooking, I.C. Engine operation and economic aspects.

#### UNIT-V:

**Geothermal Energy:** Resources, types of wells, methods of harnessing the energy  
**Ocean Energy:** OTEC, Principles utilization, setting of OTEC plants, thermodynamic



cycles. Tidal and Wave Energy: Potential and conversion techniques, mini-hydel power plants, and their economics.

**Energy Storage: Electro Chemical Storage:** lead-acid- nickel cadmium-nickel-metal-hydride and lithium type batteries-Principle of operation, Types, Advantages and disadvantages. Non-Electric Storage: Methods of Energy storage –Pumped Energy Storage – Compressed air Energy Storage, Superconducting Magnet Energy Storage.

**TEXT BOOKS:**

1. Non-Conventional Energy Sources, G. D. Rai, 6<sup>th</sup> Edition, Khanna Publishers, 2004
2. Non-Convention Energy Resources, B. H. Khan, 3<sup>rd</sup> Edition, McGraw-Hill, 2017

**REFERENCES:**

1. Renewable Energy Sources, Twidell & Weir, 3<sup>rd</sup> Edition, CRC Press, 2015
2. Solar Energy, Sukhatme, 3<sup>rd</sup> Edition, McGraw-Hill, 2008
3. Non-Conventional Energy, Ashok V. Desai, Wiley Eastern, 1990

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1ME303) 3D PRINTING AND DESIGN

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the need and know about the applications of 3D Printing
- To understand the need of liquid and solid based 3D Printing systems
- To know about the powder-based 3D Printing systems
- To understand the importance of CAD, post-processing and inspection involved in 3D Printing

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Summarize the importance of 3D Printing and its applications

**CO-2:** Explain the process involved in liquid and solid based 3D Printing Systems

**CO-3:** Explain the powder-based 3D Printing systems

**CO-4:** Apply the knowledge of CAD, post-processing and inspection involved in 3D Printing

#### UNIT-I:

**Introduction:** Introduction to 3D Printing, Classification, 3D Printing Process Chain, Materials for 3D Printing, Distinction between 3D Printing & Conventional Manufacturing.

**Applications:** Brief overview of applications in Aerospace, Automotive, Biomedical, Defense, Construction, Jewelry, Coin and Tableware Industry, Artificial Intelligence & IOT in 3D Printing

#### UNIT-II:

**Liquid Based 3D Printing Systems:** Introduction, Principle, Processes and Applications of Material Jetting and Stereolithography.

#### UNIT-III:

**Solid Based 3D Printing Systems:** Introduction, Principle, Processes and Applications of Fused Deposition Modeling (FDM) and Laminated Object Manufacturing (LOM).

#### UNIT-IV:

**Powder Based 3D Printing Systems:** Introduction, Principle, Processes and Applications of Selective Laser Sintering (SLS), Three-Dimensional Printing (3DP).

#### UNIT-V:

**CAD, Post Processing and Inspection for 3D Printing:** CAD data formats, CAD model preparation, Part orientation and support generation, Overview of 3D Printing softwares - MAGICS and MIMICS.

**Post Processing:** Introduction, Post Processing Techniques like Support material removal, Cleaning, Sanding and Polishing.

**Inspection:** Introduction, Significance, Inspection techniques like Dimensional measurement along X, Y and Z axes, visual inspection of the surface finish (overall aesthetics and intact features), flatness or warp check, and FOD (foreign objects or debris) check.

#### **TEXT BOOKS:**

1. Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing, Ian Gibson, David W. Rosen, Brent Stucker, Springer, 2010
2. Rapid Prototyping: Principles and Applications, Chua C. K., Leong K. F., and Lim C. S., 3<sup>rd</sup> Edition, World Scientific, 2010

#### **REFERENCES:**

1. Rapid Prototyping and Engineering Applications: A Toolbox for Prototype Development, Liou L. W. and Liou F. W., CRC Press, 2007
2. Rapid Prototyping: Theory and Practice, Kamrani A. K. and Nasr E. A., Springer, 2006
3. Rapid Tooling: Technologies and Industrial Applications, Hilton P. D. and Jacobs P. F., CRC Press, 2000
4. Rapid Prototyping, Gebhardt A. Hanser, Gardener Publications, 2003

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1ME304) ADVANCED MATERIALS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Basic sciences

#### **COURSE OBJECTIVES:**

- To understand basic principles of advanced materials
- To understand the applications of all kinds of Industrial materials
- To select materials for given applications.
- To identify the need of developing Nano materials, smart materials, ceramics, glasses and non-metallic materials

**COURSE OUTCOMES:** After completion of the course, the student will be able to

**CO-1:** Explain the concepts and principles of advanced materials

**CO-2:** Suggest the material for the applications of all kinds of Industrial components

**CO-3:** Understand the material selection concepts to select a material for a given application

**CO-4:** Understand the behaviour and applications of Nano materials, smart materials, ceramics, glasses and non-metallic materials

#### **UNIT -I:**

**Classification and Selection of Materials:** Classification of materials, properties required in Engineering materials, Selection of Materials; Motivation for selection, cost basis and service requirements - Selection for mechanical properties, strength, toughness, fatigue and creep - Selection for surface durability corrosion and wear resistance – Case studies in materials selection with relevance to aero, auto, marine, machinery and nuclear applications.

#### **UNIT -II:**

**Composite Materials:** Fiber reinforced, laminated and dispersed materials with metallic matrix of aluminium, copper and Titanium alloys and with non-metallic matrix of unsaturated polyesters and epoxy resins. Important properties and applications of these materials.

#### **UNIT- III:**

**Low & High Temperature Materials:** Properties required for low temperature applications, Materials available for low temperature applications, Requirements of materials for high temperature applications, Materials available for high temperature applications, Applications of low and high temperature materials.

**UNIT- IV:**

**Modern Metallic Materials:** Dual Steels, Micro alloyed, High Strength Low alloy (HSLA) Steel, Maraging Steel, Inter metallics, Ni and Ti Aluminides.

Non-metallic Materials: Polymeric materials, Fibers, Foams, Adhesives and Coatings, structure, Properties and Applications.

**UNIT- V:**

**Smart Materials:** Shape Memory Alloys, Varistors and Intelligent materials for bio-medical applications.

Nanomaterials: Definition, Types of nanomaterials including carbon nanotubes and nanocomposites, Physical and mechanical properties, Applications of nanomaterials.

**TEXT BOOKS:**

1. Materials Science and Engineering, WD. Callister Jr. Wiley India Pvt. Ltd 2010
2. Engineering Design: A Materials and Processing Approach G.E. Dieter McGraw Hill 1991

**REFERENCES:**

1. Materials Selection in Mechanical Design, M.F. Ashby Pergamon Press 1992
2. Introduction to Engineering Materials & Manufacturing Processes, NIIT Prentice Hall of India, 2003
3. Engineering Materials Properties and Selection, Kenneth G. Budinski Prentice Hall of India, 1992
4. Selection of Engineering Materials, Gladius Lewis Prentice-Hall, New Jersey, 1990

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1EC307) 5G/6G TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn air Interface for 5G as the most important elements that differentiate between 2G, 3G, 4G and 5G
- To study the concepts of 5G networks, technologies, and services
- To understand the concept of 5G NR and different 5G waveforms
- To study the basic principles of 6G networks

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the need for 5G and the Use cases for 5G

**CO-2:** Understand 5G architectural details

**CO-3:** compare different 5G radio access technologies

**CO-4:** Analyse various modulation schemes for 5G

**CO-5:** Understand the evolution of 5G to 6G

#### UNIT-I:

**Fundamentals of 5G Technology:** Introduction, Historical trends of wireless communications, Evolution of LTE technology to beyond 4G, 5G roadmap, 5G Overview, Physical Layer Changes in 5G, Protocol changes in 5G, 5G Use cases.

#### UNIT-II:

**5G Network Architecture:** Introduction, 5G RAN architecture, 5G Core, EPC to 5GC, Main Functional Entities of the 5G Core, High-Level Features of 5G Core, Network Slicing, QoS, 5G Security.

#### UNIT-III:

**Numerology and Slot Structure:** Numerology and Slot Structure in 4G LTE, SCSs for 5G NR, frequency ranges, Bandwidths, and bands for 5G NR, Symbol, Slot, Subframe, and frame for 5G NR, slot structure for 5G NR.

#### UNIT – IV

**5G radio-access Technologies:** 5G Waveform, OFDM, Modulation and coding in 5G, Propagation Characteristics of 5G Channel models, MIMO communication essentials, Massive MIMO in 5G, Small cells, D2D, M2M

#### UNIT – V

**Evolution of 6G systems:** Evolution from 5G to 6G, 6G frequency bands, Bandwidths and data rates, channel models, OTFS orthogonal Time frequency Space, Rate split multiple access, NOMA for 6G, AI/ML model for 6G network optimization.

**TEXT BOOKS:**

1. Fundamentals of 5G Communications: Connectivity for Enhanced Mobile Broadband and Beyond Wanshi Chen, Peter Gaal, Juan Montojo, Haris Zisimopoulos © 2021 | Published: July 7, 2021
2. Fundamentals of 5G Mobile Networks Jonathan Rodriguez, WILEY Publications

**REFERENCES:**

1. 5G Mobile and Wireless Communications Technology Afif Osseiran, Jose F. Monserrat, Patrick Marsch Cambridge University Press Second Edition 2011
2. 5G NR: The Next Generation Wireless Access Technology Erik Dahlman, Stefan Parkvall, Johan Sko'ld Elsevier First Edition 2016
3. Evolution of Air Interface towards 5G Prof. Suvra Sekhar Das Electrical and Electronics Engineering, IIT Kharagpur
4. Evolution of Air Interface for 5G, River Publishers 2018, Journal and conference papers, white papers

**ONLINE RESOURCES:**

1. <https://www.theiotacademy.co/5G-6G-Talks>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EC306) FUNDAMENTALS OF VLSI

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn the fabrication process of integrated circuit and operation of MOS transistors
- To understand the electrical properties of MOS transistors
- To learn the basics of digital design
- To Understand the design procedure of various digital circuits
- To study the concepts of stick diagrams and layouts with the knowledge of MOS layers

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand IC Fabrication process steps required for various MOS circuits

**CO-2:** Know the various electrical properties of MOS transistors

**CO-3:** Understand the importance of digital circuits in IC design.

**CO-4:** Design the various combinational and digital circuits.

**CO-5:** Design the circuits using CMOS logic style and understand the role of testing

#### UNIT-I:

**Introduction to Integrated Circuit Technology:** MOS Technologies, PMOS, NMOS, CMOS fabrication process. Basic MOS Transistors Operation - Enhancement and Depletion Mode Transistors, and VLSI Design Flow.

#### UNIT-II:

**Basic Electrical Properties of MOS:** Drain to Source Current equation, MOS transistor threshold voltage, Transconductance and Output conductance, figure of merit. Pass transistor, NMOS Inverter, CMOS Inverter.

#### UNIT-III:

**Introduction to Digital Design:** Binary Logic, Postulates and theorems, SOP and POS forms, digital logic gates, Karnaugh Maps –minimization using two variable, three variable and four variable K-Maps, Don't Care Conditions.

#### UNIT-IV:

Overview of Digital Circuits – Half Adder, Full Adder, Half subtractor and Full Subtractor, Ripple carry adder, fundamentals of Decoders, Encoders, Multiplexers. Combinational Vs Sequential Circuits, Functionality of Flip Flops-RS flip flop, D flip flop, JK flip flop, T flip flop, Introduction to Registers and Counters.



**UNIT-V:**

**Combinational MOS Logic Circuits:** Primitive CMOS logic gates - NOR and NAND gate, Realizing Boolean expressions using CMOS gates.

**VLSI Circuit Design Processes:** MOS Layers, Stick Diagrams- nMOS and CMOS design styles. Layout Diagram for nMOS and CMOS inverters, Need for Scaling in VLSI, Testing Philosophy and Role of Testing in IC technology.

**TEXT BOOKS:**

1. Essentials of VLSI Circuits and Systems, Kamran Eshraghian, Douglas and A. Pucknell, PHI, 2005
2. Digital Design, M. Morris Mano, 3<sup>rd</sup> Edition, Pearson Education/PHI, 2003

**REFERENCES:**

1. CMOS VLSI Design – A Circuits and Systems Perspective, Neil H. E. Weste, David Harris, Ayan Banerjee, 4th Edition, Pearson, 2015
2. Introduction to VLSI Systems: A Logic, Circuit and System Perspective, Ming-BO Lin, CRC Press, 2011
3. Essentials of Electronic Testing for Digital, Memory and Mixed Signal VLSI Circuits, M.L. Bushnell, V. D. Agrawal, Kluwer Academic Publishers, 2015

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1CS304) AI FOR BEGINNERS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand and analyze the basic concepts of artificial intelligence
- To identify, explore the complex problem-solving strategies and approaches
- To analyze the concepts of basic concepts of neural networks and learning process
- To explore and analyze the methodology used in machine learning and computer vision

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand and apply the basic concepts of artificial intelligence and its use cases. lives

**CO-2:** Explore the various search strategies and approaches for problem solving

**CO-3:** Correlate the fields related to AI, and articulate various learning paradigms

**CO-4:** Describe several issues and ethical concerns surrounding AI

#### UNIT-I:

**Introduction to AI:** What is AI-On Overview, History of AI, Applications and Examples of AI, AI Concepts, Terminology, Key fields of AI. AI Issues, Concerns, and Ethical Considerations.

#### UNIT-II:

**AI as Search Process:** On overview of Search Strategy. Types of Searches- Uninformed, Informed, Bidirectional search, Heuristic search. Local search, Local beam search, Adversarial Search.

#### UNIT-III:

**AI as Knowledge Exploration:** Introduction to Propositional Logic, Rules of Inference, First Order Logic (FOL) Syntax, Semantics, Entailment, Tools to represent knowledge.

#### UNIT-IV:

**AI as a Learning Task:** Introduction to Learning, Learning types -Supervised, Unsupervised, Reinforcement Learning, Machine learning, Deep Learning, The link between AI, ML, DL.

#### UNIT-V:

**AI as Neural Networks:** Introduction to biological neural networks. Link between biological neuron and artificial neuron. Architecture of artificial neural network, Types of Neural networks-single layer, multilayer, Back propagation networks.

**TEXT BOOKS:**

1. Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, 3<sup>rd</sup> Edition, Prentice Hall, 2010
2. Machine Learning, Tom M. Mitchell, M. C. Graw Hill Publications
3. Neural Networks-A Comprehensive Foundation, Simon Haykin, 2<sup>nd</sup> Edition, Pearson Education, 2004

**REFERENCES:**

1. Artificial Intelligence, Elaine Rich & Kevin Knight, 2<sup>nd</sup> Edition, TMH
2. Artificial Intelligence, A New Synthesis, Nils J. Nilsson, Elsevier
3. Artificial Neural Networks, Yegnanarayana B., PHI

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS309) BLOCKCHAIN TECHNOLOGY ESSENTIALS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce and get the technological overview of blockchain technologies
- To Study the foundation of Blockchain Technology and demonstrate the various types of Blockchain
- To explore the application area of Blockchain Technology
- To introduce smart contract, consensus algorithm and Security Mechanism
- Introduction to available platforms to implement Blockchain Technology

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand and explore the Blockchain Technology

**CO-2:** Describe smart contract concepts

**CO-3:** Explore different types of Blockchain

**CO-4:** Develop the platforms to implement Blockchain Technology

#### UNIT-I:

**Fundamental of Blockchain:** Introduction to Centralized, Decentralized and Distributed system, computer network peer to peer connection, understanding of blockchain, History of Blockchain, Various technical definitions of Blockchain.

**Generic elements of a Blockchain:** Block, Transaction, Node, Why It's Called "Blockchain", Characteristics of Blockchain Technology, Advantages of blockchain technology, Limitations of blockchain as a technology, Applications of blockchain technology, Tiers of blockchain technology Blockchain 0, Blockchain 1, Blockchain 2, Blockchain 3, Generation of Blockchain X, Confidentiality, Integrity, Authentication

#### UNIT-II:

**Concept of Blockchain Technology:** Component of block, Structure of Block chain, Technical Characteristics of the Blockchain, genesis block, Nonce, architecture of blockchain applications, Types of blockchain: Public blockchain, private blockchain, hybrid blockchain, examples of Public, private, hybrid blockchain and its merit and demerit, Permissioned ledger, Distributed ledger, Shared ledger, Fully private and proprietary blockchains, Tokenized blockchains, Tokenless blockchains

#### UNIT-III:

Smart Contract, Byzantine Fault Tolerance (BFT), Byzantine generals problem, byzantine agreement problems, Cryptography, Hashing, Distributed database, Consensus mechanisms, and basic of Cryptographic primitives, Technical Characteristics of Secure Hash Algorithms (SHA), Digital signature, Consensus Algorithm: Proof of work (PoW), Proof-of-Stake (PoS), Proof of authority (PoA)

**UNIT-IV:**

Cryptocurrency, Digital currency, Bitcoin, Bitcoin definition, Keys and addresses, Public keys in Bitcoin, Private keys in Bitcoin, how bitcoin currency work, Ethereum, Ethereum network, either, Ethereum Nodes, Ethereum clients, transaction fees, Ethereum Virtual Machine, Various Cryptocurrency and differences.

**UNIT-V:**

**Implementation Platforms:** Hyperledger as a protocol, Reference architecture, Hyperledger Fabric, Transaction Flow, Hyperledger Fabric Details, Fabric Membership, Fabric Membership, Blockchain in AI, Blockchain in IoT, Blockchain in Cloud Computing.

**TEXT BOOKS:**

1. Mastering Blockchain, Imaran Bashir, 2<sup>nd</sup> Edition, Packt
2. Blockchain Basic, Daniel Drescher, A Press

**REFERENCES:**

1. Blockchain For Dummies®, IBM Limited Edition, John Wiley & Sons, Inc

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EI303) FUNDAMENTALS OF ROBOTICS AND DRONES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basics of robots
- To classify various types of actuators, grippers and sensors
- To acquire knowledge on kinematics and vision systems used for different robots
- To acquire knowledge on the basics of drones

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Acquire knowledge on various types of Power Sources (actuators), sensors and grippers

**CO-2:** Acquire knowledge on different applications of various types of robots

**CO-3:** Analyze the direct and the inverse kinematic problems and calculate the manipulator Dynamics

**CO-4:** Acquire knowledge on the applications of machine vision in robotics

**CO-5:** Acquire Knowledge on the basics of drones

#### UNIT-I:

**Basic Concepts:** An overview of Robotics, classification of Robots, Robot Components, Robot degrees of freedom, Robot Joints, Robot Coordinates, Robot reference frames, Programming modes, Robot Characteristics.

#### UNIT-II:

##### Actuators, Grippers and Sensors

**Actuators:** Characteristics of activating system, comparison of activating system Hydraulic devices, Pneumatic devices, electric motors, magneto-strictive actuators.

**Sensors & Grippers:** Robot end effectors, Classification, drive system for Gripper, Mechanical Grippers, Magnetic Grippers, Vacuum Grippers, Adhesive Grippers, Hooks, Scoops. Sensor characteristics, Position sensors, velocity sensors, acceleration sensors, torque sensors, light and infrared sensors, touch and tactile sensors, proximity sensors, range finders.

#### UNIT-III:

**Kinematics:** Forward and reverse kinematics. Matrix representation of translational and Rotational motion – Homogeneous Transformation-DH representation of standard configuration Robots Inverse Kinematics. Joint space vs. Cartesian space-Basics of Trajectory planning in joint and Cartesian space.

**UNIT-IV:**

**Robot Vision:** Low level and High-level vision Image acquisition, Illumination Techniques, Imaging Geometry, Some Basic Relationships between Pixels, Segmentation, Description, Segmentation and Description of 3-D Structures, Recognition, Interpretation.

**UNIT-V:**

**Basics of Drones:** Theory behind how drones work, individual components that makeup a drone, basic concepts involved radio-controlled model flying, building a complete quad copter drone from scratch.

**TEXT BOOKS:**

1. Introduction To Robotics: Analysis, Control, Applications, Saeed B. Niku, 2<sup>nd</sup> Edition, Wiley
2. Industrial Robotics, Technology Programming and Applications, Mikell P. Groover, Nicholas G. Odrey, Mitchel Weiss, Roger N. Nagel, Ashish Dutta, McGraw-Hill, 2012

**REFERENCES:**

1. Robotics Technology and Flexible Automation, Deb S. R., John Wiley
2. Robots and Manufacturing Automation, Asfahl C. R, John Wiley
3. Robotic Engineering–An Integrated Approach, Klaffer R. D., Chimielewski T. A., Negin M., Prentice Hall of India
4. Drones for Beginners, Udemy

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1IT305) INTRODUCTION TO CLOUD COMPUTING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To impart the basics of cloud computing for business management
- To illustrate and explore the benefits of cloud storage and its applications, usage by managers
- To enable students explore cloud computing driven real time systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the characteristics and models in Cloud computing

**CO-2:** Asses Cloud services applications and the challenges associated with Cloud Computing

**CO-3:** Apply various cloud services and deployment models and virtualization techniques for business

**CO-4:** Analyze the concepts of cloud storage and demonstrate their use

**CO-5:** Evaluate various cloud programming models and apply them in virtual office management

#### UNIT-I:

**Cloud Computing Overview:** Origins of Cloud computing – Cloud components - Essential characteristics – On-demand self-service, Broad network access, Location independent resource pooling , Rapid elasticity , Measured service, Comparing cloud providers with traditional IT service providers, Roots of cloud computing.

#### UNIT-II:

**Cloud Insights:** Architectural influences – High-performance computing, Utility and Enterprise grid computing, Cloud scenarios – Benefits: scalability, simplicity, vendors, security, Limitations – Sensitive information - Application development- security level of third party - security benefits, Regularity issues: Government policies.

#### UNIT-III:

**Cloud Architecture- Layers and Models:** Layers in cloud architecture, Software as a Service (SaaS), features of SaaS and benefits, Platform as a Service (PaaS), features of PaaS and benefits, Infrastructure as a Service (IaaS), features of IaaS and benefits, Service providers, challenges and risks in cloud adoption. Cloud deployment model: Public clouds – Private clouds – Community clouds - Hybrid clouds - Advantages of Cloud computing.



**UNIT-IV:**

**Cloud Simulators- CloudSim and Green Cloud:** Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture(User code, CloudSim, GridSim, SimJava) Understanding Working platform for CloudSim, Introduction to GreenCloud.

**UNIT-V:**

**Introduction to VMWare Simulator:** Basics of VMWare, advantages of VMware virtualization, using VMware workstation, creating virtual machines-understanding virtual machines, create a new virtual machine on local host, cloning virtual machines, virtualize a physical machine, starting and stopping a virtual machine.

**TEXT BOOK:**

1. Cloud computing a practical approach, Anthony T.Velte , Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill , New Delhi – 2010

**REFERENCES:**

1. Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online, Michael Miller – Que, 2008
2. Cloud computing for dummies, Judith Hurwitz, Robin Bloor, Marcia Kaufman, Fern Halper, Wiley Publishing, Inc, 2010
3. Cloud Computing (Principles and Paradigms), Rajkumar Buyya, James Broberg, Andrzej Goscinski, John Wiley & Sons, Inc. 2011

**ONLINE RESOURCES:**

1. [https://onlinecourses.nptel.ac.in/noc24\\_cs17/preview](https://onlinecourses.nptel.ac.in/noc24_cs17/preview)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1IT306) FUNDAMENTALS OF COMPUTER VISION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
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| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the Fundamental Concepts Related to Multi-Dimensional Signal Processing
- To understand Feature Extraction algorithms
- To understand Visual Geometric Modeling and Stochastic Optimization

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Recognize the basic fundamentals of vision and describe the scope of challenges

**CO-2:** Develop algorithms to analyze feature detection and feature alignment

**CO-3:** Analyze images and videos for problems such as tracking and structure from motion

**CO-4:** Choose object, scene recognition and categorization algorithms for real time images

**CO-5:** Apply various techniques to build computer vision applications

#### UNIT-I:

**Introduction to Computer Vision and Image Formation:** Introduction, Geometric primitives and transformations, Photometric image formation, Digital Camera image formation.

**Image Processing:** Point operators, linear filtering, more neighborhood operators, Fourier transforms, Pyramids and wavelets, Geometric transformations.

#### UNIT-II:

**Featured detection and matching:** Points and patches, Edges, Lines.

**Segmentation:** Active contours, Split and merge, Mean shift and mode finding, Normalized cuts.

**Feature-based Alignment:** 2D and 3D feature-based alignment, Pose estimation.

#### UNIT-III:

**Structure from motion:** Triangulation, Two-frame structure from motion, Factorization, Bundle adjustment, constrained structure and motion.

**Dense motion Estimation:** Translational alignment, parametric motion, Spline-based motion, Optical flow, Layered motion.

#### UNIT-IV:

**Image Stitching:** Motion Models, Global alignment, Sparse and dense corresponding, Global Optimization.

**UNIT-V:**

**Recognition:** Object detection, Face recognition, Instance recognition, Category recognition, Context and scene understanding.

**TEXT BOOKS:**

1. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer-Verlag London Limited, 2011
2. Digital Image Processing, R.C.Gonzalez and R.E.Woods, Addison Wesley, 2008

**REFERENCES:**

1. Pattern Recognition: Statistical. Structural and Neural Approaches, Robert J.Schallkoff, John Wiley and Sons, 1992
2. Computer Vision: A Modern Approach, D.A. Forsyth and J.Ponce, Pearson Education, 2003
3. Multiple View Geometry, R.Hartley and A.Zisserman, Cambridge University Press, 2002
4. Multiple View Geometry in Computer Vision, Richard Hartley and Andrew Zisserman, 2<sup>nd</sup> Edition, Cambridge University Press, March 2004
5. Introduction to Statistical Pattern Recognition, K.Fukunaga, 2<sup>nd</sup> Edition, Academic press, Morgan Kaufmann, 1990

**ONLINE RESOURCES:**

1. [https://onlinecourses.nptel.ac.in/noc24\\_ee38/preview](https://onlinecourses.nptel.ac.in/noc24_ee38/preview)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1AE303) INTRODUCTION TO ADVANCED VEHICLE TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the layout of an automobile and functionality of chassis elements
- To know the working of automotive engine and electrical systems
- To provide the concepts of electric and hybrid vehicles
- To present various intelligent automotive systems and levels of vehicle autonomy

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the functionalities of automotive systems and subsystems

**CO-2:** Describe the working of automotive engine and electrical systems

**CO-3:** Summarize the concepts of electric and hybrid vehicles

**CO-4:** Discuss various intelligent automotive systems

**CO-5:** Review the levels of vehicle autonomy and communication

#### UNIT-I:

**Introduction:** Classification of automobiles, layout of an automobile and types of bodies.

**Automotive Chassis:** Introduction to chassis systems - engine, cooling, lubrication, fuel feed, ignition, electrical, driveline - clutch, transmission, propeller shaft, differential, axles, wheels and tyres, steering, suspension and braking.

#### UNIT-II:

**Engine:** Working principle of four stroke and two stroke SI and CI engines, fuel system – layout of petrol and diesel fuel systems, electronic fuel injection - multi-point fuel injection, gasoline direct injection and common rail direct injection.

**Electrical System:** Simple automotive wiring diagram, components of electrical system and functionality.

#### UNIT-III:

**Electric and Hybrid Vehicles:** Electric vehicle – layout, components, configurations, advantages and limitations. Hybrid vehicle - concepts of hybrid electric drivetrain based on hybridization and powertrain configuration, architecture of series, parallel and series-parallel hybrid electric drivetrains, modes of operation, merits and demerits.

#### UNIT-IV:

**Intelligent Vehicle Systems:** Automotive navigation, night vision, head-up display, airbag, seat belt tightening system, immobilizers, adaptive cruise control, forward collision warning, lane departure warning and anti-lock braking system.

**UNIT-V:**

**Autonomous Vehicles:** Levels of automation, research, challenges, commercial development, sensor systems, sensor suits, environmental challenges, graceful degradation, V2V and V2I communication, sharing the drive, integrity, security, verification and policy implications.

**TEXTBOOKS:**

1. Advanced Vehicle Technology, Heinz Heisler, Butterworth Heinemann, 2002
2. Intelligent Vehicle Technologies: Theory and Applications, Ljubo Vlacic, Michel Parent and Fumio Harashima, Butterworth-Heinemann, Oxford, 2001
3. Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay and Ali Emadi, CRS Press, 2004

**REFERENCES:**

1. Automotive Mechanics, Giri N. K., Khanna Publications, 2006
2. Automotive Electrical Equipment, Kohli P. L., Tata McGraw Hill Co., Ltd., New Delhi, 1975
3. Electric and Hybrid Vehicles – Design Fundamentals, Iqbal Husain, CRC Press, 2010
4. Autonomous Vehicle Technology-A Guide for Policymakers, James M. Anderson, Nidhi Kalra, Karlyn D. Stanley, Paul Sorensen, Constantine Samaras, Oluwatobi A. Oluwatola, RAND Corporation, Santa Monica, Calif., 2016

**ONLINE RESOURCES:**

1. [https://onlinecourses.nptel.ac.in/noc21\\_ee112/preview](https://onlinecourses.nptel.ac.in/noc21_ee112/preview)
2. <https://www.classcentral.com/subject/autonomous>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22OE1AE403) GREEN HYDROGEN

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the various details about energy scenarios.
- To know various details about hydrogen.
- To outline various details about green hydrogen.
- To understand the different applications of green hydrogen.
- To acquire knowledge on economic and environmental aspects of green hydrogen

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Describe various details about energy scenarios

**CO-2:** Explain various details about hydrogen

**CO-3:** Present different details about green hydrogen production, storage and transport

**CO-4:** Identify the different applications of green hydrogen

**CO-5:** Explore the various economic and environmental aspects of green hydrogen

#### UNIT-I:

**Introduction:** Global energy scenario, types of energy sources, role of fossil fuels in the transport sector, types of fuels - gaseous, liquid, combustion equation. Engine basics, working of SI and CI engine. Emissions- Types of emissions, effects on human health, environment. Need for sustainable energy solutions, renewable energy technologies- biofuels, biodiesel, solar, wind, hydro, present status.

#### UNIT-II:

**Hydrogen:** Hydrogen – properties, types – brown, grey, blue and green, applications, advantages and limitations, usage in IC engines and emissions. Production- coal, biomass, natural gas, material compatibility, standards

#### UNIT-III:

**Green Hydrogen:** Production – electrolysis - alkaline, proton exchange membrane, and solid oxide electrolysis. Storage - Compressed gas storage, storage, material-based storage - metal hydrides, chemical carriers, stationary storage. Transport - pipelines and transportation methods-tankers and shipping. Distribution networks and challenges.

**UNIT-IV:**

**Green Hydrogen Applications:** Power generation- hydrogen fuel cells, Integration into the power grid. Transportation-hydrogen fuel cell vehicles, Infrastructure for hydrogen mobility. Industrial applications- Refining, ammonia production, steelmaking, Emerging applications in various industries

**UNIT-V:**

**Economic and Environmental Aspects:** Cost analysis of green hydrogen production, Market trends and future projections, Environmental impact-life cycle assessment, Comparison with fossil fuels and other hydrogen types. Government incentives and policies, international agreements and collaborations

**TEXTBOOKS:**

1. Alternative Fuels for IC Engines, Anand Krishnaswamy, 1<sup>st</sup> Edition, McGraw Hill, 2024
2. Alternative fuels, S.S Thipse James, Jaico, 1<sup>st</sup> Edition, Publishing House, 2020
3. Hydrogen Fuel: Production, Transport, and Storage, Ram.B. Guptha, CRC press, 2008

**REFERENCES:**

1. Hydrogen Economy and Green Hydrogen Technologies, Yakson Bobby, Independently Published UK, 2024
2. The Hydrogen Revolution: a blueprint for the future of clean energy, Marco Alverá, Hodder Studio, 2021
3. Hydrogen Energy: Challenges and Solutions for a Cleaner Future, Bahman Zohuri, Springer, 2008

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS310) INTRODUCTION TO APPLICATION DEVELOPMENT WITH C#

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To create an integrated development environment for object-oriented C# programs
- To build website menus with CSS and JavaScript
- To relate programming language constructs and problem solving techniques
- To analyze and Apply modifications to C# programs that solve real-world problems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the fundamentals of HTML5 and define the styles for web pages using CSS

**CO-2:** Create web pages and add dynamic behavior to web pages using Javascript

**CO-3:** Communicate with the database using SQL

**CO-4:** Develop a simple CUI [Character User Interface] based application using C# & SQL

#### UNIT-I:

**Computer, Software Engineering Fundamentals & OOP:** Introduction to Computer Basics, Basics of Network, Networking Levels and Layers and Protocols, Protocol Stacks, Networking and Internet Service, Software Engineering Fundamentals - Overview of Requirement Analysis, Overview of Software Design, Overview of Software Implementation, Overview of Testing, Overview of Software Maintenance, Overview of Configuration management and version Control, Agile Basics, OOP - Object Oriented Concepts, Objects and Classes, Principles in Object- Oriented technology

**Usecase:** Create a class for BankAccount

#### UNIT-II:

**HTML & CSS:** Introduction to Web Technology, Introduction to HTML5, HTML5 Elements, Semantic Elements, Table, List, Working with Links, Image Handling, Form-Input Elements, HTML5 Form elements, HTML5 Attributes, Video & Audio, iframes, CSS - Introduction to CSS3, CSS Syntax, CSS Styling, Text and Fonts properties, CSS Selectors, Different color schemes, CSS Borders, CSS Margins, CSS Backgrounds

**Usecase:** Create a website for college

#### UNIT-III:

**JavaScript, RDBMS Concepts and SQL:** JavaScript basics, Functions in Javascript, Javascript validation, Events, Javascript event handling, JavaScript Strings, JavaScript Dates, Array in Javascript, Document Object Model (Window, Frame, Navigator



Objects), Working with Document Object (Its Properties and methods, Cookie handling), Introduction to RDBMS Concepts, Introduction to SQL, Creating and Managing Tables, Data Manipulation, Basic SQL SELECT Statements, Scalar & Aggregate Functions, Joins & Subqueries, Views & Index

**Usecase:** Apply validations for Telephone Complaint Registration Form

**Usecase:** Create student table for College Management System(CMS)

#### **UNIT-IV:**

**Introduction to C# Programming:** Introduction to .NET Framework 4.5 - What is .NET Framework, .NET Framework, Languages, and Tools, .NET Framework Major Components, Common Language Runtime (CLR), Compilation and Execution in .NET, Understand the .NET Framework 4.5stack, Exploring VS2017, Introduction to C# 6.0 - Features of C#, C# Compilation and Execution, General Structure of a C# Program, Creating and Using a DLL

**Usecase:** Create a Console Application (.exe) project called CalcClientApp

#### **UNIT-V:**

**Language Fundamentals of C#:** Language Fundamentals - Keywords, Value Types and Reference Types, Implicit and explicit type conversions, Boxing and Unboxing, Enum, Operators and Assignments, Variables and Literals, Flow

**Control:** C# Control Statements, Nullable, Classes and Objects, Strings, Array, Generic Collections

**Usecase:** Store employee objects using Generic Collections

**Basics of ADO.NET:** Various Connection Architectures, Understanding ADO.NET and its class library, Important Classes in ADO.NET, Connection Class, Command Class, DataReader Class, DataAdapter Class, DataSet Class

**Usecase:** Implement ADO.NET classes that belong to both Connected and Disconnected Architectures

#### **TEXT BOOKS:**

1. Web Programming, Building Internet Applications, Chris Bates, 2<sup>nd</sup> Edition, Wiley Dreamtech
2. Introduction to Database Systems, C.J.Date, Pearson Education
3. Professional C# 2012 with .NET 4.5, Christian Nagel et al. Wiley India, 2012

#### **REFERENCES:**

1. Programming World Wide Web, Sebesta, Pearson
2. Internet and World Wide Web – How to program, Dietel and Nieto PHI/Pearson Education Asia
3. Database Development and Management, Lee Chao, Auerbach Publications, Taylor & Francis Group
4. Pro C# 2010 and the .NET 4 Platform, Andrew Troelsen, 5th Edition, A Press, 2010
5. Programming C# 4.0, Ian Griffiths, Matthew Adams, Jesse Liberty, 6th Edition, O'Reilly, 2010

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS311) INTRODUCTION TO APPLICATION DEVELOPMENT WITH JAVA

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To build website menus with CSS and JavaScript
- To create a bank account using class and keep track of customers data
- To create and store employee objects using collection framework
- To analyze and apply menu driven program using JDBC to save, delete, update
- To create an integrated development environment for object-oriented Java programs

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the fundamentals of HTML5 and define the styles for web pages using CSS

**CO-2:** Create customers data who are buying products using Classes and Objects

**CO-3:** Understand the Collection Framework concepts and its implementation

**CO-4:** Communicate with the database using SQL

**CO-5:** Develop a simple CUI [Character User Interface] based application using Java & SQL

#### UNIT-I:

**HTML:** Introduction to Web Technology, HTML5 Introduction, HTML5 Elements, Semantic Elements, Table, List, Links in HTML5, Handling of Images, Form Elements, HTML5 Form elements and Attributes, Video & Audio, iframes

**Style Sheets:** Introduction to CascadingStyleSheet3, CSS Syntax, CSS Styling, Text and Fonts properties, CSS Selectors, Color schemes, CSS Borders, CSS Margins, CSS Backgrounds

**Use case:** Design a website for college

**JavaScript:** Introduction to JavaScript, JavaScript Functions, JavaScript validation, Event handling in JavaScript, JavaScript Strings, JavaScript Dates, Array in JavaScript, Document Object Model (Window, Frame, Navigator Objects), Document Object (Its Properties and methods, Cookie handling),

#### UNIT-II:

**Introduction to Java:** Java Environment, Java Fundamentals - Keywords, Primitive Data Types, Operators and Assignments, Java's Control Statements, Wrapper Classes, Using Scanner Class,

**Object Oriented Programming:** Object Oriented Paradigm, Classes and Objects, Principles in Object- Oriented technology

**Use case:** Create a class for Bank Account

**Strings:** String Handling functions, Array - One dimensional array, Array of Objects, Using Arrays class, variable length arguments

**Use case:** To keep track of customers data who are buying products from a store

#### **UNIT-III:**

**The Collection Framework:** Lists – Array List, LinkedList, Stack, Vector, Set – HashSet, Linked Hash Set, Tree Set, Map – HashMap, Linked HashMap, Hash table. Retrieving Elements from Collections – Enumeration, Iterator, List Iterator, String Tokenizer – Sorting using Comparable and Comparator.

**Use Case:** Store employee objects using collection framework

#### **UNIT-IV:**

**JDBC:** Overview of JDBC, JDBC Architecture, Types of JDBC Drivers. Process SQL with JDBC - Create Connection, Query, Update

**Use case:** Write the menu driven program using JDBC which will have following options

- a. Store
- b. Display by id
- c. Delete by id
- d. Update salary by id
- e. Exit

**RDBMS Concepts and SQL:** Introduction to RDBMS Concepts, Introduction to SQL, Creating and Managing Tables, Data Manipulation, Basic SQL SELECT Statements, Scalar & Aggregate Functions, Joins & Subqueries, Views & Index

**Use case:** Check the validations for Telephone Complaint Registration Form

**Use case:** Create student table for College Management System (CMS)

#### **UNIT-V:**

**Computer:** Computer Fundamentals, Preface to Networks, Networking Levels, Layers of Computer Networks, Protocol Stacks, Networking, and Internet Service

**Software Engineering Fundamentals:** Introduction, Requirements Collection & Analysis, Fundamentals of Software Design, Software Implementation, Types of Testing, Software Maintenance, Overview of Configuration management and version Control Tools, Basics of Agile Process

#### **TEXT BOOKS:**

1. Web Programming, Building Internet Applications, Chris Bates, 2nd Edition, Wiley Dreamtech
2. Introduction to Database Systems, C.J. Date, Pearson Education
3. Big Java, Cay Horstmann, John Wiley and Sons, 2nd Edition

#### **REFERENCES:**

1. Programming World Wide Web, Sebesta, Pearson
2. Internet and World Wide Web – How to program, Dietel and Nieto PHI/Pearson Education Asia
3. Database Development and Management, Lee Chao, Auerbach Publications, Taylor & Francis Group
4. Java How to Program, H.M.Dietel and P.J.Dietel, 6<sup>th</sup> Edition, Pearson Education/PHI

5. Core Java 2, Vol. 1, Fundamentals, CayS. Horstmann and Gary Cornell, 7<sup>th</sup> Edition, Pearson Education

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CS312) INTRODUCTION TO APPLICATION DEVELOPMENT WITH PYTHON

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To create an integrated development environment for object-oriented Python programs
- To build website menus with CSS and JavaScript
- To relate programming language constructs and problem solving techniques
- To analyze and Apply modifications to Python programs that solve real-world problems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the fundamentals of HTML5 and define the styles for web pages using CSS

**CO-2:** Create web pages and add dynamic behavior to web pages using Javascript

**CO-3:** Communicate with the database using SQL

**CO-4:** Develop a simple CUI [Character User Interface] based application using Python & SQL

#### UNIT-I:

**Concepts of Networks, Overview of Software Engineering & OOP:** Computer Basics, Network basics, Networking Levels, Layers and Protocols, Protocol Stacks, Networking and services of Internet

Software Engineering lifecycle - Overview of Requirement Analysis, Software Design, Implementation of software, Outline of Testing, Maintenance, Configuration management and version Control, Agile fundamentals

OOP - Object Oriented Concepts, OOP Principles

**Usecase:** Create a class for Employee Account

#### UNIT-II:

**Introduction to Web Technology:** Overview of Web Technology, Introduction to HTML5, HTML5 Elements, Semantic Elements, Table, List, Links, Image Handling, Form-Input Elements, HTML5 Form elements, HTML5 Attributes, Video & Audio, iframes,

CSS - Introduction to CSS3, CSS Syntax, CSS Styling, Text and Fonts properties, CSS Selectors, Different color schemes, CSS Borders, Margins, Backgrounds

**Usecase:** Create a website for an institution

#### UNIT-III:

**Outline of JavaScript, RDBMS Concepts and SQL:** JavaScript basics, Functions, validations, Events, handling events, Strings, Dates, Arrays, DOM(Window, Frame, Navigator Objects), Document Object -Properties and methods, handling of Cookies,

RDBMS Concepts, SQL, Management of Tables, Manipulation of tables, SQL SELECT Statements, Scalar & Aggregate Functions, Joins & Sub queries, Views & Index

**Usecase:** Apply validations for Telephone Complaint Registration Form

**Usecase:** Create student table for College Management System (CMS)

#### **UNIT-IV:**

**Introduction to Python:** Introduction, Features of Python, Versions, Keywords and Identifiers, Statements & Comments, Variables, Datatypes, Type Conversion, I/O and import, Language Fundamentals - Operators, Namespace, Modules in Python, Python DateTime

**Classes and Objects:** Classes and Objects in Python? Advantages of Using Classes in Python, Defining a Class in Python, Creating an Object in Python, The self, The\_init\_() function in Python, class and instance variables, Python Inheritance and its Types,

**Usecase:** Develop an application using Python for accepting your personal details and display the same

**Usecase:** Store employee objects using various data structures

#### **UNIT-V:**

**Advance Concepts in Python:** Strings, Lists, Sets, Tuples, Dictionary .

Array - What is an Array, Difference between Array and List in Python, Creating an Array, Accessing a Python Array Element, Basic Operations of Arrays, Functions - Creating a Function, Calling a Function, Pass by reference vs value, Required arguments, Keyword arguments, Default arguments, Variable-length arguments, The Anonymous Functions, The return Statement, Global vs. Local variables, Modules - What is a Module?, Create a Module, Use a Module, Variables in Module, Naming a Module, Renaming a Module, Built-in Modules, Using the dir() Function, Import From Module, Packages, NumPy

**Usecase:** Develop an application for Hospital Management System(HMS)

#### **TEXT BOOKS:**

1. Web Programming, Building Internet Applications, Chris Bates, 2<sup>nd</sup> Edition, Wiley Dreamtech
2. Introduction to Database Systems, C.J. Date, Pearson Education
3. Python Programming: A Modern Approach, Vamsi Kurama, Pearson

#### **REFERENCES:**

1. Programming World Wide Web, Sebesta, Pearson
2. Internet and World Wide Web – How to program, Dietel and Nieto PHI/Pearson Education Asia
3. Database Development and Management, Lee Chao, Auerbach Publications, Taylor & Francis Group
4. Core Python Programming, W.Chun, Pearson
5. Introduction to Python, Kenneth A. Lambert, Cengage

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1MT304) NUMERICAL METHODS AND LINEAR PROGRAMMING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PREREQUISITES:** Differentiation, Integration, Basic concepts of Linear Algebra

#### COURSE OBJECTIVES:

- Solving numerical methods for non linear equations
- Applying various methods of interpolation and their application
- Evaluating definite integrals and solve Ordinary Differential Equations
- Formulating of Linear Programming Problem and of approaches of optimization techniques
- Applying different methods to solve the transportation problem

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply the numerical methods to find roots of algebraic and transcendental equations

**CO-2:** Interpolate the polynomial function using finite difference methods

**CO-3:** Solve the ordinary differential equations using various numerical techniques

**CO-4:** Model the Linear Programming Problem and estimate the feasible solutions

**CO-5:** Apply optimization techniques in industrial optimization problems

#### UNIT-I:

**Solutions of Non-Linear Equations:** Introduction; Mathematical preliminaries; Solution of algebraic and transcendental equation –bisection method, the method of false position, Fixed point iterative method, Newton - Raphson method, and their order of convergence

#### UNIT-II:

**Interpolation:** Introduction; Finite differences; Forward differences; Backward differences; Central differences; Differences of a polynomial; Newton's formulae for interpolation; Central difference interpolation formulae; Gauss's central difference formulae and Lagrange's and Hermite interpolation formulae

#### UNIT-III:

**Numerical Integration:** Numerical integration: Trapezoidal rule, Simpson's 1/3 rule, and Simpson's 3/8 rule, Gaussian quadrature 2 & 3 point formulae.

**Numerical Solutions of Ordinary Differential Equations:** Solution of initial value problems by Taylor's series, Picard's method of successive approximations, Euler's method, Modified Euler's method and 4th order Runge – Kutta method, Predictor Corrector methods - Adams Bash forth method- Adams Moulton method (without proof).

**UNIT-IV:**

**Linear programming:** Basic concepts, problem formulation, Concept of Duality, Dual of given Linear Programming Problem, canonical and standard forms of Linear Programming Problem, simplex method, Solutions to Linear Programming Problem by Dual Simplex Method, Artificial variables technique- Big-M method

**UNIT-V:**

**Transportation problems:** Balanced and Unbalanced transportation problems- North-West corner rule, Least Cost Method, Vogel's Approximation Method (VAM) and MODI method.

**TEXT BOOKS:**

1. Higher Engineering Mathematics, B.V. Ramana; 33<sup>rd</sup> Reprint, McGraw Hill Education (India) private Limited, 2018
2. Advanced Engineering Mathematics by R.K.Jain and S.R.K.Iyengar; Narosa Publications, 2016
3. Operations Research, Kanti Swarup, P.K. Gupta and ManMohan, 13<sup>th</sup> edition, Sultan Chand and Sons, 2007

**REFERENCES:**

1. Operations Research, Theory and Applications, J.K. Sharma, 6<sup>th</sup> Edition, Trinity Press, Laxmi Publications Pvt. Ltd. 2016
2. Advanced Engineering Mathematics by Erwin Kreyszig, 8<sup>th</sup> Edition; Publisher: John Wiley
3. Elementary Numerical Analysis – an algorithmic approach -Samuel D. Conte and Carl De Boor (2006); 3<sup>rd</sup> edition; Publisher: Tata McGraw Hill
4. Operations Research: An introduction, Taha H.A., 7<sup>th</sup> edition, Pearson Prentice Hall, 2002
5. Introduction to Operations Research, Hillier and Lieberman, 8<sup>th</sup> Ed., McGraw Hill



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1MT305) OPTIMIZATION TECHNIQUES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PREREQUISITES:** Basic knowledge of Calculus and Linear Algebra

#### COURSE OBJECTIVES:

- To understanding concepts of convex sets, convex functions
- To derivative based and derivative free optimization
- To formulating LP problem, solution of LP and feasibility of solutions
- To applying different methods to solve the transportation problem
- To understanding network based optimization technique

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explore and understand role of convex functions in optimization problems

**CO-2:** Derive the Karush-Kahn-Tucker(KKT) optimality conditions for constrained optimization problem

**CO-3:** Formulating of Linear Programming Problem and of approaches of optimization techniques

**CO-4:** Apply optimization techniques in industrial optimization problems

**CO-5:** Learn the role modern tools in optimization problems

#### UNIT-I:

**Optimization Basics:** Review of calculus: Derivatives, Gradient, Convex sets, convex functions, Optimization, Basics of Optimization Problems: Representation, Local and Global minima, Conditions of Optimality, Newton's method, Gradient Method (Gradient Descent Method) and Steepest Descent Method of finding optimality with Numerical Examples

#### UNIT-II:

**Constrained Optimization:** Constrained optimization problem formulation, Equality and Inequality constraints problems with numerical examples, Tangent space and normal space, Lagrange's Theorem, Karush-Kahn-Tucker (KKT) condition, convex optimization.

#### UNIT-III:

**Linear programming:** Basic concepts, problem formulation, Concept of Duality, Dual of given Linear Programming Problem, canonical and standard forms of Linear Programming Problem, simplex method, Solutions to Linear Programming Problem by Dual Simplex Method, Artificial variables technique- Big-M method

**UNIT-IV:**

**Transportation Problems:** Balanced and Unbalanced transportation problems- North-West corner rule, Least Cost Method, Vogel's Approximation Method (VAM) and MODI method.

**UNIT-V:**

**Genetic and Evolutionary Optimization:** Introduction, Genetic Algorithms: Representation, Chromosomes and Genes, Generations; Fitness, Genetic Algorithms: selection, crossover, mutation, elitism with numerical examples, Multi-Objective Optimization

**TEXT BOOKS:**

1. Operations Research, Kanti Swarup, P.K. Gupta and ManMohan, 13<sup>th</sup> edition, Sultan Chand and Sons, 2007
2. Optimization Techniques: An Introduction, L. R. Foulds, Undergraduate Texts in Mathematics (UTM)
3. Optimization insights and applications, Jan Brinkhuis and Vladimir Tikhomirov

**REFERENCES:**

1. Optimization methods for Engineering design: Applications and Theory, Alan R. Parkinson Richard J. Balling John D. Hedengren, Brigham Young University
2. Optimization Toolbox: MATLAB User's Guide
3. Numerical Recipes in C: The Art of Scientific Computing, William H. Press, Saul A. Teukolsky, William T. Vetterling and Brian P. Flannery
4. Engineering optimization: Theory and Practice, S.S.Rao

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1PH301) NANOSCIENCE AND TECHNOLOGY FOR ENGINEERS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basic principles of quantum physics and nanoscience
- To study size dependent properties of materials at nanoscale
- To explore various fabrication techniques for nanomaterials and devices
- To ascertain different characterization techniques to analyze nanomaterials
- To extend applications of nanomaterials in engineering

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the basic principles of nanoscience

**CO-2:** Realize the importance of size dependent properties of nano materials

**CO-3:** Apply various fabrication techniques for nanomaterials and devices

**CO-4:** Identify suitable characterization techniques to analyze nanomaterials

**CO-5:** Illustrate the applications of nanomaterials in engineering

#### UNIT-I:

**Basics of Nanoscience:** Introduction to quantum mechanics, electron as wave, Schrodinger time independent wave equation, particle in a potential box, Heisenberg Uncertainty Principle, Pauli's exclusion principle, Introduction to Free electron theory, density of states for zero, one, two, and three dimensional materials (Qualitative), Distinction between metal, semiconductors and insulators from the K-P model (Qualitative), Quantum confinement, Quantum wells, wires, dots.

#### UNIT-II:

**Properties of Nanomaterials:** Surface to Volume ratio, Size dependent properties- structural, Mechanical, Thermal, Electrical, Optical, and Magnetic properties of nanomaterials, Electrical and Mechanical properties of Carbon nanotubes.

#### UNIT-III:

##### **Fabrication Techniques of Nanomaterials and Devices:**

Physical methods - Ball milling, Thermal evaporator, - electron beam evaporator, Sputtering, Laser ablation, Patterning-electron beam lithography, photo lithography. (construction, working, advantages and disadvantages).

Chemical methods - Sol-gel method, Hydrothermal, Co-Precipitation, Chemical vapor deposition. (construction, working, advantages and disadvantages).

#### UNIT-IV:

**Characterization of Nanomaterials:** XRD (Powder & GI), Secondary Electron Microscopy (SEM), FE-SEM, Transmission Electron Microscopy (TEM), Scanning probe

microscopy techniques (AFM, MFM, STM), VSM, UV-VIS spectroscopy, Raman spectroscopy.

#### **UNIT-V:**

**Nanomaterials in Engineering:** Nano sensors-gas and chemical, Introduction to Nano Electromechanical Systems (NEMS), Applications of Nanomaterials in Drug delivery, Coatings, and Renewable energy. Quantum well lasers.

#### **TEXT BOOKS:**

1. Nanotechnology: Principles & Practicals, Sulbha K Kulakarni, Capital Publishing Co. New Delhi
2. Nanostructures & Nanomaterials Synthesis, Properties & Applications. Guozhong Cao, Imperial College Press, London
3. Essentials of Nanoscience and Nanotechnology by Katta Narsimha Reddy

#### **REFERENCES:**

1. Introduction to Nanotechnology. C.P. Poole Jr. and F.J. Ownes, Wiley Student Edition
2. Nano: The Essentials. T. Pradeep, McGraw Hill Education
3. Nanoscience and Technology: Novel Structure and Phenomena-Ping and Sheng

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1PH302) ESSENTIALS FOR QUANTUM COMPUTING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To study various mathematical tools required for quantum computing
- To recall the basic principles of quantum physics
- To understand the basics of quantum computing
- To realize the concepts of various qubits for quantum computation
- To explain various types of quantum gates

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** understand various mathematical tools for quantum computing

**CO-2:** realize the importance of quantum mechanics

**CO-3:** Understand quantum computation

**CO-4:** realize the different types of qubits for quantum computation

**CO-5:** know various quantum logic gates

#### UNIT-I:

**Mathematics for Quantum Mechanics:** Linear vector space, Hilbert space, Dirac notation, Operators- Linear operator, Hermitian operators, Skew Hermitian operator, Inverse and Unitary operator (lowering and rising operators-Qualitative), Eigen value and Eigen vector of operator, Unitary transformation, and its properties. Matrix representation of Kets, Bras, Operators, Eigen values and Eigen vectors of matrices, Position and Momentum Operators and their commutation relations.

#### UNIT-II:

**Basics of Quantum Mechanics:** Introduction to quantum mechanics, Heisenberg Uncertainty Principle, Schrodinger time dependent and independent wave equation, Particle in a potential box, Pauli's exclusion principle, Quantum harmonic oscillator, Linearity and Superposition, Probability density and Expectation values..

#### UNIT-III:

**Fundamentals of Quantum Computing:** Introduction to Quantum Computing, Types and Uses of Quantum Computers, The Bloch sphere (Qualitative), Entanglement, Interference and Coherence/decoherence

#### UNIT-IV:

**Qubits for Quantum Computing:** Qubits- Superconducting Qubits, Quantum dot-based Qubit, Diamond NV center and Photonic Qubits, Representation, and measurements of qubits.

**UNIT-V:**

**Quantum logic Gates:** Basic quantum logic gates. Quantum States and Registers, Single Qubit gates, Hadamard Gate, Pauli Gates, Phase gate (S) and (T), Multi Qubit Gates, CNOT Gate. Introduction to Quantum cryptography, Error correction and communication.

**TEXT BOOKS:**

1. Quantum mechanics- concepts and applications, Nouredine Zettili, Wiley publications
2. Quantum Computing: Beginners Introduction, PARAG K. LALA, Mc Graw Hill
3. Concepts of Modern Physics by Arthur Beiser, McGraw-Hill Higher Education

**REFERENCES:**

1. Introduction to Quantum Mechanics by D J Griffiths, Cambridge University Press
2. Molecular Quantum Mechanics by Peter Atkins, Ronald Friedman, Oxford University Press
3. Quantum Computing- A gentle Introduction by Eleanor Rieffel and Wolfgang Polak, The MIT Press Cambridge, Massachusetts London
4. Quantum Mechanics by G. Arul Das, Prentice-Hall of India Private Limited, New Delhi
5. A Textbook of Quantum Mechanics by P.M. Mathews and K. Venkatesan, McGraw-Hill Education, India

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CH301) INTRODUCTION TO CLIMATE CHANGE

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce students to the concept of climate change and its historical context
- To analyze the ecological consequences of climate change Impacts
- To explore strategies for adaptation & mitigation of climate change
- To understand the global climate change models and predictions
- To examine international and national climate policies, and the challenges associated with climate change governance

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Identify natural and anthropogenic drivers of climate change

**CO-2:** Assess the impacts of climate change on environment

**CO-3:** Appraise various adaptation & mitigation options, for effective climate management

**CO-4:** Able to analyse climate models and adaptability of AI and IOT

**CO-5:** Evaluate the effectiveness of climate policies and assess the importance of technological innovations

#### UNIT-I:

**Basics of Climate Change:** Introduction & causes of climate change.

**Natural Drivers:** volcanoes, change in earth's orbit, solar variations & ocean currents.

**Anthropogenic Drivers:** greenhouse gas emissions, aerosols & land use. Key climate variables - temperature, precipitation, sea level. Human footprint on global warming.

#### UNIT-II:

**Impact and Vulnerabilities:** Climate change impacts on: water resources, wetlands, glaciers melting, biodiversity, agriculture, marine environment – (sea level rise, ocean current and circulation, ocean acidification, coastal lives) and human health - case studies.

#### UNIT-III:

**Mitigation & Adaptation Strategies:** Renewable energy technologies (solar, wind, hydro, geothermal & biomass), energy efficiency measures, carbon capture and storage (CCS). Policy frameworks (Kyoto Protocol, Paris Agreement). Adaptation strategies (resilient infrastructure, climate-smart agriculture & urban planning).

#### UNIT-IV:

**Climate Models and Projections:** Basics of climate modelling. Historical climate data analysis (past, present and future). General Circulation Models (GCMs), emission

scenarios, Representative concentration pathways (RCPs), uncertainty in climate projections, regional climate modelling, impacts on ecosystems and society. Fundamentals of AI & IOT for climate modelling.

#### **UNIT-V:**

**Climate Policy and Governance:** Global climate change governance, Climate change finance sources: challenges and opportunities to assess and manage Climate finance. Evaluate climate change policies: UNFCCC and other entities. Climate negotiations, National scenario; in NAPCC, India's commitments (INDCs) and National communication (NATCOM). Initiative policies and regulation: Important agencies and organizations,

#### **TEXT BOOKS:**

1. Climate Change: A Multidisciplinary Approach, Anindita Bhadra and Navroz K. Dubash, 1<sup>st</sup> Edition, Sage Publications India Pvt Ltd, 2019
2. Climate Change and India: Vulnerability Assessment and Adaptation, Arvind Kumar and Shalini Sharma, 1<sup>st</sup> Edition, Springer, 2019
3. Climate Change Biology, Jonathan A. Newman and Madhur Anand, 1<sup>st</sup> Edition, CABI, 2011

#### **REFERENCES:**

1. Climate Change: A Very Short Introduction, Mark Maslin, 4<sup>th</sup> Edition, Oxford University Press 2019
2. Climate Change and Biodiversity in India, M.S. Manohar Rao and P.V. Sridhar Rao, 1<sup>st</sup> Edition, Universities Press, 2019
3. Climate Change: What Everyone Needs to Know, Joseph Romm, 2<sup>nd</sup> Edition, Oxford University Press, 2018
4. Climate Change: Impacts, Adaptation, and Vulnerability, Intergovernmental Panel on Climate Change (IPCC), 1<sup>st</sup> Edition, Cambridge University Press, 2007
5. Environmental Governance: Reconsidering the Boundaries of Environmental Policy, Jens Newig and Edward Challies, 1<sup>st</sup> Edition, MIT Press, 2019

#### **ONLINE RESOURCES:**

1. Coursera: Climate Change: The Science
2. edX: Climate Action: Solutions for a Changing Planet
3. Coursera: Climate Change Policy and Public Health



## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1CH302) SUSTAINABLE CHEMISTRY FOR ENGINEERING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To explore the principles and applications of sustainable chemistry
- To emphasize the importance of green chemistry in addressing environmental challenges
- To introduce sustainable materials for green practices
- To provide comprehensive understanding of sustainable energy technologies
- To promote sustainable development practices aiming towards societal needs

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Identify the role of chemistry in promoting sustainability

**CO-2:** Familiarize with the concept of green chemistry and emphasize on LCA

**CO-3:** Interpret the importance of sustainable materials & green manufacturing

**CO-4:** Analyse and implement sustainable energy solutions

**CO-5:** Integrate technology for sustainable practices

#### UNIT-I:

**Introduction to Sustainable Chemistry:** Overview of sustainability concepts and principles, the role of chemistry in promoting sustainability, Sustainable development goals and their relation to chemistry (SDGs) -Zero hunger, good health & wellbeing, clean water and sanitation, affordable & clean energy, Industry, innovation & infrastructure, sustainable cities & communities, responsible consumption & production, climate action, life below water, life on land, partnerships for the goals. Case studies on sustainable chemistry practices.

#### UNIT-II:

**Green Chemistry Principles & Applications:** Definition & scope of green chemistry, Relation between green chemistry & sustainability, 12 Principles of green chemistry, Applications of green chemistry. Introduction to Life Cycle Assessment (LCA) methodologies - ISO14040, ISO14044, Cradle to grave, cradle to gate and cradle to cradle. Application of LCA to engineering processes and products.

#### UNIT-III:

**Sustainable Materials & Green Manufacturing:** Introduction to sustainable materials, criteria for evaluating the sustainability of materials. Impact of sustainable materials on environment. Eco friendly materials and their applications in packaging, construction, and consumer goods. Integration of smart materials into sustainable design. Green manufacturing-Definition & significance, emerging technologies for sustainable & green manufacturing. Industry 4.0 and its impact on green practices.

#### **UNIT-IV:**

**Sustainable Energy Technologies:** Definition & importance of sustainable energy, global energy consumption & environmental impact, Overview of clean energy sources that have minimal environmental impact compared to traditional fossil fuels- solar energy, wind energy, biomass energy. Sustainable transport-introduction to Electric Vehicles (EV), Technological advancements over the years in EVs, electric vehicle performance & efficiency comparison with traditional internal combustion engine vehicles, integration of autonomous technology with EVs.

#### **UNIT-V:**

**Trends towards Sustainability:** Waste minimization and resource recovery, waste reduction, and waste tracking systems. Recycling & waste management- Plastics recycling – Code of practice – Primary, secondary, tertiary, and quaternary recycling with examples. Mechanical recycling of commonly used plastics, such as PP, PE, PET, etc. Waste plastics as fillers. Advances towards clean chemistry-minimizing the use of traditional chemical catalysts by integrating biocatalysts (enzymes) into chemical processes. Integration of technology - application of artificial intelligence & machine learning for designing & optimization of sustainable chemical processes.

#### **TEXT BOOKS:**

1. Sustainable Chemistry, Stanley E. Manahan, 2<sup>nd</sup> Edition, CRC Press,
2. Green Chemistry, Theory and Practical, Paul T. Anastas and John C. Warner, Oxford University Press
3. Green Chemistry-Environment Friendly Alternatives, Rashmi Sanghi and M.M. Srivastava, Narosa Publishing House, New Delhi, 2009

#### **REFERENCES:**

1. Green Chemistry: An Introductory Text, Mike Lancaster, RSC Books
2. Sustainable Energy – Without the Hot Air, David J.C. MacKay, UIT Cambridge Ltd.
3. Green Manufacturing: Fundamentals and Applications, David A. Dornfeld and Barbara S. Linke, 1<sup>st</sup> Edition, Springer
4. Waste and Resource Management: Sustainable Handling of Waste Materials, Bernd Bilitewski, Georg Härdtle, and Klaus Marek, 4<sup>th</sup> Edition, Wiley
5. Artificial Intelligence in Chemistry and Drug Design" by Cory M. Reynolds and Alexander Tropsha, Royal Society of Chemistry, 1<sup>st</sup> Edition

#### **ONLINE RESOURCES:**

1. Coursera - Introduction to Environmental Science and Sustainability
2. edX - Applications of Artificial Intelligence in Chemistry
3. FutureLearn - Recycling: Global Practices of Sustainable Cities
4. Coursera - Plastic Waste Management

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1EN301) STORYTELLING FOR EFFECTIVE COMMUNICATION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Should have adequate knowledge of LSRW skills

#### INTRODUCTION:

Effective communication is essential in all aspects of life, whether it is in personal relationships or professional settings. One powerful tool for impactful communication is storytelling as it connects to the audience/reader quickly. A well-told story can captivate an audience/reader, evoke emotion, and convey important messages in a memorable way. Storytelling creates engagement, alongside fostering creativity and flexibility which makes it an incredibly helpful tool across all types of communication. While presenting facts, figures, and logical arguments, using a suitable storytelling strategy not only yields the best results but also helps hone negotiation, team-building, and leadership skills in addition to a host of other life skills. Storytelling is associated with both verbal and nonverbal communication.

#### COURSE OBJECTIVES:

- To develop storytelling strategies among students
- To enable students to develop stories for various contexts
- To hone the communication skills of the students
- To develop skills to apply digital technology for storytelling purposes
- To develop life skills among the students through various stories

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand the storytelling strategies for various contexts

**CO-2:** Develop digital stories for various contexts

**CO-3:** Use the elements of storytelling, understand narrative techniques, and characters, and build the stories

**CO-4:** Hone their life skills to be applicable in real-life contexts

**CO-5:** Apply storytelling techniques for placement activities

#### UNIT-I:

##### Storytelling your way to power

Why we tell stories

Significance of verbal and nonverbal communication while telling a story

Connect your message with the story

Matching your audience/readers to your story

#### UNIT-II:

##### Digital storytelling

Storytelling techniques

Storytelling strategies for presentations

### **UNIT-III:**

#### **Elements of a great story**

plot, narrative techniques, first person, second person, third person narration.

Assembling the building blocks

Characters and their development

End with a bang

Common mistakes storytellers make

### **UNIT-IV:**

#### **Life Skills through storytelling-Sample stories**

Honing leadership skills

Team skills

Change management skills

Interpersonal skills

### **UNIT-V:**

#### **Storytelling for selection process**

Self-introduction as a story

Anecdotes useful during a group discussion

Anecdotes useful during an interview

SOP -a story writing

### **TEXT BOOKS:**

1. Using Storytelling to Effectively Communicate Data, Michael Freeman Infinite Skills, 2015
2. Communication, Sharing Our Stories of Experience, Howard Kamler, Psychological Press, 1983
3. Storytelling, Michael Wilson, Emerald Publishing Limited, 2022

### **REFERENCES:**

1. Digital Storytelling for Employability, Ed by: Laura Malita, Vanna Boffo, Firenze University Press, 2010
2. Storytelling with You, Cole Nussbaumer Knaflic, Willy publishers, 2022
3. Conversation Skills, Keith Coleman, Communication & Social Skills, 2019
4. Digital Storytelling in the Classroom, Jason. B. Ohler, Corwin, 2013
5. The Instructional Value of Digital Storytelling, Patricia McGee, Routledge, 2015

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EN302) FILM ANALYSIS AND CRITICAL APPRECIATION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### INTRODUCTION:

Film Analysis and Critical Appreciation is a thought-provoking Open Elective course aimed at fostering a holistic perspective essential for future engineering and technology professionals. This course is designed to transcend the realms of entertainment, providing a unique lens through which learners can develop a robust sociocultural awareness and gain exposure to humanities, enriching their interdisciplinary skills

#### COURSE OBJECTIVES:

- To help students acquire knowledge of the history and development of cinema by studying and analysing major film movements to comprehend their impact on the evolution of cinematic language
- To train students to identify and differentiate various film genres, understanding their conventions, narrative structures, and other significant aspects
- To encourage students to understand and critically evaluate key elements such as cinematography, editing, sound, and mise-en-scène in film besides recognising the nuanced use of these elements to convey meaning
- To provide students the opportunity to explore major theories of film criticism and understand how these theories contribute to the interpretation and analysis of films
- To equip students with the critical thinking skills and appropriate strategies to conduct thematic analysis by exploring how films address and convey social, cultural, and political themes and gain insights into the interplay of narrative, visuals, and sound in conveying and shaping thematic content

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Trace the historical development of cinema and understand its evolution as a distinctive art form

**CO-2:** Identify and analyse different film genres, their characteristics and narrative conventions to examine how genres contribute to audience expectations and influence storytelling

**CO-3:** Critically evaluate key elements in films, including cinematography, editing, sound, and mise-en-scène, and analyse their impact on storytelling in selected films, providing specific examples

**CO-4:** Comprehend major theories of film criticism, applying them to analyze and interpret selected movies for enhanced understanding of a film's meaning

**CO-5:** Demonstrate skills in conducting thematic analysis, examining how movies address social, cultural, and political themes

## **UNIT-I:**

**Introduction:** A brief history of cinema; development of film as an art form; an overview of some major movements in world cinema: German Expressionism, Italian New Realism, French New Wave, British New Wave, Dogme 95

## **UNIT-II:**

### **Film Genres**

**Broad classification:** Narrative, Documentary, and Avant-garde; Major genres: action, comedy, drama, romance, fantasy, horror, science fiction, biopics, detective and mystery, adventure; awareness about sub-genres, hybrids and cross-genre films; conventions and expectations within specific genres

## **UNIT-III:**

**Elements of Film:** Delve into the cinematic experience from the audience's perspective by exploring the Big Four: Mise-en-scène: props, costumes, lighting, setting; Cinematography: camera angle, camera distance, camera movement; Editing: graphic match, diegesis, shots; Sound: diegetic and non-diegetic

## **UNIT-IV:**

### **Theories of Film Criticism**

**Four levels of meaning:** referential content, explicit content, implicit content, symptomatic content; Approaches to the study of cinema: Semiotics: Realist, Formalist, Rhetorical;

**Structuralist:** Mythic, Political, Feminist, Psychoanalytical, Sociological; Contextual: Auteur, Genre, Historical

## **UNIT-V:**

**Critical Analysis of Plot, Characters and Themes:** Story and Plot; Linear and Non-linear narrative structure; Major character archetypes; stereotypes and tropes; Interpretation of themes; Relationship between theme and character; Role of the audience in meaning-making: subjectivity and cultural context; Questions of power, identity, ethical dilemmas, and moral conflicts; Literary adaptations in film: analysing how thematic and character elements are translated.

## **SUGGESTED LIST OF FILMS FOR ANALYSIS**

### **WORLD CINEMA:**

1. The Cabinet of Dr. Caligari (1920, German: Das Cabinet des Dr. Caligari; silent)
2. Mother (1926, Russian)
3. Modern Times (1936, American)
4. The Lady Vanishes (1938, British)
5. Casablanca (1942, American)
6. Bicycle Thieves (1948, Italian)
7. Breathless (1960, French: À bout de souffle)
8. Fist of Fury (1972, Chinese)
9. The Godfather (1972, American)
10. Taxi Driver (1976, American)
11. The Celebration (1998, Danish: Festen)
12. Pulp Fiction (1994, American)

13. Charlie and the Chocolate Factory (2005, American)
14. Inception (2010, American)
15. The Social Network (2010, American)
16. Barbie (2023)

#### **INDIAN CINEMA:**

1. Harischandra (1913)
2. Pather Panchali (1955)
3. Mayabazaar (1957)
4. Sholay (1975)
5. Dilwale Dulhania Le Jayenge (1995)
6. Monsoon Wedding (2001)
7. Anniyan/ Aparichitudu (2005)
8. Tare Zameen Par (2007)
9. The Lunchbox (2013)
10. Bahubali – The Beginning (2015)
11. Awe(2018)
12. RRR (2022)
13. 12th Fail (2023)

#### **REFERENCES:**

1. It's Just a Movie: A Teaching Essay for Introductory Media Classes, Smith, Greg M. Cinema Journal, vol. 41 no. 1, 2001, p. 127-134. Project MUSE, <https://doi.org/10.1353/cj.2001.0025>
2. The Art of Watching Films, Boggs, J. M., & Petrie, D. W., 7<sup>th</sup> edition. New York: McGraw-Hill, 2008.
3. Piper, Jim. Get The Picture? US: Allworth Press, 2008

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1EN401) CROSS-CULTURAL COMMUNICATION THROUGH WORLD LITERATURE

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Should have adequate LSRW skills

#### INTRODUCTION:

In this course, we will be critically exploring literature created by writers from all over the world. We will also use the work to understand different cultures around the world with the intention to grasp cross-cultural communication cues. With globalization marching fast, it is essential for the present students to comprehend the cultures of different nations. We will embark on our journey with a mind for understanding and exploring cultures and cross-influences that show up in the literature of that place. A part of this journey will, then, include comprehending and comparing the cultural communication across these nations, and exploring the evolution of societies and nation-states. This journey will allow engineering students to comprehend the cultures, societies, and human psychology of individuals across the globe. This understanding will help them become aware of the cultures, people, and societies of different nations of the world and can shape them as global citizens and competent communicators with a good understanding of cross-cultural communication. We will make our journey through the medium of research groups, panel presentations, seminar discussions, critical writing, and exams.

#### COURSE OBJECTIVES:

- To develop open- and fair-minded thinking skills for the analysis of societies/cultures/nations
- To examine how cultures have shaped the literature of nations
- To understand different cultures through a study of literature and society there by analyzing cross cultural communication
- To equip students with cross cultural communication
- To train students to become global citizens

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Open and fair-minded thinking skills and will be able to analyse societies/cultures/nations

**CO-2:** Capacity to examine the cultures that have shaped the literature of different nations

**CO-3:** Ability to understand world cultures and societal issues with an intention to grasp cross-cultural communication

**CO-4:** Equipped to understand the nuances of cross-cultural communication

**CO-5:** Capable of applying the knowledge of cross-cultural communication strategies to real-life scenarios



#### **UNIT-I:**

##### **Introduction to cross-cultural communication**

##### **American Literature**

1. "America and I" a short story by Anzia Yezierska
2. "Notes of a Native Son" an essay by James Baldwin

#### **UNIT-II:**

##### **Basic concepts and definition of culture and communication**

##### **African Literature**

1. "A Private Experience" a short story by Chimamanda Ngozi Adichie
2. "Civilian and Soldier" a poem by Wole Soyinka

#### **UNIT-III:**

##### **Examining variation of faiths and values in relation to cross-cultural communication**

##### **Russian Literature**

1. "The Beggar Boy at Christ's Christmas Tree" a short story by Fyodor Dostoevsky
2. "The Dragonfly & the Ant" a poem by Ivan Krylov's.

#### **UNIT-IV:**

##### **Barriers in cross communication contexts**

##### **British Literature**

1. "On Education" an essay by Bertrand Russell
2. "An Irish Engagement" –a one act play by Walter Watts

#### **UNIT-V:**

##### **Aspects of communication affected by culture**

##### **Asian Literature**

1. A Matrimonial Fraud-A Chinese story.
2. The Idea of an Ever-ever Land, An Excerpt from Shashi Tharoor's Essay in 'Left, Right, and Centre

#### **TEXT BOOKS:**

1. The Handbook of Communication in Cross-cultural Perspective, Ed, Donal Carbaugh, Taylor & Francis, 2016
2. Cross cultural Communication Global Perspective, N.L. Gupta, Concept Publishing Company 1998
3. Communicating and Adapting Across Cultures: Living and Working in the Global Village, Riall Nolan · 1999, ABC-CLIO

#### **REFERENCES:**

1. How I Found America: Collected Stories, Yezierska, Anzia. Yezierska, Anzia, published by Persea, 2003
2. Notes of a Native Son, Baldwin, James. published by Beacon Press, 2012
3. The Thing around Your Neck, Chimamanda Ngozi Adichie published by The Guardian: The Observer, 2008
4. Civilian and Soldier, Wole Soyinka, Emory University, Atlanta 1996
5. The Heavenly Christmas Tree, Fyodor Dostoevsky, Creative Co. 1992

**ONLINE RESOURCES:**

1. [https://ia601609.us.archive.org/10/items/cu31924010707440/cu31924010707440.p](https://ia601609.us.archive.org/10/items/cu31924010707440/cu31924010707440.pdf)  
df- Chinese short stories
2. <https://ruverses.com/ivan-krylov/dragonfly-and-the-ant/3704/>
3. [https://dreamtopper.in/wp-content/uploads/2023/05/Broken-Images-12th-](https://dreamtopper.in/wp-content/uploads/2023/05/Broken-Images-12th-Kaleidoscope-Book.pdf)  
Kaleidoscope-Book.pdf

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1EN402) BUSINESS COMMUNICATION

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Basic Proficiency in LSRW Skills

#### **COURSE OBJECTIVES:**

- To express oneself in English with greater fluency, accuracy, and confidence
- To exhibit good team-building skills
- To develop skills and techniques (speaking & writing) for effective communication
- To make effective presentations
- To conduct oneself professionally in a workplace environment

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Communicate effectively and confidently

**CO-2:** Participate in Team building, and work efficiently in teams

**CO-3:** Comprehend and practice suitable workplace etiquette

**CO-4:** Draft business letters, emails, and memos using appropriate language

**CO-5:** Participate in and contribute to meetings constructively

#### **UNIT-I:**

##### **Effective Communication**

- a) Fundamentals of Effective Communication
- b) How to sell your ideas
- c) Types of communication networks
- d) Barriers of Communication
- e) Business vocabulary 1

#### **UNIT-II:**

##### **Effective Communication for Team Building**

- a) Importance of Groups in an Organization; Interactions in a Group, Group Decision Taking
- b) Stages of Team Building, Interaction with the Team
- c) Group Dynamics, Managing Conflicts in the Team
- d) Problem-solving, Decision Making and Interpersonal skills
- e) Business Vocabulary 2

#### **UNIT-III:**

##### **Office Etiquette**

- a) Personal Grooming
- b) Presentation Etiquette
- c) Spaces & Gender Differences, Inclusivity
- d) Business Vocabulary 3

**UNIT-IV:****Effective Writing Skills**

- a) 7 Cs of Communication
- b) Writing Process: Pre-writing, writing & Post-writing
- c) Neutral and Positive messages
- d) Writing emails, Business Letters, Circulars, Notices & Memos
- e) Communicating in the Digital world - netiquette

**UNIT-V:****Business Meetings**

- a) Conducting and participating in meetings
- b) Writing Agenda & Minutes of the meeting
- c) Etiquette in meetings

**TEXT BOOKS:**

1. Effective Business Communication, Murphy, Herta A., Herbert Hildebrandt, Jane Thomas, 7<sup>th</sup> Edition, New Delhi: Tata McGraw Hill Education Private Ltd., 2008
2. Effective Emails: The Secret to Straightforward Communication at Work, Fenning, Chris, N.p., Alignment Group Limited, 2022

**REFERENCES:**

1. Talk Like TED: The 9 Public Speaking Secrets of the World's Top Minds, Gallo, Carmine, United Kingdom, Pan Macmillan, 2014
2. Made to Stick: Why Some Ideas Survive and Others Die, Heath, Chip, and Heath, Dan, United States, Random House Publishing Group, 2007
3. Team Building: Proven Strategies for Improving Team Performance, Dyer, W. Gibb, et al, United Kingdom, Wiley, 2013

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1EN303) CREATIVE WRITING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### INTRODUCTION:

This Open Elective course on Creative Writing is designed to equip students of engineering and technology with effective written communication and the capacity to approach problem-solving and innovation with a creative mindset. The course enables students to refine their ability to convey complex ideas with clarity and creativity by delving into the intricacies of figurative language, narrative voice, and multiple perspectives. Beyond the craft of writing, the course addresses critical aspects of the writing process, from idea generation to self-evaluation, peer feedback, and ethical considerations. Students will also gain insights into current trends, emerging genres, and the impact of technology on the ever-evolving landscape of creative writing.

#### COURSE OBJECTIVES:

- To nurture creativity and develop a unique writing voice
- To equip students with the fundamental elements and techniques of effective creative writing across various genres
- To guide students through the writing process, fostering critical thinking, revision, and editing skills
- To prepare students for potential publication or sharing their work, exploring different avenues and ethical considerations
- To spark curiosity and engagement with contemporary trends and the future of creative writing

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Create original works of short fiction, non-fiction (memoir or travelogue), and/or poetry demonstrating mastery of genre conventions and narrative techniques

**CO-2:** Analyze and apply elements of creative writing, including plot, character development, setting, imagery, and voice, to enhance their writing

**CO-3:** Effectively engage in the writing process by brainstorming, outlining, drafting, revising, editing, and proofreading their work

**CO-4:** Demonstrate an understanding of different publication options and ethical considerations in creative writing

**CO-5:** Explore current trends and future potential of creative writing, experimenting with emerging genres and formats

#### UNIT-I:

**Introduction to Creative Writing and Genres:** What is Creative Writing? Different types such as fiction, non-fiction (memoir and travelogue), poetry, and drama; Role of imagination; Discovering your voice: identifying your unique writing style and

perspective; Popular genres, their key features, and conventions; Reading as a writer: analysing published works to learn from masters and discover writing techniques

#### **UNIT-II:**

**Elements of Creative Writing:** Building strong plots, developing compelling characters, crafting vivid settings, and mastering the art of dialogue; Figurative language and imagery: exploring metaphors, similes, personification, and other relevant literary devices; Point of view, narrative voice, and multiple perspectives

#### **UNIT-III:**

**Writing Process 1 - Strategies and Techniques:** Brainstorming and idea generation: techniques for overcoming writer's block and generating fresh ideas; Outlining and character development: planning the story structure and creating believable characters with depth and motivation; Scene building and description: learning to craft immersive scenes that engage the reader's senses; Genre-specific strategies; Ethical considerations in creative writing: exploring issues like plagiarism, cultural appropriation, and responsible representation

#### **UNIT-IV:**

**Writing Process 2 - Review, Revise, Edit, and Proofread:** Self-evaluation, peer feedback and incorporating constructive criticism; Revision and editing: strategies for tightening prose, fixing plot holes, and polishing writing for clarity and flow; Copyediting and proofreading: correcting typos, grammatical errors, and formatting inconsistencies; Preparing for publication: understanding different publishing methods, agent queries, and self-publishing options

#### **UNIT-V:**

**Current Trends in Creative Writing:** Emerging genres and popular trends such as flash fiction, graphic novels, and online storytelling platforms; Impact of technology on writing, readership, and publishing models; New forms of narrative and the evolving landscape of creative writing.

#### **REFERENCES:**

1. Abrams, M.H. Glossary of Literary Terms. Boston: Wadsworth Publishing Company, 2005
2. Bell, Julia and Magrs, Paul. The Creative Writing Course-Book. London: Macmillan, 2001
3. Earnshaw, Steven (Ed). The Handbook of Creative Writing. Edinburgh: EUP, 2007

#### **SUGGESTED READINGS:**

1. Atwood, Margaret. Negotiating with the Dead: A Writer on Writing. Cambridge: CUP, 2002
2. Bell, James Scott. How to Write Dazzling Dialogue. CA: Compendium Press, 2014
3. Egri, Lajos. The Art of Dramatic Writing. NY: Simon and Schuster, 1960
4. Gardner, John. The Art of Fiction. New York: Vintage, 1991
5. Johnson, Jeannie. Why Write Poetry? US: F. D. Univ. Press, 2007
6. Mezo, Richard E. Fire in the Blood: A Handbook of Figurative Language. USA: Universal Publishers. 1999

7. Strunk, William and White, E. B. *The Elements of Style*. London: Longman, 1999
8. Turabian, Kate L. *A Manual for Writers*. Chicago: Univ. of Chicago Press, 2007
9. Ueland, Brenda. *If You Want to Write*. India: General Press, 2019
10. Zinsser, William. *On Writing Well*. New York: Harper Collins, 2006

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1MG302) PERSONAL FINANCE AND TAX PLANNING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basic concepts of personal finance and describe steps in personal management process
- To create awareness on investment, speculation & gambling and discuss about security trading and settlement
- To describe investment avenues like financial assets and money market instruments for making investment
- To explain investments alternatives like mutual funds and insurance options for making investment
- To acquaint with the basic concepts of income tax and describing various sources of income and e-filing

**COURSE OUTCOMES:** After completion of the course, students should be able to

**CO-1:** Explore the various concepts of personal finance and effectively implement personal management process

**CO-2:** Differentiate between investment, speculation & gambling and able to demonstrate about security trading and settlement

**CO-3:** Assess various investment avenues and able to implement effective investment

**CO-4:** Analyze various types mutual fund schemes and insurance plans

**CO-5:** Assess various concepts of income tax and able to file income tax returns through e-filing

#### UNIT-I:

**Introduction to Personal Finance:** Definition, Importance & Need of Personal Finance, Key areas of Personal Financial Management, Personal Financial Management Processes.

#### UNIT-II:

**Introduction to Investment Decisions:** Investment, Speculation and Gambling, Features of Investment, Investment Process, the Investment Environment, Securities Market of India, Securities Trading and Settlement, Types of Orders, Margin Trading.

#### UNIT-III:

**Investment Alternatives-I:** Investment Avenues: Financial Assets, Capital market instruments- Stocks, Bonds, Commodities and Derivatives, Post office Deposits, Bank Deposits.

**Money Market Instruments:** Need and Importance, Types-Commercial Papers, Treasury bills, Certificate of Deposits, Bills of Exchange, Mortgage loans.



**Other Investment Avenues:** Gold, Real estate, PPF, NPS and SIP.

**UNIT-IV:**

**Investment Alternatives-II:** Introduction to Mutual funds: Definitions of Mutual funds, Structure of Mutual funds, advantages and disadvantages of Mutual funds, Types of Mutual funds. AMFI.

**Insurance:** IRDA; definition and characteristics of Insurance, Types of Insurance, Life Insurance, General Insurance and Health Insurance.

**UNIT-V:**

**Introduction to Taxation:** Introduction, types of taxes, direct and indirect taxes, Key terms of IT.

**Sources of Income:** Income from salaries, Income from House Property, Income and Profession and Business and Income from other sources.

Assessment of Individual Tax- Total Income – deductions – Process of E-filing.

**TEXT BOOKS:**

1. Personal Finance, Jeff Madura, 7<sup>th</sup> Edition, Pearson, 2020
2. Personal Financial Planning, Randall S. Billingsley, Lawrence J. Gitman, 15<sup>th</sup> Edition, Cengage, 2021
3. Direct Taxes Law & Practice, Vinod K. Singhania, Kapil Singhania, Taxmann, 2023

**REFERENCES:**

1. Handbook on Income Tax, CA. Raj K Agarwal, 9<sup>th</sup> edition, Bharat, 2024
2. Personal Finance, Arthur J. Keown, 8<sup>th</sup> Edition, Pearson, 2019
3. Personal Finance Planning, S. Murali, K.R. Subba Krishna, Himalaya Publishing House, 2019
4. Systematic Approach to Income Tax, Girish Ahuja, Ravi Gupta, 48<sup>th</sup> Edition, Commercial Publishers (India) Pvt. Ltd., 2023

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1MG303) CORPORATE FINANCE

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To provide basic functions of financial management and describing the concepts and techniques of time value of money and its impact on decisions of financial management
- To describe various aspects of Investment decision process and explain various techniques of capital budgeting Decisions.
- To discuss/learn long term and short-term sources of financing
- To explain capitalization and capital structures and different types of leverages and its impact on profitability or earnings of the organization
- To elucidate/ explain the basic concepts of working capital estimation & its importance and explain about various strategies of receivable and inventory management

**COURSE OUTCOMES:** After completion of the course, students should be able to

**CO-1:** Analyze the various concepts of finance function and able to distinguish between present value and future value of cash flows and its impact on financial decision

**CO-2:** Examine various capital expenditure decisions and its impact on profitability of the organization.

**CO-3:** Explore various sources of finance and optimal capital mix

**CO-4:** Assess different types of leverages and the impact of leverage on earnings of the organization

**CO-5:** Analyze various concepts of working capital and able to estimate working capital required by the organization for smooth running of its operation

#### UNIT-I:

**Introduction to Financial Management:** Nature, scope, objectives, and goals of finance function, concept of time value of money- future value and present value, the basic valuation model.

#### UNIT-II:

**Financial Planning and Strategy:** Introduction, strategic decision making and planning, interface between financial planning and strategic management, financial forecasting, basics of capital budgeting: meaning, features and capital budgeting process, capital budgeting decisions techniques- traditional methods: payback period, accounting rate of return, modern methods: net present value, internal rate of return, and profitability index.

**UNIT-III:**

**Sources of Financing:** Introduction to financial market-role of financial markets, segments of financial market, product and services, long term sources of finance-equity, preference shares, debentures, internal finance, loan financing, short term sources of finances.

**UNIT-IV:**

**Capitalization and Capital structures:** Concept of capitalization, types of capitalization, capital structure, capital structure vs. financial structure, financial leverage, operating leverage and composite leverage.

**UNIT-V:**

**Working Capital Management:** Introduction to working capital, meaning and concept of working capital, gross working capital and net working capital, operating cycle nature and characteristics of working capital, factors affecting working capital management, estimating working capital.

Receivables management- meaning and importance, strategies of receivable management. Inventory management- definition of inventory, need and techniques of inventory management.

**TEXT BOOKS:**

1. Financial Management, Prasanna Chandra, 11<sup>th</sup> edition, Mc Graw hill, 2022
2. Financial Management: Text and Problems, M.Y.Khan, P.K.Jain, 8<sup>th</sup> edition, Mc Graw Hill, 2019
3. Financial Management, I.M. Pandey, 11<sup>th</sup> edition, Vikas Publications, 2015

**REFERENCES:**

1. Financial Management, Eugene F. Brigham Michael C.Ehrhardt, 14<sup>th</sup> edition, Cengage Learning, 2015
2. Management Accounting, R.K.Sharma and Shashi K.Gupta, Kalyani Publications, 2017
3. Financial Management, S.N.Maheshwari, Sultan Chand & Sons, 2019

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1MG304) COST AND MANAGEMENT ACCOUNTANCY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To describe basic concepts of cost and management accounting and key differences between financial accounting, cost accounting and management accounting
- To provide various techniques of cost accounting for various industries to ascertain price of a product or service
- To discuss about concept of process costing and treatment of losses and ascertaining cost of by products and joint products
- To create awareness on standard costing technique and concepts of analysis of variance and its role in cost controlling
- To explain steps in budgetary control and able to prepare various types of budgets

**COURSE OUTCOMES:** After completion of the course, students should be able to

**CO-1:** Assess the importance of cost and management accounting and able to differentiate between various types of accounting

**CO-2:** Analyse various cost accounting techniques and identify cost of a product or service

**CO-3:** Asses the importance and applications of process costing and differentiate between normal and abnormal losses and treatment on products and services

**CO-4:** Apply concepts of standard costing and able to calculate various types of variances

**CO-5:** Demonstrate the steps in budgetary control and able to prepare various types of budgets

#### UNIT-I:

**Introduction to Management and Cost Accounting:** Introduction to management accounting-definition, objective, nature and scope of management accounting, cost accounting- cost, costing and cost accounting, objectives and importance of cost accounting, classification of costs; management accounting vs. cost accounting vs. financial accounting, role of accounting information in planning and control.

#### UNIT-II:

**Costing for Specific Industries & Marginal Costing:** Costing for specific Industries-Unit costing, job costing, cost sheet and tender, machine hour rate.

Marginal costing- Features, advantages and limitations of marginal costing, break-even analysis-assumptions, limitations, and applications of marginal costing; make or buy decisions, key or limiting factor, product mix decisions.

**UNIT-III:**

**Process Costing:** Process Costing-features, applications, advantages and limitations of process costing, cost of process – process losses and its treatment, accounting for joint product and by-product.

**UNIT-IV:****Standard costing and Variance analysis**

**Standard Costing:** Meaning, and importance, Variance Analysis-types-material variance, labour variance and sales variance.

**UNIT-V:**

**Budgets and Budgetary Control:** Budgetary Control- Budget, budgetary control, steps in budgetary control, different types of budgets- sales budget, Cash budget, Flexible budget, Production budget, Performance budget, Zero Based Budgeting.

**TEXT BOOKS:**

1. Cost Accounting, M.Y.Khan and P.K.Jain, 2th edition, Mc Graw Hill Education, 2017
2. Management Accounting, R.K.Sharma and Shashi K. Gupta, Kalyani Publications, 2017
3. A Text book of Cost and Management Accounting, M.N.Arora, Vikas Publishing, 2021

**REFERENCES:**

1. Cost and Management Accountancy, P.C.Tulsian, SChand, 2022
2. Cost Accounting: Theory and Practice, Bhabatosh Benerjee, 14<sup>th</sup> revised edition, PHI, 2022
3. Cost and Management Accounting, Ravi M Kishore, 6<sup>th</sup> e Reprint, Taxmann, 2021

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1MG301) FUNDAMENTALS OF MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the principles, functions, theories and practices of different management areas and to provide them with practical exposure to cases of success/failure in business
- To explain the importance of planning, decision making in an organization
- To discuss the various organizational structures, authority and responsibility, line & staff relationships in an organization
- To understand recruitment, selection process and discuss the need of directing in an organization
- To discuss the differences between coordination and cooperation and explain the controlling function

**COURSE OUTCOMES:** After completion of the course, the students will be able to

**CO-1:** Apply principles and theories to improve the practice of management

**CO-2:** Devise the plans and take the right decisions based on the environmental analysis

**CO-3:** Assess the types of organizational structures and evaluate the impact on employees for organizational effectiveness

**CO-4:** Undertake full and fair recruitment and selection systematically and compare and contrast leader vs manager

**CO-5:** Differentiate between coordination and cooperation and the different types of control in organization

#### UNIT-I:

**Introduction to Management:** Concept, nature and scope, importance of management, hierarchy/levels of management, management skills and managerial competencies, Evolution of management thought: Classical Theories- Scientific Management, Administrative Management, Neo-Classical Theories- Hierarchy of needs, Two Factor Theory, Hawthorne Experiments, Theory X & Theory Y, Modern Theories: Systems Theory and Contingency Theory, Introduction to Functions (POSDCORB) of Management.

#### UNIT-II:

**Planning and Decision Making:** Planning- Concept and importance, planning process, Essentials of a Good plan, Types of plans, planning tools and techniques, Management by Objectives, environmental analysis and diagnosis.

Decision Making-Concept and process, types of Decision-making, Decision-Making Model.

### **UNIT-III:**

**Organizing:** Concept, nature, Principles, span of Management, organizational structure, types of organizational structures- mechanistic and organic and their merits and demerits, Authority and responsibility - delegation, centralization and decentralization of authority, line and staff relationship.

### **UNIT-IV:**

**Staffing and Directing:** Staffing- Nature of staffing function, Human Resource Planning, Sources of Recruitment, recruitment process and types. Selection - process, types of selection, job offer, placement and induction

Directing-Need for directing, principles and characteristics of directing, Leader Vs Manager.

### **UNIT-V:**

**Coordination and Controlling:** Coordination-Need, types and techniques of Coordination-Distinction between coordination and co-operation-Requisites for excellent co-ordination

Controlling- Importance of controlling, characteristics of control, steps, resistance to control, design of effective control system and types.

### **TEXT BOOKS:**

1. Essentials of Management: An International, Innovation and Leadership Perspective, Harold Koontz & Heini Weihrich, 11<sup>th</sup> Edition, Tata Mc Graw-Hill Education, 2020
2. Fundamentals of Management, Stephen P. Robbins, Mary Coulter, David De Cenzo, 9<sup>th</sup> edition, Pearson Education, 2016
3. Introduction to Management, John R. Schermerhorn., Daniel G. Bachrach, 13<sup>th</sup> edition, Wiley, 2016

### **REFERENCES:**

1. Principles of Management, Anil Bhat, Arya Kumar, Oxford University Press, 2018
2. Principles and Practices of Management and Organizational Behaviour, Chandrani Singh, Aditi Khatri, Sage Publications, 2016
3. Principles of Management, T. Ramasamy, Himalaya Publishing House, 2018

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

### (22OE1MG306) MARKETING MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the basic concepts of marketing and market research process
- To explain the aspects of analysing market opportunities, customer value and marketing mix
- To elucidate on designing a customer driven strategy through marketing segmentation, targeting and positioning
- To describe the significance of distribution decisions, pricing, promotion and communication strategies
- To discuss the trends in contemporary marketing

**COURSE OUTCOMES:** After completion of the course, students should be able to

**CO-1:** Impart knowledge on marketing process, mix and environment

**CO-2:** Analyse the market opportunities and customer value with the use of marketing mix elements

**CO-3:** Create the segmentation, targeting and positioning of the products and services

**CO-4:** Find the right mix of pricing, promotion and marketing channels and evaluate the communication strategies

**CO-5:** Assess the trends in contemporary marketing

#### UNIT-I:

**Introduction to Marketing:** Introduction to Marketing, Core Marketing Concepts, Marketing environment, Marketing Strategies and Plans, Market research, market research process, Marketing vs Selling.

#### UNIT-II:

**Analyzing Marketing Opportunities, Customer Value and Marketing Mix:** Consumer Decision-making, Building Customer Value, Analyzing Consumer Markets, Consumer Behaviour, Cultural, Social & Personal Factors, Developing Products & Brands, Product Levels; Classifying Products, Product Range, Product Line & Product Mix, Product Life Cycle, New Product Development, New Service Development, Stages of Product/Service innovation development, The Process of Adoption, Branding.

#### UNIT-III:

**Designing a Customer Driven Strategy:** Market Segmentation, Targeting, Positioning Process, Segmentation of Consumer Market, Business Market, Requirement for Effective Segmentation, Market Targeting, Evaluating Market Segmentation,



Selecting Target Market Segmentation, Positioning and Repositioning, Positioning Maps, Product Positioning Strategies.

#### **UNIT-IV:**

Distribution Decisions, Promotion, Pricing & Communication Strategies Marketing Channels, Channel Intermediates and Functions, Channel Structure, Channel for Consumer Products, Business and Industrial Products, Alternative Channel, Channel Strategy Decisions.

The Promotional Mix, Advertising, Public Relations, Sales Promotion, Personal Selling, Direct and Online Marketing.

Pricing- Factors affecting pricing, pricing methods and strategies

Marketing Communication: Communication Process, Communication Promotion Mix, Factors affecting the Promotion Mix.

#### **UNIT-V:**

**Contemporary Marketing:** Current trends in Marketing, Customer Relationship Management, Trend Markets: Digital Marketing and platforms, Rural Marketing and its trends, International Marketing and its importance, Green Marketing-Principles & Practices, concept of Trail Marketing and services, Search Engine Optimisation.

#### **TEXT BOOKS:**

1. Principles of Marketing, Philip Kotler, Gray Armstrong, Prafulla Agnihotri, 18<sup>th</sup> edition, Pearson Education, 2020
2. Marketing: An Introduction, Rosalind Masterson, Nichola Phillips, David Pickton, 5<sup>th</sup> edition, Sage Publications, 2021
3. Principles of Marketing: A South Asian Perspective, Lamb, Hair, Sharma, Mc Daniel, Cengage Learning, 2016

#### **REFERENCES:**

1. Marketing 4.0: Moving from Traditional to Digital, Philip Kotler, Hermawan Kartajaya, Iwan Setiawan, Wiley, 2016
2. Marketing Management, Arun Kumar & N. Meenakshi, 3<sup>rd</sup> edition, Vikas Publications, 2016
3. Marketing Management, Ramaswamy, Nama Kumari, 6<sup>th</sup> edition, Sage Publications, 2018

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. V Semester

#### (22OE1MG305) HUMAN RESOURCE MANAGEMENT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To explain the concepts, significance and role of human resource management in an organization
- To describe the aspects of job analysis, job evaluation and manpower planning
- To understand the techniques of recruitment and process of selection
- To analyze the significance of training & development and succession planning
- To learn the concepts of performance management, appraisals and compensation management

**COURSE OUTCOMES:** After completion of the course, the student should be able to:

**CO-1:** Appreciate the concepts, role and functions of HRM and the need of HR to act as a strategic business partner of the organization

**CO-2:** Relate the need for job analysis and job evaluation and, their significance in human resources planning

**CO-3:** Apply the practices of recruitment and the process of selection in relevance with needs of the organization

**CO-4:** Design the suitable methods of employee training & development in accordance with the emerging trends and necessities

**CO-5:** Analyze the system of performance management and job evaluation in an organisation and suggest applicable changes where necessary, for the improvement of the organisation and employee efficiency

#### UNIT- I:

**Introduction to HRM:** Definition, Nature & scope of Human Resource Management - Evolution of Human Resource Management, Objectives, Functions, Importance of Human Resource Management, Challenges of Human Resource Management, Contemporary role of Human Resource Manager, Concept of e-HRM.

#### UNIT-II:

**Job Analysis, & Human Resource Planning:** Concept of Job Analysis - Job Description and Specifications, Introduction to Human Resource Planning, Importance, Objectives of HR Planning, Human Resource Planning Process.

#### UNIT-III:

**Recruitment & Selection:** Introduction to Recruitment and Selection, Concepts of Recruitment – Recruitment Techniques, Sources of Recruitment. Selection - Types of Selection Methods, Selection Process, on-boarding and Induction, Exit Process.

**UNIT-IV:**

**Training and Development:** Nature and Scope of Training and Development - Objectives, Methods, Process, Analysis of Training Needs & Designing the Training Program, Implementation, Feedback.

Development- Training Vs Development, Management Development, Process, Evaluation of Development Programs, Succession Planning.

**UNIT-V:**

**Performance and Compensation Management:** Concept and process of Performance Management, Performance Appraisal-Definition, Objectives, Process and Methods, Mentoring and Coaching.

**Compensation Management:** factors influencing and challenges, Job Evaluation - Definition, Objectives, Process and Methods, Components of CTC.

**TEXT BOOKS:**

1. Human Resource Management: Text and Cases, K. Aswathappa, TMH, 8<sup>th</sup> Edition, 2017
2. Essentials of Human Resource Management, P. Subba Rao, Himalaya Publishing, 6<sup>th</sup> Edition, 2021
3. Human Resource Management, Gary Dessler, Biju Varkkey, Pearson, 4<sup>th</sup> Edition, 2017

**REFERENCES:**

1. Human Resource Management, Biswajeet Pattanayak, 5<sup>th</sup> Edition, 2018
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